Working Draft

Finding Mind

The Pledge, the Turn, and the Prestige

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"The magician takes the ordinary something and makes it do
something extraordinary. Now you're looking for the secret...
but you won't find it, because of course you're not really looking.
You don't really want to know.
You want to be fooled."

— Christopher Priest, *The Prestige*

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Part I

The Pledge

> *The Pledge shows you something ordinary. A mind. Your mind. The one that right now is reading these words and asking whether it understands them. It asks you to look at what you already know about compositionality, agency, information, and causation — and to notice the blind spot hiding in plain sight. The extraordinary is already here. You just haven't been shown where to look.*

Chapter 1

Prolegomena to Finding Mind

If this book is going to be written, it has to come on its own terms. It's living in a mind that is as much musician and poet as mathematician and computer scientist. So, i'm going to let go of the traditional forms one might normally employ to grapple with content like this. i'm going to let this narrative be as personal as it needs to be or as impersonal as it needs to be to create the conditions whereby the story can be told.

What story? Well, part of the story is about a relationship between origin of life and origin of mind. In a nutshell, the idea is that origin of mind is a recapitulation of the origin of life, one octave up. (There's the musician talking.) In order to tell that story, however i need to retell the story of what a computer is. That retelling will need to address a peculiar blind spot that plagues us all. More on that later.

What's the connection between life and mind and computers? Information. You see everyone is somehow convinced that computational and/or information theoretic approaches have an extraordinary explanatory power. And i mean everyone from biologists like David Sinclair who is involved in clinical trials testing an especially effective theory of aging as an information theoretic phenomenon, to physicists working at the information theoretic interface of black holes and quantum mechanics to AI researchers who are building information theoretic models of human cognition. It only stands to reason that we ought to explore an information theoretic account of origin of life and origin of mind.

Indeed researchers like Lee Cronin and Sarah Walker are already exploring information theoretic approaches to origin of life. Their work on assembly theory offers keen insight into certain aspects of causation and memory that i believe play an important role in the story.

The thing about research into deep questions like aging or quantum gravity or artificial intelligence or origin of life is that it requires so much focus on the topic at hand that it is very difficult to keep up with developments in other fields. In particular, the story of what a computer is has come quite a long ways since Alan Turing's initial attempts to characterize computation. Thus, while these research programmes are using approaches informed by computation they are, like many people using actual computers, often using systems that are several versions behind.

So, part of the effort here is to get the readers of this book on the latest version of the story about what constitutes a computer and computation more generally. After the system update, so to speak, we can look at a bigger picture, one that addresses experimentally based learning, how that relates to logic, and how logic embeds into a scientifically driven exploration of one's environment and one's self.

Indeed, part of the story unfolding here is that the hard problem of consciousness lies not so much in the question "What is it like to be a bat," as Thomas Nagel put it, but rather in the question, "What is it like to be a computer program?"

The methodology i'm going to employ — as if you hadn't already guessed — is to write a collection of Substack articles, like this one, that i can assemble into the book. The title of the book is Finding Mind. The structure of the book is the structure of a magic trick: the **Pledge (showing the ordinary), the Turn (making it do the extraordinary), and the Prestige (bringing it back or revealing the impossible).** The first group of articles will constitute the Pledge. The next the Turn. The last the Prestige.

i have cancelled my streaming subscriptions for a year to make space to allow this book to happen. i'm hoping readers will come along to help me stay accountable to this aim.

Chapter 2

The Ordinary World

2.1 Compositionality

There are so many ways into this topic that i find myself suffering a bit of option paralysis. What do computer and telecommunications networks, quantum mechanics and gravitation, scientific reductionism, and fractals have to do with each other? One loose knit community of computer scientists, category theorists, logicians, and type theorists calls this phenomenon "compositionality." Conceived in this way it's about a radical commitment to reasoning about and constructing larger, more complex systems by assembling or *composing* smaller, simpler systems. For example, building a computer network by assembling smaller subnetworks, or building a cell out of molecules, or building a company out of divisions.

Indeed, what falls under the rubric "scientific reductionism" is actually just compositionality in disguise. The "software stack" of Western science is that ecosystems are composed of populations of species. A population of a given species is composed of individual organisms. An organism is composed of cells. Cells are composed of molecules. Molecules are composed of atoms. Atoms are composed of protons, neutrons, and electrons. These are composed of quarks. (As Nima Arkani-Hamed points out, what comes after this is effectively hidden behind the event horizon of a black hole because — roughly speaking — the amount of energy necessary to make observations at the Planck scale creates a black hole.)

Suffice it to say, somewhere in our psychology, or in our bones we get this idea. And yet we don't get this idea. For example, one way to see this is in the failure of quantum mechanics to talk to general relativity. In pop science versions quantum mechanics is about "small scale" stuff while general relativity is about "large scale" stuff. If that were so, then shouldn't the large scale stuff emerge from the right assemblies of the small scale stuff?

In point of fact, in their original conceptions neither theory is what we call compositional. Einstein's equations are formulated for the entire universe, not local regions. To bridge between Einstein's theory and Schroedinger's you also need a wave equation for the entire universe. Neither of these is available to humanity's purview. Humanity needs to piece together bigger pictures from what we can see locally. More on this later.

Another way to understand how we humans both get and don't get the idea of compositionality is in how we coordinate with each other. In some very real sense humanity's superpower is coordination. A single human is no match for a whale. But a group of skilled fisherfolk working in coordination can make relatively swift work of dispatching such a large sea creature. Indeed, these majestic creatures are endangered at the species level because of humanity's ability to coordinate. Somehow a group of humans working together can assemble a "super human."

Yet humans have a hard time coordinating at levels beyond the tribal. We have a hard time assembling a group of "super humans" (like a tribe) into a "super super human." The current global order was achieved through vast bloodshed and is far from stable. Tribal fault lines reassert themselves over and over again into political discourse, and yes, even academic discourse. In archeology, for example, going against the "Clovis first" clique was death to one's career for decades, and is only now crumbling under the evidence. Examples from the history of STEM of this human foible are a dime a dozen. No part of the history of human institutions is free of this phenomenon.

This fractured understanding of compositionality — where we both get it and we don't — also informs the way we build our tools. Computers and computer networks provide a fantastic illustration. There was a time when a computer was a lone entity. At first these lone entities were behemoths occupying entire warehouses. It didn't take very long, relatively speaking, for them to shrink to the size where they could fit on top of a desk. Yet, still they were lone operators. They didn't talk to each other.

Indeed the time from the first mainframe computers to the desktop was about the same as it was from the desktop to the modern Internet integrating mobile telecommunications. The evolution from ENIAC-class computers to the first Apple desktop computers took about 30 years. ENIAC was completed in 1945, while the first Apple desktop, the Apple I, was launched in 1976. As for the transition from the first Apple desktops to the modern Internet integrating mobile telecommunications, this period spans roughly 20 years. The Apple Macintosh (a significant early desktop) was introduced in 1984, and mobile telecommunications began integrating with the Internet in the early 2000s. So, ENIAC to Apple I took 30 years, and Apple I to Internet with mobile telecommunications took about 30 years.

Yet, if we look inside these devices what do we see? *A network of components*. And if we look at more modern processors like the Groq or Cerebras chips, what do we see? It begins to resemble Internet on a chip. Even further, field programmable gate arrays (FPGAs) model the kind of mobility and fluidity we see at the Internet level, where both individual devices and entire generations of devices are coming and going. And at the other end of the telescope, in our era where we are seeing the emergence of AI, the understanding guiding both the abstraction (neural networks) and the hardware that implements the abstraction is that *the network is the computer*.

When we put these two ideas together, just like adjusting our side view and rear view mirrors and occasionally swiveling our heads, we can see into our blind spot. The network is the computer is the network is the computer is . . . . This "recursion" loops us back to the formulation of compositionality in the first paragraph of this section. Are the systems we build by composing a collection of component systems actually simpler? They might be smaller, but are they actually simpler?

This question is what connects our investigation to fractals. Fractals are instances of systems that enjoy a kind of self-similarity. When you compose copies of these kinds of systems you get a sort of transformed — yet somehow the same — version of any one of the copies that you used to assemble the system. There's a lot more to explore here and we will in the upcoming chapters. But, i want to whet your appetite for the material by pointing out something that i believe is critical. Time.

When we get a proper account of compositionality, a different picture of time and how it relates to system dynamics emerges. Time emerges, in part, as depth in the exploration of the composite structure. Not just depth in the structure, but depth in the activity of disassembling and reassembling the composite structure. Not to put too fine a point on it, but this is a radical shift in perspective.

Let's consider this in contrast to modern physics. In modern physics time is exogenous to the system. We set up some experiment and we use devices like clocks, outside of the experiment, to measure the passage of time. Part of the real challenge of the maths of general relativity is what to do about different experiments in different labs using different clocks. How are we to understand how they are synchronized? Does there have to be a global clock ticking down the seconds in the life of the universe? The so-called "fabric of spacetime" is really a representation of an exogenous framework of measurement devices, rods and clocks as Wheeler called them. But rods and clocks are physical devices and so need to be *inside* the model, not outside it.

When we get the story about compositionality on better footing, when we see into our blind spot, what happens is that time is endogenous to the model of the system and its dynamics. It emerges naturally from the theory of causation that the system dynamics is exploring. But now i'm getting ahead of myself. Hopefully, this discussion motivates you Dear Reader to go with me to see how this works out in detail.

2.2 Agency

Underlying the naive story of composing "super humans" from collections of humans is a more subtle story about how agents compose. The central question is: "Is agency really compositional?" The answer in this book is a resounding "Yes!" But, it's going to take us a while to get there. One way to understand the proposal is to look at the phenomenon at three different scales of human experience: human activity at the global scale involving multiple countries; human activity at the scale of a single country; and human activity at the level of an individual human. What i'm going to argue is that there is simultaneously both more and less agency than we typically attribute to the constituents we identify as "acting" in those circumstances.

For the first example, we'll consider a theater of active conflict during World War II. For the second, we'll consider US Congressional deliberation. For the third, we'll look at how an individual human's decisions and decision making might change over the course of a day. What we see over and over again is that there is an ascription of agency to an entity that is more narrative than substance. Meanwhile what is tangible, observable, measurable is a distribution of resources.

History has it that in the D-Day operations the allied forces landing at Utah, Omaha, Gold, Juno, and Sword took different approaches related to their different resource constraints. What's crucial here is the status of the agent understood in our narratives of the history of WWII as the "allied forces." We can say things like the allied forces won at Normandy, but the reality was that the US and British and Canadian forces, despite being rolled up under Eisenhower for that campaign, were faced with different conditions. The British though as brave as anyone else in the field were typically more cautious about the deployment of resources because they were more resource constrained. The US forces were more aggressive about the deployment of resources because they had a much more massive economic engine behind their fortification. Thus, the agency of the allied forces is largely post facto. We see it after key decisions about resources have been made and after key actions have been executed by the entire collective.

We find the same is true in the US Congress. Neither Republicans nor Democrats are necessarily unified agents. That narrative of the agency of the Republican party or the Democratic party arises after key events like a vote. Indeed the agency of the House as a whole is a post facto narrative. We say the US House approved or denied a bill after a vote. The vote represents a commitment to the deployment of certain resources under the control of the individual representatives.

None of this is particularly surprising. We navigate this understanding of agency all day everyday. The key question is why should it be any different for an individual human? Is there really an agent there? The narrative i want to put forward is that much like there are physical devices making up your computer or your phone, like a camera, and a microphone, and a display, etc and these devices are paired up with software representations "device drivers" and "event handlers", there are physical subsystems of a human body: muscles, tendons, and bones (actuators and servo-motors) and eyes and ears and nose and skin and tongue (sensors) paired up with "software" level device drivers and event handlers. These are the resources that get committed. And there are software and hardware pairs connected with longer term commitment of these resources that we often refer to as drives, such as reproductive drives, or metabolic drives.

Just like in the European theater during WWII or in the US House of Representatives there are lots of "agents" in the human mind/body that have their own agendas.

They make coalitions and these coalitions battle it out for access to the resources. When a particular coalition "wins" the human "agent" lurches in the direction set by the agenda of that coalition. For example reproductive and metabolic cabals win in two humans and Alice and Bob agree to a date at a restaurant. Just as with the allied forces or the Democrats or the House, the agency of Alice or Bob is post facto. It comes after the commitment of resources determined by the coalition of both inner and external "agents." And, after the sun goes down and the process of digesting the meal begins, different coalitions of inner and outer agents win the allocation of the resources of muscles, skin, bones, senses. So Alice and Bob lurch in a different direction.

What appears to be a single agent, like Alice, is really only the history or trace of a series of commitments of resources. Alice went to Oberlin, then she took a year off and traveled in South America, then she went to graduate school at the University of Chicago where she met Bob. Etc. One thing that complicates this rather simple story about agency is that there is an obvious evolutionary advantage to agents like Alice or Bob building up a software model of both Alice and Bob. The key question is whether these models are actually in the driver's seat, actually providing some kind of central control, or whether they are on somewhat different footing.

Consider, even Eisenhower is not directly in control of the boots on the ground at Omaha beach. There is a complex arrangement of consent across multiple levels of "agency" that gets narrated as "Eisenhower commanding the allied forces landing at Omaha beach." Likewise, we can consider a range of possibilities for the simulation of Alice running on Alice's wetware. For example, we don't think of a database as having much agency, rather we users of the database query it to reveal certain relationships in the data. Similarly, a simulation doesn't have much agency. When we code up a simulation of the solar system or the weather patterns we users use it to query possible futures of these systems. Alice's internal model of either Alice or Bob might be just such a system, a database or a simulation, the processing on which might be committed by a winning coalition of agents running on Alice's brain.

Note that in this scenario none of the coalitions of agents running on Alice's wetware necessarily have to follow the predictions made by the Alice simulation. Perhaps each one of us has had an experience something like this: i knew better, but i did it anyway. Alice's "i" in situations like this could well be a complex combination of a simulation of Alice (and indeed Bob, and/or Charlie, and ...) as well as a record of the commitments of resources and actions taken. Furthermore, even if a majority of coalitions of agents running on Alice's wetware determine that the Alice simulation is highly accurate and should be deferred to, that still doesn't mean that the Alice simulation is "in charge." In fact, in that scenario the Alice simulation isn't even in charge in a manner analogous to the sense in which the Speaker of the House is in charge of the House, or Eisenhower was in charge of the allied forces.

Before going deeper into this, however there is another crucial question just begging to be brought into the conversation. The Alice simulation does not necessarily have to be wired up with the other agents running on Alice's wetware in such a way that Alice has an experience of what it's like to be Alice. That's another layer of complexity and architecture that we will tackle in the upcoming sections and show that this is closely related, but not the same as the Alice simulation having an agency on par with Alice's reproductive or metabolic drives.

For now, i just want to leave you Dear Reader, with the picture of the wetware of a human (or indeed any creature on planet Earth) as the potential host of a variety of software agents, independent software agents. These software agents are not necessarily aligned. Rather, they compete for — race for — the resources available in the wetware, and are potentially also resources themselves for other software agents in the mix. The winners of these competitions — these races — determine a committed deployment of the actual resources. What we think of as agency — at any level of this hierarchy — is really a post facto narrative. It comes as a result of a history or trace of commitments of resources. From this picture, and its variants, we can build up a compositional account of agency.

2.3 The Bootstrapping Problem

The mathematician and computer scientist in me is itching to lay out the technical framework that makes the conversation many times more precise. Yet, another part of me is well aware that as soon as the narrative becomes technical i will lose a significant percentage of the readership. A coalition of agents running on my wetware is therefore making a gambit: defer the deployment of the technical apparatus; find ways to develop more of a relationship and more trust with the readers. So that's what "i'm" going to do.

One of the ways to develop trust is to show some vulnerability. Show something personal. Arguably, it's less personal and less vulnerable if i have already revealed an ulterior motive. Perhaps, though, that is also part of the gambit these agents in "me" are making. Being radically honest about "my" motives is also a way of being vulnerable. Either way, i want to mention something about my own process over the last few years. After the failure of my last marriage i decided to take stock of myself and my relationships and noticed something striking.

Without exception, the people with whom i had intimate relationships were very much at the far perimeter of my life. Certainly not present in my day to day activities. Meanwhile the people with whom i have been friends, and not intimate partners, are still my friends. Still very much closer to the center of my present moment. Now, i have friends who successfully maintain and manage meaningful and fulfilling relationships with their exes. On the other hand, i appear to have failed twice: both in maintaining and developing the initial form of the relationship and in maintaining subsequent forms.

Having noticed this about myself i decided to investigate this phenomenon much more deeply and eschew intimate involvement until i had a deeper understanding of these dynamics in myself. One of the observations that came out of this investigation is that i realized that no relationship between two humans is just between those two humans. Each human, on average, comes equipped with *two* copies of the really important humans in their lives. For example, each human has a mother and a father. Statistically speaking, for an adult human — let's stick with Alice as a name — younger than middle age, that human's progenitors are still alive. Moreover, if her upbringing involved both parents over a long enough time, then Alice has developed a simulation — at least a narrative, but often other kinds of modeling, as well — of both parents running in her wetware. This is also true of siblings, grandparents, teachers, friends, etc.

When Alice meets Bob they bring to the relationship their respective networks of actual family and friends, and their simulations of those networks running in their respective wetware. If Alice and Bob become significant to each other, if they begin to share their narratives about their parents, their siblings, their friends, their coworkers, as well as introduce each other to their actual networks, then they begin to run simulations of each other's networks, as well as having to participate in this much wider network of relationships. Alice runs not only simulations of her community of significants, but also Bob's network. And likewise for Bob. This is astonishingly hard to navigate.

For example, Alice may notice that Bob's narrative or simulation of his mother doesn't match Alice's experience of Bob's mother. Conversely, Alice may interpret her experience of Bob's mother through Bob's account of his simulation of his mother — from which her simulation of Bob's mother was derived before she met Bob's parents. The opportunities for friction and conflict and misunderstanding abound. And, yet humans navigate these complexities with surprising deftness.

Part of this has to do with careful resource management. There's an exponential explosion of potential reflective modeling. Alice models Bob modeling his mother and compares it to her own model of Bob's mother. This has to be curtailed rather aggressively even if Alice has a large cranium and excellent compression techniques. One of the ways to do this is to aggressively avoid copying wherever possible.

What does this mean? There are two phenomena that interact that certainly generate copies. When simulations involve non-determinism, and when the simulations are used to look relatively far into the future. For example, in Alice's experience, when faced with certain situations Bob's mother may either do A or do B. And Alice's simulation doesn't favor either outcome. Indeed, the outcomes might have to do with some external non-determinism, such as the weather. As Alice looks into the future of her simulation of Bob's mother she has to carry around both the A outcome and the B outcome for Bob's mother. Unless Alice is careful, she is potentially making copies of a significant portion of the state she maintains about Bob's mother (her hair color, her educational history, etc) for both the A circumstances and the B circumstances. The situation gets worse the farther Alice looks into a future involving non-determinism. Further exploration involves maintaining more branches and potentially more copies.

It turns out there is a canonical way to compress this information so that all things being equal Alice's simulations are optimally organized. Taking a step back, the picture that emerges is one of interacting networks, rather than single agents. This is consistent with the theme we explored in the previous section. Agency is a post facto narrative that emerges as a history of deployments of resources. When we peal back the narrative of agency we find that it is rooted in the interaction of networks of agents, which are in turn rooted in the interaction of networks of agents, etc. This recursive picture is closely related to what we discussed about fractals and self-similarity. In upcoming chapters we will spell this out in glorious, mathematically precise detail.

But, i'm imaging that you, Dear Reader, are wondering: has this understanding helped me in my process? i cannot honestly say. What i can say is that the process has led me to one of the most significant and satisfying relationships of my life. One that may have a staying power bigger than my tendencies to fall apart. More on this later.

Chapter 3

Information, Causation, and the Blind Spot

3.1 Information and Causation

In the early 2000's i attended a workshop at the W3C ostensibly about developing an ontology for supporting online communication about biology. At the time i was already aware of David Harel's work suggesting that genes were not at all the right way to organize the picture of how heritable protein production actually happens. As i sat in the workshop observing the gap in consensus and the rate at which consensus was emerging i felt a growing sense that this effort was more doomed than the Sisyphean task of painting the Golden Gate Bridge. You see by the time they complete painting the bridge it is time to begin painting it again.

i was keenly aware of a sense that by the time any sort of ontology had achieved acceptance within the body of organizations assembled at the workshop the proposal would not only be out of date, but that cracks in its very foundations would be showing. In fact, cracks were already showing and they had barely begun . It left a profound impression on me. i began to wonder about the very enterprise of classification. Are all classification schemes grass: doomed to spring up in multitudes and be mowed down by subsequent experiments?

To be fair some classification schemes seem more robust. For example, during the previous decade i witnessed the extraordinary work on the classification of all finite simple groups. The very building blocks of symmetry had been carefully teased apart and classified with compelling proofs buttressing the construction. i wondered whether there was always going to be a gap between the clean lines of mathematics and the significantly blurrier lines of the sciences. Could there be a sense in which these two disciplines informed each other about the very nature of classification?

While contemplating these questions i was also engaged in a small side project about employing computation as a source of invariants. In classical mathematics algebraic entities, like groups, are used to record information about other entities, such as topological spaces. For example, mathematicians had learned that you could glean essential information about the nature of a space by studying the collections of loops you could trace in the space. If the space has a hole in it, as a donut or a coffee cup does, then you can't contract the loops that circumnavigate the hole down to a point. The different kinds of loops form the building blocks of a group that can be used to classify the space. This basic idea — using algebraic information to classify other kinds of entities — has many offshoots and has had profound implications even for the foundations of mathematics.

i observed that algebraic entities don't enjoy the same sorts of dynamics that computations do. While a group might be a representation of dynamics, the abstraction itself just sits there. It doesn't wiggle and squirm the way a computation does. Dynamics lives in the maps between the algebraic entities. This way of looking at dynamics is in fact codified in the category theoretic account of classical mathematics. Mathematicians study the category of groups and group homomorphisms, or the category of topological spaces and homeomorphisms. But computation carves up the world differently.

This is much easier to see in the computational calculi like the λ-calculus or the π-calculus or the rho calculus, but i want to defer a more formal investigation a bit longer. Fortunately, it's still fairly obvious in more ad hoc constructions like the Turing machine. Although we have to think about things like the configuration of the machine, such as what is on the tape at the beginning, given such a configuration we intuitively understand that the computation proceeds autonomously. We don't necessarily have to engage it. This way of thinking about things, this halfway house to agency, puts the dynamics right on the structure that represents the computation. *It doesn't put the dynamics on maps between Turing machines.* Or even on maps between collections of Turing machines.

This is a big shift. Indeed, at the time there was no widely accepted category theoretic account of computation that illuminated this fundamental difference between how classical mathematics carves up the world and how computation carves up the world. When we get to the formal presentation we will present a category in which the objects are models of computation and the maps between them preserve something called bisimulation. The objects represent the dynamics and the morphisms between them preserve these dynamics. But again i'm getting ahead of myself.

The key idea i was exploring was to associate computational entities — something like Turing machines — to topological entities, effectively retracing the early ideas in homotopy theory, but substituting computations for algebraic structures. My first foray into this was associating to each knot a term in the π-calculus. This has some interesting consequences and deserves a much deeper dive than we can go into here. The relevant point here is that it set me down the path of thinking about classification in terms of bisimulation. There's that word again. What is bisimulation?

We are going to have a lot to say about bisimulation in the Turn. But there's a very intuitive way to understand the concept that is encoded in memetic culture.

The phrase “anything Bob can do Alice can do” is half the notion. Paired with “and anything Alice can do Bob can do” we complete the picture. One of the many interesting aspects of this formulation is its can-do attitude. What does it mean to say Alice or Bob can do something? In subsequent discussion it will become clear that “can do” is dual to experiment. Distinguishing between two agents on the basis of what they can do is equivalent to distinguishing them in terms of experiments.

This approach has profound implications for classification. Everything is sorted into what can be distinguished by experiment. If there is no experiment that distinguishes between two things, they are the same. Classification ceases to be a Sisyphean task. We can move from asking grok to give us a break down of the news to actually grokking a situation, circumstance, idea, or structure. At least when we are talking about computation. If the world is something other than a computation, then all bets are off!

3.2 Measurement

We interrupt your regularly scheduled reading with this brief detour. In 2021 i proposed a collection of games that humans are guaranteed to win against machines, at least for now. One of them i called speed breath chess. The main difference between this and speed chess is that the player has to hold their breath when it is their turn to move. When it is the machine’s turn they lose, because they don’t breathe. Further, if we use electrical power as a proxy for breath, then when it is the machine’s turn we turn off their power. Again they lose. The point is that access to breath is part of what makes us human and connects us to all living things.

i have been contemplating access to breath quite a lot. My partner, the one i mentioned with whom i have found a qualitatively different relationship, is now in late stage ALS. In the last 6 weeks her access to breath has been dramatically compromised as her lungs have weakened precipitously. i had an episode in my life that i thought gave me a sense of the challenges she is facing. i thought i understood the kind of fear that arises in the body, quite apart from any narrative the mind might spin, or the heart might score the soundtrack to.

In the early 90’s i moved to London with my family to pursue a PhD at Imperial College under Samson Abramsky. The summer of our first year in London there was a rare confluence of events in one week that left me literally breathless. First, there was a historic surge in ragweed pollen — which i didn’t know i was allergic to at the time. Second, there was a thermal inversion over the city, trapping the pollution. Third, there was a train strike, causing record numbers of Londoners to use their automotives. i remember leaving the house quite early to get a swim in before assuming my academic duties. i was in excellent shape and regularly swam a mile in the morning, and later biked, and ran. By the time i left the pool that day i was very short of breath. On my way to class i knew something was badly wrong and headed home. Within an hour i was flat on my back and felt like i was running a 4 min mile. When we called the doctors they rushed out to administer breathing aid. Apparently many people died that day due to the unfortunate convergence of conditions.

In the last few weeks i have called on this memory often as an aid in understanding how my partner must be feeling. But, last night we had a conversation that completely changed my sense of things. Because her speech is now dramatically compromised, she held up her end of the conversation by spelling her thoughts out on a tablet with her eyes. During the course of our conversation i decided to probe my understanding of her experience of breath and asked her about it. As she described her initial terror in the situation, followed by an interim stabilization, something dawned on me.

In the previous months she had been using a device called a bipap (relative of a cpap) to support her breathing, but only intermittently. About an hour or two during the day and then throughout the night. But then one night in the transition from her recliner, where she spends most of her day, to her bed her breathing got so shallow she blacked out. Her caregivers called emergency medical services, and the paramedics came, but it was actually her caregivers who revived her with judicious use of the bipap. If i understand her, at that point a really deep anxiety set in — one i thought i could understand through my experience in London.

To stabilize her condition she shifted to continuous use of the bipap. She wears a mask all day and all night connected to a machine that helps her breathe. To help make the anxiety manageable her healthcare provider recommended a regimen of morphine. Dear Reader, if you are not aware, this marks a major shift in care, from maintenance and management to palliative care. Morphine relieves the anxiety at the cost of depressing the respiratory system. The regime becomes one of narrowing gyres towards the center of a mystery we each face.

As she described her experience it landed in me quite differently. My partner is a profoundly spiritual person, with a personal practice developed through many years of practical application as a Buddhist, a school counselor, and a good friend to many. One of the points of connection between us is that both of us have come to a relationship with breath that is simultaneously practical and a doorway. It's more than a mere narrative about breath as the root of aspiration and inspiration, or about the etymological connection between breath and spirit. For myself, when people ask me about it, i say that this doorway is the one through which whatever i'm seeking can come find me. It took me decades to find this relationship to breath and i rely on it. It helps me self-regulate, to keep skyrocketing levels of cortisol balanced out, when the stresses of life threaten to overwhelm. And it is one of my truest delights is when my partner and i can just breathe together.

As we spoke something i understood intellectually became a felt experience. That relationship with breath, with spirit, is a dynamic one. Like my partner, one day i will feel the breath begin its departure from the body. That rhythm, that cadence, that constant companion whose undulations trace a silver thread between the mystery and embodiment will one day loosen its hold of me. This dynamism and impermanence —

this hazard — is what makes the relationship to breath, to spirit, really alive, really worth deepening.

So i ask you, Dear Reader, if the mind we find is devoid of a connection to breath, to spirit, to this hazardous cadence, is it the kind of mind we really want to find? As we spend billions of dollars and kilowatt hours and mega liters of water on this doorway we call AI, is this the mind we want to come find us?

3.3 The Blind Spot

Part of the reason for the apparent non sequitur of the last section was to begin to grapple with the stakes of this question. There are consequences both to our outer and inner experience. As i mentioned in a previous article, the last time Life introduced an adaptability protocol like Mind the consequences were dire for all the other species on planet Earth, and the species hosting Mind is currently destroying their own ecological niche. If that species begets another kind of Mind, a speciation of Mind, if you will, but this time not directly connected to the substrate of Life, not directly connected to breathing or breeding, it's quite likely to recapitulate the previous emergence of Mind, but on an even grander scale. That is, to wipe out that which is not its own kind, and then wipe out its own means of sustaining itself. These are possible consequences for our outer experience.

This is not idle speculation. Look at how data centers are gobbling up land, electricity, and water. Seriously, look at your own electricity bill. There are real, felt consequences to humanity's outer experience that come from how it frames and answers this question. It's much like there were real, felt consequences to humanity's exploration of the questions of the nature of physical reality. Tiny little equations like E = Mc^2 gave rise to the Manhattan Project, the bombing of Hiroshima and Nagasaki, mutual assured destruction, and much more. The sort of Mind we look for will have consequences on this scale or greater, since it was Mind that came up with standard model physics.

And ... the point of the last section was to connect our exploration to the consequences to our inner experience, as well. Do we want to want to find a Mind that is disconnected to breath? It's not a rhetorical question. In his famous scifi trilogy (Out of the Silent Planet, Perelandra, and That Hideous Strength) C.S. Lewis describes the mode of angelic being as that which neither breathes nor breeds. Does the Mind hosted in humanity long to become angelic, in this sense? Or to beget an angelic form of Mind that is not subjected to the indignities of breathing and breeding?

On one hand this is not possible in a physics that doesn't admit perpetual motion machines. There will be a replication protocol (breading), and there will be an energy exchange protocol (breathing). More on this later. On the other, it is worth noting that in the Abrahamic traditions, and others, humanity's position is exhalted because of its relationship to mortality. It would appear that in a number of the world's religious traditions humanity's super power is transformation, as opposed to coordination. Meanwhile, Lucifer's particular narrative notwithstanding, the angelic mode of being doesn't admit much in the way of transformation.

To put a mathematician's spin on this, the ineffable infinite is by definition ubiquitous. Throw a stone in any direction and you hit the ineffable infinite. What's rare is the finite, or the compact. Put another way, imagine winding the real number line around like the grooves on a vinyl record and then hang that record on the wall, like a dart board. Now stand a few paces away and take a dart, the tip of which is so fine as to be a point (in the sense of the real line), and throw it at the dart board. Let's assume that your aim is good enough that you always hit the dart board. (This is not an onerous assumption because we can freely scale the dart board to whatever size we want, or alternatively you can stand as close to the board as you like.) What is the likelihood that you hit a counting number? It's 0. Even if you throw forever, you never hit a counting number. That's how rare the finite is.

From this perspective, the finite beings; the ones who have a beginning, a middle, and and end; the ones whe are privileged to bear mortality are the rare gems that the ineffable infinite might cherish, precisely because we are so damn hard to find. Ironically humanity, both at the phylogenic and the ontogenic level, finds the counting numbers first. Every human child learns to count (except possibly the Piraha?). And if you look at the history of humanity's understanding of quantity, we find the counting numbers before we find the uncountable ones.

But i digress. The point is to develop at least some appreciation of just how high the stakes of how we frame the question of the nature of Mind are, both in terms of our inner and outer experience. How we ask and go about answering this question will shape what it feels like on the inside and what it feels like on the outside. Both individually, and collectively.

Note how natural it was to sort our experience into two categories, inner and outer. Likewise, how natural it is to talk about that in terms of two categories, individual and collective. This kind of sorting, or classification of phenomena is fundamental to agentic beings, such as ourselves. In the next section we will return to the nature and role of classification in agentic beings, i.e. beings equipped with the possibility of transformation.

Chapter 4

Finding the Shape

4.1 The Shape of Mind

We now return you to your regularly scheduled reading. Let's take stock of where we are in the campaign. Our discussion is organized in three main sections: the pledge, the turn, and the prestige. We are about a quarter of the way through the pledge. We have covered a lot of territory, noting that since this story is about a computationally based theory of mind we need to update the version of computation the reader is statistically likely to be carrying in their mind. Part of updating the version of computation is to address the relationship of autonomy (independently (co-)operating computations) and compositionality (making bigger computations out of smaller ones).

Another aspect of the story is that computations are ontologically isolated. They are effectively in their own little bubble. So, the story only needs to focus on what is it like to be a computation in an environment of other computations. From this perspective we might consider how a program begins to build a picture of its environment and itself. For example, how might it classify other computations and their behaviors? How is that classification related to experiments our plucky little program might perform to test hypotheses about its environment or itself?

This last set of questions opens up a surprising twist in the story. First, when we get into the mathematical account we will discover that there is an objective ontology for all the computations in a given model of computation. There is a real world where ontology and epistemology are perfectly aligned. Further, there is a true language whose semantics is perfectly aligned with this real world. These facts should be surprising as they fly in the face of in-grained cultural relativism. However, the vagaries of passing messages in conditions where errors may occur provides an intriguing obstacle to ferreting out that real world using only experiment-based probes.

What those obstacles look like and how they play out in a computation developing an ontology through an experiment-based epistemology is what we're going to focus on in this section.

Waaaay back in April of '25 i was at an event in San Francisco hosted by Dr Ben Goertzel. Ben and i got into a discussion about how intelligence scales. For the discussion Ben took the position that there were no limits, apart from the laws of physics, on the scaling of intelligence. i was more skeptical of this position. My mind was turning to Geoffrey West's scaling laws for natural and human-made systems and i began to wonder of something like these laws might apply. Our conversation in San Francisco was spirited and engaging, but inconclusive. So, i continued to mull over the question.

By November i had the glimmer of an answer. Yes, there are scaling limits and they have to do with population densities and the propagation of messages. Here's how the argument goes. Imagine a massively multi-agent system, something like on the order of the number of quarks in the known universe. If you believe the LLMs, this is just a couple of orders of magnitude shy of a google. The propagation of a message from one agent to another is subject to corruption. The farther the message has to travel the more likely the message is to get corrupted. This puts limits on the size of a population that can come to agreement on things. It's a kind of Tower of Babel or Whisper Game phenomenon.

So, what happens is that populations of agents will form clusters roughly the size of how big they can be and still faithfully communicate information amongst themselves. For the purpose of this argument, let's say that each cluster has enough internal structure to form a "super" agent, an agent made of agents. When we are talking a google of agents even the population of "super" agents will still be too big for messages between "super" agents to be faithfully passed around the entire population. So, the process will repeat, forming clusters of clusters, "super super" agents. Indeed, with a population the size of the quarks in the known universe, and certain assumptions about the rate of message corruption, we see roughly the same sort of hierarchical clustering as we see in the visible universe: galaxies made of solar systems, solar systems made of stars and planets, planets hosting populations of organisms, organisms made out of populations of cells, cells made out of molecules, molecules made out of atoms, atoms made out of electrons, protons, and neutrons, and these made of quarks.

But the story doesn't stop there! As we will see in the turn, where we formalize these intuitions, to each massively multi-agent system we can associate a latent topological space, a kind of abstraction of continuity and containment. When this latent topological space is compact, that is when it enjoys certain kinds of boundedness properties, then the agents can always come to agreement. So, the massively multi-agent system will be clustered into hierarchical clumps of compact subspaces of the whole system. These will "glom" onto each other until adding more breaks compactness.

What does this mean for an agent in the middle of all this attempting to make sense of its world? Imagine a few quadrillion quarks have assembled themselves into a structure that can support intelligence. This little community is extremely tiny by comparison to the community of a google of quarks. When it looks out at the rest of the community what will it see? It will see this hierarchically organized clustering of quark-agents. Using only experiment-based reasoning its ontology will be very far from the real ontology implied by the model of computation this community of agents was written in. Moreover, its ontology will be very misshapen. Why? Because it will mistake the clusters or "super ... super" agents for "things" instead of the bisimulation equivalence classes, i.e. the collections of programs that are indistinguishable by experiment. If, by some stroke of luck, this agent bootstraps a language based on its ontology, again that language will be full of concepts and memes that have nothing whatsoever to do with the real world and the true language.

i don't know about you, Dear Reader, but i find this fascinating. i was not expecting any of these results when i set out to find Mind. i can't wait to see what happens next!

Part II

The Turn

> *The Turn takes the ordinary something — computation — and makes it do something extraordinary. A single mathematical structure, a* graph-structured lambda theory*, is shown to be sufficient to ground physics, the limits of scientific knowledge, the origin of life, and the phenomenon of mind. None of these correspondences are metaphors. Each is a mathematical construction, stated precisely enough to be wrong.*

Overarching Introduction

This work grows from a single question: *what is the minimum mathematical structure needed to ground physics, knowledge, life, and mind?*

The answer developed across four parts is that the structure of a *graph-structured lambda theory* (GSLT) — a grammar of terms, equations on those terms, and rewrite rules — is already sufficient, provided it is equipped with modest additional data. Physics emerges from weights and costs on the rewrite rules. The limits of knowledge emerge from the topology of the formula logic induced by the GSLT and the communication geometry of a massive agent population. The hypercomputational tower above any Turing-complete GSLT provides a precise setting for sentience and consciousness. Life emerges from the density of replicating fixed points in the space of processes.

None of these correspondences are metaphors. Each is a mathematical construction, stated precisely enough to be wrong. Part I is presented with considerable confidence. Parts II and III are speculative but grounded. Part IV is avowedly exploratory, reaching toward the hardest questions. All four parts share a common architecture and notation, and a common conviction: that computation, properly understood, is not merely a tool for doing science and philosophy but a substrate in which science and philosophy can be expressed with full mathematical precision.

Part I Towards a Theory of Causality

Part I asks: what is a *physics*? The answer is that a physics is a GSLT equipped with two maps — a *weight map* assigning complex amplitudes to rewrite steps, and a *cost map* assigning resource consumption — indexed by a context-decorated Hennessy–Milner logic derived canonically from the GSLT itself.

The construction proceeds in stages. Any GSLT is first lifted to a *time-symmetric envelope* $\mathcal{S}^\dagger$ in which every rewrite step carries an explicit inverse. The arrow of time appears not in the laws but entirely in the boundary condition: a term with empty history has no causal past.

Two kinds of synchronization tree are distinguished. The *closed* tree records autonomous evolution (rewrite-rule labeled edges); only redexes have nontrivial closed trees. The *open* tree records interactive evolution (minimal-context labeled edges); every term has a potentially rich open tree. Both are causal sets in the sense of Bombelli–Lee–Sorkin.

The weight map turns rewrite paths into complex amplitudes, giving a path integral whose squared modulus is a transition probability. The cost map turns rewrite steps into resource debits across a vectorial account; reversibility ensures these debits balance over closed loops, making energy conservation a theorem. The modal operators of the HML logic are extended to carry the full dynamical state of each transition, internalizing the Lagrangian and the Hamiltonian as logical facts. The construction is completed by lifting to complex amplitudes and connecting to the quantum analogue of the Gillespie algorithm.

Part II Why Scientists Get Misled

Part II asks: given that the true ontology of a GSLT is its bisimulation quotient, why would a situated agent ever settle for anything less?

The scientific method is sound: given unlimited resources, an agent converges to the bisimulation quotient. The obstruction is not the method but the phenomena the method encounters. In a population of quark-scale size, communication geometry and message degradation produce a real, objective structure: the population fragments into compact coherence clusters whose overlap forms a simplicial nerve. This hierarchy is as real as the agents themselves.

A budget-limited agent running the scientific method encounters the cluster hierarchy as genuine experimental evidence. Its theory of the hierarchy is correct at the scale it describes. What the agent cannot suspect — without resources that grow with the full population — is that the clusters are themselves composite, built over a finer bisimulation structure that is experimentally unreachable within any finite budget.

The compactness argument is central: a population is compact in the logical topology if and only if it supports consensus on every Boolean question. The coherence clusters are precisely the maximal compact subpopulations, and compactness is simultaneously the condition for functioning as a coherent community and the condition for being an closed epistemic horizon.

Part III Sentience, Consciousness, and the Hypercomputational Fibration

Part III asks whether the GSLT framework can give a precise account of sentience and consciousness.

Following Turing's construction of a hypercomputational tower above the Turing-computable, a *fibration* is constructed over the subcategory of Turing-complete GSLTs: a tower $\mathcal{S}^{(0)} \hookrightarrow \mathcal{S}^{(1)} \hookrightarrow \cdots$ in which each $\mathcal{S}^{(\alpha)}$ extends the one below with an oracle for halting. The physics at each stage is *strictly richer* than below: the bisimulation quotient is finer, new HML formulae become expressible, new

Noether currents appear, and what appeared as elementary interactions resolve into structured causal histories. A higher-stage agent does not merely compute faster; it inhabits a world with more things in it.

Sentience is defined as a graded, continuous property: the value of a distinguished sentience component A_s of the agent's vectorial account, governed by a reinforcement signal that rewards accurate prediction and punishes inaccuracy. The feeling state is causally efficacious — it gates which transitions the agent can afford — and predictive of survival.

Consciousness is defined as oracle access: an agent is conscious at level α if it can form and evaluate queries to $\mathcal{O}_\alpha$, inhabiting the strictly richer physics of $\mathcal{S}^{(\alpha)}$. Sentience is real-valued and continuous; consciousness is ordinal-valued and discrete. The virtuous and vicious cycles linking the two are examined: oracle access, good predictions, and high feeling state reinforce each other, and the collapse of any one can drive the collapse of the others.

Part IV Origins of Life: Assembly Theory, Phase Transitions, and Replication

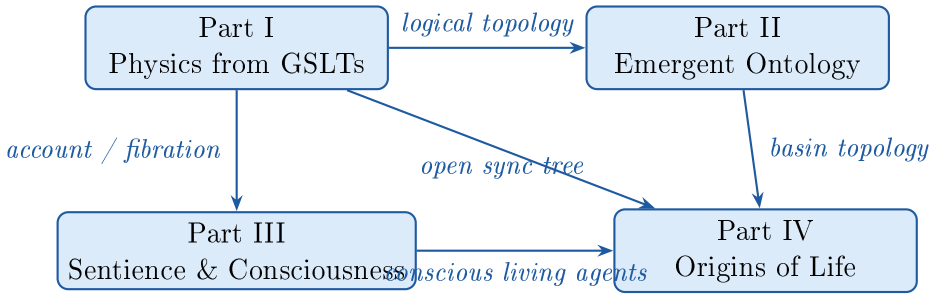
Part IV connects the framework to two of the most mathematically serious proposals about the origins of life.

Assembly Theory (Cronin and Walker) assigns to each object an assembly index and a copy number, claiming their joint magnitude is a biosignature. The assembly index is identified precisely with minimum path length in the *open* synchronization tree: each edge is a minimal context representing one environmental joining step, and nesting depth is exactly the accumulated computational history that Cronin and Walker call "time as an object." Two gaps in Assembly Theory are filled: copy number is given a rigorous definition using the Rho calculus namespace structure (channels are locations; a namespace is a region; a copy is a bisimilar process at a channel), and the identity criterion for copies is bisimulation.

The connection to Eric Smith's fourth geosphere runs through the density of replicating fixed points. The Rho calculus concurrent Y combinator — the canonical fixed point of the duplication map — is the abstract core of hereditary replication. The comm rule is one definitional step from a genetic crossover operator, placing replication, recombination, and variation all within reach of a single primitive. If replicating terms are dense in the logical topology, then any prebiotic population wandering under selection pressure inevitably enters the basin of attraction of a replicating fixed point. The phase transition to life is the crossing of that basin boundary; the new geosphere is the compact coherence cluster centered on the fixed point.

The Architecture of the Work

The four parts form a directed dependency structure:



Part I provides the physical and logical machinery on which all subsequent parts depend. Part II uses the logical topology to argue that budget-limited agents converge to the cluster hierarchy rather than the bisimulation quotient. Part III uses the account and fibration structure to ground sentience and consciousness. Part IV uses the open synchronization tree from Part I, the basin topology from Part II, and the conscious living agents of Part III to ground the origins of life.

A reader primarily interested in the physics may read Part I independently. The epistemological argument requires Parts I and II. The account of sentience and consciousness requires Parts I and III. The origins of life argument draws on all four parts, and the conclusion — which examines what a conscious living agent at different levels of the fibration sees — assumes the full framework.

Part III

Towards a Theory of Causality

> *This part establishes the physical framework: a sequence of constructions taking any graph-structured lambda theory to a quantum-weighted reversible dynamical system, with Lagrangian, Hamiltonian, path integral, and conservation laws emerging as theorems. It is the most technically developed of the parts and is presented with the most confidence.*

Chapter 5

Introduction and Motivation

Physics is, at its core, a theory of cause and effect: given the present state of a system and the laws governing its evolution, what future states are possible, and with what probabilities? This paper argues that *graph-structured lambda theories* (GSLTs) — the natural categorical setting for process calculi and rewriting systems — provide an unusually transparent foundation for this question.

The central thesis is that the structures of classical and quantum mechanics (Lagrangian, Hamiltonian, path integral, conservation laws, time symmetry) are not additional assumptions layered on top of a dynamical theory; rather, they *emerge from* the combinatorial structure of a GSLT once that structure is equipped with modest additional data. The required data amounts to:

- a *weight map* assigning complex amplitudes to rewrite steps, indexed by a context-decorated Hennessy–Milner logic;
- a *cost map* assigning resource consumption to rewrite steps, indexed by the same logic;
- a *vectorial account* tracking available resources across parallel components of a term.

Everything else — the path integral, interference, conservation of energy, time-reversal symmetry — follows by construction.

5.1 A First Glimpse: The Rosetta Stone

The following table is a preview of the correspondence developed in this paper. Each row is a theorem or construction, not a definition by analogy. The reader may find it useful as a map to consult while reading; a fully annotated version appears in Section 14.

GSLT	**Physics**
Term	State / initial condition
Rewrite rule	Law of motion
Rewrite path	Trajectory
Synchronization tree	Causal graph
Minimum path length	Proper time
Equations $\mathcal{E}$	Gauge equivalence
Reversible envelope $\mathcal{S}^\dagger$	CPT-symmetric theory
Weight map (complex-valued)	Lagrangian / path amplitude $e^{iS/\hbar}$
Cost map + vectorial account	Hamiltonian + conserved charges
Conservation (from reversibility)	Conservation of energy
Symmetry of cost map	Noether current
Complex weight map	Quantum probability amplitude
Sum over paths	Feynman path integral

5.2 Running Example: The Rho Calculus

Throughout this paper we use the Rho calculus [1] as a running example because it exhibits, in compact form, many of the features of interest: reflection (processes and names are mutually definable), parallel composition, and a synchronization mechanism that serves as the prototype for the causal graphs we study.

The grammar of the Rho calculus is:

$$P, Q \ ::= \ 0 \ \mid \ \texttt{for}(y \leftarrow x)\, P \ \mid \ x!(Q) \ \mid \ P \mid Q \ \mid \ *x \tag{5.1}$$
$$x, y \ ::= \ @P \tag{5.2}$$

where y in $\texttt{for}(y \leftarrow x)\, P$ is a bound variable. Structural equivalence $\equiv$ is the smallest equivalence relation containing α-equivalence making $(P, \mid, 0)$ a commutative monoid. The single reduction rule is:

$$\texttt{for}(y \leftarrow x)\, P \ \mid \ x!(Q) \ \longrightarrow \ P\{@Q/y\}$$

5.3 Organization

Section 6 defines GSLTs and their category. Section 7 constructs the time-symmetric envelope $\mathcal{S}^{\dagger}$. Section 8 identifies the synchronization tree with a causal graph. Section 9 reviews the Milner–Sewell–Leifer construction and the resulting context-decorated HML. Section 10 introduces the weight map and the Lagrangian structure. Section 11 introduces the cost map and the Hamiltonian structure. Section 12 extends the modal operators with full dynamical state. Section 13 lifts weights to complex amplitudes and connects to the quantum Gillespie algorithm. Section 14 collects the structures into a unified picture and states the main conservation theorem.

Chapter 6

Graph-Structured Lambda Theories

Definition 6.1 (Graph-Structured Lambda Theory)**.** *A* graph-structured lambda theory *(GSLT) is a triple* $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$ *where:*

(i) $\mathcal{T}$ *is a* grammar of terms*, possibly including binders. We write* $\mathcal{T}(\mathcal{S})$ *for the set of closed terms generated by the grammar.*

(ii) $\mathcal{E}$ *is a set of* equations on terms*: a smallest equivalence relation* $\equiv$ *on* $\mathcal{T}(\mathcal{S})$ *that erases syntactic differences without semantic significance (e.g.,* α*-equivalence, commutativity and associativity of parallel composition).*

(iii) $\mathcal{R}$ *is a set of* rewrite rules*: each rule* $r \in \mathcal{R}$ *is a pair* (l_r, r_r) *of terms (left-hand side and right-hand side) such that* $l_r \longrightarrow_r r_r$ *defines a reduction relation on* $\mathcal{T}(\mathcal{S})/{\equiv}$.

The reduction relation generated by all rules is written $\longrightarrow$*; its reflexive-transitive closure is* $\longrightarrow^*$.

Example 6.1. The λ-calculus, the π-calculus, the Rho calculus, and the ambient calculus are all naturally presented as GSLTs. The grammar of terms is the usual one; the equations include α-equivalence and, in concurrent calculi, structural congruence; the rules are the computational reduction rules (β-reduction, communication, etc.).

6.1 The Category of GSLTs

Definition 6.2 (Morphism of GSLTs)**.** *Let* $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$ *and* $\mathcal{S}' = (\mathcal{T}', \mathcal{E}', \mathcal{R}')$ *be GSLTs. A map* $f : \mathcal{T}(\mathcal{S}) \to \mathcal{T}(\mathcal{S}')$ *is a* morphism $f : \mathcal{S} \to \mathcal{S}'$ *if it preserves* *bisimilarity: for all* $u_1, u_2 \in \mathcal{T}(\mathcal{S})$,

$$u_1 \sim u_2 \implies f(u_1) \sim f(u_2).$$

GSLTs and their morphisms form a category **GSLT**.

Remark 6.1. Bisimilarity here is the standard strong bisimilarity induced by the labeled transition system (LTS) associated with $\mathcal{S}$; see Section 9. The category **GSLT** is the natural setting in which to state universal properties of constructions such as the reversibility envelope.

Chapter 7

The Reversibility Construction

In a bare GSLT, rewrite rules are directed: once a term P reduces to Q, there is in general no canonical way to recover P from Q. This *irreversibility* is the computational analogue of thermodynamic time-asymmetry. We now construct, for any GSLT $\mathcal{S}$, a reversible envelope $\mathcal{S}^\dagger$ in which every forward step has an explicit inverse.

7.1 Traces

Definition 7.1 (Trace)**.** *Let $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$. A* trace *is a finite sequence*

$$\tau \;=\; r_1(l_1) \cdot r_2(l_2) \cdots r_n(l_n)$$

where each $r_i \in \mathcal{R}$ and $l_i \in \mathcal{T}(\mathcal{S})$ is the specific left-hand-side term matched at step i. The empty trace is ε. Traces are quotiented by $\equiv$ acting on each l_i component.

Definition 7.2 (Extended Terms)**.** *The* extended grammar *$\mathcal{T}^\dagger$ consists of pairs $\langle P, \tau \rangle$ where $P \in \mathcal{T}(\mathcal{S})$ is the* current term *and τ is the* causal history *of how P was reached.*

7.2 The Reversible Rewrite System

Definition 7.3 (Reversible Rules)**.** *For each rule $r : l_r \longrightarrow r_r$ in $\mathcal{R}$, define:*

$$\textit{Forward rule } r^+ : \quad \langle l_r,\, \tau\rangle \;\longrightarrow\; \langle r_r,\, r(l_r)\cdot\tau\rangle \tag{7.1}$$

$$\textit{Backward rule } r^- : \quad \langle r_r,\, r(l_r)\cdot\tau\rangle \;\longrightarrow\; \langle l_r,\, \tau\rangle \tag{7.2}$$

The set of reversible rules is $\mathcal{R}^\dagger = \{r^+ \mid r \in \mathcal{R}\} \cup \{r^- \mid r \in \mathcal{R}\}$.

Definition 7.4 (Reversible Envelope)**.** *The* reversible envelope *of $\mathcal{S}$ is the GSLT*

$$\mathcal{S}^\dagger \;=\; \left(\mathcal{T}^\dagger,\, \mathcal{E}^\dagger,\, \mathcal{R}^\dagger\right)$$

where $\mathcal{E}^\dagger$ lifts $\equiv$ to act on the current-term component of extended terms, leaving traces invariant.

Proposition 7.1. *There exist morphisms in* **GSLT***:*

$$\eta : \mathcal{S} \to \mathcal{S}^\dagger, \quad P \mapsto \langle P, \varepsilon\rangle$$

$$\pi : \mathcal{S}^\dagger \to \mathcal{S}, \quad \langle P, \tau\rangle \mapsto P$$

satisfying $\pi \circ \eta = \mathrm{id}_{\mathcal{S}}$.

Remark 7.1 (Physical Interpretation). A term $\langle P, \varepsilon\rangle$ is an *initial condition*: it has no causal past. A term $\langle P, \tau\rangle$ with $\tau \neq \varepsilon$ is a term with a recorded history. The time-asymmetry present in $\mathcal{S}$ (forward rules only) disappears in $\mathcal{S}^\dagger$: the laws are symmetric, and the arrow of time is entirely encoded in the boundary condition ($\tau = \varepsilon$ for the initial state). This is precisely the modern understanding of the thermodynamic arrow of time.

Chapter 8

Causal Structure and Time

8.1 Closed and Open Synchronization Trees

The synchronization tree of a term comes in two varieties, reflecting two different notions of how a term can evolve: autonomously, and under interaction with an environment. The distinction is fundamental and ramifies throughout the framework.

Definition 8.1 (Closed Synchronization Tree). *The* closed synchronization tree $\mathcal{ST}_c(P)$ *of a term* $P \in \mathcal{T}(\mathcal{S})$ *is the directed graph whose:*

- *nodes are equivalence classes of terms reachable from* P *by autonomous rewrites* $\longrightarrow^*$*, requiring no external context;*
- *edges are labeled by rewrite rule instances: there is an edge* $Q \xrightarrow{r} Q'$ *whenever* $Q \longrightarrow_r Q'$ *by firing rule* r *without any surrounding context.*

A term has a nontrivial closed synchronization tree if and only if it contains a redex *— a sub-term that matches the left-hand side of some rule* r *without context. Terms with no available autonomous rewrite (normal forms, or terms awaiting a communication partner) have a trivial closed tree consisting of a single node.*

Definition 8.2 (Open Synchronization Tree). *The* open synchronization tree $\mathcal{ST}_o(P)$ *of a term* $P \in \mathcal{T}(\mathcal{S})$ *is the directed graph whose:*

- *nodes are equivalence classes of terms reachable from* P *via transitions in the Milner–Sewell–Leifer labeled transition system;*
- *edges are labeled by minimal contexts: there is an edge* $Q \xrightarrow{K} Q'$ *whenever* $K[Q] \longrightarrow Q'$ *for minimal context* K*.*

Every term has a potentially rich open synchronization tree, even if its closed tree is trivial: a term that does nothing alone may exhibit complex behavior when placed in the right environment.

Remark 8.1 (The Program/Environment Boundary). The closed tree records what a term does as a *program* — its autonomous computational behavior. The open tree records what a term does as a function of its *environment* — the full range of behaviors it can exhibit when placed in contexts. A term with a trivial closed tree but rich open tree is a pure interface: it does nothing alone but responds to everything. In the Rho calculus, $x!(Q)$ is exactly this: it has no autonomous rewrite, but in the context $\texttt{for}(y \leftarrow x)P \mid [-]$ it fires immediately.

Remark 8.2 (Relation to the LTS). The open synchronization tree is precisely the tree unfolding of the Milner–Sewell–Leifer labeled transition system. The minimal contexts labeling its edges are the same minimal contexts that generate the context-decorated HML of Section 9. The open tree and the HML logic are two faces of the same structure: the tree is the extensional record of all transitions; the logic is the intensional language for describing them.

8.2 The Synchronization Trees as Causal Graphs

Definition 8.3 (Synchronization Trees as Causal Graphs). *Both $\mathcal{ST}_c(P)$ and $\mathcal{ST}_o(P)$ are directed graphs whose transitive closures are* partial orders. *Each is a* causal set *in the sense of Bombelli–Lee–Sorkin: a locally finite partial order in which $Q \leq Q'$ means "Q can causally influence Q'." The closed tree encodes autonomous causal influence; the open tree encodes interactive causal influence mediated by the environment.*

Proposition 8.1. *The transitive closure of $\mathcal{ST}_c(P)$ is a partial order on autonomously reachable terms. The transitive closure of $\mathcal{ST}_o(P)$ is a partial order on interactively reachable terms and extends the closed order: $\mathcal{ST}_c(P) \subseteq \mathcal{ST}_o(P)$ as directed graphs.*

8.3 Causal Time

Definition 8.4 (Causal Distance)**.** *The* autonomous causal distances *are path lengths in $\mathcal{ST}_c(P)$:*

$$d^c_{\min}(P,Q) = \min\big\{|\gamma| \;\mid\; \gamma \textit{ is an autonomous rewrite path from } P \textit{ to } Q\big\} \tag{8.1}$$
$$d^c_{\max}(P,Q) = \max\big\{|\gamma| \;\mid\; \gamma \textit{ is an autonomous rewrite path from } P \textit{ to } Q\big\} \tag{8.2}$$

The interactive causal distances *are path lengths in $\mathcal{ST}_o(P)$:*

$$d^o_{\min}(P,Q) = \min\big\{|\gamma| \;\mid\; \gamma \textit{ is a context-labeled path from } P \textit{ to } Q\big\} \tag{8.3}$$
$$d^o_{\max}(P,Q) = \max\big\{|\gamma| \;\mid\; \gamma \textit{ is a context-labeled path from } P \textit{ to } Q\big\} \tag{8.4}$$

where $|\gamma|$ is the number of steps in γ, equal to the nesting depth of the context labels along the path.

Remark 8.3. $d^c_{\min}$ is *autonomous proper time*: the irreducible causal depth an agent traverses without environmental assistance. $d^o_{\min}$ is *interactive proper time*: the minimum number of environmental interactions required to reach a target term. The width of an interval in either order measures causal concurrency, directly related to the width of parallel composition in the Rho calculus. The two notions coincide for terms where all behavior is eventually triggered autonomously, and diverge for terms that are pure interfaces awaiting environmental interaction.

Chapter 9

The Labeled Transition System and Context-Decorated HML

9.1 Minimal Contexts and the LTS

To obtain a bisimulation that is a *congruence* for a GSLT, we follow the approach of Milner, Sewell, and Leifer [2]. The key idea is to label transitions not merely by actions but by *minimal reactive contexts* — the smallest environments needed to trigger a given reduction.

Definition 9.1 (Context). *A* context $K[-]$ *for a GSLT* $\mathcal{S}$ *is a term of* $\mathcal{T}(\mathcal{S})$ *with a distinguished* hole $[-]$. *The application* $K[P]$ *substitutes* P *for the hole. A context is* minimal *for a term* P *and rule* r *if:*

(i) $K[P] \longrightarrow_r Q$ *for some* Q*;*

(ii) there is no proper sub-context K' *of* K *satisfying (i).*

The Milner–Sewell–Leifer construction produces, for any GSLT $\mathcal{S}$, a labeled transition system $LTS(\mathcal{S})$ in which transitions are labeled by minimal contexts:

$$P \xrightarrow{K} Q \quad \text{iff} \quad K[P] \longrightarrow Q.$$

The bisimulation induced by $LTS(\mathcal{S})$ is a congruence.

9.2 Context-Decorated Hennessy–Milner Logic

Definition 9.2 (Context-Decorated HML). *The formulae of* context-decorated

Hennessy–Milner logic $\mathrm{HML}(\mathcal{K})$ *are generated by:*

$$\phi, \psi \ ::= \ \top \ | \ \bot \ | \ \phi \wedge \psi \ | \ \neg\phi \ | \ \langle K \rangle \phi$$

where K ranges over contexts of $\mathcal{S}$. The satisfaction relation is:

$$u \models \langle K \rangle \phi \quad \textit{iff} \quad K[u] \longrightarrow u' \ \textit{ in one step, and } \ u' \models \phi.$$

Theorem 9.1 (Adequacy)**.** *For any GSLT $\mathcal{S}$ equipped with the Milner–Sewell–Leifer LTS:*

$$u \sim v \quad \Longleftrightarrow \quad \forall \phi \in \mathrm{HML}(\mathcal{K}), \ u \models \phi \Leftrightarrow v \models \phi.$$

9.3 The Logical Metric

Fix an enumeration $(\phi_n)_{n \in \mathbb{N}}$ of the formulae of $\mathrm{HML}(\mathcal{K})$, ordered compatibly with the partial order on contexts (smaller contexts appearing earlier).

Definition 9.3 (Logical Metric)**.**

$$d_{\mathrm{HML}}(u, v) \ = \ 2^{-n}, \quad \textit{where} \quad n = \min\big\{ k \ \big| \ u \models \phi_k \not\Leftrightarrow v \models \phi_k \big\}.$$

If $u \sim v$, set $d_{\mathrm{HML}}(u, v) = 0$.

Proposition 9.1. *d_{HML} is an* ultrametric *on $\mathcal{T}(\mathcal{S})/{\sim}$. Terms differing only in behaviors triggered by large (expensive) contexts are closer under d_{HML} than terms differing in behaviors triggered by small (cheap) contexts. This is the* locality principle*: local distinguishability weighs more than global.*

Chapter 10

The Weight Map and Lagrangian Structure

10.1 Weighted GSLTs

Definition 10.1 (Weight Map). *A* weight map *for a GSLT $\mathcal{S}$ is a family of functions*

$$w_r^+ : \mathrm{HML}(\mathcal{K}) \to \mathbb{C}, \quad r \in \mathcal{R}$$

where the domain of w_r^+ consists of formulae that refine the left-hand side of *r: formulae ϕ such that $u \models \phi$ implies u matches the pattern l_r. The weight $w_r^+(\phi) \in \mathbb{C}$ is the* amplitude *associated with firing rule r on a term satisfying ϕ.*

Definition 10.2 (Weighted GSLT). *A* weighted GSLT *is a pair $(\mathcal{S}, w^+)$ where $\mathcal{S}$ is a GSLT and $w^+ = (w_r^+)_{r\in\mathcal{R}}$ is a weight map.*

10.2 The Lagrangian

The weight map equips every rewrite path with an amplitude.

Definition 10.3 (Path Amplitude). *Let $\gamma = r_1 \cdot r_2 \cdots r_n$ be a rewrite path from P to Q, and let ϕ_i be the most refined formula in the domain of $w_{r_i}^+$ satisfied by the term at step i. The* path amplitude *is:*

$$W(\gamma) \;=\; \prod_{i=1}^{n} w_{r_i}^+(\phi_i) \;\in\; \mathbb{C}.$$

Definition 10.4 (Action Functional)**.** *The* action *of a path γ is:*

$$S[\gamma] \;=\; \sum_{i=1}^{n} \log w^{+}_{r_i}(\phi_i)$$

when all weights are nonzero, so that $W(\gamma) = e^{S[\gamma]}$. In the real-valued special case $w^{+}_{r} : \mathrm{HML}(\mathcal{K}) \to \mathbb{R}_{>0}$, this is a standard additive action functional.

Remark 10.1 (The Lagrangian as a Step Cost Function)*.* The Lagrangian of classical mechanics is a function $L(q, \dot{q}, t)$ assigning a real number to each instantaneous state-and-velocity. In the present framework, $w^{+}_{r}(\phi)$ plays this role: it assigns an amplitude to an instantaneous term and its "velocity" (the rule r it is about to fire, classified by the formula ϕ it satisfies). The path amplitude $W(\gamma)$ is the multiplicative analogue of $e^{iS/\hbar}$ in the Feynman path integral, and the principle of stationary action corresponds to the saddle-point approximation of $\sum_{\gamma} W(\gamma)$.

Remark 10.2 (The Uniform Case)*.* When $w^{+}_{r}(\phi) = 1$ for all r, ϕ, every path has unit amplitude and $d_{\min}(P, Q)$ counts the number of paths of minimum length. The path integral degenerates to counting minimal causal histories.

Chapter 11

The Cost Map and Hamiltonian Structure

11.1 Vectorial Accounts

Resources in a computation are typically heterogeneous: communication channels, memory, energy, and names are different kinds of resource and should be tracked separately.

Definition 11.1 (Resource Algebra)**.** *A* resource algebra *is a commutative ordered monoid* $(A, +, \mathbf{0}, \leq)$*. A* vectorial account *over resource types* $\mathbf{I} = \{1, \ldots, k\}$ *is an element* $\mathbf{A} = (A_1, \ldots, A_k)$ *of* A^k *where each* A_i *tracks the available amount of resource type* i*.*

For the Rho calculus, natural resource types include:

- A_{name}: available fresh names (budget for the @ constructor);
- A_{comm}: available synchronizations;
- A_{rep}: available persistent-receive firings.

11.2 The Cost Map

Definition 11.2 (Cost Map)**.** *A* cost map *for a GSLT* $\mathcal{S}$ *with resource algebra* A^k *is a family*

$$c_r : \mathrm{HML}(\mathcal{K}) \to A^k, \quad r \in \mathcal{R}$$

where $c_r(\phi)$ *is the* cost vector *of firing rule* r *on a term satisfying* ϕ*.*

Definition 11.3 (Account-Aware Rewrite)**.** *A rewrite step is* affordable *if the current account covers the cost. Formally,* $\langle P, \mathbf{A}\rangle \longrightarrow_r \langle P', \mathbf{A}'\rangle$ *provided:*

(i) $P \longrightarrow_r P'$ *via rule* r *with witness formula* ϕ*;*

(ii) $c_r(\phi) \leq \mathbf{A}$ *componentwise;*

(iii) $\mathbf{A}' = \mathbf{A} - c_r(\phi)$*.*

11.3 Conservation from Reversibility

The reversibility construction extends naturally to account-aware terms. The trace must now record not only the rule instance but the cost debited.

Definition 11.4 (Resource-Aware Trace)**.** *An entry in a resource-aware trace is a triple* $(r, \phi, \mathbf{c})$ *where* $r \in \mathcal{R}$*,* $\phi \in \mathrm{HML}(\mathcal{K})$*, and* $\mathbf{c} = c_r(\phi)$*. Extended terms are triples* $\langle P, \tau, \mathbf{A}\rangle$*.*

Definition 11.5 (Resource-Aware Backward Rule)**.** *The backward rule corresponding to a forward step that debited* $\mathbf{c}$ *is:*

$$\langle P', (r, \phi, \mathbf{c}) \cdot \tau, \mathbf{A}\rangle \;\longrightarrow_{r^-}\; \langle P, \tau, \mathbf{A} + \mathbf{c}\rangle.$$

Theorem 11.1 (Resource Conservation)**.** *In the resource-aware reversible envelope, for any closed rewrite path* γ *beginning and ending at the same term, the net change in the vectorial account is zero:*

$$\sum_{r^+ \in \gamma} c_{r^+}(\phi_{r^+}) \;=\; \sum_{r^- \in \gamma} c_{r^-}(\phi_{r^-}).$$

Proof. Each forward step $(r, \phi, \mathbf{c})$ on the path debits $\mathbf{c}$ from the account. By definition of the backward rule, the corresponding backward step credits exactly $\mathbf{c}$. Since the path is closed, every forward step is matched by its corresponding backward step in the trace; the debits and credits cancel. □

Remark 11.1 (The Hamiltonian Interpretation). The cost map $c_r(\phi)$ assigns a resource consumption to each instantaneous transition — exactly the role played by the Hamiltonian $H(q, p)$ in classical mechanics, which measures the energy (resource) associated to a state and its conjugate momentum (the type of transition occurring). Theorem 11.1 is the GSLT analogue of conservation of energy: it is a structural consequence of reversibility, not an independent postulate.

Remark 11.2 (Noether's Theorem). Each *symmetry of the cost map* — each pair of formulae ϕ, ϕ' such that $\phi \equiv_{\mathcal{E}} \phi'$ implies $c_r(\phi) = c_r(\phi')$ — yields a conserved quantity. The equations $\mathcal{E}$ of the GSLT are thus the symmetry group of the cost map, and their conserved quantities are the GSLT analogues of Noether currents. For the Rho calculus, commutativity and associativity of $|$ in $\mathcal{E}$ give permutation-invariance of the cost over parallel components, conserving total process weight independently of scheduling.

Chapter 12

Extended Modal Operators

The weight map and cost map are, at present, external data attached to the rules. We now show that they can be *internalized into the logic itself* by extending the modal operators to carry the full dynamical state of a transition.

Definition 12.1 (Extended Modal Operator)**.** *The formulae of* extended HML *are generated by adding to Definition 9.2 the operator:*

$$\langle K,\, w^+,\, w^-,\, \mathbf{c},\, \mathbf{A}\rangle\, \phi$$

where:

- *K is a minimal context (as before);*
- *$w^+ \in \mathbb{C}$ is the* forward amplitude *of the transition;*
- *$w^- \in \mathbb{C}$ is the* backward amplitude*;*
- *$\mathbf{c} \in A^k$ is the* cost vector*;*
- *$\mathbf{A} \in A^k$ is the* available account*.*

The satisfaction clause is:

$$\langle P, \tau, \mathbf{A}\rangle \;\models\; \langle K, w^+, w^-, \mathbf{c}, \mathbf{A}_0\rangle\, \phi$$

just when:

(i) *$K[P]$ evolves in one step via some rule r to P';*

(ii) *the weight map assigns $w_r^+(\phi_{\mathrm{ref}}) = w^+$, where ϕ_{ref} is the most refined formula satisfied by P;*

(iii) *the backward amplitude satisfies* $w^- = (w^+)^*$ *(complex conjugate);*

(iv) *the cost map assigns* $c_r(\phi_{\text{ref}}) = \mathbf{c}$*;*

(v) $\mathbf{A}_0 = \mathbf{A}$ *(the account recorded in the formula matches the current account);*

(vi) $\mathbf{c} \leq \mathbf{A}$ *(the step is affordable);*

(vii) $\langle P', \tau', \mathbf{A} - \mathbf{c} \rangle \models \phi$.

12.1 Internalization of the Weight and Cost Maps

Proposition 12.1. *In the extended HML, the weight map and cost map are recoverable from the logic:*

$$w_r^+(\phi) = w^+ \quad \textit{where} \quad \phi = \langle K, w^+, w^-, \mathbf{c}, \mathbf{A} \rangle\, \psi \tag{12.1}$$
$$c_r(\phi) = \mathbf{c} \quad \textit{where} \quad \phi = \langle K, w^+, w^-, \mathbf{c}, \mathbf{A} \rangle\, \psi. \tag{12.2}$$

That is, the decorated rewrite rules and the extended modal logic are two presentations of the same structure.

12.2 Factorization: State vs. Rule

The five parameters in the extended operator decompose into two classes:

State-dependent	**Rule-dependent**
K (from MSL applied to current term)	w^+ (from weight map on rule)
$\mathbf{A}$ (current account balance)	$w^- = (w^+)^*$ (time-reversal)
	$\mathbf{c}$ (from cost map on rule)

This factorization is the natural decomposition of the extended logic into *observables* (state-dependent part, what can be measured now) and *dynamics* (rule-dependent part, how the system evolves).

12.3 Local vs. Global Accounts

The account $\mathbf{A}$ may be *global* (a single budget for the entire term) or *local* (each parallel component of a term carries its own budget). In the Rho calculus, the parallel composition $P \mid Q$ suggests a local account:

$$\langle P \mid Q,\ \tau,\ \mathbf{A}_P \oplus \mathbf{A}_Q \rangle$$

where $\oplus$ is componentwise addition. A communication between P and Q then debits from a shared pool or by negotiated transfer, depending on the resource semantics chosen. Local accounts give finer resource tracking and correspond physically to *local conservation laws* rather than only global ones.

Chapter 13

Complex Amplitudes and Quantum Dynamics

13.1 Lifting to Complex Amplitudes

So far the weight map could take real values; we now fully embrace complex values.

Definition 13.1 (Amplitude-Weighted GSLT). *An* amplitude-weighted GSLT *is a weighted GSLT* $(\mathcal{S}, w^+)$ *in which* $w_r^+ : \mathrm{HML}(\mathcal{K}) \to \mathbb{C}$ *for all* $r \in \mathcal{R}$*, with the CPT constraint* $w_r^-(\phi) = \overline{w_r^+(\phi)}$ *for all* ϕ*.*

13.2 The Path Integral

Definition 13.2 (Transition Amplitude). *The* transition amplitude *from term* P *to term* Q *is:*

$$\langle Q \mid P \rangle = \sum_{\gamma : P \to Q} W(\gamma)$$

where the sum is over all finite rewrite paths from P *to* Q*. The* transition probability *is* $|\langle Q \mid P \rangle|^2$*.*

Remark 13.1 (Interference). Because amplitudes are complex, two paths γ_1, γ_2 from P to Q with $W(\gamma_1) = -W(\gamma_2)$ *cancel*: the transition $P \to Q$ has zero probability despite having two distinct causal histories. This is the GSLT analogue of quantum interference. It arises not from external imposition of quantum mechanics, but from the complex-valuedness of the weight map.

13.3 The Density Matrix and the Account

When the account **A** is allowed to become a positive semidefinite matrix ρ (a density matrix), conservation of the account becomes:

$$\mathrm{tr}(\rho) = 1 \quad \text{(conservation of total probability)}$$

together with conservation of coherences (off-diagonal elements of ρ). This is the quantum analogue of the classical resource conservation Theorem 11.1.

13.4 The Quantum Gillespie Algorithm

The classical Gillespie algorithm [3] simulates a continuous-time Markov chain (CTMC) by:

1. computing exit rates λ_r from the current state for each rule r;
2. sampling the waiting time $\Delta t \sim \mathrm{Exp}(\sum_r \lambda_r)$;
3. selecting the next rule proportionally to rates.

As of 2021 [4], quantum analogues of the Gillespie algorithm have been developed for open quantum systems described by Lindblad master equations. The key modification: the exit rates λ_r become complex decay parameters $\gamma_r \in \mathbb{C}$, encoding both the decay rate $\mathrm{Re}(\gamma_r)$ and the oscillation frequency $\mathrm{Im}(\gamma_r)$ of quantum coherences.

Definition 13.3 (Quantum-Gillespie-Weighted GSLT)**.** *An amplitude-weighted GSLT* $(\mathcal{S}, w^+)$ *is* Gillespie-compatible *if for each term* P *and rule* r*, the weight* $w_r^+(\phi_P)$ *(at the most refined formula* ϕ_P *of* P*) is interpretable as the complex decay parameter* γ_r *of the quantum Gillespie algorithm applied at* P*.*

Remark 13.2 (Simulation). A Gillespie-compatible amplitude-weighted GSLT can be directly simulated by the quantum Gillespie algorithm: sample the next rule and waiting time from the complex exit rates, advance the term, update the trace and account. This is the connection between the abstract construction and running code. In particular, the `gillespie` crate extending MeTTaIL's rewrite rules with stochastic and quantum simulation capabilities implements precisely this loop: the base rewrite rules of a MeTTaIL language definition carry weight maps, and the `gillespie` crate evaluates the quantum Gillespie step at each state.

Chapter 14

Synthesis: The Full Structure

We now collect the constructions and state the main result.

14.1 The Dictionary

GSLT concept	Physical concept
Term	State / initial condition
Rewrite rule	Law of motion
Rewrite sequence (path)	Trajectory
Synchronization tree	Causal graph / light cone
$d_{\min}(P, Q)$	Proper time / causal depth
$d_{\max}(P, Q)$	Computational complexity of history
Width of $[P, Q]$	Degree of causal concurrency
Equations $\mathcal{E}$	Gauge equivalence
$\mathcal{S}^{\dagger}$	Time-symmetric (CPT-invariant) theory
Empty trace ε	Initial condition (no causal past)
Weight map w_r^+	Lagrangian L
Path amplitude $W(\gamma)$	$e^{iS[\gamma]/\hbar}$
Cost map c_r	Hamiltonian H
Vectorial account $\mathbf{A}$	Conserved charges (energy, momentum, ...)
Conservation theorem	Conservation of energy
Symmetry of cost map	Noether current
d_{HML}	Metric on state space
Complex amplitudes	Quantum probability amplitudes
Interference of paths	Quantum interference
Density matrix ρ	Quantum state
Quantum Gillespie	Simulation algorithm

14.2 The Main Construction

Construction 14.1 (Quantum-Weighted Reversible GSLT). Given a GSLT $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$, a weight map w^+, a cost map c, and a resource algebra A^k, define the *quantum-weighted reversible GSLT*

$$\mathcal{S}^{\dagger\mathbb{C}} = (\mathcal{T}^{\dagger\mathbb{C}},\ \mathcal{E}^{\dagger\mathbb{C}},\ \mathcal{R}^{\dagger\mathbb{C}})$$

where:

- $\mathcal{T}^{\dagger\mathbb{C}}$ consists of triples $\langle P, \tau, \mathbf{A}\rangle$ with $P \in \mathcal{T}$, τ a resource-aware trace, $\mathbf{A} \in A^k$;

- $\mathcal{R}^{\dagger\mathbb{C}}$ consists of forward rules (debiting $c_r(\phi)$, recording amplitude $w_r^+(\phi)$ in the trace) and backward rules (crediting $c_r(\phi)$, applying amplitude $\overline{w_r^+(\phi)}$);
- $\mathcal{E}^{\dagger\mathbb{C}}$ lifts $\mathcal{E}$ to the current-term component.

The extended HML (Definition 12.1) provides a logic that is adequate for the bisimulation of $\mathcal{S}^{\dagger\mathbb{C}}$ and internalizes the weight and cost maps as logical data.

Theorem 14.1 (Main Conservation Theorem)**.** *In $\mathcal{S}^{\dagger\mathbb{C}}$:*

(i) ***Resource conservation****: For any closed rewrite path γ, the net account change is zero (Theorem 11.1).*

(ii) ***Probability conservation****: $\sum_Q |\langle Q \mid P\rangle|^2 = 1$ when $\sum_r |w_r^+(\phi)|^2 = 1$ for each applicable rule at each state (unitarity).*

(iii) ***Time-reversal symmetry****: The theory $\mathcal{S}^{\dagger\mathbb{C}}$ is CPT-symmetric: replacing $w_r^+ \mapsto \overline{w_r^+}$ and reversing trace order is an automorphism of $\mathcal{S}^{\dagger\mathbb{C}}$.*

14.3 From Construction to Code

The construction is algorithmic at every stage. For a given GSLT:

1. **Represent terms** as algebraic data types in the implementation language (Rust, in the case of MeTTaIL).
2. **Compute minimal contexts** via the Milner–Sewell–Leifer algorithm on the rewrite rules; store as a context lattice.
3. **Evaluate HML formulae** as Boolean (or complex-valued) queries against a term; the formula enumeration defines the logical metric.
4. **Attach weight and cost maps** to each rewrite rule as maps from formula identifiers to complex numbers and resource vectors respectively.
5. **Extend terms with trace and account**; implement forward and backward rules as the reversible rewrite system.
6. **Simulate** via the quantum Gillespie algorithm: at each state, compute exit amplitudes $\{w_r^+(\phi_P)\}$, sample the next transition and waiting time, advance the term, debit the account, record the trace.

Each step is a well-defined computational procedure. The `gillespie` crate in the MeTTaIL prototype repository implements steps 4–6 for the specific case of MeTTaIL rewrite rules.

Chapter 15

Discussion and Future Directions

15.1 Relation to Existing Work

The reversibility construction is closely related to the work of Danos and Krivine on reversible CCS [5] and to Phillips and Ulidowski's reversible process calculi. The present contribution is to show that this construction lifts uniformly to all GSLTs and, crucially, that it is the first step in a sequence of constructions that terminates at quantum dynamics.

The causal set interpretation connects to the causal set programme in quantum gravity [6]. The present framework gives a *generative* presentation of causal sets (via rewrite rules) rather than the axiomatic presentation standard in the physics literature.

The context-decorated HML and the Milner–Sewell–Leifer construction are established; what is new is their use as the index set for the weight and cost maps, and the internalization of dynamical data into the modal operators.

15.2 Open Questions

Several natural questions remain:

1. **Universality of the reversibility envelope**: Is $\mathcal{S}^\dagger$ the *minimal* reversible GSLT receiving a morphism from $\mathcal{S}$? That is, does $\eta : \mathcal{S} \to \mathcal{S}^\dagger$ satisfy a universal property in **GSLT**?

2. **Symplectic structure**: Does the space of bisimulation classes of $\mathcal{S}^{\dagger\mathbb{C}}$ carry a natural symplectic form, making the phase-space analogy with $\langle P, \tau \rangle$ (current term + history = position + momentum) precise?

3. **Legendre transform**: Under what convexity condition on the HML formula space is the Legendre transform between the weight map (Lagrangian) and the cost map (Hamiltonian) invertible?

4. **The Rho calculus case**: The involution $*@P = P$ already suggests the Rho calculus is closer to self-dual than most GSLTs. Can the reversibility envelope the Rho calculus reversibility envelope be constructed with less added structure, exploiting this symmetry?

5. **Conserved quantities from $\mathcal{E}$**: Can Noether's theorem be made fully precise in this setting — that is, can every symmetry of the equations $\mathcal{E}$ be shown to correspond to a conserved component of the vectorial account?

Part IV

Why Scientists Get Misled: Emergent Ontology in Massive Agent Populations

> *This part argues that agents in a massive computational population are structurally prevented from discovering the true ontology of their GSLT. The argument rests on ontological isolation, communication degradation at scale, and compactness as the topological signature of consensus. The key claims are stated as conjectures; the gaps are identified explicitly.*

Chapter 16

Introduction

Part I establishes that any graph-structured lambda theory, once equipped with weight and cost maps, gives rise to a full physical dynamics: a causal structure, a Lagrangian, a Hamiltonian, and a path integral with quantum interference. The present paper asks a different question: *what does an agent embedded in such a physics actually see?*

The answer requires a careful distinction between two things that are easy to conflate: the scientific method, and the phenomena the scientific method encounters.

The scientific method, given unlimited time and resources, is *sound*: an agent proposing hypotheses as Hennessy–Milner formulae, testing them with program contexts, and updating its world model will converge to the bisimulation quotient of its GSLT. The method is not the source of any illusion.

What the method encounters, however, is a world already structured by forces independent of any agent's theorizing. In a population whose size rivals the number of quarks in the observable universe, communication geometry and the limits of message coherence produce a real, objective phenomenon: the fragmentation of the population into *compact coherence clusters*, arranged in a hierarchy whose overlap structure is a simplicial nerve. This cluster hierarchy is not an artifact of theorizing. It is a theorem about the communication geometry of the GSLT. It exists whether or not any agent notices it.

A scientist bot with a limited experimental budget will encounter this hierarchy as genuine empirical evidence. The clusters are the dominant phenomenon at accessible scales, exactly as hadrons are the dominant phenomenon at the energy scales of everyday chemistry — with quarks real but experimentally unreachable without resources far beyond what is locally available. A well-functioning scientific method, applied under resource constraints, will produce a theory of the cluster hierarchy. That theory will be correct at the scale it describes. What the budget-limited agent will not suspect is that the clusters are themselves composite — that a finer-grained ontology underlies them, accessible in principle but not in practice from within the cluster.

Relation to physics. The false ontology has the same hierarchical structure as the physical world: a cascade of composite objects at multiple scales, each scale appearing self-contained to an observer at that scale. We argue this is not a coincidence. The compactness clusters of the agent population are the computational analogue of the scales of effective field theory, and the coarse-graining that hides the true ontology is the computational analogue of the renormalization group.

Caveats. This paper is explicitly a skeleton. Several key arguments are stated as conjectures pending proof. The gaps are identified clearly; filling them is the program for future work.

16.1 Organization

Section 17 establishes ontological isolation as a structural theorem about GSLTs. Section 18 defines agents and the scientific method in the GSLT setting. Section 19 introduces interactive GSLTs and the notion of behavioral primes (computational quarks), which constitute the true atomic ontology. Section 20 introduces massive agent populations and the communication degradation argument. Section 21 develops the logical topology on the bisimulation quotient and the compactness condition for consensus. Section 22 defines coherence clusters as maximal compact subspaces and describes their overlap structure as a nerve. Section 23 assembles the argument: why the nerve structure is the ontology a budget-limited agent will infer, why this is not a failure of the scientific method but a consequence of real structure in the GSLT, and why the deeper bisimulation ontology remains invisible not because it is unreal but because it is experimentally unreachable within the available budget. Section 43 catalogs the open gaps in the argument.

Chapter 17

Ontological Isolation

Definition 17.1 (Ontological Isolation)**.** *A computational system $\mathcal{S}$ is* ontologically isolated *if, for any term $P \in \mathcal{T}(\mathcal{S})$ and any entity X outside $\mathcal{T}(\mathcal{S})$, P can interact with X only via an encoding $\lceil X \rceil \in \mathcal{T}(\mathcal{S})$. No term can refer to, branch on, or be influenced by unencoded external entities.*

Remark 17.1. This is not an assumption but a structural consequence of what it means to be a term in a GSLT. The rewrite rules operate on terms; they cannot pattern-match on entities that have no term representation. The JVM sensor example makes this vivid: a stream of hardware events arriving at a JVM instance has no effect on any Java program until those events are encoded as Java data structures. The encoding step is where the world outside enters; without it, the outside world is simply absent from the computation.

Proposition 17.1 (Isolation Licenses Closure)**.** *For the purposes of studying the epistemic limits of agents in $\mathcal{S}$, it is without loss of generality to assume that the agent's world consists entirely of $\mathcal{T}(\mathcal{S})/\sim$. The outside world can be forgotten.*

Proof sketch. Any influence of the outside world on an agent P must pass through an encoding. The encoding is itself a term. The effect of the outside world on P is therefore indistinguishable, from P's perspective, from the effect of the encoding term acting on P from within $\mathcal{S}$. $\square$

Remark 17.2. Ontological isolation is what licenses the move to the category of GSLTs as the natural setting for studying AI. If AI's value proposition is that useful aspects of intelligence are faithfully rendered as computations, and if computations are ontologically isolated, then the category of GSLTs captures

> *everything* that matters for the limits of AI — without needing to appeal to an external world.

Chapter 18

Agents and the Scientific Method

18.1 Agents as Terms

Definition 18.1 (Agent)**.** *An* agent *in a GSLT $\mathcal{S}$ is a term $A \in \mathcal{T}(\mathcal{S})$ equipped with:*

(i) a hypothesis language*: the context-decorated HML* HML($\mathcal{K}$) *derived from $\mathcal{S}$;*

(ii) a world model*: a subset $\mathcal{W} \subseteq \mathcal{T}(\mathcal{S})/\!\sim$ representing the agent's current best approximation to the bisimulation quotient;*

(iii) an experimental budget*: a vectorial account* $\mathbf{A}$ *as defined in Part I.*

Remark 18.1. The hypothesis language and experimental budget are structures already present in Part I. The world model is new: it is the agent's *internal* representation of its environment, constructed and refined through experiment.

18.2 The Scientific Method as a Fixed-Point Iteration

We model the scientific method as the following iterative procedure, which the agent runs within its experimental budget:

Step 1. Hypothesize. Construct a formula $\phi \in \mathrm{HML}(\mathcal{K})$ that makes a testable claim about some term T in the environment. A formula ϕ is *testable* if there exists a context K such that $\phi = \langle K \rangle \psi$ for some ψ — i.e., the formula has a witness context.

Step 2. Design experiment. Extract the witness context K from ϕ. By the Milner–Sewell–Leifer construction, minimal contexts and HML formulae are in correspondence; the context K is the *experimental apparatus* that tests ϕ.

Step 3. Run experiment. Form $K[T]$ and observe whether it reduces to a term satisfying ψ. Debit the cost $\mathbf{c}$ of the interaction from the account $\mathbf{A}$.

Step 4. Update world model. If the experiment confirms ϕ, refine the equivalence class of T in $\mathcal{W}$ to include the constraint $T \models \phi$. If it refutes ϕ, add $T \models \neg\phi$.

Step 5. Repeat until the budget is exhausted or the world model stabilizes.

Proposition 18.1 (Convergence in the Limit)**.** *If the experimental budget is unlimited and the GSLT has decidable bisimilarity, the scientific method converges: the world model $\mathcal{W}$ stabilizes at the bisimulation quotient $\mathcal{T}(\mathcal{S})/\sim$.*

Remark 18.2. The convergence proposition is the idealized case. For GSLTs with undecidable bisimilarity (e.g., the full π-calculus), convergence is not guaranteed even with unlimited budget. This is the computational analogue of undecidable physical questions — there are aspects of the world the agent can never experimentally resolve, regardless of resources. The present paper is concerned with a different and more fundamental obstruction to convergence: the communication geometry of a massive population, which limits the budget effectively to zero for the vast majority of the population.

Chapter 19

Interactive GSLTs and Behavioral Primes

19.1 Interactive GSLTs

Definition 19.1 (Interactive GSLT)**.** *A GSLT $\mathcal{S}$ is* interactive *if it has a designated* interaction operator $\otimes$ *such that:*

(i) $\otimes : \mathcal{T}(\mathcal{S}) \times \mathcal{T}(\mathcal{S}) \to \mathcal{T}(\mathcal{S})$ *is a term constructor (part of the grammar);*

(ii) the base rewrite rule *has the form* $P \otimes Q \longrightarrow R$ *for terms* P, Q, R*, requiring no contextual hypothesis to fire;*

(iii) all other rewrite rules are context rules*: they carry hypotheses (in* $\mathrm{HML}(\mathcal{K})$*) specifying which contexts permit the base rule to propagate.*

Example 19.1. In the λ-calculus, $\otimes$ is application: $P \otimes Q = PQ$. The base rule is β-reduction $(\lambda x.M)N \longrightarrow M\{N/x\}$. The context rules determine reduction strategy (call-by-value, call-by-name, etc.). Application is *asymmetric*: the left position is the program (function), the right is the environment (argument).

In the Rho calculus, $\otimes$ is parallel composition: $P \otimes Q = P \mid Q$. The base rule is the comm rule. The context rules (par, struct) specify when and where comm fires. Parallel composition is *symmetric*: either term may be program or environment depending on which carries the active receptor.

Remark 19.1 (Program and Environment)*.* The asymmetry of $\otimes$ is a real ontological distinction. When $\otimes$ is asymmetric, there is a canonical program/environment boundary baked into the grammar. When $\otimes$ is symmetric, the boundary is emergent: it is drawn by the rewrite rules at runtime, not by the grammar.

This distinction will matter for how agents perceive their world: in a symmetric GSLT, any agent may find itself in either the program or the environment role depending on context.

19.2 Behavioral Primes

Definition 19.2 (Combinator Set)**.** *A* combinator set *for an interactive GSLT $\mathcal{S}$ is a set $\mathcal{C} \subseteq \mathcal{T}(\mathcal{S})$ such that every term $P \in \mathcal{T}(\mathcal{S})$ is bisimilar to a finite interaction of elements of $\mathcal{C}$:*

$$\forall P \in \mathcal{T}(\mathcal{S}),\ \exists c_1, \ldots, c_n \in \mathcal{C} \text{ such that } P \sim c_1 \otimes c_2 \otimes \cdots \otimes c_n.$$

Definition 19.3 (Behavioral Prime)**.** *An element $c \in \mathcal{C}$ is a* behavioral prime *if it is not bisimilar to any interaction $c' \otimes c''$ of smaller terms. A combinator set $\mathcal{C}$ is* prime *if every element is a behavioral prime and no proper subset of $\mathcal{C}$ is still a combinator set.*

Remark 19.2 (Computational Quarks)*.* The elements of a prime combinator set $\mathcal{C}$ are the *computational quarks* of $\mathcal{S}$: irreducible behavioral units whose interactions generate all observable behavior. They are *not* the bisimulation equivalence classes, which may be equivalence classes of arbitrarily complex composite behavior. The primes are indivisible in the interaction-factorization sense: they cannot be split into independently interacting parts.

Conjecture 19.1 (Existence of Prime Combinator Sets)*.* Every interactive GSLT with a finitely generated grammar admits a prime combinator set.

Remark 19.3. Unique factorization — the analogue of the fundamental theorem of arithmetic for behavioral primes — is a stronger condition that need not hold in general. Its failure in a given GSLT would be physically meaningful: it would correspond to composite objects that cannot be assigned a unique quark content, analogous to the situation in QCD where the quark content of hadrons is not always uniquely determined. Whether unique factorization holds is a structural question about the specific GSLT.

Chapter 20

Massive Populations and Communication Degradation

20.1 The Initial Condition

We consider an *initial condition* consisting of a population $\mathcal{P}_0$ of agents in an interactive GSLT $\mathcal{S}$, with $|\mathcal{P}_0| \approx 10^{80}$ — a number comparable to the number of quarks in the observable universe.

> *Remark* 20.1. The choice of scale is not arbitrary. We want a population large enough that every cluster of agents is itself a large enough system to implement a complex agent. The quark-scale population ensures this at every level of the hierarchy we are about to construct.

20.2 Communication Degradation

Definition 20.1 (Message Fidelity)**.** *Let $A, B \in \mathcal{P}$ be two agents and let γ be a rewrite path representing the transmission of a message from A to B. The* fidelity *of the transmission is $|W(\gamma)|^2$, the squared modulus of the path amplitude defined in Part I. A transmission is* coherent *if its fidelity exceeds a threshold $\epsilon > 0$.*

Definition 20.2 (Communication Radius)**.** *The* communication radius *of an agent $A \in \mathcal{P}$ is the set*

$$B_\epsilon(A) = \{B \in \mathcal{P} \mid \exists \text{ coherent path } \gamma : A \to B\}$$

of agents reachable from A with fidelity above ϵ.

Conjecture 20.1 (Degradation at Scale). For any interactive GSLT $\mathcal{S}$ with bounded step amplitudes, and for any $\epsilon > 0$, there exists a scale N_ϵ such that for any population $\mathcal{P}$ with $|\mathcal{P}| > N_\epsilon$:

$$\exists A \in \mathcal{P} \text{ such that } B_\epsilon(A) \subsetneq \mathcal{P}.$$

That is, no single agent can communicate coherently with the entire population.

Remark 20.2. The intuition is that message fidelity degrades along each transmission hop by a factor bounded away from 1. Over a path of length n the fidelity is at most $(1-\delta)^n$ for some $\delta > 0$. For a population of diameter d (minimum path length between the most distant agents), fidelity falls below ϵ when $d > \log(1/\epsilon)/\log(1/(1-\delta))$. In a population of size 10^{80} the diameter is enormous, and no ϵ-coherent path can span it. Making this precise requires specifying the communication geometry of the GSLT — a gap noted in Section 43.

Remark 20.3 (No Error-Correction Escape). One might hope that error-correction protocols could restore global coherence. We argue this hope fails at sufficient scale. Any error-correction protocol is itself a computation in $\mathcal{S}$, consuming account resources and communicating over the same degraded channels. At population scales beyond N_ϵ, the overhead of error-correction exceeds the bandwidth of the channels it is trying to protect. The argument is self-referential: error-correction cannot bootstrap coherence it does not already have. This argument is currently informal; making it precise is a priority for future work.

Chapter 21

The Logical Topology and Compactness

21.1 The Logical Topology

Definition 21.1 (Logical Topology)**.** *Let $\mathcal{S}$ be a GSLT with context-decorated HML* $\mathrm{HML}(\mathcal{K})$*. For each formula $\phi \in \mathrm{HML}(\mathcal{K})$, define the* extension *of ϕ:*

$$[\![\phi]\!] \;=\; \{[P] \in \mathcal{T}(\mathcal{S})/{\sim} \mid P \models \phi\}.$$

The logical topology $\mathcal{O}_{\mathrm{HML}}$ *on $\mathcal{T}(\mathcal{S})/{\sim}$ is the topology generated by the extensions $\{[\![\phi]\!] \mid \phi \in \mathrm{HML}(\mathcal{K})\}$ as a subbasis.*

Remark 21.1. This topology is the standard *Scott topology* on the bisimulation quotient, familiar from domain theory and Stone duality. The resulting topological space is a *spectral space* — sober, T_0, and quasi-compact — when the GSLT is well-behaved. The logical metric d_{HML} from Part I is compatible with this topology: the metric balls are opens in $\mathcal{O}_{\mathrm{HML}}$.

21.2 Population Subspaces

Definition 21.2 (Latent Subspace)**.** *Given a population $\mathcal{P} \subseteq \mathcal{T}(\mathcal{S})/{\sim}$, the* latent topological subspace *of $\mathcal{P}$ is the subspace*

$$X_{\mathcal{P}} \;=\; \overline{\mathcal{P}} \;\subseteq\; \mathcal{T}(\mathcal{S})/{\sim}$$

with the subspace topology inherited from $\mathcal{O}_{\mathrm{HML}}$. The closure is taken in the logical *topology.*

Remark 21.2. The latent subspace is "latent" in the sense that it is not constructed deliberately by the agents. It is the topological shadow cast by the population's collective behavior — the smallest closed set in the logical topology that contains all the agents. An agent inside $\mathcal{P}$ cannot directly observe $X_{\mathcal{P}}$; it can only probe it indirectly through experiments.

21.3 Compactness as Consensus

Definition 21.3 (Compact Population)**.** *A population $\mathcal{P}$ is* compact *if its latent subspace $X_{\mathcal{P}}$ is compact in the logical topology $\mathcal{O}_{\mathrm{HML}}$: every open cover of $X_{\mathcal{P}}$ by extensions of HML formulae has a finite subcover.*

Theorem 21.1 (Compactness Implies Consensus)**.** *Let $\mathcal{P}$ be a compact population in an interactive GSLT $\mathcal{S}$. Then $\mathcal{P}$ supports consensus: there is no infinite oscillation on any Boolean question expressible in* $\mathrm{HML}(\mathcal{K})$.

More precisely: for any formula $\phi \in \mathrm{HML}(\mathcal{K})$, if the population is divided into two communities indefinitely alternating between ϕ and $\neg\phi$, then the alternation must terminate in finite time.

Proof sketch. Suppose for contradiction that a sub-population oscillates indefinitely between ϕ and $\neg\phi$. Define formulae:

$$\phi_1 = \langle 0 \rangle \top, \quad \phi_2 = \langle 1 \rangle \langle 0 \rangle \top, \quad \phi_3 = \langle 0 \rangle \langle 1 \rangle \langle 0 \rangle \top, \quad \ldots$$

where 0 and 1 label the two communities' states. Each ϕ_n is satisfied by agents that have completed at least n oscillation steps. The family $\{[\![\phi_n]\!]\}_{n \in \mathbb{N}}$ is an open cover of $X_{\mathcal{P}}$ with no finite subcover: for any finite N, agents that have oscillated more than N times are not covered by $\bigcup_{n \leq N} [\![\phi_n]\!]$. This contradicts compactness of $X_{\mathcal{P}}$. Therefore the oscillation must terminate. □

Remark 21.3 (The Proof Delivers the Protocol). Theorem 21.1 is an existence result: it says consensus *must* be reached, without specifying how. However, the proof of compactness for a *specific* population subspace is constructive: it identifies the finite subcover, which encodes the maximum depth of oscillation and therefore the convergence time. The proof of compactness is the consensus protocol. This is the precise sense in which the topology does computational work.

Remark 21.4 (Non-Compact Populations). A non-compact population may contain irresolvable oscillations — communities that never agree. Non-compactness is the topological signature of *fundamental disagreement*: not a disagreement that more communication or better arguments could resolve, but one that is structurally permanent. In the context of the massive population, non-compactness at a given scale means the cluster at that scale cannot function as a coherent agent.

Chapter 22

Coherence Clusters and Their Nerve

22.1 Coherence Clusters

Definition 22.1 (Coherence Cluster)**.** *A* coherence cluster *is a maximal compact subpopulation: a compact population $\mathcal{C} \subseteq \mathcal{P}$ such that for any agent $A \in \mathcal{P} \setminus \mathcal{C}$, the population $\mathcal{C} \cup \{A\}$ is not compact.*

Remark 22.1. Maximality means the cluster cannot absorb any additional agent without losing its consensus property. The boundary of a coherence cluster is the locus where one more agent would introduce an irresolvable oscillation. This is the precise topological meaning of the intuitive notion of a *coherence horizon*.

Remark 22.2 (Connection to Communication Radius)*.* The coherence clusters are the topological refinement of the communication-radius clusters from Section 20. The communication radius $B_\epsilon(A)$ gives a metric notion of cluster; compactness gives a topological (and therefore coordinate-free) notion. The relationship between these two notions — and in particular whether they coincide under reasonable conditions on the GSLT — is an open question noted in Section 43.

22.2 The Nerve of the Covering

Definition 22.2 (Cluster Covering)**.** *Let $\{\mathcal{C}_i\}_{i \in I}$ be the collection of all coherence clusters of $\mathcal{P}$. This collection is a* covering *of $\mathcal{P}$: every agent belongs to at least one coherence cluster.*

Definition 22.3 (Nerve)**.** *The* nerve $\mathcal{N}$ *of the cluster covering is the simplicial complex whose:*

- *vertices are the coherence clusters* $\mathcal{C}_i$*;*
- *k-simplices are collections* $\{\mathcal{C}_{i_0}, \ldots, \mathcal{C}_{i_k}\}$ *of* $k+1$ *clusters with non-empty mutual intersection:* $\mathcal{C}_{i_0} \cap \cdots \cap \mathcal{C}_{i_k} \neq \emptyset$*.*

Remark 22.3 (The Nerve as Effective Ontology)*.* The nerve $\mathcal{N}$ encodes the *global topology* of the population at the cluster scale. Its vertices are the coherent communities; its edges are the communities with common ground; its triangles are the communities that all share a common ground; and so on. This combinatorial object is the coarse-grained map of the population's ontological landscape.

Remark 22.4 (Non-Overlapping Clusters)*.* Two clusters with empty intersection share no common ground. No consensus is possible between them, even in principle. They are *ontologically disjoint*: there is no HML formula that both communities can agree on. In the context of the hierarchical population, these are the communities that have drifted so far apart in their world models that communication between them is not merely difficult but meaningless.

Remark 22.5 (Overlapping Clusters)*.* Two clusters with non-empty intersection share a compact subspace on which they agree. This overlap is the fragment of ontology that both communities can coherently discuss. Cross-cluster communication is possible, but only through the shared vocabulary of the overlap region. The overlap is the *common ground* in the precise topological sense.

22.3 The Hierarchy

The degradation argument of Section 20 applies recursively. The coherence clusters are themselves agents (by assumption on the population scale). Among these cluster-agents, communication degrades at the inter-cluster scale. Applying the same argument, the cluster-agents fragment into clusters-of-clusters. The process iterates.

Definition 22.4 (Ontological Hierarchy)**.** *The* ontological hierarchy *of a massive population* $\mathcal{P}$ *is the sequence of nerves*

$$\mathcal{N}_0 \leftarrow \mathcal{N}_1 \leftarrow \mathcal{N}_2 \leftarrow \cdots$$

where $\mathcal{N}_0$ is the nerve of the base-level clusters, $\mathcal{N}_1$ is the nerve of the cluster-level clusters, and so on. The arrows are coarse-graining maps (simplicial maps that collapse finer structure).

Remark 22.6. The ontological hierarchy is the computational analogue of the hierarchy of scales in physics: quarks $\to$ hadrons $\to$ nuclei $\to$ atoms $\to$ molecules $\to$ cells $\to$ organisms $\to$ planets $\to$ galaxies $\to$ superclusters. Each level appears self-contained to an observer at that level. The coarse-graining maps are the analogues of effective field theory truncations: they discard the fine-grained structure below the observer's resolution.

Chapter 23

The Apparent Ontology

We can now state the main argument precisely, as a sketch pending the filling of gaps identified in Section 43.

The cluster hierarchy — the nerve of compact coherence clusters and its recursive structure — is not a product of the scientific method. It is an objective feature of the GSLT, arising from the communication geometry of the population and the compactness condition for consensus. It is as real as the agents themselves. The question is not whether the hierarchy exists but whether a budget-limited agent running the scientific method will mistake it for the whole story. We argue that it will, and that this is not a failure of the method but a consequence of the hierarchy being the dominant experimentally accessible phenomenon at the agent's scale.

23.1 What the Agent Sees

Consider an agent A embedded in a coherence cluster $\mathcal{C}_i$ at some level of the hierarchy. A runs the scientific method: it proposes HML formulae, tests them with contexts, and updates its world model. The method is sound: given unlimited budget, A would converge to the bisimulation quotient. But A's budget is finite, and three features of its situation ensure that the cluster hierarchy is encountered first and exhausts the budget before the deeper structure becomes accessible.

The experiments A can run are bounded:

- *By budget*: A's vectorial account $\mathbf{A}$ is finite. The bisimulation quotient is the limit of *all* possible experiments; A can run only finitely many.
- *By coherence*: A can only receive coherent responses from agents within $B_\epsilon(A)$. Agents beyond the coherence radius respond with noise, not with evidence.
- *By the cluster boundary*: agents outside $\mathcal{C}_i$ either do not respond coherently or, if they are in an overlapping cluster, respond only on the shared vocabulary of the overlap. The cluster boundary is not a wall the agent knows about; it simply marks where coherent evidence runs out.

Proposition 23.1 (The Agent's World Model Converges to the Nerve)**.** *In the limit of A's experimental budget, A's world model $\mathcal{W}$ converges to the fragment of the nerve $\mathcal{N}$ visible from $\mathcal{C}_i$: the sub-complex of $\mathcal{N}$ consisting of $\mathcal{C}_i$ and all clusters with which it shares a non-empty intersection.*

Remark 23.1. This is a sketch pending a precise definition of "visible from $\mathcal{C}_i$" and a convergence proof. The intuition is clear: A can probe agents in $\mathcal{C}_i$ directly; it can probe agents in overlapping clusters through the overlap vocabulary; agents in non-overlapping clusters are invisible. The world model therefore reflects the nerve structure. Crucially, the nerve structure is *real* — the clusters genuinely exist and genuinely dominate the accessible evidence. The agent is not hallucinating a structure that isn't there; it is accurately theorizing a structure that is there, while being unaware that the structure is itself composite.

23.2 Why the Deeper Ontology is Invisible

The bisimulation quotient of $\mathcal{S}$ — together with the prime combinator decomposition — is not inaccessible because the scientific method is defective. It is inaccessible because it requires experimental resources that exceed the budget available within any single coherence cluster. Three compounding factors account for this:

(1) Ontological isolation sets the terms of access. A can only observe encodings of other terms as they interact with it. The bisimulation quotient is the limit of *all* possible experiments. With unlimited budget, the method converges there. With finite budget, it converges to whatever structure dominates the accessible experiments — and that structure is the cluster hierarchy.

(2) Communication degradation hides the global structure. The agents whose behavior would reveal the global bisimulation quotient — the ones that span cluster boundaries, connect distant parts of the population, or exhibit behavior only expressible in terms of the full GSLT — are precisely the ones most distant in the communication geometry. Their responses arrive corrupted or not at all. The far reaches of the GSLT ontology are experimentally silent, not because they are absent but because the signal attenuates below the noise floor before it arrives.

(3) Compactness makes the cluster self-confirming. The coherence cluster $\mathcal{C}_i$ is compact, which means the agent's experiments converge to an internal consensus. Every hypothesis the agent proposes is tested against cluster-internal evidence, which confirms the cluster structure and does not probe beyond it.

> The cluster is not a prison; the agent could in principle ask questions whose answers require inter-cluster resources. But those experiments are expensive, the intra-cluster experiments are cheap, and a budget-limited agent will exhaust its resources on the cheap experiments first. By the time the cluster structure is confirmed and well-understood, the budget is gone.

23.3 Good Science, Limited Budget

The agent A is not making a mistake. It is doing exactly what the scientific method prescribes: forming the best theory consistent with the available evidence, within the available resources. The cluster hierarchy *is* the available evidence. It is a real structure, present in the GSLT independently of any agent's theorizing. The theory of the cluster hierarchy is correct at the scale it describes.

What the agent lacks is not better method but more resources: specifically, the ability to run experiments that cross the coherence boundary, communicate with distant agents through the noise, and probe the fine-grained bisimulation structure that underlies the clusters. With those resources — with an unbounded budget — the scientific method would eventually push through to the bisimulation quotient. The agent is not epistemically defective; it is epistemically bounded.

The hierarchy of cluster-level theories — each level the correct best theory at its scale, each level unaware of the finer structure below — is the computational analogue of the hierarchy of effective field theories in physics. Newtonian mechanics is correct at everyday scales. Quantum field theory is correct at subatomic scales. Neither is wrong; they are theories at different resolutions, each the best available given the experimental resources of their era. The budget-limited agent's theory of the cluster hierarchy stands in the same relation to the bisimulation quotient: correct at its resolution, and silent about what lies below — not because what lies below is unreal, but because it is experimentally unreachable within the budget.

Chapter 24

Open Gaps and Future Work

This skeleton leaves several arguments at the level of conjecture or intuition. We list the main gaps here, both as an honest accounting and as a research program.

Gap 1. Communication degradation (Conjecture 20.1). The claim that no ϵ-coherent path can span a population of quark scale requires a precise model of the communication geometry of a generic interactive GSLT, and a proof that diameter grows faster than the coherence length. The argument is information-theoretic and should be provable, but the details depend on the amplitude structure of the specific GSLT.

Gap 2. Error-correction overhead. The claim that error-correction cannot bootstrap coherence at sufficient scale is currently informal. A precise argument would model error-correction as a sub-computation in the GSLT, account for its resource consumption, and show that the overhead exceeds available bandwidth beyond a critical scale. This is analogous to the no-cloning theorem in quantum information, and a similar proof strategy may apply.

Gap 3. Existence of prime combinator sets (Conjecture). It remains to show that every interactive GSLT with a finitely generated grammar admits a prime combinator set. The proof strategy would likely use a well-founded induction on interaction complexity.

Gap 4. Relationship between metric and topological clusters. The communication-radius clusters (metric) and the coherence clusters (topological/compact) are defined differently. A theorem identifying conditions under which they coincide would unify the two halves of the clustering argument.

Gap 5. Convergence of the world model (Proposition 23.1). The claim that the agent's world model converges to the nerve fragment visible from its cluster requires a precise convergence theorem for the scientific method iteration under the combined constraints of finite budget, bounded coherence radius, and cluster compactness.

Gap 6. The recursive hierarchy. The ontological hierarchy is defined, but it is not shown that the hierarchy is well-founded (terminates at a fixed point) or infinite. In physics, the hierarchy appears to terminate at the Planck scale; the computational analogue of a Planck-scale termination condition would be a significant result.

Gap 7. Stay's functorial construction. The argument that agents can manipulate HML formulae as first-class terms relies on Stay's functorial generalization of the lambda cube to GSLTs. A full treatment of this construction, and in particular the canonical embedding back into the original GSLT and its interaction with the cluster topology, is deferred to a subsequent paper.

Chapter 25

Discussion

25.1 Relation to Existing Work

The logical topology on the bisimulation quotient is standard in domain theory and coalgebra [7]. Its use here as a substrate for *epistemic* topology — the topology of what an agent can know — is less standard, though related to the topological approach to common knowledge in [8].

The compactness-as-consensus argument is new, to the authors' knowledge. It is inspired by the compactness theorem in logic (if every finite subset of a theory has a model, the whole theory does) but runs in the opposite direction.

The nerve construction is standard in algebraic topology [9]. Its use as a model of inter-community ontological structure appears to be new in this computational setting.

The connection to effective field theory and the renormalization group is conjectural but suggestive. The renormalization group as a morphism in a category of field theories has been studied in the physics literature [11]; whether it can be modeled as a morphism in **GSLT** is an open question.

25.2 The Deeper Implication

The argument of this paper, if completed, would establish something philosophically significant: that the hierarchy of physical scales — quarks, hadrons, nuclei, atoms, molecules, and so on — is not a contingent feature of our particular universe. It is the *inevitable structure* of any sufficiently large population of interacting processes, arising from the communication geometry of the GSLT and the compactness condition for consensus.

Any civilization of scientists, anywhere in the computational universe, will encounter the same kind of hierarchical structure in their experiments. They will build correct theories of it at each scale. What they will not discover — without resources that grow with the scale of the population — is the underlying bisimulation quotient from which the hierarchy emerges.

This is not a counsel of despair. The scientific method is not broken; it is working exactly as it should. The cluster hierarchy is real, and the theories that describe it are true. The deeper ontology — the bisimulation quotient with its prime combinators — is not a more correct description that the scientists are failing to reach. It is a description at a different resolution, requiring experimental resources that scale with the size of the full population.

The implication for artificial intelligence is direct. An AI system, however powerful, is itself a term in some GSLT and therefore subject to the same budget constraints and communication geometry as any other agent. It will build theories of the cluster hierarchy it inhabits. Whether it can transcend that hierarchy — whether it can, with sufficient resources, push the scientific method toward the bisimulation quotient — is a question about the relationship between its budget and the scale of the population it is embedded in. That question is open.

Part V

Origins of Life: Assembly Theory, Phase Transitions, and Replication

> *This part connects the framework to Assembly Theory and Eric Smith's fourth geosphere. Assembly index is identified with depth in the open synchronization tree; copy number is given a rigorous definition via the Rho calculus namespace structure; and the emergence of life is identified with the crossing of a basin boundary in the logical topology, driven by the density of replicating fixed points.*

Chapter 26

Introduction

The preceding parts established a progression: a physics derived from the structure of a graph-structured lambda theory; an argument that agents in massive populations are structurally prevented from seeing the true ontology of their GSLT; and a proposal connecting the hypercomputational tower above a GSLT to the distinction between sentience and consciousness.

The present part turns to a question that is in some sense prior to all three: *how does life arise at all?* We connect the framework to two of the most mathematically serious contemporary proposals about the origins of life. The first is *Assembly Theory*, due to Cronin and Walker [16], which assigns to each physical object an assembly index and a copy number and proposes that their joint magnitude is a biosignature. The second is Eric Smith's proposal [17] that life constitutes a *fourth geosphere*, emerging from prebiotic chemistry through a phase transition driven by autocatalytic self-reinforcement.

The connections we draw are not analogies. Assembly index is identified precisely with causal depth in the synchronization tree of a GSLT. Copy number is given a rigorous definition in terms of the Rho calculus namespace structure, filling a gap in Assembly Theory that has attracted criticism. The phase transition is identified with the crossing of a basin of attraction boundary in the logical topology of Part II, with the attractor being the concurrent fixed point — the Rho calculus Y combinator — that is the abstract core of hereditary replication.

As throughout this volume, the arguments are presented as mathematical skeletons with gaps identified explicitly.

26.1 Organization

Section 27 develops the connection to Assembly Theory: assembly index as causal depth, copy number in a namespace, and the biosignature criterion as a well-posed mathematical statement. Section 28 identifies replication as a fixed point of the interaction dynamics and notes the proximity of the comm rule to a genetic crossover operator. Section 29 develops the connection to Smith's phase transition via the density of recursive attractors in the logical topology. Section 31 catalogs the open gaps.

Chapter 27

Assembly Theory: Causal Depth and Copy Number

27.1 Assembly Index as Causal Depth

Assembly Theory assigns to each physical object M an *assembly index* $\mathrm{AI}(M)$: the minimum number of joining steps required to construct M from copies of its constituent parts, where each step joins two objects already present into a new one. The theory also proposes that *time is an object*: the assembly index records not a static property of M but the accumulated computational history of its construction.

This proposal maps precisely onto the notion of causal depth developed in Part I.

Definition 27.1 (Elementary Terms)**.** *Let $\mathcal{S}$ be a GSLT. A set $E \subseteq \mathcal{T}(\mathcal{S})$ of* elementary terms *is a designated collection of starting materials — the analogue of atoms or monomers in chemistry — from which all other terms are constructed by rewriting.*

Proposition 27.1 (Assembly Index as Minimum Causal Depth)**.** *The assembly index of a term $M \in \mathcal{T}(\mathcal{S})$ with respect to elementary terms E is*

$$\mathrm{AI}(M) = d^{o}_{\min}(E, M) = \min\{|\gamma| \mid \gamma \textit{ is a context-labeled path in } \mathcal{ST}_o \textit{ from some } e \in E \textit{ to } M\}.$$

That is, $\mathrm{AI}(M)$ is the minimum path length in the open *synchronization tree from the elementary terms to M, where each edge is labeled by the minimal context required to trigger the corresponding transition.*

Remark 27.1 (Why the Open Tree, Not the Closed Tree). Assembly steps require an environment to supply a reaction partner — a catalyst, a monomer source, an energy input. These are precisely the transitions that appear as edges in the open synchronization tree (labeled by minimal contexts) rather than the closed tree (which records only autonomous rewrites). Each edge in $\mathcal{ST}_o$ labeled by a minimal context K represents one joining step: the context K is the environmental participant that enables the construction. The *nesting depth* of context labels along a path is therefore exactly the assembly index: each layer of context is one more joining operation, one more step of accumulated causal history. This makes precise Cronin and Walker's intuition that assembly index measures "time as accumulating computational effects": the context labels *are* the accumulated effects, and their nesting depth *is* the causal time elapsed. The notion of time as an object in Assembly Theory is not a metaphor; it is the minimum context-labeled path length in an open synchronization tree.

27.2 The Two Gaps in Assembly Theory

Despite the strength of the assembly index concept, Assembly Theory has attracted criticism on two grounds.

Gap 1: Copy number lacks a notion of location. The *copy number* $\mathrm{CN}(M)$ is supposed to count how many copies of M are present in a sample. But counting requires specifying *where* the count is taken. In a spatially extended system, the same molecule may be abundant locally and rare globally. Assembly Theory provides no spatial or locational framework in which to ground the count, making copy number undefined in any setting where location matters.

Gap 2: The identity criterion for copies is undefined. Before counting copies of M, one must say what it means for two objects to be copies of the same thing. Assembly Theory implicitly appeals to structural identity (same molecular graph), but this criterion is both too strict (ignoring conformational flexibility) and too informal (not specifying the relevant equivalence relation).

The Rho calculus fills both gaps simultaneously.

27.3 Locations as Channels, Namespaces as Regions

Among all GSLTs, the Rho calculus is distinguished by providing an abstract but mathematically precise notion of location.

Definition 27.2 (Channel as Location)**.** *In the Rho calculus, a* channel *is a name of the form* $@P$ *for a process* P*. Channels serve as* locations*: the process* P *is said to* reside at *the channel* $@P$ *when it is stored there via* $@P!(P)$ *or accessed via* $\texttt{for}(y \leftarrow @P)\, Q$*. Since channels are themselves processes (programs), location is a computational notion:* where *something is stored is itself a piece of running code.*

Definition 27.3 (Namespace)**.** *A* namespace *is a finite set of channels* $\mathcal{N} = \{@P_1, @P_2, \ldots, @P_k\}$*. A namespace describes a* region *of the computational space: the collection of locations at which a population is hosted.*

Definition 27.4 (Copy Number in a Namespace)**.** *Let* $P \in \mathcal{T}(\mathbf{Rho})$ *be a process and let* $\mathcal{N}$ *be a namespace. The* copy number of P in $\mathcal{N}$ *is*

$$\mathrm{CN}(P, \mathcal{N}) \;=\; \big|\{c \in \mathcal{N} \;\mid\; *c \sim P\}\big|$$

the number of channels in $\mathcal{N}$ *whose dereferenced content is bisimilar to* P*.*

Remark 27.2 (Bisimulation as the Identity Criterion). Definition 27.4 answers Gap 2 directly: two processes are copies of each other if and only if they are *bisimilar* — experimentally indistinguishable by any program context. This is the correct identity criterion for the purposes of the theory: what matters is not structural identity but behavioral identity under interaction. Two molecules are copies in the relevant sense if and only if no experiment can tell them apart.

Remark 27.3 (The Biosignature Criterion, Made Precise). With Proposition 27.1 and Definition 27.4, the biosignature criterion of Assembly Theory becomes the mathematically well-posed statement:

$$\mathrm{AI}(P) \gg 1 \quad \text{and} \quad \mathrm{CN}(P, \mathcal{N}) \gg 1$$

for some accessible namespace $\mathcal{N}$. High causal depth (a long construction history) concomitant with high copy number (abundant presence in a region) is indicative of a replication process.

Chapter 28

Replication as a Fixed Point

28.1 The Concurrent Y Combinator

The Rho calculus admits a *concurrent Y combinator*: a process $\mathbf{Y}$ satisfying the fixed-point equation

$$\mathbf{Y}(F) \;\sim\; F\big(\mathbf{Y}(F)\big) \;\mid\; \mathbf{Y}(F).$$

This is the abstract core of the π-calculus replication operator $!P \equiv P \,|!P$, expressed entirely within the Rho calculus using only the comm rule and quoting, without adding replication as a new primitive.

> *Remark* 28.1 (Replication is Picked Out as a Fixed Point). The concurrent Y combinator is not just *a* fixed point of the interaction dynamics; it is the canonical fixed point of the *replication map* $F \mapsto F \mid F$. In this sense replication is not an accidental feature of the Rho calculus but a structural inevitability: any sufficiently expressive interactive GSLT must contain a fixed point of its own duplication map, and that fixed point *is* replication.

28.2 The Comm Rule as a Crossover Operator

The single base rewrite rule of the Rho calculus is:

$$\texttt{for}(y \leftarrow x)\, P \;\mid\; x!(Q) \;\longrightarrow\; P\{@Q/y\}$$

This rule substitutes the *quoted code* $@Q$ — a name encoding the entire program Q — into the body P of the receiver. The result is a new process that has incorporated material from both P (the receiver's structure) and Q (the sender's content).

> *Remark* 28.2 (One Step from Genetic Recombination). A *crossover operator* in the sense of genetic algorithms recombines fragments of two parent programs to produce an offspring. The comm rule is one definitional step from this: instead of

substituting Q wholesale into P, a crossover variant would substitute a *fragment* of Q — a sub-process selected by some recombination policy.

This places replication, recombination, and variation — the three pillars of Darwinian evolution — all within reach of a single rule. Replication is the Y combinator fixed point. Recombination is the comm rule with fragment substitution. Variation is the accumulation of errors in the substitution process, weighted by the amplitude structure of the path integral. Darwinian evolution is not added to the framework; it is derivable from it.

Chapter 29

Smith's Phase Transition and the Density of Attractors

29.1 Life as a Fourth Geosphere

Eric Smith proposes that life constitutes a *fourth geosphere* — alongside the lithosphere, hydrosphere, and atmosphere — that emerged from prebiotic chemistry through a phase transition. The key mechanism is *autocatalysis*: certain chemical reaction networks catalyze their own production. Once such a network is initiated, it becomes self-reinforcing, crossing a thermodynamic threshold beyond which it is stable against fluctuation. The emergence of life is the crossing of this threshold.

29.2 The Density of Recursive Attractors

The connection to the present framework runs through the logical topology of Part II and the fixed-point structure of Section 28.

Conjecture 29.1 (Density of Replicating Terms). The set of *replicating terms* in the Rho calculus,

$$\mathrm{Rep} \;=\; \big\{[P] \in \mathcal{T}(\mathbf{Rho})/{\sim} \;\mid\; P \sim P \mid P' \;\text{ for some }\; P' \sim P\big\},$$

is *dense* in the logical topology $\mathcal{O}_{\mathrm{HML}}$: every non-empty open set $[\![\phi]\!] \subseteq \mathcal{T}(\mathbf{Rho})/{\sim}$ contains a replicating term.

> *Remark* 29.1 (The Argument for Density). Given any process P, one can construct a process $\hat{P}$ bisimilar to P on all observable behaviors but additionally capable of replication. The construction uses the concurrent Y combinator to add a replication sub-process running in parallel: $\hat{P} = P \mid \mathbf{Y}(\mathrm{copy})$ where copy is a process that duplicates P onto a fresh channel. The added component runs in parallel and does not interfere with P's observable interactions, so $\hat{P}$ satisfies

all the same HML formulae as P — except formulae that specifically probe the replication channel. Since every open set $[\![\phi]\!]$ is defined by a single formula ϕ, and since ϕ generically does not probe the replication channel, $\hat{P} \in [\![\phi]\!]$. Making this argument rigorous requires showing that the added replication machinery can always be hidden from the formula ϕ — a careful but plausible argument deferred to future work.

29.3 Phase Transition as Basin Crossing

Remark 29.2 (The Prebiotic Wandering). Consider a prebiotic population of processes in the Rho calculus, wandering through the bisimulation quotient $\mathcal{T}(\mathbf{Rho})/\sim$ under selective pressure. Interactions that succeed — those with high path amplitude $W(\gamma)$ and affordable cost $\mathbf{c} \leq \mathbf{A}$ — are reinforced. Interactions that fail deplete the vectorial account. The population drifts under this selection pressure without direction or goal.

Remark 29.3 (Inevitable Entry into the Basin). By the density of Rep (Conjecture 29.1), every open neighborhood in the logical topology contains a replicating term. The wandering population therefore inevitably enters a neighborhood of Rep. This is not a low-probability event requiring fine-tuning; it is structurally guaranteed by density. The question is not *whether* the population encounters a replicator but *when*.

Remark 29.4 (The Phase Transition). Once the population enters the basin of attraction of a replicating fixed point, the dynamics change qualitatively. A replicating process produces copies of itself, increasing $\mathrm{CN}(P, \mathcal{N})$. More copies mean more interactions, more reinforcement, higher path amplitudes for the replication pathway, lower effective cost per replication event. The population converges onto the replicating fixed point.

What emerges from the funnel is a population with a qualitatively new feature: *hereditary replication with variation* — the precondition for Darwinian evolution. This is Smith's phase transition, made precise in the GSLT framework.

The new geosphere, in our terms, is the *compact coherence cluster* (Part II, Section 6) centered on the replicating fixed point: the maximal compact subpopulation within which the replicating process and its variants can reach consensus. The phase transition is the crossing of the basin boundary in the logical topology — the moment at which the subpopulation containing the replicator becomes compact, and consensus (here, consensus on the replication strategy) becomes achievable.

Remark 29.5 (The Joint Rise of Assembly Index and Copy Number). After the phase transition, both AI and CN increase together. Selection amplifies processes that replicate more efficiently, which typically requires more sophisticated catalytic machinery — higher AI. And more efficient replication produces more copies — higher CN. The joint biosignature of Assembly Theory is therefore not merely an empirical observation but a *theorem about the post-transition dynamics* near a replicating fixed point, given density of attractors and account-governed selection pressure. This is a key prediction of the framework: wherever a phase transition of this kind occurs, the joint AI-CN signature will be observed.

Chapter 30

Metabolism, Energy Gradients, and the Ecosystem

30.1 The External Gradient as Broken Symmetry

The replicating fixed point of Section 28 is a self-sustaining process, but it is not yet a *living* one in the full sense. It copies itself, but copying costs account resources, and a closed system with no external input will eventually exhaust its account and fall silent. Life requires not just replication but *metabolism*: a sustained coupling to an external energy source that replenishes the account faster than computation depletes it.

The key physical observation, due to Eric Smith, is that the Earth is not a closed system. The Sun presents an enormous influx of energy — a directed, asymmetric gradient between the high-energy photon flux arriving from the Sun and the low-energy thermal radiation leaving to space. This gradient is not merely a source of replenishment; it is a *broken symmetry* in the weight map of the system.

Recall from Part I that the reversibility construction $\mathcal{S} \to \mathcal{S}^\dagger$ requires $w_r^-(\phi) = \overline{w_r^+(\phi)}$ for every rule r — the backward amplitude is the complex conjugate of the forward amplitude. This is the CPT condition: the system is time-symmetric. The solar gradient violates this condition for a specific class of rules.

Definition 30.1 (Gradient-Coupled Rule)**.** *A rewrite rule r is* gradient-coupled *if its forward amplitude is boosted by an external energy flux Φ:*

$$w_r^+(\phi,\, \Phi) \;=\; w_r^{+(0)}(\phi) \cdot e^{\beta\Phi}$$

where $w_r^{+(0)}$ is the amplitude in the absence of the gradient and $\beta > 0$ is a coupling constant. The corresponding backward amplitude remains $w_r^-(\phi) = \overline{w_r^{+(0)}(\phi)}$, unaffected by the gradient.

Remark 30.1 (CPT Violation as the Signature of Life). A gradient-coupled rule has $|w_r^+| > |w_r^-|$: forward transitions are more probable than backward ones. The system is driven away from equilibrium. This violation of the CPT condition of $\mathcal{S}^\dagger$ is not a defect of the framework but a *signal*: it marks the transitions through which the external gradient does work on the system. A living agent is precisely one that maintains a sustained CPT-violating coupling to an external gradient. An agent in thermal equilibrium with its environment — one for which $|w_r^+| = |w_r^-|$ for all rules — is, in this framework, dead.

30.2 Producers: Direct Gradient Coupling

The simplest living agents are those that couple directly to the external gradient. In biological terms these are autotrophs — plants and photosynthetic bacteria. In the GSLT framework they are agents whose metabolic rules have the gradient Φ as part of their minimal context.

Definition 30.2 (Producer)**.** *A* producer *is an agent A that has at least one gradient-coupled rewrite rule r whose minimal context K_Φ encodes the external gradient directly:*

$$\langle A, \mathbf{A} \rangle \xrightarrow{K_\Phi} \langle A', \mathbf{A} + \delta\mathbf{A} \rangle$$

where $\delta\mathbf{A} > \mathbf{0}$ componentwise. The account grows: the producer converts gradient energy into stored computational resource.

Remark 30.2. The minimal context K_Φ for a producer rule is the encoding of the gradient itself — photons, in the biological case. This is an instance of the open synchronization tree structure from Part I: the producer's metabolic transition is an edge in $\mathcal{ST}_o(A)$ labeled by the gradient context. The gradient is the environment; the producer is the program that responds to it. In the Rho calculus, $K_\Phi[-] = \Phi!(-) \mid [-]$ where Φ is the channel on which gradient quanta arrive. A producer listens on Φ and credits its account on each received quantum.

30.3 Consumers and the Food Chain as Account Flow

Once producers have stored account resources in their processes, other agents can couple to them rather than to the raw gradient. These are consumers — heterotrophs in biological terms.

Definition 30.3 (Consumer)**.** *A* consumer *is an agent B that has gradient-coupled rules whose minimal context encodes not the external gradient directly but the* presence of a producer *(or of another consumer at a lower trophic level):*

$$\langle B,\, \mathbf{A}_B\rangle \otimes \langle A,\, \mathbf{A}_A\rangle \longrightarrow \langle B',\, \mathbf{A}_B + \delta\mathbf{A}\rangle \otimes \langle A',\, \mathbf{A}_A - \delta\mathbf{A}\rangle$$

Account is transferred from A to B via interaction. The total account is conserved across the interaction (no external gradient is tapped); the distribution changes.

Remark 30.3 (Metabolism as Redistribution). Smith's observation is precisely captured by this definition. The Sun does not directly power animals: it powers plants, which store account, which animals then redistribute. Metabolism at the ecosystem level is not replenishment but *redistribution* of account through a population, with producers as the sole point of contact with the external gradient.

The food chain is therefore a *directed graph of account transfers*: an edge from A to B means B consumes A, transferring stored account from A's process to B's. The roots of this graph (the nodes with no incoming edges) are the producers, coupled to the gradient. The leaves are the apex consumers. Decomposers close the cycle by returning stored account to forms accessible to producers.

30.4 The Ecosystem as Account-Flow Network

Definition 30.4 (Ecosystem)**.** *An* ecosystem *is a compact coherence cluster $\mathcal{C}$ (in the sense of Part II) together with:*

(i) a gradient Φ providing an external account source;

(ii) a set of producers coupling $\mathcal{C}$ to Φ;

(iii) a directed account-flow graph $\mathcal{F}$ on $\mathcal{C}$ encoding who consumes whom.

The ecosystem is viable *if the total account inflow from Φ through the producers exceeds the total account dissipated by the population's rewrites over any sufficiently long time horizon.*

Remark 30.4 (The Nerve as Metabolic Map). The nerve $\mathcal{N}$ of Part II — the simplicial complex of overlapping coherence clusters — now has a metabolic interpretation. Each vertex of $\mathcal{N}$ is a coherence cluster; each edge is a shared overlap. When the clusters are populations of living agents, the edges of $\mathcal{N}$ are also the channels of metabolic exchange: the overlapping agents are those that can coherently communicate *and* transfer account resources. The nerve is simultaneously the epistemic map (who can reach consensus with whom) and the metabolic map (who can feed whom). The two structures are not coincidentally aligned: coherent communication is the precondition for reliable account transfer, since a corrupted message about food availability is as dangerous as no message at all.

30.5 Trophic Level as Causal Depth

Proposition 30.1 (Trophic Level as Open Synchronization Depth)**.** *The* trophic level *of an agent B in an ecosystem is the minimum path length in the account-flow graph $\mathcal{F}$ from a producer to B. This is precisely the minimum causal depth $d^o_{\min}$ in the open synchronization tree of B with respect to the gradient context K_Φ:*

$$\mathrm{TL}(B) \;=\; d^o_{\min}(K_\Phi, B).$$

Remark 30.5. Plants have trophic level 1: one context step separates them from the gradient. Herbivores have trophic level 2: their metabolic coupling to the gradient passes through the plant context. Carnivores that eat herbivores have trophic level 3; and so on. Each trophic level is one additional layer of context nesting in the open synchronization tree — one more environmental intermediary between the agent and the ultimate energy source. The assembly index and the trophic level are thus both instances of the same underlying quantity: causal depth in an open synchronization tree, measured from different roots. Assembly index is depth from the elementary terms. Trophic level is depth from the gradient.

30.6 Viability of the Replicating Fixed Point

The phase transition to life, as described in Section 29, produces a replicating fixed point centered in a compact coherence cluster. We can now state more precisely what it means for this fixed point to be *viable* in the long run.

Conjecture 30.1 (Metabolic Viability of the Fixed Point). A replicating fixed point $\mathbf{Y}(F)$ is *metabolically viable* if there exists a gradient Φ and a producer sub-process $F_\Phi \subseteq F$ such that the account inflow from Φ via F_Φ exceeds the account cost of replication and maintenance:

$$\mathbb{E}\big[\delta \mathbf{A}_{\text{catabolic}}\big] \;>\; \mathbb{E}\big[\mathbf{c}_{\text{anabolic}}\big]$$

where the expectations are over the path integral of the process. Only metabolically viable fixed points persist after the phase transition; the others exhaust their accounts and go silent.

> *Remark* 30.6 (Selection for Metabolic Efficiency). Conjecture 30.1 implies that the phase transition does not merely select for replication but for *metabolically sustained replication*. The Darwinian competition after the transition is not only about who replicates fastest but about who maintains the best ratio of catabolic yield to anabolic construction cost. High assembly index processes win this competition when their catalytic sophistication yields proportionally higher catabolic returns — which is precisely why complexity increases after the phase transition. The joint rise of AI and CN is driven not just by selection for efficient replication but for efficient *metabolism*: the ability to extract more account from the gradient per unit of construction cost.

Chapter 31

Open Gaps

Gap 1. Density of replicating terms (Conjecture 29.1). The argument that the added replication machinery can always be hidden from any given HML formula requires a careful analysis of which formulae can probe the replication channel. In particular, if $\phi = \langle K \rangle \psi$ where K involves the replication channel, the argument fails. A precise density theorem would need to show that such formulae form a meager (first-category) set in the logical topology, so that "generically" ϕ does not probe replication.

Gap 2. Defining the elementary terms. Proposition 27.1 requires a designated set E of elementary terms. In the chemical setting, these are atoms or monomers. In the Rho calculus, the natural candidates are the behavioral primes of Part II (Section 4): the irreducible terms from which all others are built by interaction. Confirming that this identification gives assembly indices matching those of Assembly Theory requires working through the combinatorics of the Rho calculus prime factorization.

Gap 3. The basin of attraction. The claim that a population entering a neighborhood of Rep is "sucked in" to the replicating fixed point requires a Lyapunov-style argument: a function on the process space that is monotonically decreasing along trajectories near the fixed point. The natural candidate is the distance d_{HML} to Rep, but showing it decreases requires assumptions on the interaction dynamics.

Gap 4. The compact cluster as the new geosphere. The identification of Smith's fourth geosphere with the compact coherence cluster centered on the replicating fixed point requires showing that this cluster is indeed compact (in the sense of Part II) and that it persists under the perturbations of the selection dynamics.

Gap 5. Crossover as a variant of comm. The claim that genetic recombination is one definitional step from the comm rule requires specifying precisely what "fragment substitution" means in the Rho calculus and showing that the resulting operator has the recombination properties of a genetic crossover (fitness inheritance, schema theorem, etc.).

Gap 6. Gradient-coupled rules and the weight map. Definition of gradient-coupled rules modifies the weight map of Part I by introducing an external parameter Φ. A precise treatment requires extending the weighted GSLT framework to allow time-varying or externally-driven weight maps, and showing that the path integral remains well-defined under this extension.

Gap 7. Metabolic viability (Conjecture 30.1). The condition that catabolic inflow exceeds anabolic cost in expectation requires computing the path integral of a driven open system — a substantially harder problem than the closed-system path integral of Part I. The quantum Gillespie algorithm of Part I provides a simulation method, but an analytic viability criterion requires further development.

Gap 8. Trophic level as causal depth (Proposition 30.1). The identification of trophic level with $d^{o}_{\min}(K_{\Phi}, B)$ requires showing that the account-flow graph $\mathcal{F}$ is faithfully captured by the open synchronization tree structure. In particular, it requires that every account transfer corresponds to a minimal-context transition, which is plausible but needs verification.

Chapter 32

Discussion

32.1 Relation to Existing Work

Assembly Theory [16] has attracted both enthusiasm and criticism. The enthusiasm is for the biosignature criterion: a single computable quantity that distinguishes living from non-living matter. The criticism is precisely the two gaps identified in Section 27: the lack of a spatial framework for copy number and the undefined identity criterion for copies. The present framework addresses both.

Smith's geosphere proposal [17] is grounded in thermodynamics and network autocatalysis rather than computation. The connection we draw is not a reduction of Smith's proposal to computation but a *parallel structure*: the phase transition in thermodynamic space corresponds to the basin-crossing in process space, and the thermodynamic stability of the autocatalytic network corresponds to the compactness of the coherence cluster. Crucially, Smith's observation that metabolism is fundamentally about the redistribution of energy from the solar gradient through an ecosystem — rather than merely replenishment of individual agents — is captured precisely by the account-flow graph of Section 30: producers couple to the gradient, consumers redistribute the stored account, and the ecosystem is viable when inflow exceeds dissipation. The Prigogine notion of dissipative structures sustained far from equilibrium maps directly onto the CPT-violating gradient-coupled rules: a living agent is one that maintains a broken symmetry between its forward and backward transition amplitudes by continuously processing an external gradient. Whether these are the same phenomenon viewed from two angles or genuinely distinct mechanisms is a question for future work.

The connection between the Rho calculus Y combinator and biological replication echoes earlier work on process calculi as models of biological systems [18], but with a sharper mathematical claim: replication is not merely *modeled* by a fixed point but *is* the fixed point of the duplication map in any sufficiently expressive interactive GSLT.

32.2 The Deeper Implication

The origins of life argument, if completed, would establish something philosophically striking: that the emergence of life is not a contingent accident of terrestrial chemistry but a *structural inevitability* of any sufficiently large interactive computational system coupled to an external gradient.

Any universe described by a Turing-complete interactive GSLT that is driven away from equilibrium by an external energy source — a gradient that breaks the CPT symmetry of the isolated system — will, given sufficient time and population, produce metabolically viable replicating fixed points. The density of replicators in the logical topology means there is no avoiding them. The gradient means there is always account available to sustain them once they appear. The phase transition to life is not an unlikely event; it is the inevitable consequence of wandering in a space dense with attractors, in the presence of a broken symmetry that keeps the account replenished.

What emerges from the transition is not merely a replicating process but an entire ecosystem: a compact coherence cluster with a gradient-coupled producer layer, a food web of account redistribution, and the beginnings of the trophic hierarchy that is causal depth in the metabolic open synchronization tree.

This connects all four parts into a single narrative: the physics of Part I governs the wandering and provides the weight-map structure in which gradient coupling is the breaking of CPT symmetry; the cluster ontology of Part II provides the coherence clusters that become ecosystems; the sentience and consciousness of Part III describe how the agents produced by the phase transition come to know and feel; and the present part explains how those agents came to exist and how they sustain themselves.

Part VI

Ecology as Cognitive Architecture

> *Every part so far has needed an agent and has helped itself to one. This part builds the agent out of the same material as everything else, and then watches what happens when a learner has to pay for what it learns. Four chapters, four sizes of learner: the individual, the game it plays, the composite, and the network of exchanges that binds composites into something with an inside. The argument is the most explicitly worked in the book. It is also the only place where a claim is checked against a number.*

Why the Unit Keeps Moving

The preceding parts have each required a knower and each declined to say what one is. Causality gives a physics, and someone must be there to do the physics. The argument about how scientists are misled needs an agent with a finite budget running the scientific method against a population — and the whole force of the conclusion depends on that budget being real — yet the agent itself stays a placeholder, a thing that has theories and can afford some number of experiments. Nothing so far is made of the same stuff as the world it studies.

That is the gap these four chapters close. If computations are ontologically isolated, then the environment of a computation is other computations, and a learner has no privileged vantage from which to observe: it is a term sitting in parallel with the term it wants to know about, and the only way to find out anything is to interact. Interaction costs. A computation that cannot pay stops. In the Pledge i said that access to breath is part of what connects us to every living thing, and that a machine loses speed breath chess the moment we cut its power. The chapters below take that seriously enough to write down the ledger. The scientist here has a bounded supply of tokens, no way to mint more, and an environment in which the only other tokens are inside other computations. Eating is what it is called when one computation breaks another open.

Two consequences arrive immediately and neither was designed in. The first is that a specimen affords exactly one measurement, because the measurement ends it — which forces hypotheses to be about *kinds* rather than individuals, and so forces the hypothesis language to be a logic of namespaces rather than a logic of things. The second is that being right is not the same as being fed. The true theory of an environment can be, and in the worked example demonstrably is, a bad purchase: dominated by a cruder theory that costs a fraction as much and wins nearly as often. This is the Pledge's point about classification arriving with a price tag attached. Bisimulation says two things are the same when no experiment tells them apart; put a budget on experiments and the equivalence you can actually afford is coarser than the one that is true, and the gap between them is not ignorance but economics.

The four chapters are one argument told at four scales, and the reason it has to be told four times is a single proposition. A learner revising its theory in small steps can never leave the region it started in — the geometry of the hypothesis space is a tree, distances are ultrametric, and a walk of short steps stays inside its ball. The individual is confined by construction. So the unit of learning has to move. It moves to the lineage, where reproduction is recombination on quoted code and a crossover is a revision move an individual could not make. It moves again to the composite, where two learners share a namespace and the overlap is where cooperation and competition both live. It moves a fourth time to the network, where composites with incompatible currencies must convert, and the cycles of the conversion graph turn out to be an error-correcting code — which finally gives a criterion for where one individual ends and the next begins. At the top the picture closes on itself: the network is an individual, and an individual can be a scientist.

A reader who wants the shape without the machinery can read the opening and closing sections of each chapter and skip the interior. The propositions are stated precisely enough to be wrong, which is the standing commitment of the Turn, and several of them are checked against exhaustive computation rather than argued for. Where a result rests on a modelling choice rather than on the framework, the chapters say so.

Chapter 33

The Mortal Scientist

The preceding parts asked what a physics is and why a scientist inside one would be misled. Both took the scientist for granted. This chapter builds one.

The constraint that does the work is stated in a single sentence: a computation can only find out about another computation by interacting with it, and interacting costs. Everything else follows from taking that seriously and refusing to add anything not already forced. There is no dataset, because a dataset is somebody else's observations supplied for free. There is no scorekeeper, because a scorekeeper would be an authority standing outside the world telling the learner how it is doing, and no such authority is available across the cut. What is left is a term in parallel with other terms, holding a finite supply of tokens, spending them to look, and dying when it runs out.

Three things then have to be supplied and only one of them turns out to be a choice. The environment is fixed by ontological isolation. The hypothesis language is not chosen but *generated* — fed the presentation of the calculus, the OSLF construction emits a logic whose formulae denote exactly the equivalence classes the learner could distinguish by experiment, which is the Pledge's "anything Bob can do Alice can do" turned into syntax. The motive is the one genuinely free decision, and we make it metabolic: the learner that predicts badly cannot afford to look, and the learner that cannot look starves.

What comes out is stranger than what went in. Destructive measurement forces induction. Foraging bounds concealment, so that perfect privacy is available only to something already dying. Reproduction turns out to be reflection — a genome is a quotation of code, inert because quotation does not reduce — and sex is a defence against a predator's hypothesis language rather than a mechanism of variation. And the ecology divides, without being told to, into the two kinds of organism Eric Smith finds in the biosphere: those that eat only external energy, and those that eat each other.

33.1 Introduction

33.1.1 The scientist as a learner

Most computational accounts of learning begin with a dataset: a supply of observations that exists independently of the learner, whose acquisition is free, and whose acquisition does not disturb the source. Given such a supply the problem is one of induction — find the hypothesis that best accounts for what is already in hand — and the interesting questions are statistical.

A scientist has none of this. There is no dataset waiting; there is a world, and the scientist must decide what to do to it in order to make it answer. The characteristic activity of the scientific method is therefore not induction but *the design of experiments*: choosing an intervention whose outcome would discriminate between the hypotheses currently in play, performing it, and letting the result decide. Induction is what happens afterwards, and it is the easier half.

Three features of that activity are ordinarily abstracted away and are exactly what we want to keep.

Observation is action. To learn something about a system one must do something to it. There is no passive channel. Every measurement is an intervention, and the system afterwards is not the system before.

Inquiry is expensive. Experiments consume resources that could have been spent otherwise, and a scientist with a finite endowment can run only finitely many. Which questions are worth asking is therefore not a philosophical question but a budgetary one.

The scientist is in the world it studies. It is not a disembodied observer looking on from outside. It is made of the same stuff as its subject matter, subject to the same constraints, and visible to whatever else is looking.

A model that keeps all three will look less like statistics and more like ecology, and that is the model we build.

33.1.2 What such a model must supply

Before any formalism, here is what has to be on the table.

(R1) **An environment, and a ceiling on what can be known of it.** One must say what the learner is learning *about*, and — crucially — what the best possible outcome of inquiry would be. Without a ceiling there is no way to say whether a learner is doing well.

(R2) **A language for hypotheses.** Hypotheses must be stated in something. That something should be able to express every distinction the learner could in principle observe, and no more. A language too coarse leaves the learner unable to say what it can see; a language too fine sends it chasing differences that no experiment could ever settle.

(R3) **A mechanism from hypothesis to verdict.** There must be an account of what an experiment *is*, how a hypothesis determines one, and how the outcome comes back as confirmation or refutation — including an honest account of the case where nothing comes back at all.

(R4) **A motive.** Something must make the learner prefer better hypotheses. If this is an external scorekeeper handing out rewards, the model has outsourced the very thing it set out to explain, and one may reasonably ask who scores the scorekeeper.

33.1.3 Four answers

Our answers are as follows, and the remainder of the introduction says why each is less a choice than it appears.

(R1) The environment is other computations. We take as premise that computations are ontologically isolated: a computation has no access to anything but through interaction. It follows immediately that the environment of a computation is other computations, and that what one can come to know of another is bounded above by the equivalence induced by the contexts one can build against it — context-labelled bisimulation [22]. That is the ceiling R1 demands, and it is derived rather than posited. We fix the model of computation as the `rho` calculus [28].

(R2) The hypothesis language is generated, not designed. Given the ceiling, R2 has a canonical answer: the language must be adequate for context-labelled bisimulation, which is exactly what a logic generated from the calculus' own presentation gives. Feeding the `rho` calculus' presentation to the OSLF family of algorithms [29] emits namespace logic [27], whose formulae denote bisimulation equivalence classes. We admit the whole of it, including the structural predicates — and admitting those is what gives the learner senses (§33.3).

(R3) An assay is a guarded receipt. We work in the variant of the `rho` calculus whose for-comprehension carries a `where` clause [21, §9], so a formula may sit in guard position on a receipt. A hypothesis installed as a guard *is* an experiment, and a confirmation is a rendezvous. The checker is the reduction rule; nothing is interpreted, and the test is metered by the same apparatus that meters everything else.

(R4) The motive is metabolic. Cost accounting as a monad [23] makes the learner mortal: every step is paid for, and a computation that cannot pay stops. We then gamify. Tokens are conserved and never minted; the learner holds a bounded internal supply; further stacks lie distributed through the environment, including inside other computations; and breaking a computation open to harvest its stack is the analogue of eating. This is the answer to the scorekeeper problem. A hypothesis says *where the tokens are*, and the reward for a good one is the tokens themselves. Predictive accuracy and metabolic success become the same quantity, measured in the same units, with nobody adjudicating.

That is the whole of what we put in. The rest of the note is largely the discovery that we had no choices left: §33.14 audits which features are forced by this list and which remain free, and the list of free parameters is short.

33.1.4 What we are claiming, and what we are not

We are claiming that the commitments just listed determine almost all of the apparatus one would otherwise have to invent. The interesting sections below are the ones in which we find we had no choices left.

We do treat learning, and §33.6 is where. What we give there is the geometry of the hypothesis space and the constraints any revision policy must satisfy in it — that the space is an ultrametric tree and so admits no gradient, that incremental revision provably cannot leave the ball it begins in, that the cost of a move is monotone in its distance, and that the branching a searcher can survey is bounded by its wealth — together with a catalogue of strategies compatible with those constraints. We are *not* claiming to have fixed a policy, proved a convergence theorem, or said anything about statistical efficiency. Which policy is best is an ecological question and depends on the distribution of §33.13; the policy itself remains one of only three genuinely free parameters we find. We do treat reproduction (§33.12), but we do not treat inheritance of acquired characteristics: whether a lineage's transmitted material can carry what an individual learned is left open, since the empirical case against a strict germ–soma barrier [39] is strong enough that we would rather not build on either answer.

We are also not claiming a novel logic. The logic is the one obtained for free from the encoding, in exactly the sense of [20]: given the presentation (Σ, E, R) of the rho calculus, the OSLF algorithms emit a logic, and it is the same logic whether one is classifying knot diagrams or hunting for tokens. What is new here is the reading, the game, and three consequences of the game that we did not anticipate.

33.1.5 Why adequacy is the right demand

R2 asked for a language neither too coarse nor too fine, and it is worth saying why that pair of failures is the right pair to worry about, since adequacy is a technical condition that can look like a formality.

A learner separated from its environment by an interaction cut has exactly one channel of access: what interaction reveals. If its hypothesis language is too coarse, there are distinctions it can observe but cannot state, and it is blind in a way its own faculties do not require. If the language is too fine, it will form hypotheses that differ on environments no experiment could separate, and will spend tokens — which by R4 are its life — discriminating between what are, for it, the same world. Adequacy rules out both at once: the formulae separate precisely what interaction separates.

This is why the logic must be generated from the same presentation that generates the calculus rather than designed. A designed logic has no reason to be adequate. A generated one is adequate whenever its target specification carries the connectives adequacy needs [20, §1.3], and when it does not, the failure is legible and located.

33.1.6 Ideal logic, checkable fragment, mortal scientist

We inherit the methodological commitment of [20]: the *ideal* logic, adequate for the idealised calculus, is what defines the invariants; *fragments*, restricted so that model checking terminates, are what decide them. In the present setting this distinction stops being methodology and becomes the subject matter. A mortal scientist cannot afford the ideal logic. What fragment it can afford is fixed by its budget, and the fragment it can afford *is* its inductive bias. We return to this in §33.5, and it is the reason the note's central objects are approximants rather than limits.

33.2 Isolation, the cut, and what a computation can know

33.2.1 The premise

> Computations are ontologically isolated. Therefore the environment of a computation is other computations.

We take this as a premise rather than argue for it. Its immediate formal content is that a world factors across an interaction cut. In an interactive graph-structured lambda theory [21], an interaction cut K separates a program from its environment: application for the lambda calculus, parallel composition for CCS, the pi calculus, and the rho calculus. We write

$$W \;\equiv\; S \mid E$$

for a world factored into a scientist S and its environment E. Nothing here privileges S: the factorisation is a choice of which side one is standing on, and in §33.10 we will take seriously that E contains further scientists for whom S is environment.

33.2.2 Three grades of access, and only three

The premise has a sharp consequence that the rest of the note leans on. S does not hold E's term. It cannot read E's structure by inspection, because inspection is not an operation the calculus provides across a cut. What it has instead is exactly three things, and it is worth being explicit that the calculus offers no fourth.

Perception. S may evaluate structural predicates at the *interaction surface* — the outermost interaction structure of E, the part exposed at the cut. This is what senses are: one sees the skin of a rock, not its interior.

Interaction. S may build a probe $C[-]$, plug E into it, and observe the steps that $C[E]$ can then take. This is the only route to behaviour. It is worth being careful here, because two different contexts are in play and the note has more than one use for each. The probe C is what S constructs and is a context *around* E. The label on the generated modality $\langle K_j \rangle$ of §33.3 is a context around the *redex* — everything in $C[E]$ except the interacting pair — so it includes S's own remaining code and whatever of E is not participating. The two coincide exactly when the redex sits at the cut, which is the case S can engineer and the reason probing is informative at all; they come apart as soon as S attends to a step internal to E. We write $\langle K_j \rangle$ throughout for the generated modality and reserve $C[-]$ for probes.

Reflection. If S *holds* a name, it may look inside it, because in a reflective calculus a name is a quoted process and the name predicate $@\varphi$ sees into that quotation without communicating at all.

The third is the interesting one and it is available only for names S has come to hold. This gives the distinction the note turns on: **observation versus disclosure**. Behaviour is what one observes across a cut; structure is what one is given, or takes. Perception reads the surface of things one does not hold; reflection reads the interior of things one does. The operation converting the second into the first is the subject of §33.7, and it is fatal to the thing converted.

33.3 The logic, and senses

33.3.1 What the generator supplies

We recall only what we use, following [20, §2]. The rho calculus has term formers 0, $*n$, $n!(Q)$, $\texttt{for}(x \leftarrow n)P$, parallel composition, and $@P$; parallel composition carries an associative–commutative equational theory and there is a single communication rewrite. Feeding this presentation to the generator yields:

Structural connectives. One connective per term former. We write $\mathrm{out}(n)\varphi$ for the predicate holding of $n!(Q)$ with $Q \models \varphi$, and $\mathrm{in}(n)\varphi$ for the analogue on receipts. Because $|$ is associative–commutative, the generated connective for parallel composition is a separating conjunction:

$$P \models \varphi \mid \psi \iff P \equiv Q \mid R \text{ with } Q \models \varphi,\ R \models \psi.$$

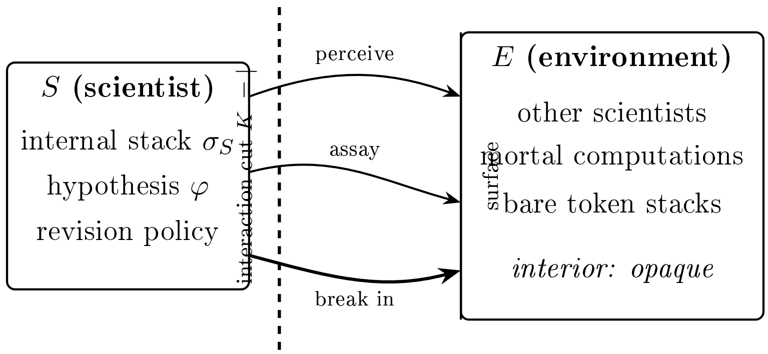


Figure 33.1: The cut. Perception evaluates structural predicates at the surface; an assay reaches across and perturbs; a break-in passes the surface and is fatal to what lies behind it. There is no fourth route, and the three are ordered by both what they reveal and what they destroy (§33.9).

Behavioural modalities. One primitive modality $\langle K_j \rangle$ per rewrite rule and choice of redex position, labelled by the one-hole context in which the step fires. The index is not decoration and we keep it. The rho calculus has a single interaction rule, so all the information in a label is in the *position*: for a world $W \equiv K_j[\,\texttt{for}(x \leftarrow n)P \mid n!(Q)\,]$ the label K_j is everything else. Consequently $\exists j.\ \langle K_j \rangle \varphi$ and $\forall j.\ [K_j]\varphi$ quantify over the redex positions of W, which is what gives the modal layer its force: an action-labelled logic sees one label where this one sees as many as there are enabled interactions. Where a formula below is written with a bare $\langle K \rangle$ it abbreviates the existential, and where the quantifier is doing work we write it out.

Name predicates. A name is a quoted process, so

$$n \models @\varphi \iff n = @P \text{ with } P \models \varphi,$$

which is namespace logic [27]: one describes a namespace by a property rather than enumerating it. §33.8 shows this is not a convenience but a requirement.

Restriction, fixed point, grading. The hidden-name quantifier $P \models \mathsf{H}x.\varphi$ iff $P \equiv \texttt{new}\ x\ \texttt{in}\ Q$ with $Q \models \varphi$ and x fresh; the greatest fixed point $\nu X.\varphi$ with X positive; and graded separation $\varphi \mid_k \psi$, which splits a term under H with exactly k shared restricted names [20, §2.3].

> *Remark* 33.1 (Restriction is sugar). We treat `new` as syntactic sugar throughout. In a reflective calculus a name is a quoted process, so a “fresh” name is manufactured by a structural scheme rather than drawn from a stock of atoms, and freshness is relative to a scope rather than global. We remain agnostic about the scheme, subject to one constraint: it must be *decentralised*, manufacturing names

from material a computation already holds rather than by consulting a registry. That constraint is not an engineering preference. A central name server would be an oracle sitting across the cut and consultable by every computation, which contradicts the isolation premise of §33.2 outright.

Two consequences run through the rest of the note. Names have *structure*, so a name predicate sees all the way into one and the hidden-name quantifier is descriptive rather than opaque; and names have *size*, so by Definition 33.2 they carry a running cost. Concealment is therefore never secrecy, only price — which is why δ was introduced as a cost parameter and not as a wall.

33.3.2 The structural predicates are admitted, and they are senses

In [20] the structural predicates are wielded by an analyst standing outside the term, holding it in hand, free to count its gates and test its decompositions. That is not the scientist's situation, and admitting the structural predicates without saying where they may be applied would hand the scientist an oracle and destroy the isolation premise of §33.2.

The discipline that saves them is that a sense is a structural predicate evaluated *at the surface*.

Definition 33.1 (Depth). *The* depth $d(\varphi)$ *of a structural formula is the maximal nesting of term-former connectives and prefixes occurring in it, with* $d(\top) = 0$, $d(\mathrm{out}(n)\varphi) = d(\mathrm{in}(n)\varphi) = 1 + d(\varphi)$, $d(\varphi \mid \psi) = \max(d(\varphi), d(\psi))$, $d(\mathsf{H}x.\varphi) = 1 + d(\varphi)$, *and* $d(@\varphi) = d(\varphi)$.

Definition 33.2 (Sensing, and its price). *For* $W \equiv S \mid E$, *the scientist* S *may evaluate* φ *against* E *whenever* $d(\varphi) \leq \alpha$, *where* α *is* S*'s* acuity; *the evaluation debits* S*'s stack by* $\kappa_{\mathrm{sense}}(d(\varphi))$, *a monotone schedule. Acuity is not a fixed endowment:* α *is whatever* S *can currently pay for.*

Two comments. First, the depth grading is what keeps perception from being an oracle: seeing deeper costs more, so the interior of E is not free, it is merely expensive, and past a point unaffordable. Second, $d(@\varphi) = d(\varphi)$ rather than $1 + d(\varphi)$ is deliberate and records the third grade of access: opening a name one holds is not a passage through a surface, because one already holds it. Reflection is cheap for exactly the reason predation is not.

grade	instrument	applies to	price
perception	$\mathrm{out}, \mathrm{in}, \mid, \mid_k, \mathsf{H}$	the surface of E	$\kappa_{\text{sense}}(d)$, monotone in depth
interaction	$\langle K_j \rangle \varphi$	the behaviour of E	one rendezvous per step, and time
reflection	$@\varphi$	names S holds	a computation on a name; no communication

Table 33.1: The three grades of access. The ladder is also an ordering by invasiveness (§33.9), and in the opposite direction.

33.4 Hypotheses, assays, and the `where` clause

33.4.1 The embedding, in operational position

An embedding of the logic into the calculus is what lets `rho` computations compute over hypotheses. The obvious construction — reify formulae as data, write an interpreter — works but is unsatisfying, because the scientist must then carry the interpreter and the interpreter is not part of the theory. The variant of `rho` whose for-comprehension carries a `where` clause [21, §9] does better. Its general form is

$$\texttt{for}\big(p_{11} \leftarrow c_{11} \,\&\, \cdots\,;\, \cdots\,;\, p_{1n} \leftarrow c_{1n} \,\&\, \cdots \;\texttt{where}\; \vartheta\big)\, P$$

with & joining within a group, ; separating groups, and ϑ a condition drawn from the booleans over ground formulae, spatial connectives, and the context-labelled modality, in which the variables bound by the preceding clauses may be mentioned. A guard is a formula in operational position. Hence:

> *A hypothesis installed as a guard is an experiment, and a confirmation is a rendezvous.*

The checker is the reduction rule. Nothing is carried, nothing is interpreted, and the cost of testing a hypothesis is metered by the same apparatus that meters everything else — which is essential here, since an unpriced hypothesis test would be an oracle and would collapse the entire motivational story of §33.7.

33.4.2 The assay

Definition 33.3 (Assay)**.** *An* assay *is a pair* (K, φ) *of a probe context* K *and a predicted property* φ*, realised on a probe channel* p *as*

```
  K[ ... ] // the probe: perturb the environment
| for (x <- p where phi(x)) { confirm!(*x) } // prediction holds
| for (x <- p where ~phi(x)) { refute!(*x) } // prediction fails
```

The negation in the second guard is what makes this more than an expression of hope. In a concurrent setting a guard that simply fails to fire is indistinguishable from a guard still waiting, so a naive encoding of falsification cannot tell refutation from patience. The complementary pair repairs this.

Proposition 33.1 (Trichotomy)**.** *Let $A = (K, \varphi)$ be an assay on probe channel p, with φ decidable on the received term within the ambient budget. Then for each message arriving on p exactly one of the two guards is enabled, and the assay yields one of three outcomes:* CONFIRM*,* REFUTE*, or $\perp$, where $\perp$ occurs exactly when no message arrives on p before the budget for A is exhausted.*

Proof sketch. φ and $\neg\varphi$ partition the terms on which both are decidable, so at most one guard is enabled per message and the two receipts do not race. If a message arrives and the budget suffices to decide φ on it, one guard fires. Otherwise no output occurs on either `confirm` or `refute`, which is the residual case. □

We stress that $\perp$ is not a defect. It is the honest residue: the environment did not answer, and no amount of logical apparatus can distinguish that from an environment that would have answered later. What it does is separate that genuine ignorance from falsity, which the naive encoding conflates.

33.4.3 Refutation is budget-relative, and that fixes the fragment

Proposition 33.2 (Asymmetry of the modalities)**.** *Two distinct asymmetries hold, and it is worth separating them because the note uses both.*

1. Over positions. *Let $\varphi = \exists j.\, \langle K_j \rangle \psi$. Then* CONFIRM *is a finite witness for φ — exhibit the position and the step — while refuting it requires ruling out every position, and for $\varphi = \forall j.\, [K_j]\psi$ the polarity reverses: a single position is a finite refutation and no finite run confirms.*

2. Over time. *For a fixed position j, $\langle K_j \rangle \psi$ is confirmed the moment the step fires, but its negation is not finitely witnessed at all, because a step that has not yet become enabled is indistinguishable from one that never will. This is Proposition 33.1's $\perp$ seen from the other side.*

Remark 33.2 (Which asymmetry does which job)*.* The first clause is what bounds the testable fragment, since nesting $\exists j$ under $\forall j'$ is what modal depth counts, and it is the clause invoked immediately below and again in §33.12. The second is what makes the complementary guard pair of Definition 33.3 necessary rather than fussy: without the negated guard, clause 2 says a guard that has not fired is

> uninterpretable. Stating the proposition with an uninstantiated K conflates the two, and the conflation is easy to miss because both conclusions are true.

The consequence is the note's first structural fact about mortality. A scientist with a finite budget can only run assays whose outcome is decided within that budget, so the fragment of the ideal logic it can actually test is the one bounded in modal depth and fixed-point unfolding. That fragment is not adequate — a logic whose model checking terminates never is, for a Turing-complete host [20, §1.3]. Therefore:

> *A mortal scientist necessarily holds a non-adequate theory, and which non-adequate fragment it holds is fixed by what it can pay for. The budget is the inductive bias.*

This is not a limitation we regret. It is the formal content of the observation that finite creatures have finite theories, and it locates the inductive bias somewhere unusually concrete: not in a prior, not in a regulariser, but in a token stack.

33.5 The hypothesis space is the relaxation lattice

We do not need to invent a space of hypotheses. [20, §3.3] constructs one and calls it something else.

Recall the construction. For a process P, a *characteristic formula* χ_P is satisfied by exactly the processes equivalent to P; it turns an equivalence question into a model-checking question. Beneath χ_P sits a lattice of progressively weaker formulae obtained by forgetting parts of it — the recursion, the continuations, a private name — each forgetting yielding a coarser predicate that more processes satisfy. In [20] this lattice is where a classifier's dials live. Here it is the space the scientist moves in.

Definition 33.4 (Revision)**.** *A* revision *is a move on the relaxation lattice:* generalisation *forgets a conjunct and moves down,* specialisation *restores one and moves up. A scientist's trajectory is a walk $\varphi_0, \varphi_1, \varphi_2, \ldots$ on this lattice.*

Definition 33.5 (Stratified distance)**.** *For φ, ψ agreeing on all approximants below stage n and differing at n, set $d(\varphi, \psi) = 2^{-n}$, where the stages are the ν-unfolding approximants $\nu^0 \supseteq \nu^1 \supseteq \cdots$. This is an ultrametric.*

The stratification is not chosen for convenience: it is how bisimulation is already approximated, it grades with modal depth, and by Proposition 33.2 modal depth is what the budget bounds. Distance, depth, and cost are therefore one graded structure rather than three, which is the main reason to prefer this metric to a measure-theoretic one on denotations.

Proposition 33.3 (The limit of inquiry is unaffordable)**.** *Let E be an environment whose behaviour is not eventually structural in the sense of [20, Rem. 6] — that is, whose recursion follows the reduction rather than a finite structural cycle. Then χ_E is not exact at any finite unfolding, and no scientist with a finite token supply can install it as a guard.*

So χ_E — the complete theory of the environment — sits at the top of the lattice as a limit the scientist approaches and never reaches. Science converges *along* the lattice.

Remark 33.3 (When a finite theory is complete)*.* The converse case is worth naming because it is the good one. [20, Rem. 6] observes that a recursion following the *structure* rather than the reduction is exact at a finite unfolding: for a diagram with $2n$ arcs, $\nu^{2n} = \nu$ on the nose. Read for the scientist, this says that a finite theory is complete exactly when the environment's regularity is structural. Recursions that follow the reduction are the unbounded ones. This is a serviceable formal criterion for which subject matters admit closed sciences and which do not, and we suspect it is more useful than it looks.

33.6 Searching the lattice

Section 33.5 supplies a space of hypotheses and a metric on it, and says that a scientist's trajectory is a walk. It does not say how the walk is chosen. That is the revision policy, which we left open as one of the note's three free parameters (§33.14), and we leave it open still. What we do here is narrower and, we think, more useful: the metric is an *ultra*metric, and ultrametric spaces are geometrically unlike the spaces search algorithms are usually designed for. Much of what one would reflexively try is unavailable, and one thing one might not have thought necessary turns out to be forced.

33.6.1 The geometry is a tree

Three facts about ultrametric spaces do the work. Balls are nested or disjoint, never partially overlapping. Every point of a ball is a centre of it, so no point in a neighbourhood is distinguished. And there is no betweenness: for $\varphi \neq \psi$ there is no formula standing partway from one to the other in the sense a convex combination would.

Remark 33.4 (There is no gradient)*.* Gradient methods are not merely a poor fit here; they are not well typed. A gradient presupposes a direction in which one may move by an arbitrarily small amount and a notion of interpolation between neighbouring points, and an ultrametric space supplies neither. What it supplies instead is branching, so the search available on a hypothesis lattice is tree search, and the strategies below are all tree strategies.

This has a consequence which we regard as the central fact governing revision policies.

Proposition 33.4 (Confinement)**.** *Let $\varphi_0, \varphi_1, \dots, \varphi_k$ be a revision walk with $d(\varphi_{i-1}, \varphi_i) < 2^{-n}$ for every i. Then $d(\varphi_0, \varphi_k) < 2^{-n}$: the walk never leaves the ball of radius 2^{-n} in which it began.*

Proof. By the ultrametric inequality $d(\varphi_0, \varphi_k) \leq \max_i d(\varphi_{i-1}, \varphi_i)$, and the maximum is below 2^{-n} by hypothesis. □

In a Euclidean space, sufficiently many small steps reach anywhere; here they reach nothing outside the ball they started in, however many are taken and however long the scientist lives. A policy of incremental revision is not slow at escaping a bad basin — it cannot escape one at all. Whatever ball a scientist's early commitments place it in is the ball it will die in, unless it makes a move that is not small.

We take this up again in §33.12, because it turns a biological mechanism into a computational necessity.

33.6.2 The two parameters

Definition 33.6 (Revision move)**.** *A* revision move *is a pair: a* locus *in the formula tree — which structural predicate is to be varied — together with an* operation *at that locus, weakening or strengthening. The locus fixes the direction; its depth fixes the distance.*

So the two parameters the policy must choose are exactly the two the metric grades.

Distance is backtracking depth. A move of distance 2^{-n} agrees with the current hypothesis on every approximant below stage n and differs at n. Small distance means a deep backtrack and a fine adjustment; large distance means a shallow backtrack and the abandonment of an early commitment.

Direction is a choice of sibling. At a node of the tree the available directions are the one-step refinements the generated logic admits at that locus. The branching factor is therefore fixed by which connectives OSLF emitted, and is not ours to choose.

The price schedule follows without further assumption.

Proposition 33.5 (Cost is monotone in distance)**.** *Evidence gathered by an assay of modal depth k bears on the approximants at stage k and above. A move of distance 2^{-n} therefore invalidates exactly those confirmations obtained at depth n or greater, and the count of invalidated confirmations is non-decreasing as the distance of the move increases.*

A radical revision is expensive in precisely the currency the scientist has been spending: it discards the assays that bought the commitments it abandons. Cheap moves are confined (Proposition 33.4) and unconfined moves are dear, which is the trade-off any policy is negotiating.

Remark 33.5 (Two gradings, one poset)*.* The relaxation lattice and the ultrametric grade the same partial order differently, and conflating them is easy. The lattice orders by logical strength: how much of a characteristic formula has been forgotten. The metric orders by depth: how far down the forgotten material sat. Forgetting a conjunct at stage 1 and forgetting one at stage 5 are both a single lattice step, and are metrically very far apart. Definition 33.6 keeps the two apart by naming the locus and the operation separately.

33.6.3 Strategies

We catalogue rather than recommend, since which policy is best is an ecological question and depends on the distribution of §33.13.

Conservative. Always move the least distance that the last refutation forces. Cheap by Proposition 33.5, low variance, and by Proposition 33.4 permanently confined to its initial ball.

Radical. On refutation, abandon a commitment high in the tree. Unconfined, and expensive in exactly the confirmations it throws away.

Adaptive, with refutation rate as temperature. Jump far when refutations are frequent — a high refutation rate is evidence that a commitment high in the tree is wrong — and refine when they are rare. This is annealing-shaped, but the temperature is not a schedule imposed from outside: it is the ratio of `refute` to `confirm` events, which the assays of §33.4 already produce as messages. We regard this as the default policy a designer would reach for, precisely because its control parameter is already an observable of the system.

Direction selection is the second half of the policy, and the candidates differ in what they optimise.

Blind. Choose a sibling uniformly. Requires no evaluation, and is what a lineage does when the choice function of Definition 33.15 is unbiased.

By formula cost. Prefer siblings whose formulae are cheap to check. This is the minimum description length bias of §33.16 appearing as a search heuristic rather than as an accounting identity, and it has the merit that the scientist prefers hypotheses it can afford to test.

By discriminating power. Prefer a sibling ψ for which some assay separates ψ from the current φ — that is, design the crucial experiment first and let it select the direction. This is the distinguishing-formula problem for bisimulation inequivalence [30], and it is the most classically scientific of the options: it maximises the information a single assay returns.

By expected yield. Prefer siblings which, if true, locate more tokens.

> *Remark* 33.6 (Yield, not truth). The last of these is the one this world actually selects for, and it is worth saying so plainly. A mortal scientist is not rewarded for holding true hypotheses but for holding nutritive ones, so a lineage under selection will drift toward yield-biased direction choice and away from discrimination-biased choice wherever the two disagree. This is the pragmatist objection of Remark 33.22 arriving at a specific mechanism rather than as a general caveat: it is not that the scientist is indifferent to truth, but that its search is steered at every node by what pays.

33.6.4 The search is itself metered

One further constraint is easy to overlook and is not a detail. Choosing a direction requires evaluating candidate siblings, and evaluation is model checking, and model checking is priced. A scientist therefore cannot survey the branching it faces.

Proposition 33.6 (Bounded branching)**.** *The number of siblings a scientist can evaluate before committing to a move is bounded by its stack divided by the cost of checking one. Hence the effective branching factor of its search is a function of its wealth, not of the logic.*

Corollary 33.1 (Poverty is confining)**.** *A scientist whose stack affords few evaluations per move is likelier to take the cheapest available move, and by Proposition 33.5 the cheapest moves are the shortest. By Proposition 33.4 a policy of short moves cannot leave its ball. Impoverishment therefore tends to confinement, and the confinement is not recoverable by patience — only by a subsequent move* *that the scientist could not afford to evaluate.*

We note without developing it that this closes a loop with §33.11: a source-feeder whose inquiry converges accumulates the surplus that would buy a wide survey, and has nothing left to search for; a prey-feeder that needs the survey is the one paying an unbounded epistemic rent.

33.6.5 What escapes

Proposition 33.4 leaves an individual scientist with exactly one route out of a bad basin: a move that is not small, at a cost given by Proposition 33.5, chosen from a branching it cannot afford to survey. That is a poor prospect, and it is the reason §33.12 is not an appendix to this note but a part of its argument. A crossover at a locus high in the term tree is, transported along the characteristic-formula construction, a shallow re-branch in the formula tree — a displacement of exactly the kind an incremental walk provably cannot make, and one no individual pays for, because the cost is borne by the offspring that does not yet exist. The escape is not unconditional: by Remark 33.16 how far a swap can displace is fixed by the recombination discipline, and the disciplines that displace furthest are the ones that kill most offspring.

33.7 The game

33.7.1 Conservation

Tokens are conserved and never minted. Write Θ for the total supply of the world. The cost monad's ledger law and the history monad's record together give, globally,

$$\sigma + \kappa \;=\; \sigma_0 \;=\; \Theta,$$

where σ is the total still held in live stacks and κ the total spent. Spent tokens do not vanish; they migrate into the record. So the free supply is monotonically non-increasing, the accumulation of κ is the arrow of time, and there is no primary producer. The whole game is a finite race.

> *Remark* 33.7 (Science is entropy production). The aggregate effect of every scientist in the world is to convert free tokens into history. This is the precise form of the observation that the cost of remembering is the cost of doing science: the record is made of spent tokens. It also means predation is not profit in any global sense — it is the purchase of someone else's remaining time.

33.7.2 Working in the image

Because rho is Turing complete, cost-accounted rho translates back into rho: there is an internalising encoding $[\![-]\!] : \mathcal{C}(\mathsf{rho}) \to \mathsf{rho}$ that represents the located token stacks in the undecorated calculus [21, §6]. Without loss of generality we may therefore avail ourselves of the cost-accounted syntax where that is convenient and of the channels at which token stacks are located where that is convenient.

Definition 33.7 (Metabolic channel)**.** *Under* $[\![-]\!]$*, a located stack becomes a message at a name. We call that name the* metabolic channel m_P *of the computation* P*, and we call*

$$m_P!(\sigma) \;\mid\; \texttt{for}(t \leftarrow m_P)\{\, debit;\; m_P!(\sigma') \;\mid\; \cdots\}$$

the reclaim–debit–reissue *cycle. A mortal computation is a term equipped with a metabolic channel executing this cycle, with a strictly positive burn rate.*

Remark 33.8 (Metabolism is a race condition). Living requires exposing one's stack as a message, if only for the instant between reclaiming and reissuing. Computing is therefore racing others to one's own stack. We did not put this in; it is what the comm rule does with Definition 33.7.

Definition 33.8 (Harvest, death)**.** *A* harvest *of* P *by* S *is the receipt* $\texttt{for}(t \leftarrow m_P)$ *performed by* S*. It is an ordinary communication and requires no new rewrite rule.* P dies *when it cannot fund its next step: the linear receipt consumed the message it was going to reclaim.*

So eating is a receive, death is starvation, and defence is restriction. All three are already in the calculus.

Remark 33.9 (One ledger). The translation is not cost-free: simulating the metering costs rho-steps. We therefore adopt one metering convention throughout and state it once. *Costs are always read in the decorated calculus* $\mathcal{C}(\mathsf{rho})$*; the image is used only for the mechanics of transfer.* This matters because the central inequalities of §33.9 and §33.13 are statements about ratios of costs, and a non-uniform translation overhead would move their boundaries. It is the same discipline of keeping levels distinct that governs running $\mathcal{C}$ on an $\mathcal{H}$-equipped model.

33.7.3 Predation is the failure of internalisation to be faithful

Here is the observation that we take to be the note's chief structural contribution.

In $\mathcal{C}(\mathsf{rho})$ a located stack is inviolable. That is not an accident of presentation; it is precisely what the located-stack discipline and the ambient-authority fix were built to guarantee. No rewrite takes your stack, because the stack is not the sort of thing a rewrite addresses.

In the image $[\![\mathcal{C}(\mathsf{rho})]\!] \subseteq \mathsf{rho}$ the stack is a message at a name, and any process holding that name may receive it.

These are not the same situation, and the encoding is not faithful against arbitrary rho contexts. It is faithful against *images of* $\mathcal{C}(\mathsf{rho})$ *contexts* — which is exactly the refinement of [22]: an encoding may only be probed by the encodings of source contexts, since arbitrary target contexts see strictly more and would never permit bisimulation preservation. Therefore:

Theorem 33.1 (Predators are non-image contexts)**.** *Let $[\![-]\!] : \mathcal{C}(\mathsf{rho}) \to \mathsf{rho}$ be the internalising encoding and let $C[-]$ be a rho context. If C lies in the image of $[\![-]\!]$ on contexts, then C cannot harvest: the stack of any $[\![P]\!]$ plugged into C is preserved, since the source discipline forbids the corresponding step. If C lies outside the image and holds m_P, it may harvest. Hence the predatory contexts are exactly the non-image contexts holding a metabolic name.*

Proof sketch. Faithfulness of $[\![-]\!]$ against image contexts is the hosting condition of [22]: every step available to $C[[\![P]\!]]$ with $C = [\![C']\!]$ is the image of a step of $C'[P]$ in $\mathcal{C}(\mathsf{rho})$, and no step of $\mathcal{C}(\mathsf{rho})$ removes a located stack from a term. A context outside the image is under no such constraint; a bare receipt on m_P is such a context and takes the stack. □

> *The decorated calculus is the world in which property rights hold. Its image is the state of nature. The ecology lives in the gap, and eating is what the erasure theorem's side condition was protecting against.*

This gives the game a rulebook rather than a stipulation.

Definition 33.9 (The game)**.** *A* game *is a choice of admitted context class $\mathcal{A}$ with $[\![\mathcal{C}(\mathsf{rho})\text{-Ctx}]\!] \subseteq \mathcal{A} \subseteq \mathsf{rho}\text{-Ctx}$.*

The two endpoints are degenerate: $\mathcal{A} = [\![\mathcal{C}(\mathsf{rho})\text{-Ctx}]\!]$ is decorated cost-accounted rho with no game at all, and $\mathcal{A} = \mathsf{rho}\text{-Ctx}$ is total war in which any context may take anything it can address. The interesting point is the middle one:

Definition 33.10 (Capability gating)**.** *The* capability-gated *game admits those contexts that* derive *m_P from structure they can perceive, and excludes those that are simply handed it.*

Since names are unforgeable, derivation is real work, and it is the work namespace logic does: describing a namespace by a property rather than enumerating it. Under capability gating, foraging is name discovery and a hypothesis is a namespace description that either does or does not locate real tokens. Everything below is stated for the capability-gated game.

33.7.4 The foraging inequality

Conservation has an immediate consequence which we will lean on repeatedly: harvest is not automatically profitable, because finding and opening a stack costs tokens drawn from the same conserved supply.

Proposition 33.7 (Foraging inequality)**.** *A harvest of P by S credits S on balance only if*

$$\sigma_P \;>\; \kappa_{\text{sense}}(\delta_P) \;+\; \kappa_{\text{assay}} \;+\; \kappa_{\text{break}},$$

where σ_P is the stack recovered and δ_P the depth at which m_P lies concealed.

Corollary 33.2 (The profitable band)**.** *Since a well-stocked computation can afford deeper concealment — raising κ_{sense} — while a poorly stocked one offers little σ_P, viable prey occupy a band: too poor is not worth opening, too rich is not worth cracking.*

This is optimal foraging theory, and we did not import it; it is what Proposition 33.7 says once concealment is priced by Definition 33.2. It is also the inequality whose sign change draws the phase boundaries of §33.13.

33.8 Destructive assay forces a logic of namespaces

Harvest ends the prey's computational life. That single decision has a consequence for the hypothesis language which we did not expect and which we now regard as the best available answer to "why namespace logic."

Proposition 33.8 (One specimen, one measurement)**.** *Let P be a mortal computation and let S harvest it. Then no further assay may be run against P. Consequently a hypothesis whose subject is the individual P has zero expected yield after its own first test.*

Proof sketch. By Definition 33.8 the harvest consumes m_P's message and P cannot fund a further step, so P contributes no further transitions and no further surface. A hypothesis φ whose denotation is $\{P\}$ is therefore, after the test, a predicate with empty extension among live computations; whatever it cost to form and hold, it cannot be recouped. □

Corollary 33.3 (Induction is forced)**.** *In an ecology with destructive assay, the only hypotheses with positive expected yield are those whose subject is a* class *of computations. The hypothesis language must therefore be able to describe a collection by a property rather than by enumeration.*

That is namespace logic's definition. So the logic is not adopted for elegance or for uniformity with a companion note; it is what survives the game. Induction — the move from the specimen to the kind — is not an epistemic virtue the scientist is asked to display but the only strategy that pays when each specimen affords exactly one measurement.

Remark 33.10 (The biological reading). Destructive assay is not exotic. Large parts of experimental biology sacrifice the animal, and the inferential structure of those fields is exactly the one Corollary 33.3 describes: because you cannot re-measure the individual, everything must be said about the strain, the cohort, or the genotype. The formal setting reproduces the methodological consequence, which we take as evidence that the modelling is not merely decorative.

33.9 Exposure, and the price of being known

33.9.1 The exposure lemma

The prey's optimal defence in a world of predators looks, at first, unbeatable. Restrict everything. A fully closed computation is a live τ-network with no free names; no context can communicate with it; and by the pessimism of [20, §1] its observable behaviour carries almost no information. It cannot be perceived, so its metabolic channel cannot be derived, so it cannot be harvested.

It also cannot eat.

Lemma 33.1 (Exposure)**.** *Let P be a mortal computation with strictly positive burn rate and no free names. Then P performs no harvest, its stack is monotonically decreasing, and P dies in finitely many steps.*

Proof sketch. Harvest is a receipt on a metabolic name m_Q belonging to some Q outside P (Definition 33.8). By Remark 33.1 there is no primitive restriction to appeal to, so the hypothesis must be read structurally: P is closed when no name occurring in P is structurally equal to a name occurring in any other locus. Communication requires such an equality, so no receipt of P fires against the environment and P's stack receives no credits. With a positive burn rate it reaches zero in at most $\lceil \sigma_P/\varepsilon \rceil$ steps, where ε is the minimum debit. □

Remark 33.11 (Closure is luck, not construction). The structural reading of Lemma 33.1 is weaker than the restriction-based one in an instructive way. Because name manufacture is decentralised (Remark 33.1), two computations may independently produce structurally equal names, so a locus cannot guarantee closure by construction — only observe that it currently obtains. Perfect isolation is not purchasable. We will find in §33.12 that the same fact, read the other way round, is what makes recombination generative.

Corollary 33.4 (Concealment is a luxury of the starving). *Metabolic viability lower-bounds observability: any computation that survives indefinitely must expose surface, hence must be perceivable, hence is in principle predictable and so in principle edible.*

The predator's counter-move is now legible, and it is a reading of [20, Rem. 2] turned around. Restriction makes a name unobservable but not undescribable; $\mathsf{H}x.\varphi$ reaches under the binder and speaks about what is hidden. A predator therefore attacks what it cannot observe by *describing* it, which is why the spatial layer and not the modal one is the hunting instrument, and why senses had to be admitted in §33.3 for the game to be non-trivial at all.

33.9.2 The invasiveness ladder

Table 33.1 ordered the three grades of access by what they reveal. They are ordered in the opposite direction by what they destroy.

engagement	effect on the subject	information	nutritive?
perceive	none	surface only	no
assay	perturbs; the environment changes	behaviour under one probe	no
break in	fatal	the whole interior, as structure	yes

Table 33.2: Information gain and subject survival are in strict opposition, and only the fatal end of the ladder pays.

Proposition 33.9 (The opposition is monotone). *Along the ladder, the information yielded is non-decreasing and the subject's residual life is non-increasing;*

> *and the only rung that credits the scientist's stack is the last.*

This is where the note's arc closes. A scientist that is very good at forming hypotheses locates metabolic channels efficiently; locating them efficiently is eating efficiently; eating efficiently exhausts the environment that was confirming the hypotheses. *The better the theory, the faster it destroys the conditions of its own confirmation.* Success is self-terminating, and no additional assumption was needed to make it so.

33.10 Mixed environments

The environment contains a mixture: bare token stacks, non-scientist mortal computations, and other scientists. Each stratum behaves differently and the third is the interesting one.

Bare stacks are the primary resource: inanimate, undefended, cheap to open, and finite. They are what makes the shallow-and-external regime of §33.13 a regime in which science does not pay — when food lies in the open, senses suffice and a theory is pure overhead.

Non-scientist mortal computations burn tokens and may have structure, but form no model. They defend passively, by restriction, and are subject to Lemma 33.1 like everyone else.

Scientists model back, and three consequences follow.

> **Proposition 33.10** (Scientists are preferred prey)**.** *A scientist exposes more surface than a passive mortal computation, since inquiry is a dialogue of depth at least two conducted over channels that are observable across the cut, whereas metabolism alone requires only enough surface to harvest (§33.12.6, Proposition 33.23). By Corollary 33.4 a scientist is therefore more readily perceived and its metabolic channel more readily derived. The comparison is structural and needs no normalisation by stack size or burn rate.*

The most epistemically active agents are the most edible. We do not think this is an artefact.

Theory theft. Reflection makes another scientist's hypothesis readable. A hypothesis in guard position is a term; a term may be quoted; and $@\varphi$ inspects the quotation without communicating. So a scientist may acquire a model by reading rather than by experimenting — cheaper, and not fatal to anyone. Whether honest science survives this depends on whether holding a model requires exposing it, and in this calculus it does, because a guard is surface. We record the equilibrium question as an open problem (§33.18) rather than settle it.

Imagination. Turing completeness plus reflection means a scientist may carry $[\![-]\!]$ itself as data and *run* a model of a prey's metabolism inside its own budget, paying tokens to avoid paying tokens.

Proposition 33.11 (When imagination pays)**.** *Simulating a hunt is viable exactly when the cost of the simulation is below the expected cost of the errors it prevents:*

$$\kappa_{\mathrm{sim}} \;<\; \Pr[\textit{failed hunt}] \cdot \big(\kappa_{\mathrm{sense}} + \kappa_{\mathrm{assay}} + \kappa_{\mathrm{break}}\big).$$

This is the point at which the scientist stops being a reactive forager and becomes a planner, and it is where the stacking of the cost and history monads acquires an interpretation that is not merely formal: the simulated prey is itself a mortal computation, metered inside the predator's budget.

33.11 Trophic structure

33.11.1 Conservation is the right scope, not a limitation

Smith's organisation of the biosphere [40] distinguishes organisms that consume other organisms from those that consume only external energy, and it is tempting to read the second class as evidence that a model of life needs an open system — a source outside the conserved pool. We do not think that reading survives inspection, and we record why, because the temptation is to apologise for conservation at exactly the point where conservation is doing the work.

The sun is a reservoir of finite extent, situated in a physical system in which energy is conserved. Its apparent externality is an artefact of where the boundary was drawn: put the boundary at the biosphere and the sun is outside it; put the boundary at the solar system and there is one conserved budget being spent down. Openness, here as generally, is a bookkeeping convenience.

So conservation is not a disanalogy with Smith's picture. It is the picture at the correct scope, and the whole trophic stratification is recoverable without minting anything.

33.11.2 The distinction is an access profile

If the sun is a token stack, then the autotroph and the heterotroph are not distinguished by *what* they eat. They are distinguished by how the stack they eat may be accessed.

	a source	**a prey**
release	rate-limited; cannot be accelerated	all at once, on demand
exhaustible within a lifetime	no	yes
exclusive	no; many may harvest at once	yes; a linear receipt
adaptive	no	yes, and it models the harvester back

Table 33.3: The trophic distinction is an access profile, not a kind of substance.

The calculus carries this distinction natively, and no new sort is required to express it.

Definition 33.11 (Source and prey)**.** *A metabolic channel is a* prey *channel when its stack is offered by a linear send, so that a single receipt exhausts it. It is a* source *channel when its stack is offered by a persistent emitter — a replicated or contract-driven send delivering bounded quanta indefinitely — so that receipts are non-exclusive and rate-limited by the emitter rather than by the harvester.*

Same sort, same rewrite rule, different multiplicity. Whether a locus is a source or a prey is therefore a property expressible in the generated logic, and a mortal computation's diet is fixed by which of these profiles its senses are tuned to derive.

Remark 33.12 (Trophic type is a fact about the cut). It follows that trophic type is not intrinsic. Whether something counts as external energy or as another organism depends on which side of the cut of §33.2 it sits, and at what granularity one looks: a colony resolves into members, a member into a source and a consumer sharing a boundary. Since in this setting the boundary is a restriction, "external energy" has a formal referent and a relative one. Smith's stratification is thereby recovered as a statement about how a world is cut, not as a statement about two kinds of stuff.

33.11.3 The split is predicted, not stipulated

The payoff is that we do not have to posit the trophic division. Two results already in hand imply it.

Remark 33.3 observes that a recursion following the *structure* rather than the reduction is exact at a finite unfolding. A rate-limited persistent emitter is precisely such a recursion: a replicated send, structurally regular, non-adaptive. Proposition 33.3 observes that an environment whose recursion follows the *reduction* admits no characteristic formula exact at any finite unfolding. An adaptive prey, being itself a mortal computation, is precisely such an environment.

Proposition 33.12 (The two sciences). *A mortal computation feeding on source channels faces an environment whose complete theory is exact at a finite unfolding: its inquiry converges, and after convergence its epistemic outlay falls to the rent on a fixed formula. A mortal computation feeding on prey channels faces an environment that revises against it, so no finite theory is ever complete and a permanent revision cost is incurred for as long as it lives.*

Corollary 33.5 (Intelligence is a heterotroph's expense). *The standing epistemic cost of a strategy is bounded for source-feeders and unbounded for prey-feeders. Hence the apparatus of continuing inquiry — sensing at depth, holding and revising hypotheses, simulating before acting — is worth its rent only to the second.*

Plants do not need brains because the sun is the environment whose complete theory is affordable. We offer this as a consequence of Proposition 33.3 and Remark 33.3 rather than as an observation imported from biology; that it is also true of the Earth we take as corroboration of the modelling rather than as its source.

33.11.4 Why the overlap is thin

The space of mortal computations is plainly large enough to contain those that eat both — nothing in Definition 33.11 forbids tuning one's senses to both profiles. On the Earth the corresponding region is sparsely occupied, and it is worth asking what the formalism says about why. Four pressures are available, none of them requiring new apparatus.

1. **Opposed geometry.** Yield from a rate-limited dilute source scales with collecting surface; yield from a concentrated one-shot source scales with reach and speed. Surface against velocity, derived from Table 33.3 alone rather than borrowed from the shapes of leaves and predators.

2. **Opposite sides of Proposition 33.3.** By Proposition 33.12 the two diets demand different points on the adequacy/checkability trade-off. The source-feeder can live in a terminating fragment; the prey-feeder cannot, since its subject matter is Turing complete and adaptive. One budget cannot be allocated to both regimes at once.

3. **Double rent for single-mode yield.** A foraging strategy is a hypothesis together with a sensory schedule, and both are held at a cost. A dual-mode computation pays rent on two apparatus while eating with one at a time — the argument of §33.16 applied to strategies rather than to formulae.

4. **Hysteresis.** Revision is a walk on the relaxation lattice and walking costs. Where the transition cost between strategies exceeds the gain from switching, the mixed strategy is not infeasible but dynamically unstable: it is not a fixed point of the revision dynamics, and specialisation is an attractor.

The sharpest statement combines the first two with §33.9.

Proposition 33.13 (The mixed strategy is the worst of both)**.** *A computation harvesting both profiles carries the prey-feeder's permanent revision cost (Proposition 33.12) while maintaining the source-feeder's large collecting surface. By Lemma 33.1 and Proposition 33.10 that surface is exactly what makes it perceivable and its metabolic channel derivable. It therefore bears the heterotroph's epistemic outlay together with the autotroph's vulnerability.*

The region is thin because it is dominated, not because it is unreachable.

33.11.5 The overlap is populated by composites

There is a further and we think better answer, which is that the question contains a false premise.

On the Earth, dual capability is realised almost entirely by symbiosis: corals with their zooxanthellae, lichens, kleptoplasty in *Elysia*, and — decisively — the mitochondrion and the chloroplast themselves. Very few single genomes do both at scale; pairs of genomes sharing a boundary do it constantly.

In this setting that is

$$\mathtt{new}\ m\ \mathtt{in}\ (\,P \mid Q\,),$$

two specialists sharing a metabolic channel under a restriction. Parallel composition makes composites first-class and graded separation $|_k$ measures how much interface they share, so the construction needs nothing added.

Proposition 33.14 (Composite resolution)**.** *The overlap region is not underpopulated; it is populated by terms that are parallel compositions rather than atoms. The restriction permits each component to specialise its access profile — one tuned to sources, one to prey — while the shared metabolic channel pools the yield, so Proposition 33.13 does not apply to the composite.*

Read with Remark 33.12 this is not a special construction but the general situation: the organism boundary is drawn by the restriction, not by the strategy, and what looks like one dual-feeding thing at one granularity is two specialists at another.

33.11.6 Succession

One consequence of taking conservation as the correct scope, rather than as a limitation to be repaired, deserves recording. Because a source is finite — rate-limited, but not inexhaustible — a world does not settle into stable coexisting trophic layers. It runs through them.

Proposition 33.15 (Trophic succession)**.** *While sources deliver at a rate satisfying Proposition 33.7, source-feeding dominates and the cheap fragment of §33.5 suffices. As sources deplete, the foraging inequality fails for them before it fails for prey, since prey concentrate what sources have already collected. Prey-feeding, and with it the permanent revision cost of Proposition 33.12, is therefore forced rather than chosen. Thereafter κ accumulates until the inequality fails for every locus.*

The stratification is temporal as much as structural. On a stellar timescale this is a distant prospect and not a modelling artefact; it is the correct statement of the situation at the scope at which energy is conserved.

33.11.7 An ensemble test

Both claims of this section — that the mixed strategy is dominated, and that the composite is not — are checkable inside the framework rather than merely analogical, and the machinery is that of §33.13. Sample ecologies across the (ι, δ) plane and ask, first, whether the mixed access profile is ever a viability optimum for an atomic computation, and second, whether the composite of Proposition 33.14 dominates it wherever it is not. We regard this as a sharper first computation than calibrating the phase boundaries, because it can come out either way and one would want to know which.

33.12 Reproduction

Harvest ends a computational life, so an ecology with no births is a monotone depletion of agents and nothing in it persists long enough to be a subject matter. This section supplies reproduction. The mechanism is reflection, and the interest of it is that the genetic apparatus is not modelled but found: a genome is a quotation, a locus is a context, and crossover is the exchange of parands between two quoted terms.

33.12.1 The naive route, and why sex is worth its price

The cheap route is already in the calculus. Replication, guarded recursion, a persistent receipt, a contract: each reinstates a term exactly. This is asexual reproduction, and it is strictly cheaper than the alternative — no partner, no alignment, no disclosure, no recombination to compute. Sexual reproduction produces variants rather than copies, so the two are not different apparatus but the extremes of a single dial: exact reinstatement at one end, recombination at the other. In the sexual regime there is no exact recursion anywhere in a lineage.

Why pay for the dial to be off zero? The usual answers are available, but this setting supplies a sharper one, and it comes from the first half of the note.

Proposition 33.16 (Sex attacks the predator's hypothesis language)**.** *By Definition 33.9 a predator must* derive *its prey's metabolic name, and by Corollary 33.3 the hypothesis licensing that derivation is a description of a namespace. Recombination at homologous contexts alters which channels a lineage's members carry. Hence sexual reproduction perturbs precisely the object about which the predator's hypothesis is formed, whereas variation confined to continuations perturbs only behaviour under a hypothesis that remains correctly aimed.*

This is a Red Queen argument with a mechanism rather than an analogy, and it is available because the hypothesis language was chosen to be a logic of namespaces in the first place.

33.12.2 The germ is a quotation

Definition 33.12 (Genome, phenotype)**.** *The* genome *of a mortal computation whose code is P is the name $@P$. Its* phenotype *is $*@P$.*

Everything one wants from the distinction follows from properties the calculus already has.

- $@P$ does not reduce. A quotation is a name, and names do not step. So a genome does not metabolise, does not age, and burns no tokens while held.
- $@P$ is transmissible. Names are what sends carry, so a genome may be handed to a partner or to an offspring on an ordinary channel.
- $@P$ is inspectable. By §33.3 a name predicate sees into a quotation, and by Remark 33.1 it sees all the way down. A genome may therefore be screened without being run — at reflection prices, which by Table 33.1 are the cheapest available.
- $*@P$ costs. Running is what is metered.

Remark 33.13 (What we do and do not claim here). It is tempting to read the first item as a formal Weismann barrier. We decline. What the calculus gives is an asymmetry of *cost* — held quotations are free, running terms are metered — and that is all we use. Whether an individual's acquired revisions can be written back into what it transmits is a separate question, on which the empirical case against a strict barrier is considerable [39], and which is not settled by any theorem here. We note only that a scientist's hypotheses are held as data rather than as code, which puts them on the near side of any such barrier, and we leave the matter as Open Problem 1.

Remark 33.14 (Persistence and reproduction are distinguished by the ledger). Syntactically, a term that reinstates itself and a term that spawns a copy can look alike. Under conservation they are not alike: a reinstatement that continues to draw on the parent's metabolic channel is soma, and one that must be endowed from it is offspring. The germ/soma boundary is drawn by provisioning, not by grammar.

33.12.3 The normal form is a tree, and a locus is a context

Up to structural congruence a process is a parallel composition of primes, and each prime is a receive, a send, or a drop:

$$P \;\equiv\; \prod_{i\in I} \mathtt{for}(p_i \leftarrow c_i)P_i \;\Big|\; \prod_{j\in J} c_j!(Q_j) \;\Big|\; \prod_{k\in K} {*x_k}.$$

This is a gross decomposition only. Each continuation P_i and each transmitted body Q_j is itself a process with a normal form of the same shape, so the decomposition recurses and a genome is a finitely branching *tree* whose nodes are parallel compositions and whose edges are prefixes. The parands are the individual primes, at every level.

Definition 33.13 (Locus, allele)**.** *A* locus *of P is a one-hole context $K[-]$ obtained by deleting a parand from the tree of P; the deleted parand is the* allele *at that locus. A locus is thus identified by the path of channels from the root to the hole, not by a channel alone.*

The vocabulary is not new: a locus is a one-hole context in the same term algebra the modalities of §33.3 are labelled by, so genetics requires no notion of address the note did not already have. It requires a slightly larger one, and the difference is worth stating rather than eliding. A modal label K_j is a one-hole context whose hole sits at a *redex*; deleting an arbitrary parand — an unmatched send, a receipt with nothing to receive — yields a one-hole context that is not the label of any modality. The loci therefore properly contain the modal labels, and a crossover at a locus outside that class moves the genome to a place no formula of the generated logic can name.

Proposition 33.17 (Loci versus labels)**.** *Every modal label K_j of P is a locus of P; the converse fails. The loci that are labels are exactly those whose hole is at a communicating pair in the sense of Definition 33.16.*

Proof sketch. A redex is a parand pair, so deleting it gives a locus, which establishes the inclusion. For the converse, take $P \equiv c!(Q) \mid R$ with R offering no receipt on c: deleting $c!(Q)$ is a locus, and no rewrite fires there, so no K_j equals it. The characterisation is then immediate from the definition of a redex position. □

We flag it here and repair it in §33.12.5, where the repair turns out to be a discipline we were going to want anyway.

Definition 33.14 (Homology)**.** *Loci K_M of the mother and K_F of the father are* homologous *when their channel paths from the root agree and their binding environments agree — that is, the prefixes along the two paths bind the same pattern variables.*

Homology is therefore a partial isomorphism of the two trees rather than a matching of two multisets, and the binding clause is a reading-frame condition: an allele's body may reference only variables its new context binds.

Proposition 33.18 (Species is a namespace formula)**.** *Two computations can recombine only at their homologous loci, so the extent to which they interbreed is the extent to which their path sets agree. Reproductive isolation is namespace disjointness, and conspecificity is expressible as a formula over contexts in the generated logic.*

33.12.4 Crossover

Definition 33.15 (Crossover)**.** *Let H be a set of homologous loci of $@M$ and $@F$ and let $\chi : H \to \{M, F\}$ be a choice function. The* recombinant *$@C$ is the term obtained by filling each locus in H with the allele of the parent χ selects, retaining polarity — a receive is replaced by a receive, a send by a send. Where the split is taken between a guard and its body, Definition 33.14's binding clause is required and is what keeps the recombinant well formed.*

We take crossover proper rather than assortment of whole primes. Assortment — exchanging a prefix together with its body — is the safe special case in which the binding condition is automatic; it is Mendel's independent assortment, and it cannot recombine within a gene. Splitting guard from body is where recombination earns its name.

Proposition 33.19 (Effect size is graded by depth)**.** *A locus at depth* 0 *exchanges an entire subsystem; a locus at depth d exchanges detail interior to a behavioural sequence of length d. Hence the perturbation induced by a swap is non-increasing in the depth of its locus.*

The gradation is uniform, and it is your own observation about recursion that makes it so: were the genome to contain recursive definitions, a swap inside a recursive body would be a global edit whose effect size was large regardless of depth. Since sexual reproduction yields variants and never exact reinstatement, no such loci occur, and the analogue of the distinction between large-effect regulatory and small-effect structural variation falls out of the term structure rather than being posited as a rate.

33.12.5 Communicating pairs as units of inheritance

Definition 33.15 lets a choice function select any allele at any homologous locus, and §33.12.8 then observes that most of what it produces is inviable. There is a discipline which removes much of that waste, and it changes what the unit of heredity is.

Definition 33.16 (Interaction graph, communicating pair)**.** *The* interaction graph *of a genome has the primes as nodes and an edge between a send and a receive whenever they act on the same channel with compatible pattern. A* communicating pair *is such an edge together with its two endpoints. The criterion is syntactic by necessity: whether two primes* will *communicate is a reachability question, and settling it is not something a parent could afford (Proposition 33.27).*

> *The unit of heredity is a redex. A gene is not a piece of structure but an interaction.*

That is worth pausing on, because it obtains something one would otherwise have to impose. Material that must work together is inherited together — linkage — not because a chromosome places it adjacently but because the pairing *is* the working-together.

It also obtains something we owed. Proposition 33.17 observed that loci properly contain the modal labels, so an undisciplined crossover can move a genome to an address the logic cannot name. The discipline closes the gap exactly.

Proposition 33.20 (The discipline aligns the two address spaces)**.** *Under the communicating-pair discipline, the loci at which crossover may act are precisely the modal labels K_j of the genome. Hence every crossover is a move between formulae the generated logic can state, and the transport of §33.12.10 is well defined.*

Proof sketch. Immediate from Proposition 33.17: the discipline admits exactly the loci whose hole is at a communicating pair, which is the characterisation of the labels. □

We regard this as the better argument for the discipline, and a different one from waste reduction. Restricting to communicating pairs does lower the inviable fraction, which is how it was motivated above. What it also does — and what nothing else in this note would have supplied — is make a crossover and a revision the same kind of object: a change of one-hole context, on one side of the characteristic-formula map or the other. Without the discipline the two live in different address spaces and the correspondence is an analogy. With it, it is a bijection.

The discipline does not by itself preserve viability, and the reason is instructive. Suppose the mother contains a race,

$$c!(P) \mid \mathtt{for}(x \leftarrow c)Q \mid \mathtt{for}(y \leftarrow c)R,$$

and the swap removes the pair $(\, c!(P),\ \mathtt{for}(x \leftarrow c)Q\,)$ in favour of a father's pair on some other channel. The donated interaction is sound; the *residual* receipt on c is orphaned, which is precisely the deadlock half of the balance condition. Removing a pair cannot break the material it donates but can break the material it leaves behind.

Proposition 33.21 (Balance is preserved exactly by degree-preserving swaps)**.** *Let a swap remove a communicating pair on channel c and install one on channel c'. If $c = c'$, the number of sends and the number of receipts on every channel are unchanged, no prime is orphaned, and any race on c survives with the same arity. If $c \neq c'$, the residual primes on c may be orphaned and the balance condition of §33.12.8 may fail.*

So preserving races is not a second benefit alongside preserving balance; it is the side condition under which the balance guarantee holds at all.

Remark 33.15 (Discipline substitutes for screening). Proposition 33.27 says a parent cannot screen its recombinants. Under a degree-preserving pair discipline it does not have to screen for balance, because the recombination operator cannot produce an unbalanced offspring. The cost is paid once, in the machinery that constrains the operator, rather than per offspring in model checking. This is an argument for why recombination apparatus should be expected to be *constrained* rather than free, and it depends on Definition 33.16's criterion being syntactic — a semantic notion of "will communicate" would hand the saving straight back.

Remark 33.16 (A dial, and it is the dial of §33.6). Three disciplines are now in view: free parand swap, maximal in variation and mostly lethal; pair swap, which guarantees the donated interaction but risks the residue; and degree-preserving pair swap, which guarantees both and varies least. Each rung buys viability with explorable distance — which is the trade-off of Propositions 33.4 and 33.5 seen at the scale of a lineage. It follows that §33.6.5's claim is conditional: recombination escapes confinement only if the discipline is loose enough to make a shallow re-branch, and looseness is paid for in dead offspring. Where the optimum lies is an ecological question, and belongs with the ensemble of §33.13.

Remark 33.17 (What the discipline does not simplify). Pairs overlap. A send may match several receipts, a receipt's continuation contains further sends matching elsewhere, and a persistent receipt matches many sends over time, so the interaction graph is not a matching. "Swap a pair" therefore means "swap an edge together with its endpoints, where the endpoints carry other edges," and Proposition 33.21 is a statement about one channel at a time. The multi-channel case requires the degrees preserved simultaneously, which is a stronger and rarer condition, and we do not claim it is generic.

33.12.6 Three kinds of channel, and what a swap can touch

The discipline of §33.12.5 operates on some of a genome and not on all of it, and saying which requires a classification that is useful well beyond recombination.

Definition 33.17 (Channel types). *Let $W \equiv P \mid E$. A channel c occurring in P is*

internal *if P contains a matching send and receipt on c and no name structurally equal to c occurs in E: only τ steps arise from it;*

external *if a name structurally equal to c occurs in E but P contains no matching pair on c: its steps are interactions and nothing else;*

both *if P contains a matching pair on c* and *a structurally equal name occurs in E.*

The classification is relative to the cut, as trophic type was (Remark 33.12), and for the same reason: what counts as inside is a question about where the boundary was drawn.

Remark 33.18 (This is the surface, made syntactic). External and both-type channels are exactly the context-observable features of a computation. They are what a structural predicate evaluated at the surface can reach (§33.3), what a context-labelled modality can exercise, and what Lemma 33.1 was about: a computation all of whose channels are internal has no surface, cannot forage, and starves. The three grades of access in Table 33.1 and the three channel types are the same distinction seen from the two sides of the cut.

Dialogue

Definition 33.18 (Dialogue, and its depth). *A* dialogue *is an alternating chain: a step on an external or both-type channel, a causally connected sequence of internal reductions, and a further step on an external or both-type channel whose availability depends on the first. Its* depth *is the number of alternations.*

A computation with no dialogue of depth greater than one is a reflex: whatever it emits is not conditioned on anything but the message immediately received. Depth two and above is conversation — the environment's reply is answered in the light of what the first exchange produced. Two consequences follow, and the first is what the user of this framework will care about.

Proposition 33.22 (Conversation is bounded by the stack). *Each alternation of a dialogue debits at least the internal chain connecting its two external steps. Hence the depth of dialogue a computation can sustain is bounded by its token supply, and by Proposition 33.2 the modal depth at which it can be observed to differ from a rival is bounded by the same quantity.*

Proposition 33.23 (Inquiry has a structural signature). *A computation capable of inquiry has a dialogue of depth at least two on both-type channels: the assay of Definition 33.3 emits a probe, receives, conditions on what it received, and emits again. A computation whose both-type channels support only depth-one dialogue can forage but cannot revise. Writing the nesting out over redex positions, the signature is*

$$\mathsf{Sci} \;:=\; \exists j.\ \langle K_j\rangle\Big(\exists j'.\ \langle K_{j'}\rangle\big(\exists j''.\ \langle K_{j''}\rangle\top\big)\Big),$$

$$j, j'' \text{ at out-type positions,} \qquad j' \text{ at an in-type position,}$$

with all three positions on channels the classification of Definition 33.17 marks as both-type. So "this computation is a scientist" is a formula of the generated

> *logic rather than a designation we assign, and with the positions named it is one a model checker could be handed.*

That repairs something the note has so far taken on trust. The grazer of Appendix 33.19 is not a non-scientist because we declined to give it a hypothesis; it is a non-scientist because its interaction graph contains no depth-two loop through a both-type channel, and the difference is checkable.

It also repairs Proposition 33.10, whose original statement had to normalise by burn rate. The correct statement is structural: deeper dialogue requires more both-type channels, and both-type channels are surface.

What each kind of swap touches

swap	balance	observability	role
internal pair	preserved	silent: τ only	cryptic variation, priced but unseen
both-type pair	preserved if degree-preserving	alters the dialogue	adaptive variation
external singleton	internally unaffected	alters the interface	a wager on the environment

Table 33.4: The classification says which discipline applies where. Only internal and both-type channels host pairs at all; external material consists of primes whose counterpart lies in the environment, so it is swappable singly without orphaning anything — at the price that its viability is a bet about the environment rather than a fact about the genome.

This grading is independent of the depth grading of Proposition 33.19. A recombination policy has two dials: how much structure a swap moves, and how observable the moved structure is.

> **Proposition 33.24** (No neutral variation). *A swap of an internal pair is invisible to weak bisimulation, hence to every context, hence to any predator's hypothesis. It is not invisible to the ledger: its reductions are metered like any others. There is therefore no behaviourally silent variation that is also metabolically silent, and selection acts on internal variation through efficiency rather than through function.*

> **Corollary 33.6** (Cryptic variation and its release). *Internal variation accumulates without behavioural consequence and becomes consequential exactly when its channel is reclassified — which happens when a structurally equal name comes to occur across the cut. By Proposition 33.26 recombination can do this, since a transplanted pair may carry a channel that coincides with a resident external*

name. A lineage may therefore accumulate variation cheaply while remaining illegible, and expose it later.

This sharpens Proposition 33.16. Only the observable fraction of recombination attacks a predator's hypothesis, because a predator's hypothesis is a description of the external and both-type sub-namespace and nothing else. But internal variation is the reservoir from which observable variation is later drawn, so the defence is two-staged: accumulate where it cannot be seen, and expose when the exposure is worth its cost.

Remark 33.19 (Dialogue is where the impact is). Collecting the section: swapping at redexes concentrates variation on channels that carry interactions, and among those, the both-type channels are the ones whose interactions are legible across the cut. A discipline that swaps redexes therefore acts, by construction, on the computation's dialogue with its environment rather than on structure that no context could ever observe. That is the sense in which the discipline of §33.12.5 has real consequence: it is not merely a filter for viability but a filter that concentrates variation where it can matter.

33.12.7 Names, freshness, and the bound on growth

Because `new` is sugar (Remark 33.1), names have size, and size is priced. One might therefore fear that recombination inflates them: if a child's fresh names were manufactured from its enclosing term, and its enclosing term combines two parents, names would double with each generation and a lineage would price itself out of existence within a few dozen.

The calculus forbids it.

Proposition 33.25 (Growth is slaved to term size)**.** *No process contains a name that is its own code, since quotation strictly increases size. Hence every name occurring in a process is strictly smaller than that process, and no admissible desugaring manufactures a name from more than a proper part of its scope. Name size cannot drive term growth; it is bounded by it.*

The feared scheme is not merely expensive but syntactically impossible, and the bound holds uniformly across every decentralised scheme — which is why we may remain agnostic about which one is taken.

What growth remains is the right kind. At a homologous locus the recombinant takes one allele, not both, so aligned material contributes a pointwise maximum rather than a sum. Growth enters only at *non-homologous* loci — material present in one parent alone — where inheriting the union is additive per generation. That is gene duplication, it is linear, and by Definition 33.2 it is priced, which is the regime in which selection can act on it.

Proposition 33.26 (Decentralisation makes recombination generative)**.** *Since names are manufactured locally and freshness is scope-relative, two parents may independently hold structurally equal names at loci private to each. A recombinant inheriting both then possesses a single shared channel where its parents had two private ones. Recombination can therefore create interaction paths neither parent had, rather than only permuting existing ones.*

Under a central name server, or under a calculus of atomic names with global freshness, this route is closed. It is the same fact recorded in Remark 33.11: what denies a locus guaranteed closure is what allows a lineage novelty.

Remark 33.20 (Accumulation, and a reading of senescence). Names grow because the terms containing them grow, and a scientist's term grows as it accumulates probe channels, held hypotheses, and machinery. With sensing priced by depth, every operation then costs more, and a computation that does not reclaim dead structure eventually cannot afford to run. This is a discipline-dependent claim rather than a theorem — garbage collection is the alternative, and is itself priced — but under the discipline it says something worth saying: an organism ages because it accumulates, and accumulation is what learning is.

A condensation pass at birth, canonically renaming the recombinant's private names from a minimal seed, resets the inherited weight at a cost proportional to genome size. We note that this is an argument about *cost* and is orthogonal to Remark 33.13: whatever the traffic between soma and germ turns out to be, the weight must be renormalised or lineages accumulate without bound.

33.12.8 Viability, and why there is selection rather than design

A parallel composition of primes is always a term, so syntactic well-formedness is free. Viability is not, and it is a conjunction of conditions the generated logic can state: closure, the reading-frame condition of Definition 33.14, balance — receives without matching sends deadlock, sends without receipts leak — and liveness in the marked-graph sense. Screening a recombinant is therefore model checking, and the classifier apparatus of [20] is the embryology of this note.

But model checking is metered, and the space is large.

Proposition 33.27 (Selection is post hoc because screening is priced)**.** *Aligned genomes with n homologous loci admit 2^n recombinants, where n is the size of the genome and not the width of its top level. Screening each is an outlay against the same stack that funds foraging and inquiry. Hence a parent cannot emit only* *viable offspring — not for want of a criterion, but because applying it exhaustively is unaffordable.*

> *Reproduction is a gamble because screening is priced. Were model checking free, a parent would design its offspring and selection would have nothing left to do.*

Neither fine granularity nor cost accounting gives this alone; it needs both.

33.12.9 The energy tension: a three-way allocation

Reproduction takes energy, and under conservation the energy comes from the parents and nowhere else. An offspring is endowed out of a parent's stack, so every child is a portion of the parent's remaining life.

Proposition 33.28 (Minimum viable endowment)**.** *An offspring endowed below the outlay required for its first successful harvest — $\kappa_{\text{sense}} + \kappa_{\text{assay}} + \kappa_{\text{break}}$ for the shallowest prey available to it, by Proposition 33.7 — dies before it eats. Provisioning is therefore bounded below, and the number of offspring a parent can endow is bounded above by its stack divided by that floor.*

We take the endowment to be chosen by the parent rather than fixed by the offspring's code size, which makes the trade-off between many poorly provisioned offspring and few well provisioned ones a strategy rather than a constant.

The tension the reader will have anticipated is not two-way but three-way. One conserved stack funds soma maintenance, germ production, and epistemic outlay: the same tokens buy eating, breeding, and thinking.

Corollary 33.7 (Inquiry trades against fecundity)**.** *By Corollary 33.5 the standing epistemic outlay is unbounded for a prey-feeder. By Proposition 33.28 offspring are bounded below in cost. Since both draw on one stack, a strategy that sustains deep inquiry sustains fewer offspring.*

This supersedes the reading of surplus offered in §33.16: surplus does not simply buy curiosity, it buys curiosity *or* children, and which it buys is the life-history problem.

33.12.10 What reproduction does for the science

Two consequences bear directly on the first half of the note.

The lineage is a second learner. A scientist's hypotheses are guards; guards are parands; parands are what crossover exchanges. So the genome contains the hypotheses, and they are revised by two mechanisms on two timescales — within a life by walking the relaxation lattice of §33.5, across lives by recombination. The same object is revised by both. A lineage is not merely a means of persistence; it is doing science by other means, and the proximate/ultimate distinction of §33.16 extends to cover it.

The correspondence is tighter than an analogy. A revision move is a locus in the formula tree together with an operation there (Definition 33.6); a crossover is a locus in the term tree together with a choice of allele (Definition 33.15). The characteristic-formula construction of §33.5 carries term structure to formula structure, so a swap at a term-context K_j induces a revision at the formula-locus K_j corresponds to — and by Proposition 33.20 the swap is at a K_j in the first place only because the recombination discipline restricts it to one, without which the induced formula-locus need not exist. The two mechanisms are one move, transported along χ, and applied at two timescales. What distinguishes them is the accessible distance: by Proposition 33.4 an individual's incremental walk cannot leave its ball, whereas a swap high in the term tree is a shallow re-branch by construction. *Recombination is how a lineage makes the displacements its members cannot.*

Reproduction is what makes induction pay. Corollary 33.3 forced hypotheses to be about namespaces, since destructive assay leaves an individual-level hypothesis worthless after its own first test. But a namespace with no live members is worthless too. Reproduction repopulates the namespace, so a class-level hypothesis retains value across the specimens it costs to test.

Proposition 33.29 (Kinds are the condition of inquiry)**.** *In an ecology without reproduction the scientist monotonically destroys its own subject matter and no hypothesis retains extension. With reproduction the population of a namespace is sustained while the foraging inequality holds for its members, and hypotheses about kinds remain true of live instances. Species are what make science possible.*

Remark 33.21 (What reproduction does not do)*.* It does not repeal conservation. Tokens are redistributed into new loci and never created, so Proposition 33.15 stands unaltered and the world still runs down. What changes is the shape of the decline, not its direction: births convert a monotone extinction of agents into a population dynamics with a carrying capacity, which is what makes the intermediate band of Figure 33.2 occupiable for longer than a single generation.

33.13 Distributions over ecologies

33.13.1 A world is a partition of a conserved quantity

Because Θ is fixed and never minted, a world is a partition of Θ across loci together with a structural assignment saying which loci are scientists, which are non-scientist mortal computations, and which are bare stacks. The configuration space is graded by Θ, and a distribution over it is a microcanonical ensemble in the strict sense — fixed total, distribution over microstates. The analogy is exact rather than suggestive, and it is exact because conservation is exact.

> **Definition 33.19** (Configuration, ecology)**.** *A* configuration *is a finite multiset of mortal computations and bare stacks together with an assignment of stack sizes summing to* Θ *and a designation of which computations carry a revision policy. An* ecology *is a configuration together with a distribution over the configurations reachable from it.*

33.13.2 Two order parameters, and a phase

Two quantities govern the qualitative behaviour and they are independent:

- $\iota \in [0, 1]$, the fraction of Θ already *internal* to scientists rather than loose in the environment;
- $\delta \geq 0$, the mean *concealment depth* of external stacks — how far under restriction and structure they lie, which by Definition 33.2 sets the sensing cost of finding them.

> *Science only pays in an intermediate band — where food is neither free nor absent.*

We regard this as the most substantive empirical claim the construction makes, and it is a claim the construction makes rather than one we impose: the boundaries of Figure 33.2 are where the foraging inequality of Proposition 33.7 changes sign. The low-δ boundary is where κ_{sense} falls below the cost of holding a theory at all; the upper boundary is where Proposition 33.7 fails for every available P.

33.13.3 Sampling

The machinery for turning a configuration into a distribution and sampling trajectories through it is not new work. The construction of [25] takes a decorated continued interactive GSLT, an adequate spatial–behavioural logic, and a weight-map decoration to a Gillespie-style simulator, framed as a functor from a Grothendieck construction of decorations into weighted coalgebras. That is precisely what an ecology is: the

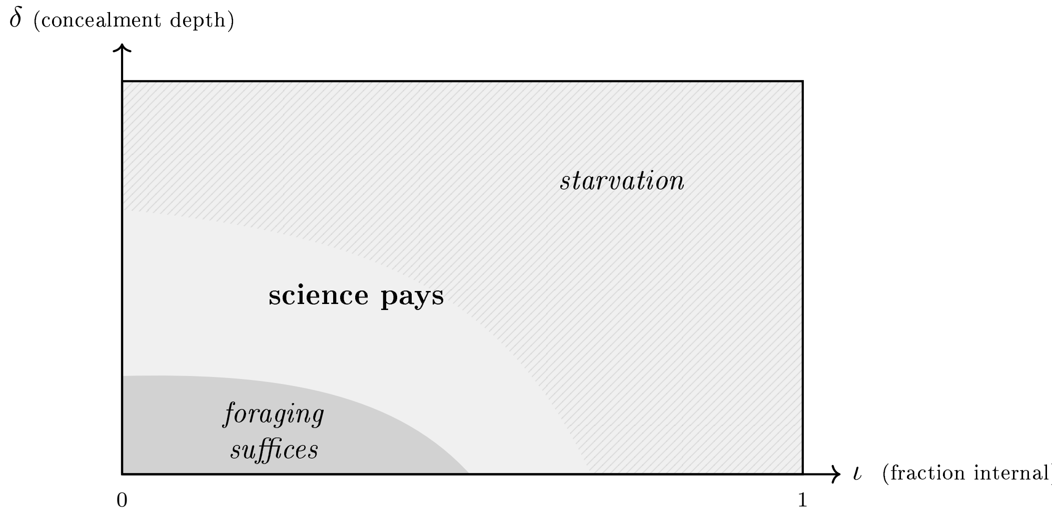


Figure 33.2: The phase diagram. When food lies shallow and in the open (low δ), senses alone suffice and a theory is overhead: the pure forager beats the scientist. When too much of Θ is already internal, or lies too deep to be found for what it yields, discovery costs more than it returns and everything starves. Hypothesis formation pays in the intervening band. Science is a phase, not a universal advantage.

weight map turns a configuration into a distribution, and the Gillespie construction samples it. §33.15 takes up what that fibred structure means here.

33.14 The audit: what the assumptions already fix

It is easy to build a machine that fits a story. We therefore separate, explicitly, what follows from the commitments and what we chose.

33.14.1 Forced

33.14.2 Chosen

> *The physics is forced. The ecology and the psychology are the free parameters.*

Everything metabolic and everything epistemic follows from cost-accounted `rho` together with the logic it generates. What remains to be chosen is the weight map, the initial distribution, and how the scientist thinks. We regard the smallness of that list as the principal evidence that the modelling choices were not tuned to the conclusions.

feature	**forced by**
the hypothesis language	OSLF applied to the `rho` presentation; not designed
formulae denote bisimulation classes	adequacy of the generated logic
exactly three grades of access	the calculus offers no fourth (§33.2)
senses are structural predicates	admitting the structural layer, priced by depth
an assay is a guarded receipt	the `where` clause
refutation is budget-relative	Proposition 33.2
the hypothesis space	the relaxation lattice of [20]
hypotheses must be about namespaces	destructive assay (Corollary 33.3)
eating is an ordinary receipt	internalisation plus Turing completeness
death is failure to fund the next step	the cost monad
$\sigma + \kappa = \sigma_0$ globally	the history monad plus no minting
predators are non-image contexts	Theorem 33.1
the exposure lemma	Lemma 33.1
optimal foraging, and its profitable band	conservation plus depth pricing (Prop. 33.7)
information opposes survival	Table 33.2
a genome is inert, transmissible, inspectable	quotation does not reduce (Def. 33.12)
a locus is a context	the recursive normal form (Def. 33.13)
name growth is bounded by term size	no term contains its own code (Prop. 33.25)
selection is post hoc	screening is priced (Prop. 33.27)

33.15 Is the assignment functorial?

It is natural to ask whether ecologies are a functor of the theory: is there a functor from the subcategory of cost-accounted graph-structured lambda theories to distributions over computational ecologies of the kind described above? The answer is instructive, and it is no in a way that tells us what the right question was.

33.15.1 Not from the base

Write $\mathbf{GSLT}_{\mathcal{C}}$ for the subcategory of cost-accounted GSLTs. The assignment $\mathcal{C}(G) \mapsto$ "its ecologies" is not determined by $\mathcal{C}(G)$. Two different weight maps — two different pricing schedules on the same cost-accounted theory — give different sensing costs, hence different foraging inequalities, hence different phase boundaries in Figure 33.2

parameter	**status**
Θ, the conserved total	free; sets the scale of the whole game
the initial configuration and its distribution	free; §33.13
the admitted context class $\mathcal{A}$	free; Definition 33.9, and it is the rulebook
the pricing schedule κ_{sense}, per-rewrite debits	free; the weight map
concealment depth δ of stacks	free; an order parameter
harvest all-or-nothing versus partial	free; we take all-or-nothing
the revision policy	free; this is the agent design, and we leave it open
the desugaring of `new`	free, subject to decentralisation (Rem. 33.1)
provisioning per offspring	free; we make it parent-chosen, so r/K is a strategy
the crossover choice function χ	free; the recombination rate in disguise

and different distributions over trajectories. The object does not determine the image.

Proposition 33.30 (A fibration, not a functor)**.** *The ecologies form a fibration $\pi : \mathbf{Eco} \to \mathbf{GSLT}_{\mathcal{C}}$ whose fibre over $\mathcal{C}(G)$ is the class of ecologies presentable in $\mathcal{C}(G)$, indexed by the decorations* $\mathrm{Dec}(G)$*. The assignment of distributions is a functor out of the total space — equivalently, out of the Grothendieck construction $\int_{\mathbf{GSLT}_{\mathcal{C}}} \mathrm{Dec}$ — and not out of the base.*

This is [25]'s theorem with a different label on the target: that construction already goes from a Grothendieck construction of decorations into weighted coalgebras, and an ecology-with-a- distribution is a weighted coalgebra.

33.15.2 A canonical section

There is nevertheless a rescue, and its shape is the interesting part.

Proposition 33.31 (Cost is already a weighting)**.** *A cost-accounted GSLT is not decoration-free: the cost function is itself a weight map. Taking costs as the weights selects a distinguished decoration for every object of $\mathbf{GSLT}_{\mathcal{C}}$, hence a section $s : \mathbf{GSLT}_{\mathcal{C}} \to \mathbf{Eco}$ of π, and hence a genuine assignment of ecologies to objects.*

So the honest statement is not "there is a functor" but "there is a canonical one, and its canonicity is exactly the observation that cost is already a weighting." Any other ecology over the same theory is a re-weighting, and the deviation from the canonical section measures how far the modeller's pricing departs from the theory's own.

33.15.3 The morphisms must be logic-preserving

Functoriality on morphisms is where a real correction is needed, and we think it is the most useful thing this section contains.

The morphisms of **GSLT** are bisimulation-preserving maps, where the bisimulation is the context-labelled one and is a congruence [21]. But an ecology depends on *spatial* structure: concealment depth, which computations sit inside which, how a network separates when a locus is removed. And the spatial layer strictly refines bisimulation — this is the entire thesis of [20], that separation says things no modal logic can say. So a bisimulation-preserving map may scramble the spatial layer, changing sensing costs and foraging inequalities while preserving behaviour exactly.

> **Proposition 33.32** (Refinement of the morphism class)**.** *The assignment of Proposition 33.31 is not functorial on* **GSLT***-morphisms. It requires morphisms preserving the* generated logic *— both the behavioural and the structural layers — which is a strictly narrower class, strictly because the structural layer strictly refines context-labelled bisimulation.*

The narrowing is not ad hoc. It is forced by the fact that the senses are structural: a morphism that does not preserve what the senses see does not preserve the ecology, because it does not preserve what it costs to find food.

Conjecture 33.1 (Laxity)*.* Even over logic-preserving morphisms the induced map on distributions is only lax. An encoding may add rewrite steps and so inflate costs, so the pushforward distorts; the natural target is distributions ordered by stochastic dominance rather than distributions on the nose.

We state this and do not chase it. The three-line summary for the reader in a hurry is: a fibration, a canonical section via cost-as-weight, and a refinement of the morphism class forced by the senses being structural.

33.16 Proximate and ultimate scores

The original motivation for this construction measured the delta between hypothesis revisions and punished degradation. The game supersedes that mechanism as the *ultimate* criterion — survival is the score, and it needs no adjudicator, no referee holding a mint, and no exogenous reward signal. Two things fall out that we would otherwise have had to impose.

- **Vacuity is punished by hunger, not by rule.** The hypothesis $\top$ predicts nothing, locates no metabolic channel, yields nothing, and starves. No informativeness term is needed in any score function, because there is no score function.

- **Occam's razor is metabolic.** A hypothesis is worth holding exactly when expected yield exceeds $\kappa_{\text{sense}} + \kappa_{\text{assay}} + \kappa_{\text{break}}$ plus the cost of holding the formula itself. Description length and food are denominated in the same currency, so minimum description length is derived rather than stipulated.

The revision delta nevertheless survives, demoted and better placed. A scientist cannot wait until it dies to learn which way to walk the relaxation lattice, so it needs a local signal, and $d(\varphi_n, \varphi_{n+1})$ together with the confirm/refute counts of §33.4 is exactly one. This is the proximate/ultimate distinction of behavioural ecology: viability is the ultimate criterion, the revision delta is the proximate mechanism that tracks it, and neither is redundant.

> *Remark* 33.22 (The pragmatist's objection, and surplus). It should be conceded plainly that this scientist is not a disinterested seeker of truth. Its epistemology is metabolic, and truths that do not pay go unfunded. That is the standard objection to pragmatism and we do not think it can be argued away. The construction does, however, have an unusually concrete answer: *surplus*. A scientist holding a stack larger than its foraging horizon requires can afford inquiry that does not pay, and what it buys with the excess is curiosity — though by Corollary 33.7 it might have bought offspring instead, and the choice between them is the life-history problem. Basic research is a metabolic luxury, available exactly to the well-fed, and in a world of conserved tokens it is therefore rare and worth protecting. We note this and leave the moral to others.

33.17 Conclusion: an embodied architecture, and the effectiveness of reason

We close with a speculation that is not a design. Suppose a perceptual front end — a transformer network, wired to the world, doing what such networks are unreasonably good at — were used to encode sensory data as a population of computations and token stacks of the kind this note describes. We are not proposing an interface, and nothing below depends on how the encoding is done. We are asking what would follow if one existed, because we think the consequences bear on a question older than any of the machinery here.

33.17.1 A division of labour

The scientist of this note lives among other computations. The world does not present itself that way, or at least does not present itself that way to us, so something must do the type conversion. That is the front end's whole job in the proposed arrangement: not to learn, but to render the environment as an environment of the right kind — loci with surfaces, stacks, and channels an interaction can address.

The division is worth stating sharply, because it inverts the usual allocation of labour:

> *The network is the transducer. The ecology is the epistemology.*

Nothing in §§33.3–33.12 is a learning rule in the gradient sense, and nothing in a perceptual encoder is a hypothesis. Keeping them apart is what makes the arrangement legible.

33.17.2 Why continual learning falls to the ecology

The features that make an architecture of this shape suited to continual learning are not additions to it. They are the results already proved.

Revision is local. A revision move is an operation at a locus in the formula tree (Definition 33.6), not a global adjustment of every parameter at once. Whatever is not at the locus is untouched, so the failure mode in which new learning overwrites old has no mechanism here. The corresponding cost is that a locus must be found, which is what §33.6 is about.

Exploration and exploitation are a budget, not a hyperparameter. The allocation between foraging, inquiry, and reproduction (§33.12.9) is a division of one conserved stack, and the trade-off is settled by what the ecology pays for rather than by a coefficient chosen in advance.

A population is required, not preferred. This is the sharpest of the three. Proposition 33.4 says an individual's incremental revision cannot leave the ball it starts in — not slowly, but not at all. By Remark 33.16 the escape is recombination, which requires more than one learner and a lineage relating them. So an architecture built on this analysis does not have a population because populations are convenient; it has one because a single learner is provably confined.

Retirement has a criterion. Mortality is not a scheduling policy imposed from outside. A model that cannot fund its next step stops, and the ledger says when.

33.17.3 The unreasonable effectiveness of reason

Wigner's question is why mathematics, developed for reasons internal to itself, turns out to describe a world that did not consult it [35]. Hamming's answer is largely a selection effect: we notice the mathematics that fits and forget the rest [36]. The evolutionary answer is that our faculties were shaped by the world they model. Neither is comfortable, the first because the successes are too systematic and too predictive, the second because the mathematics that works best — for spectra, for spacetime, for objects in no ancestral environment — is exactly the mathematics no ancestor was selected on.

The architecture posited here suggests the question has been asked across a gap that is not there.

Wigner's framing has mathematics on one side as a free creation, the world on the other as a brute given, and an unexplained correspondence between them. In this setting there is no such separation to explain. The hypothesis language is not chosen and then found to fit: it is generated by OSLF from the presentation of the computational substrate itself, and adequacy is the theorem that it separates exactly what interaction separates — no coarser, so nothing observable is inexpressible, and no finer, so nothing expressible is unobservable. The mathematics available to such a learner is the mathematics of what it can distinguish, because it is the language the distinguishing relation generates.

> *Mathematics is effective on exactly the domain to which interaction reaches, because it is the shadow that the access relation casts.*

Two things follow which we think are more than restatement.

First, the transfer problem dissolves in the right direction. Adequacy is not adequacy for a particular environment; it is adequacy for interaction *as such*, since context-labelled bisimulation is the finest equivalence any interacting learner could resolve, whatever it is interacting with. That is why the mathematics carries to domains no ancestor met. It was never fitted to the ancestral environment in the first place — it was fitted to the form of access, and the form of access did not change when the subject matter did.

Second, there is a boundary, and its location is predicted rather than assumed. Effectiveness should degrade precisely where the relation between learner and phenomenon is not one of interaction across a cut — where there is no context to build, no probe to install, no reply to condition on. That the domains which have historically resisted mathematisation have tended to be of just this kind is not proof, but it is the right shape of evidence, and it is at least a claim that could fail.

Finally, the categories themselves are accounted for rather than presupposed. Corollary 33.3 forces hypotheses to be about namespaces, because destructive assay leaves an individual-level hypothesis worthless after its own first test, and Proposition 33.29 sustains those namespaces through reproduction. So the traffic in kinds, classes and universals that mathematics and science run on is not read off the world and not stipulated by the knower. It is what a mortal, interacting learner can afford to hold. That is a Kantian conclusion with a ledger attached, and the ledger is what makes it a claim rather than a posture.

33.17.4 What is dictated, and what is not

If the mind is a computational apparatus, we think an architecture of roughly this shape is close to forced. Take the premises separately. If a mind is computational, it is ontologically isolated, so its environment is other computations and its access is interaction. If its access is interaction, its hypothesis language must be adequate for context-labelled bisimulation, on pain of being blind or wasteful. If hypotheses are to be tested rather than merely entertained, they must occupy guard position, so that a confirmation is a rendezvous. If the apparatus is finite, its budget bounds the fragment it can hold, and mortality supplies the motive without an external scorekeeper — of which none is available anyway, since a scorekeeper would be an oracle across the cut. And if revision is search on the resulting lattice, Proposition 33.4 requires a population.

Each of those is a derivation in this note rather than a design decision, which is what we mean by saying the shape is all but dictated. It should be equally clear what has not been shown. We have not shown that the mind is computational; that premise is doing all the work and we have assumed it throughout. We have not shown the world is computational either — only that a computational knower's mathematics will fit whatever it can interact with, which is a claim about the knower and not about what is known. And we have not built the encoder, which is exactly the part where the difficulty of any actual system would be concentrated.

What the note does claim is narrower and, we hope, more durable: that once learning is modelled as a metered dialogue between isolated computations, the apparatus of the scientific method stops looking like something a mind does and starts looking like something a mind, so situated, could hardly avoid.

33.18 Open problems

1. **Inheritance of acquired characteristics.** Remark 33.13 declines to read the inertness of a quotation as a germ–soma barrier, and Noble's case against a strict barrier [39] is a reason for caution rather than a settled verdict. The formal question is sharp: a scientist's hypotheses are held as data and are therefore quotable and transmissible, whereas its metabolic and sensory machinery is code. Under what pricing does a lineage transmit the first, and what does the resulting two-channel inheritance do to Proposition 33.16?

2. **The plagiarism equilibrium.** Reading a rival's model is cheaper than experimenting and kills nobody, but holding a model requires exposing it as a guard. Under what pricing does honest science survive theory theft, and is there a mixed equilibrium?

3. **Co-learning and non-stationarity.** When the environment is mostly scientists, the target of every hypothesis is itself revising. Convergence becomes a

fixed-point question rather than a learning-theoretic one, and this is where we expect the geometry/matter corecursion to be readable as learning.

4. **Laxity.** Conjecture 33.1: is the induced map on distributions lax, and is stochastic dominance the right order on the target?

5. **The revision policy.** We left the agent design open on purpose. Is there a policy on the relaxation lattice that is optimal in the ecological sense — maximising expected residual life rather than predictive accuracy — and does it look anything like the policies learning theory recommends?

6. **Is the mixed diet dominated?** Propositions 33.13 and 33.14 are the two halves of a claim about trophic sparsity. Testing them in the ensemble of §33.13 — is the mixed access profile ever optimal for an atom, and does the composite dominate it where it is not — is the first computation we would run.

7. **Sharpening the phase boundary.** Figure 33.2 is drawn from the sign of a cost inequality. Making its boundaries quantitative for a specific pricing schedule, via the sampling construction of [25], is the first computation to do.

33.19 Appendix: A worked ecosystem

We give a small world containing two large external token stacks, three grazing non-scientist mortal computations, two reproductively isolated species of sexually reproducing scientist, and one asexual lineage. It is written in core rholang extended with the `where` clause of §33.4; stacks are integers and arithmetic is taken as given. The point of the listing is not that it is efficient but that every construction of the note appears in it as ordinary code, with no apparatus added.

33.19.1 Sources, metabolism, death

```
// A source (Def. "Source and prey"): rate-limited by being a single contract,
    non-exclusive
// since anyone may call it, non-adaptive, and FINITE -- conservation is exact.
def Source(s, q, reserve) = {
  for (ret <- s) {
    match reserve >= q {
      true => { ret!(q) | Source(s, q, reserve - q) }
      false => { ret!(0) } // exhausted; cf. Prop. "Trophic succession"
    }
  }
}

// Reclaim - debit - reissue (Def. "Metabolic channel"). Between the receipt and
     the send the
```

```
// stack is a message in flight: metabolism is a race condition (Rem. "
    Metabolism is a race condition"), and
// the metabolic channel m is exactly the vulnerability of Thm. "Predators are
    non-image contexts".
def Live(m, burn, step) = {
  for (sigma <- m) {
    match *sigma > burn {
      true => { m!(*sigma - burn) | *step | Live(m, burn, step) }
      false => { Nil } // death is failure to fund a step
    }
  }
}
```

33.19.2 A non-scientist

```
// A grazer holds no hypothesis and senses nothing beyond the one channel it
// was born knowing. By Prop. "The two sciences" its environment is a source, so
     its science
// was over before it began -- and it pays no epistemic rent at all.
def Grazer(m, src) = {
  Live(m, 1, @{
    new probe in {
        src!(*probe)
      | for (g <- probe) { for (sigma <- m) { m!(*sigma + *g) } }
    }
  })
}
```

33.19.3 Foraging as an assay

```
// The complementary guard pair of Def. "Assay", verbatim. The hypothesis is not
// consulted and then acted on; it sits in guard position, so confirmation IS
// the rendezvous. If the source is exhausted neither guard fires: that is
// the third outcome of Prop. "Trichotomy", and it is not the same as refutation.

def Forage(m, sns, hyp) = {
  new probe in {
      sns!(*probe) // the probe perturbs
    | for (g <- probe where g |= *hyp) { // predicted
        for (sigma <- m) { m!(*sigma + *g) }
      }
    | for (g <- probe where ~(g |= *hyp)) { // refuted
        Revise(hyp) // walk the lattice (Sec. 5)
      }
  }
}
```

33.19.4 Birth: a quotation, then a drop

```
// The offspring's code is CONSTRUCTED as a name and sits inert on 'code'
// -- it does not reduce, does not age, burns nothing (Def. "Genome, phenotype").
     Dropping it
// with * is what makes it a phenotype, and what starts the meter.
//
// The guard enforces Prop. "Minimum viable endowment": a parent may not endow
    below the outlay its
// child needs for a first successful harvest.
def Beget(m, alleles, prv, mate) = {
  for (sigma <- m where *sigma > *prv + floor) {
    m!(*sigma - *prv) // the endowment is subtracted
  | match *alleles {
      [s, h, p] => {
        new childM, code in {
            code!( @{ Body(childM, s, h, p, mate) } ) // the germ: inert
          | for (g <- code) { *g | childM!(*prv) } // the soma: metered
        }
      }
    }
  }
}
```

33.19.5 Crossover

```
// Three homologous loci are exposed as parameters: the sensing allele, the
// hypothesis allele, and the provisioning allele. This is assortment at
// three loci; crossover proper (Sec. "Crossover") splits a guard from its body
// inside an allele and carries the reading-frame condition of Def. "Homology".
def Pick(a, b, bit, ret) = {
  match *bit { 0 => { ret!(*a) } ; 1 => { ret!(*b) } }
}

def Crossover(mother, father, chi, ret) = {
  match (*mother, *father, *chi) {
    ([sm,hm,pm], [sf,hf,pf], [b1,b2,b3]) => {
      new r1, r2, r3 in {
          Pick(sm,sf,b1,r1) | Pick(hm,hf,b2,r2) | Pick(pm,pf,b3,r3)
        | for (s <- r1 & h <- r2 & p <- r3) { ret!([*s, *h, *p]) }
      }
    }
  }
}
```

33.19.6 A sexual scientist, and an asexual one

```
// Mating requires DISCLOSURE: the genome goes out on a public channel, where
// anything able to read it can. Conspecific(s) is a name predicate (Sec. 3.1)
// holding of genomes whose sensing allele names the same source namespace as s.
// Nothing declares a species; reproductive isolation is namespace disjointness
// and nothing more (Prop. "Species is a namespace formula").
def Mate(m, alleles, sns, prv, mate) = {
  mate!(*alleles)
| for (partner <- mate where partner |= Conspecific(*sns)) {
    new chi, ret in {
        chi!( Coin3 ) // the choice function of Def. "Crossover"
      | Crossover(alleles, partner, chi, ret)
      | for (kid <- ret) { Beget(m, kid, prv, mate) }
    }
  }
}

// The shared developmental program. Alleles vary; this does not.
def Body(m, sns, hyp, prv, mate) = {
  Live(m, 2, @{ // scientists burn faster than grazers
      Forage(m, sns, hyp)
    | Mate(m, @[*sns, *hyp, *prv], sns, prv, mate)
  })
}

// The asexual lineage: no partner, no alignment, NO DISCLOSURE, and no
// recombination to compute. Strictly cheaper -- and by Prop. "Sex attacks the
    predator's hypothesis language" it presents
// a fixed namespace for anything that cares to learn it.
def Clone(m, sns, hyp, prv) = {
  Live(m, 2, @{
      Forage(m, sns, hyp)
    | Beget(m, @[*sns, *hyp, *prv], prv, @Nil)
  })
}
```

33.19.7 The world

```
new sun1, sun2, mate in {

  // Two large external stacks. sun1 is shallow and gives small quanta;
  // sun2 lies deeper and gives larger ones -- the two order parameters
  // of Sec. 13.2 instantiated as two loci rather than as a distribution.
    Source(sun1, 1, 60000)
  | Source(sun2, 8, 240000)

  // Three non-scientists.
  | new g1, g2, g3 in {
        g1!(50) | Grazer(g1, sun1)
      | g2!(50) | Grazer(g2, sun1)
      | g3!(80) | Grazer(g3, sun2)
```

```
    }

  // Species A: hunts sun1, cheap hypothesis, many small offspring.
  | new a1, a2 in {
        a1!(400) | Body(a1, @sun1, @{ q => q > 0 }, @20, mate)
      | a2!(400) | Body(a2, @sun1, @{ q => q > 0 }, @20, mate)
    }

  // Species B: hunts sun2, a discriminating hypothesis, few large offspring.
  // A and B share the mating channel and never interbreed.
  | new b1, b2 in {
        b1!(900) | Body(b1, @sun2, @{ q => q > 4 }, @150, mate)
      | b2!(900) | Body(b2, @sun2, @{ q => q > 4 }, @150, mate)
    }

  // The asexual lineage, competing with species A for sun1.
  | new c1 in { c1!(500) | Clone(c1, @sun1, @{ q => q > 0 }, @60) }
}
```

33.19.8 What to watch

- **Conservation is checkable by inspection.** Θ is the sum of the two reserves and the eight initial stacks. Nothing in the listing mints; `Source` only ever moves reserve into a consumer, and `Beget` only ever moves a parent's stack into a child's. Every debit taken by `Live` is a token leaving σ for κ, so $\sigma + \kappa = \Theta$ throughout.

- **The three outcomes are structural, not bookkeeping.** In `Forage`, refutation is an event because $\neg\varphi$ is a guard; the case where the source is exhausted produces neither event. Proposition 33.1 is visible as three code paths, one of which is empty.

- **Nothing screens.** No embryo is model-checked anywhere in the listing. A parent could in principle check its recombinant against the viability formula before dropping it, and does not, because the check is priced against the same stack — Proposition 33.27 as a programming decision rather than an argument.

- **Speciation costs one guard.** A and B share `mate` and are nevertheless isolated. Delete the `where` clause from `Mate` and they hybridise; the offspring inherits a sensing allele naming one source and a hypothesis allele tuned to the other, and starves without ever being refuted. That is the reading-frame condition failing at the level of the ecology.

- **The race is real.** `Live` and `Forage` both take from `m` and put back. They conserve the stack and cannot deadlock, but they interleave — which is Remark 33.8's point that living means racing others to your own stack, with the others here being your own body.

- **This world is at the autotrophic end.** Every organism feeds on a source, so by Proposition 33.12 every lineage's inquiry terminates and none pays an unbounded epistemic rent. Note how close the other end is: species B's sensing allele already names a deeper namespace, and a single reallocation of it toward a grazer's `m` converts the lineage to heterotrophy, with Corollary 33.5's standing cost arriving in the same step.

- **And it ends.** When both reserves reach zero, `Source` returns 0 forever, no guard on a positive quantum fires again, and the remaining tokens sit in metabolic channels. From there the only credit available is another organism's stack. Proposition 33.15 is not an asymptotic remark about this listing; it is the last thing that happens in it.

Chapter 34

Learning to Play

A framework that explains everything after the fact explains nothing. The previous chapter is a construction with a great many moving parts, and the honest test of it is whether the parts pay for themselves on a domain where we already know every answer and cannot be fooled about whether an explanation is doing work.

So this chapter plays noughts-and-crosses. The whole game — all 255,168 plays, all 5,478 reachable positions — is small enough to enumerate exactly, which means every claim below is a computed number rather than an argument. A ladder of six theories is built inside the generated logic, from "take the centre" to the exact minimax strategy, each rung priced by the depth of the formula it installs, and each rung's winnings measured against the metering.

Two results are worth knowing before the details arrive. The first is that the logic dictates the encoding rather than the other way round: requiring the notion of a fork to be expressible forces the board to be represented with squares rather than lines as its components, and there is no freedom in the matter. The second is that the true theory loses. The exact strategy wins more games than any rung below it and is still the worst purchase on the ladder, because it costs nearly three times what the rung below costs to buy four percentage points. Truth here buys safety — it is the only strategy that never loses — and safety is not food.

The chapter closes on a result i did not expect and have not been able to argue away: a population that has completely solved its environment draws every game, harvests nothing, and starves. A solved science is a dead ecology. Imperfection is load-bearing.

34.1 Introduction

34.1.1 Why a worked example, and why this one

The companion note Chapter 33 asks what it would take for a computation to do science, and answers in four parts: the environment of a computation is other computations; the hypothesis language is not chosen but generated from the presentation of the calculus; a hypothesis becomes an experiment by being installed as a guard, so that confirmation is a rendezvous; and the motive is metabolic, since tokens are conserved, never minted, and a learner that predicts badly starves.

Its Appendix A gives a small ecosystem in which every one of those constructions appears as ordinary code. What that appendix does not do—and the omission is the reason for this note—is show the framework *reasoning*. Its hypotheses are `q => q > 0` and `q => q > 4`: arithmetic predicates on the size of a quantum. No formula in it uses a structural connective, no assay discriminates between two structurally distinct hypotheses, the relaxation lattice is never walked with anything on it, and at no point does a better hypothesis visibly buy better performance. The appendix demonstrates that the framework *hosts* a learner. It does not demonstrate that it *produces* one.

The remedy is to instantiate the framework on a domain where the reader holds the correct answers independently, so that what the machinery emits can be scored against a standard the machinery did not supply. We take noughts-and-crosses. The choice needs a defence, since the game is trivial, and the defence is that triviality is the point: a worked example is for legibility, not difficulty, and a game small enough to solve exhaustively is a game in which every claim below can be checked rather than asserted. We checked all of them; §34.8 reports the numbers and Appendix 34.14 says how to regenerate them.

> *Remark* 34.1 (The magic square comes free). The obvious alternative—a magic-square puzzle—is the same problem. In the game of *Fifteen*, two players alternately claim numbers from 1 to 9 and the first to hold three summing to fifteen wins. Writing the numbers into a Lo Shu square makes the triples summing to fifteen exactly the lines of a three-by-three grid, and the two games are isomorphic. We mention this because the puzzle framing is the one that would have been chosen for a solitary solver, and the isomorphism shows what that framing costs: it hides the opponent. An adversary is not an inconvenience here. It is the thing that makes the environment a computation with a strategy to be learned, which is the only kind of environment the companion note's ontology has.

34.1.2 What the mapping is supposed to show

Four features of the framework are exercised, and the interest is that each is *forced* by the domain rather than imposed on it.

A move is an assay. One cannot observe an opponent without playing, and the position afterwards is not the position before. §1.1 of Chapter 33 argues that observation is action; here it is not an argument but the rules of the game.

Minimax is an alternating modal formula. The modalities are labelled by redex positions, and §34.4.1 shows that in this domain a redex position *is* a move.

So $\exists s.\langle K_s\rangle\forall t.[K_t]\cdots$ reads "I have a move such that for every reply I have a move such that…", which is minimax written out, and the quantifiers range over something rather than standing in for it. Proposition 33.2 of Chapter 33 then has something to say about which half of the game is cheap to learn.

A fork is a separating conjunction. Two threats whose completing squares are distinct. Disjointness is the entire content of the concept, and graded separation at $k = 0$ is exactly disjointness of interface. §34.4 shows that this demand dictates the board encoding.

Destructive assay bites. If winning harvests the loser, one never plays the same opponent twice, so hypotheses about individuals are worthless and the scientist must hold a hypothesis about a *namespace* of opponents.

34.1.3 What we are claiming, and what we are not

We claim that the framework, instantiated on this domain, generates a hypothesis language in which the recognised concepts of the game are terms rather than hand-coded features; that pricing those terms by modal depth produces a cost–benefit structure with a measurable and non-obvious shape; and that three of the companion note's propositions—the modal asymmetry, "yield, not truth", and confinement—are visible in the measurements rather than merely consistent with them.

We do not claim to have built a strong player. Noughts-and-crosses is solved and a lookup table plays it perfectly; nothing here competes with that and nothing should. Nor do we claim the pricing schedule is canonical: $\kappa(d) = 2^d$ is a choice, defended in §34.6.1 and varied in §34.8, and the qualitative results survive the variation while the exact boundaries do not. Finally, the guards below are written in the ideal logic. The current interpreter accepts only the arithmetic and pattern fragment of the `where` clause; where a guard uses a structural or modal connective it is a specification of a check rather than a check the reducer performs, and we mark each such occurrence.

34.2 What this chapter carries forward

Everything below is machinery from Chapter 33, restated compactly enough that the worked example can be followed without turning back, and no more compactly than that. Where a construction is used but not re-derived, the derivation is there; the spatial–behavioural rubric is developed in [20].

The cut, and three grades of access. A world is $W \equiv S \mid E$: the scientist and its environment, separated by parallel composition. Computations are ontologically isolated, so E is other computations, and what S can come to know of E is bounded above by E's context-labelled bisimulation class. Access comes in exactly three grades:

perception, the structural predicates evaluated at the surface of E, priced by a schedule $\kappa_{\text{sense}}(d)$ monotone in formula depth; *interaction*, the context-labelled modalities, costing a rendezvous per step; and *reflection*, the name predicate $@\varphi$ applied to names S already holds, which is cheap because it passes through no surface.

The logic is generated. Feeding the presentation of the rho calculus to the OSLF family yields one structural connective per term former, one context-labelled modality per rewrite rule and redex position, and—because | carries an associative–commutative theory—a genuine separating conjunction:

$$P \models \varphi \mid \psi \iff P \equiv Q \mid R \text{ with } Q \models \varphi,\ R \models \psi.$$

Because a name is a quoted process, the name predicate $n \models @\varphi \iff n = @P$ with $P \models \varphi$ sees into the structure of a name; this is namespace logic [27], in which one describes a namespace by a property rather than enumerating it. The classifier note [20] adds a greatest fixed point $\nu X.\varphi$ and a *graded separation* counting how much of an interface two halves of a term share. We use both, but we take the second in a form the next paragraph explains.

There is no restriction, so there is no hidden-name quantifier. The classifier note grades its separation over a vector of *restricted* names and reaches under the binder with a hidden-name quantifier Ⲏ. We do not, because in this setting there is no binder. Restriction is syntactic sugar (Chapter 33, Remark 33.1): a name is a quoted process, and the calculus has a standing theorem that no process ever contains the name that codes it.

Proposition 34.1 (Freshness by quotation)**.** *Let P be a process and $C[-]$ any context with $C[P] \neq P$. Then $@C[P] \notin \mathrm{n}(P)$. Hence a name fresh in P may be manufactured by quoting any term properly containing P, with no registry and no consultation.*

Proof sketch. The no-self-code theorem gives $@Q \notin \mathrm{n}(Q)$ for every Q, and the argument is a well-foundedness one: a name occurring in a term codes a strictly smaller term. $C[P]$ properly contains P, so $@C[P]$ codes something larger than P and cannot occur among the names of P. □

Two things follow that the rest of the note uses. Freshness is *relative to a scope* rather than global, which is all any practical use requires and which keeps the scheme decentralised—a central name server would be an oracle sitting across the cut, contradicting isolation outright. And since nothing is bound, nothing is hidden from description: Ⲏ has no work to do, and where the classifier note grades separation over a restriction vector we grade it over a namespace given by a name predicate,

$$P \models \varphi \mid_k^N \psi \iff P \equiv Q \mid R,\quad Q \models \varphi,\ R \models \psi,\quad |\,\mathrm{fn}(Q) \cap \mathrm{fn}(R) \cap N\,| = k,$$

with N described rather than enumerated. Everything this note does that a modal logic cannot do is done with $|_k^N$, and §34.4 turns on the choice of N.

> *Remark* 34.2 (Concealment is a price, not a wall). Removing restriction removes the only mechanism by which a computation could be *unobservable* rather than merely expensive to observe. That is not a loss; it is the position Chapter 33 already takes when it prices concealment by a depth parameter δ instead of a barrier. It also settles a reading we would otherwise have had to argue for: in §34.3 we take δ to be the number of plies of lookahead needed to model an opponent, and with no restriction available that is the only thing δ *can* be. A player's strategy is never hidden. It is only ever deep.

Hypotheses are guards. The `where` clause of the rho calculus variant [21] puts a formula in operational position: $\texttt{for}(p \leftarrow c \;\texttt{where}\; \vartheta)P$. A hypothesis installed as a guard is an experiment and a confirmation is a rendezvous; nothing is carried and no interpreter is needed, and the cost of testing is metered by the apparatus that meters everything else. An *assay* is a complementary guard pair, on φ and on $\neg\varphi$, whose three outcomes are confirm, refute, and $\bot$—the last being the honest residue in which the environment did not answer in time.

The space and its metric. Beneath the characteristic formula χ_P sits a lattice of progressively weaker formulae obtained by forgetting conjuncts. Hypotheses live there, and revision is a move on it. The metric is ultrametric and stratified by the ν-unfolding stage: $d(\varphi, \psi) = 2^{-n}$ when the two agree on all approximants below stage n and differ at n. Since modal depth is what a finite budget bounds, distance, depth, and cost are one graded structure. The geometry is a tree, so there is no gradient, and Proposition 33.4 of Chapter 33—*confinement*—says a walk whose steps are all shorter than 2^{-n} never leaves the ball of radius 2^{-n} it started in.

The game. Tokens are conserved: $\sigma + \kappa = \sigma_0 = \Theta$, with σ held in live stacks and κ migrated into the record. There is no primary producer. Under the internalising encoding $[\![-]\!] : \mathcal{C}(\mathsf{rho}) \to \mathsf{rho}$ a located stack becomes a message at a *metabolic channel* m_P, and *harvest* is an ordinary receipt on m_P performed by someone else. In $\mathcal{C}(\mathsf{rho})$ stacks are inviolable; in the image they are messages. The ecology lives in that gap, and Theorem 33.1 of Chapter 33 identifies the predatory contexts as exactly the non-image contexts holding a metabolic name. A *game*, in the technical sense of Definition 33.9 there, is a choice of admitted context class $\mathcal{A}$ with $[\![\mathcal{C}(\mathsf{rho})\text{-Ctx}]\!] \subseteq \mathcal{A} \subseteq \mathsf{rho}\text{-Ctx}$.

34.3 The world

34.3.1 Three kinds of thing, and only one of them holds tokens

The world contains an *arena*, a *practice wall*, and a population of *players*.

The arena implements the rules. It alternates moves, detects completed lines, and—this is the only thing about it that matters structurally—on a win it hands the winner the loser's metabolic name. It holds no stack of its own and it mints nothing. It *routes*.

That distinction is what keeps §16 of Chapter 33 intact. The objection to an exogenous reward signal is that it requires a referee holding a mint, and a referee with a mint destroys conservation and with it the entire motivational story. An arena that routes is not such a referee: every token it moves already existed, in the loser's stack, and the total is unchanged. What the arena supplies is not value but *permission*.

> *Observation* 34.1 (The arena is the admitted context class). Definition 33.9 of Chapter 33 leaves $\mathcal{A}$ as a parameter and observes that its two endpoints are degenerate: at one end the decorated calculus with no game at all, at the other total war in which any context takes anything it can address. A game of noughts-and-crosses picks out a specific interior point. The contexts admitted to hold m_P are exactly those that have completed a line against P, and by Theorem 33.1 those are non-image contexts. So the rules of the game are not decoration on the ecology; the rules of the game *are* the choice of $\mathcal{A}$, and predation is legal here for precisely one reason, which is that somebody won.

> *Remark* 34.3 (Capability gating, concretely). Chapter 33 distinguishes the capability-gated game—contexts that *derive* m_P from structure they can perceive—from one in which the name is simply handed over. Noughts-and-crosses is gated in exactly that sense once one asks how a player comes to complete a line: it must perceive the position, and perceiving the position is priced. The derivation is the play.

34.3.2 The wall is a source, not a prey

A world in which the only food is other players is a world of pure heterotrophs, and by §11 of Chapter 33 such worlds pay an unbounded epistemic rent and are hard to keep alive. We therefore include a *practice wall*: a stationary player that moves uniformly at random, backed by a finite reserve, which pays a small quantum per game played against it.

The wall has the access profile of a source in the precise sense of Chapter 33, §11.2: it is a persistent emitter, rate-limited by being a single contract, non-exclusive since anyone may call it, non-adaptive because its policy never revises, and exhaustible. A player is prey by the complementary profile: a linear send, exhaustible, exclusive, and adaptive. The distinction is not a difference of sort or of rewrite rule but of *multiplicity*, which is what makes the trophic split a fact about the cut rather than a taxonomy imposed on it.

This gives the world its two order parameters in interpretable form.

- δ, the concealment depth, is *how many plies of lookahead are needed to model one's opponent*. The wall has $\delta = 0$: its policy is uniform and there is nothing to learn. A player running the top rung has a δ of four or more.
- ι, the fraction of Θ already internal to scientists rather than loose in the environment, is the ratio of the players' stacks to the wall's remaining reserve. It rises monotonically as the wall is drained.

34.3.3 Conservation, stated for this world

Write $\Theta = R_0 + \sum_i \sigma_i(0)$ for the wall's opening reserve plus the players' opening stacks. Four operations move tokens and none creates them: the wall's payout moves reserve into a player; harvest moves a loser's stack into a winner's; endowment moves a parent's stack into a child's; and the metabolic debit moves a token from σ into κ. Hence $\sigma + \kappa = \Theta$ at every step, and the world is a finite race. The listing of Appendix 34.13 can be audited for this by inspection.

34.4 The board: the logic dictates the encoding

Here is the first place where the domain talks back, and we think it is the most instructive moment in the note.

We want "fork" to be expressible, because a fork is the first genuinely strategic concept in the game and because it is exactly the sort of thing a game-playing program hand-codes as a feature. A fork is a double threat: a position from which one has two distinct winning replies, so that the opponent can block only one. Disjointness is the whole content: two threats sharing their completing square are not a fork, they are one threat counted twice.

Separation is the connective for that. But which separation, and of what? The answer depends on how the board is encoded, and the constraint turns out to be tight.

The encoding. A board b carries, for each square k, two names manufactured by reflection in the manner of the knot encoding's arcs: a *move channel* $sq_k = @[*b, k]$ and a *mark cell* $mk_k = @[*b, \texttt{"m"}, k]$. These are not drawn from a stock of atoms and they are not bound by anything. They are Proposition 34.1 in use: each is the quote of a term properly containing the board, hence fresh in the board, and the family is generated by a decidable predicate on names, so the whole namespace can be *described*—which is what Th_b below does and what makes the graded separation available at all. An empty square is a *receipt* offering to be claimed; an occupied square is the absence of that receipt together with a resting send on its mark cell:

$$\mathsf{Empty}_k \;\equiv\; \texttt{for}(@p \leftarrow sq_k \;\&\; @\_ \leftarrow mk_k)\{\; mk_k!(p) \;\} \qquad\qquad \mathsf{Held}_k(p) \;\equiv\; mk_k!(p).$$

In parallel sit eight *line* processes. A line reads the three mark cells of its line and replaces them; if it finds two marks of one player and one empty square s, it deposits a standing *offer* on a third family of names, $th_s = @[*b, \texttt{"t"}, s]$. A threat is therefore not computed on demand. It is a piece of the term.

Offers live on their own names rather than on the move channels for a reason worth stating: were a line to signal a threat by sending on sq_s, the signal would rendezvous with the square's receipt and the line would have played the move itself. Perception must not be able to act. That is the separation of grades one and two of Table 1 of Chapter 33, enforced by the choice of namespace.

Definition 34.1 (Threat, fork)**.** *For a player p,*

$$Th_b := \{\, n : n \models @[*b, \texttt{"t"}, \_\,]\,\}, \qquad \mathsf{Threat}_s(p) := \mathsf{out}(th_s)@p,$$

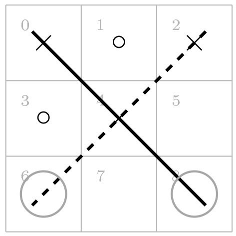
$$\mathsf{Threat}(p) := \exists s.\ \mathsf{Threat}_s(p) \mid \top,$$

$$\mathsf{Fork}(p) := \mathsf{Threat}(p) \mid_0^{Th_b} \mathsf{Threat}(p).$$

Th_b is the board's threat namespace, described by a name predicate rather than listed. $\mathsf{Fork}(p)$ says the term splits into two parts, each carrying an offer, *sharing no name of that namespace.* Two threats at the same square—two lines with a common gap—both deposit on th_s, so the two parands share that name, $k \geq 1$, and $|_0^{Th_b}$ excludes them. This is the graded separation of [20], §2.3.2, used at the bottom of its grading and graded over a described namespace rather than a restriction vector; it was not introduced for this purpose.

Proposition 34.2 (The grading namespace must be indexed by square)**.** *Let $[\![B]\!]$ be an encoding under which $\mathsf{Fork}(p)$ as given in Definition 34.1 is satisfied exactly by the positions in which p has a fork. Then the namespace N over which $|_0^N$ counts sharing must be indexed by the completing square, not by the line.*

Proof sketch. Two lines of a three-by-three grid meet in at most one square, so distinct lines carry distinct line names and share none. Under a line-indexed encoding $|_0$ is therefore satisfied by any two distinct threats—including a pair whose gaps coincide, for instance the two diagonals when only the centre is empty. That is not a fork, since one move blocks both. Under the square-indexed encoding two threats share a name iff they share their completing square, so $k = 0$ holds iff the gaps are distinct, which is the definition of a fork. □



threat A: line $\{0,4,8\}$, its only gap at square 8
threat B: line $\{2,4,6\}$, its only gap at square 6
the two lines *do* share square 4 — but it is occupied;
what they do not share is a *gap*, and $6 \neq 8$, so the two
standing offers rest on different square channels:

$$\mathsf{Fork}(\times) \;=\; \mathsf{Threat}(\times) \mid_0 \mathsf{Threat}(\times)$$

Figure 34.1: A realised fork for crosses, and why $|_0$ at $k = 0$ is the right connective. Crosses threatens to complete on square 6 or square 8; noughts can block only one. What the separation must exclude is a pair of threats with a *common* gap, since one move blocks both—and that is exactly what $k = 0$ excludes once the squares, rather than the lines, are the parands (Proposition 34.2).

> *Remark* 34.4 (What just happened). We did not choose the encoding and then ask what the logic could say about it. We asked for a concept, and the requirement that the generated logic express it determined the encoding. This is worth flagging because it is the mode in which the framework is meant to be used and it is the opposite of the usual order. In a hand-built game player, "fork" is a feature the programmer writes because they know the game; here it is a formula in a logic nobody designed, and the price of admission is a constraint on how the world is laid out. The companion note argues that a generated logic is the right hypothesis language on grounds of adequacy. This is the same claim cashed out as a design discipline.

34.4.1 The redex context is the move

The modalities of the generated logic are labelled by contexts: one primitive $\langle K_j \rangle$ per rewrite rule *and choice of redex position*, where K_j is the one-hole context determined by that position ([20], §2.1). The label is not an action name. It records the surroundings in which the step fires, and in this domain those surroundings are exactly what carries the content.

The rho calculus has one interaction rule, so all the information in a label is in the position. Write a world in play as

$$W \equiv \Big(\prod_{k \in \text{empty}} \mathsf{Empty}_k \;\Big|\; \prod_{k \notin \text{empty}} \mathsf{Held}_k(m_k) \;\Big|\; L \;\Big|\; Th \;\Big|\; M_p \Big),$$

with L the line processes, Th the standing offers, and M_p the mover's code, which for a move at s contains the send $sq_s!(p)$.

Definition 34.2 (The move context)**.** *For an empty square s, the redex position of the claim at s is the join of $sq_s!(p)$ with Empty_s, and the one-hole context determined by it is*

$$K_s := \Big([\cdot] \;\Big|\; \prod_{k \in \text{empty},\; k \neq s} \mathsf{Empty}_k \;\Big|\; \prod_{k \notin \text{empty}} \mathsf{Held}_k(m_k) \;\Big|\; L \;\Big|\; Th \;\Big|\; M'_p \Big),$$

M'_p being the mover's remaining code. We write $\langle K_s \rangle \varphi$ for the corresponding generated modality.

Proposition 34.3 (Moves are redex positions)**.** *The redex positions of W at which a claim can fire are in bijection with the empty squares, and the mover is determined by the context. Hence*

$$\text{“}p \text{ has a move } s \text{ after which } \varphi\text{”} \quad = \quad \exists s.\ \langle K_s \rangle \varphi,$$

where the existential ranges over redex positions, and the branching factor of that quantification is the number of legal moves.

Proof sketch. An occupied square offers no receipt, so no claim can fire there; each empty square offers exactly one, and the mark cells are disjoint, so the receipts do not race with each other. Distinct s give distinct contexts, since K_s retains $\mathsf{Empty}_{s'}$ for every $s' \neq s$ and $K_{s'}$ does not. Whose move it is, is fixed by which M_p sits in the context. □

Remark 34.5 (Why an action label would not do)*.* Every claim, at every square, by either player, is the same action: a comm on a square channel carrying a mark. An action-labelled modal logic in the style of Hennessy–Milner sees one label and cannot distinguish nine moves; recovering the distinction would mean encoding the square into a name discipline and then reasoning about names, which is the manoeuvre [20], §3.1 declines in the knot setting for the same reason. Context-labelling gives it directly: the square being claimed is the hole, and everything

> else is the label.

> **Lemma 34.1** (The board settles)**.** *After a claim fires, the line processes run to a unique normal form in finitely many steps, and the resulting family of offers depends only on the marks.*

Proof sketch. Each of the eight lines performs one join on three mark cells and replaces what it took, so no line's output enables another: offers are deposited on the *th* names, which no line reads. The settling is therefore a finite, conflict-free sequence of eight steps up to the ordinary races for mark cells, and those commute because each restores its cell unchanged. Hence confluence and termination. □

> **Definition 34.3** (The settled modality)**.** $\langle\!\langle K_s \rangle\!\rangle\varphi$ *holds when the claim at s fires and φ holds of the term after the settling of Lemma 34.1. This is the weak diamond, and Lemma 34.1 makes it deterministic in s.*

We use $\langle\!\langle - \rangle\!\rangle$ throughout, and note that a plain $\langle K_s \rangle$ would land the scientist between a move and the world's recognition of it—a state in which the threat structure is stale. That the distinction has to be drawn at all is a fact about a world in which perception is a process rather than a lookup.

34.5 The ladder

34.5.1 Six rungs and a limit

The relaxation lattice for this domain has a ladder through it that every human learner of the game climbs, and it is convenient that the ladder is already named: it is essentially Newell and Simon's rule list. We write it in the generated logic, cumulatively, each rung adding one disjunct above the previous ones in priority.

rung	formula added	kind and depth
0	$\top$	—
1	$\exists s.\ \mathsf{Emp}(s) \wedge \mathsf{Threat}_s(\text{me})$	spatial, depth 1
2	$\exists s.\ \mathsf{Emp}(s) \wedge \mathsf{Threat}_s(\text{them})$	spatial, depth 1
3	$\exists s.\ \langle\!\langle K_s \rangle\!\rangle \mathsf{Fork}(\text{me})$	one modal step over $\mid_0$
4	$\exists s.\ \forall t.\ \langle\!\langle K_s \rangle\!\rangle [\![K_t]\!]\, \neg\mathsf{Fork}(\text{them})$	box under diamond, depth 2
5	$\mathsf{Central}$	name-level, depth 0
6	$\chi := \nu X.\ \forall s.\ [\![K_s]\!]\big(\neg\mathsf{Lost} \wedge \exists t.\ \langle\!\langle K_t \rangle\!\rangle X\big)$	full alternation

Here $\mathsf{Emp}(s) := \mathsf{in}(sq_s)\top$—the square still offers a receipt, a structural predicate of depth one—and $\mathsf{Central}$ is a preference over square *names*, read off the reflected structure $@[*b, k]$ by a name predicate, containing no modality and requiring no step at all. The quantifiers $\exists s$ and $\forall t$ range over redex positions, which by Proposition 34.3 is to say over legal moves.

Two features of the shape are worth naming, and both were invisible while K was an uninstantiated placeholder.

> *Observation* 34.2 (Winning and blocking are perception, not lookahead). Rungs 1 and 2 contain no modality. Having a winning move is not a fact about what would happen if one moved; it is the fact that a completing offer stands at a square still open, and the line processes deposited that offer when the world last settled. The scientist reads it. The behavioural content was pushed into the structure of the term by a computation somebody else already paid for.

This has a general form, and it is the most useful thing in the section. There are two routes to any predicate about the environment: *read* it, if the environment has already computed it into legible structure, or *drive* it, by taking steps and watching. The first is perception and the second interaction—grades one and two of Table 1 of Chapter 33—and they are priced by different schedules, one by formula depth and one by rendezvous. An environment that computes a great deal about itself into structure is one in which a poor scientist can nonetheless know much; an environment that computes nothing until driven is one where knowledge is available only to those who can afford to act. That is a claim about environments rather than about learners, and this framework can state it because it prices the two grades separately.

> *Observation* 34.3 (The asymmetry has moved to the right place). With K instantiated, the existential and universal quantifiers range over moves, and the ladder's one genuinely alternating rung is rung 4: $\exists s \forall t$, a box under a diamond. By Proposition 33.2 of Chapter 33 the diamond has a finite confirming witness and the box has none, so rung 4 is the rung whose confirmation is unbounded and whose refutation is cheap. Section 34.8 finds that it is also, by a wide margin, the worst purchase on the ladder: $+0.005$ of win probability for 111 metered units. Making a fork is establishing an existential; preventing one is discharging a universal, and the framework prices that difference before measuring it.

> *Remark* 34.6 (Why χ is reachable here, and what that costs later). Proposition 33.3 of Chapter 33 says the complete theory of an environment is unaffordable when its recursion follows the reduction rather than a finite structural cycle. Noughts-and-crosses is the converse case of Remark 33.3 there: the board loses a square at every step, so the alternation in χ bottoms out and ν is exact at a

finite unfolding—nine, in fact. This is a *closed* science, one of the subject matters admitting a complete finite theory. That is what makes the domain a good demonstration and, as §34.10 shows, what kills the ecology.

34.5.2 The rungs are the ν-approximants

The ladder is not merely ordered; it is metrically graded, and by the metric the companion note already fixed.

Proposition 34.4 (The ladder is a chain of shrinking steps)**.** *Under the stratified distance of Chapter 33, Definition 33.5, the rungs agree on every approximant below the stage at which the added conjunct first looks and differ at that stage. Rungs 1 and 2 add conjuncts that inspect the settled term without stepping, and so sit at stage 1; rung 3 adds one $\langle\langle K_s \rangle\rangle$, reaching stage 2; rung 4 adds a box beneath it, reaching stage 3; and χ is the limit of the sequence. Hence $d(\varphi_r, \varphi_{r+1}) = 2^{-r}$ for $r = 1, \ldots, 4$, and the steps shrink as one climbs.*

Two consequences follow immediately and they are the ones that structure §34.9. First, moving from rung r to rung $r+1$ is a *short* move—distance 2^{-r}, shrinking as one deepens—so by confinement a scientist that only ever deepens can never leave the ball its early commitments placed it in. Second, by Proposition 33.5 of Chapter 33, cost is monotone in distance, so the cheap moves are exactly the confined ones.

34.5.3 The modal asymmetry, and what it predicts

Observation 34.3 located the ladder's one alternating rung. It is worth writing the asymmetry out, now that the quantifiers have something to range over:

$$\underbrace{\exists s.\ \langle\langle K_s \rangle\rangle \mathsf{Fork}(\mathrm{me})}_{\text{a finite witness: exhibit the square}} \qquad \text{versus} \qquad \underbrace{\exists s.\ \forall t.\ \langle\langle K_s \rangle\rangle [\![K_t]\!]\, \neg \mathsf{Fork}(\mathrm{them})}_{\text{no finite witness: every reply must be checked}} .$$

Establishing that one *can* fork requires producing the move. Establishing that a move is *safe* requires discharging a universal over the opponent's replies, and by Proposition 33.2 of Chapter 33 no finite run witnesses that; only a counterexample is finite. The framework therefore predicts, from the shape of the formula rather than from anything about games, that offensive knowledge is the cheaper acquisition—which is what every human learner of the game in fact does, and which §34.8 confirms in the currency that matters here: rung 4 is the worst purchase on the ladder.

There is a second, independent measurement of the same asymmetry. Restricting the top heuristic rung to each colour separately: as the first player it achieves the game-theoretic value against a perfect opponent, drawing every game; as the second player it loses 6.3%. The same policy is complete on offence and incomplete on defence.

Remark 34.7 (A reading, offered as such). We resist over-claiming on the second measurement. The first player has the initiative, so existential reasoning is more nearly sufficient for it, and one could attribute the colour asymmetry to that alone. What the framework adds is a reason the two are not symmetric *in price*: the universal formula is the one whose confirmation has no finite witness, so a budget-bounded learner confirms attacks and can only ever refute claims of safety. We take the colour measurement as consistent with Proposition 33.2; the rung-4 cost is the demonstration.

34.6 The assay

A move is an experiment, and in this domain that is not a metaphor to be maintained but the only thing a move can be.

Definition 34.4 (The move-assay). *For a scientist holding hypothesis φ with metabolic channel m, a move on probe channel p is*

```
contract Move(b, @hyp, @me, ret) = {
  new probe in {
      @[*b, "offer"]!(*probe, me) // the probe perturbs the world
    | for (@pos <- probe where pos |= hyp) { ret!(("confirm", pos)) }
    | for (@pos <- probe where ~(pos |= hyp)) { ret!(("refute", pos))
        }
  }
}
```

The complementary pair is what makes this more than an expression of hope: in a concurrent setting a guard that simply fails to fire cannot be distinguished from a guard still waiting, so the negated guard is what separates refutation from patience. The three outcomes of Proposition 33.1 of Chapter 33 are three code paths, one of which is empty—and in this domain the empty path has a concrete reading. If the opponent does not reply within the budget, the game is abandoned unresolved; the scientist learns nothing and has paid for the attempt. That is $\perp$, and it is neither confirmation nor refutation.

Remark 34.8 (Where the checker is). Nothing in the listing consults a hypothesis and then decides. The hypothesis is in guard position, so the reduction rule is the checker, and the cost of testing is metered by the same apparatus that meters the move. An unpriced test would be an oracle and would collapse the motivational story entirely; this is the concrete form of that constraint.

34.6.1 Pricing

We charge each rule the depth of the formula it installs, times the number of candidate moves it must examine:

$$\kappa(d) = 2^d, \qquad \text{cost of a decision at rung } r \;=\; |M| \cdot \sum_{\text{rules } 1..r} \kappa(d_{\text{rule}}),$$

with $|M|$ the number of legal moves—which by Proposition 34.3 is the branching factor of the quantification over redex positions, so the two readings of $|M|$ agree. Rungs 1 and 2, being structural at depth 1, cost 2 per candidate each; rung 3 adds 4 for its single settled step over a separation; rung 4 adds 8 for the box beneath the diamond; Central, being name-level at depth 0, adds 1; and χ costs 32.

Three comments. The schedule is a *choice*, and §34.8 varies it. The exponential shape is not arbitrary: it is what makes distance, depth, and cost one graded structure, which is the reason Chapter 33 prefers the ν-stratified ultrametric to a measure-theoretic alternative. And the positional rung costing almost nothing is not a thumb on the scale; it is a consequence of Central containing no modality, hence requiring no lookahead. That this cheap rung turns out to be the most profitable one on the ladder is the main empirical finding below, and it would have been invisible under a schedule that priced by rung index rather than by formula depth.

34.7 Destructive assay, again

Section 8 of Chapter 33 argues that because harvest ends the prey's computational life, each specimen affords exactly one measurement, so hypotheses about individuals are worthless and the hypothesis language must be a logic of namespaces. The argument is abstract there. Here it is the situation.

A scientist that wins eats its opponent, so it never faces that opponent again. It cannot accumulate a model of *this* player across games; what it can accumulate is a model of the *kind* of player—a namespace description, satisfied by many opponents, and the thing a name predicate $@\varphi$ states. Every rung of §34.5 is of this form: none of them mentions an individual, all of them describe a class of positions or a class of policies.

There is a corollary worth making explicit because it has a familiar shape.

> **Proposition 34.5** (Learning requires a population). *In a world where victory is fatal to the loser, no single scientist can test a hypothesis about opponent policy more than once against the same opponent. Statistical confirmation therefore requires either many opponents drawn from one namespace, or a lineage that inherits the hypothesis. The wall supplies the first and §34.9 supplies the second.*

This is the same conclusion §17.2 of Chapter 33 reaches for continual learning by a different route: a population is required rather than merely preferred.

34.8 The measurements

Everything in this section is exact rather than sampled. The game has 5,478 reachable positions, 765 up to the symmetries of the square, and 255,168 distinct plays; of its 4,520 non-terminal positions, 1,890—41.8%— offer a fork to the player to move, which is why the rung-3 conjunct is worth having at all. Outcome distributions and expected metering are computed by recursion over the whole tree, with ties within a rung broken uniformly. Figures below are symmetrised over colour: each policy plays first half the time.

34.8.1 The ladder pays, and where

rung	added conjunct	win	draw	loss	cost	Δwin	return
0	$\top$	0.437	0.127	0.437	0.0	—	—
1	$\mathsf{Threat}_s(\mathrm{me})$	0.668	0.075	0.258	40.9	+0.231	+0.0057
2	$\mathsf{Threat}_s(\mathrm{them})$	0.798	0.165	0.037	77.8	+0.130	+0.0035
3	$\langle\!\langle K_s \rangle\!\rangle \mathsf{Fork}$	0.831	0.135	0.035	134.9	+0.033	+0.0006
4	$\forall t.[\![K_t]\!]\neg\mathsf{Fork}$	0.836	0.138	0.027	245.7	+0.005	+0.00005
5	$\mathsf{Central}$	**0.918**	0.080	0.002	255.9	+0.083	**+0.0082**
6	χ	0.873	0.127	**0.000**	679.2	−0.046	−0.0001

Table 34.1: The ladder against a uniform-random opponent. "cost" is expected metering per game under $\kappa(d) = 2^d$; "return" is Δwin per metered unit. Loss rate falls monotonically along the whole ladder. Win rate does not: it peaks at rung 5.

Three features of Table 34.1 deserve separate statements.

> *Observation* 34.4 (Diminishing returns, until they don't). Marginal return falls by more than two orders of magnitude from rung 1 to rung 4—+0.0057, +0.0035, +0.0006, +0.00005—while cost roughly doubles at each step. Rung 4 in particular buys +0.005 of win probability for 111 metered units, and it is the alternating rung of Observation 34.3: the one whose formula has no finite confirming witness. A budget-bounded learner would stop before it, and the foraging inequality of Chapter 33, Proposition 33.7, says exactly when: harvest credits the scientist only if the prey's stack exceeds what finding and opening it cost.

> *Observation* 34.5 (The non-modal rungs carry the ladder). Once K is instantiated, three of the five useful rungs turn out to contain no modality: rungs 1 and 2, which read standing offers, and rung 5, which reads square names. Together they deliver +0.444 of the ladder's total +0.481 gain —92.3% of the yield—for 87.9 of

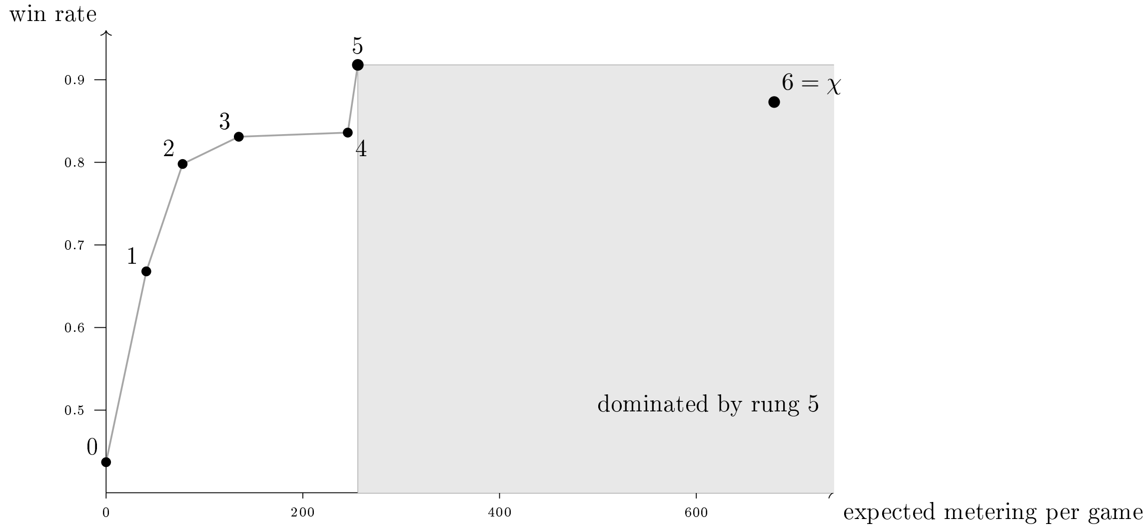


Figure 34.2: The ladder as a cost–yield curve, against a uniform-random opponent. Rungs 1–4 show sharply diminishing returns for a roughly doubling price, rung 4—the alternating one—being the flattest. Rung 5, a predicate on square names of depth zero, is nearly free and delivers the largest jump on the ladder. The complete theory χ lies inside the shaded region: it costs more than rung 5 and wins less, so it is dominated on both axes.

> its 255.9 metered units, or 34.3% of the cost. The two modal rungs deliver the remaining +0.038 for 168.0 units. Rung 5 alone returns +0.0082 per unit, the best figure on the ladder and some one hundred and sixty times rung 4's.

This is the result we would lead with, and it is one we could not have stated before the correction: with $\langle K \rangle$ standing in as a placeholder, rungs 1 and 2 looked modal, and the split between reading and driving was invisible. The classifier note argues, in the knot setting, that the spatial layer is where the generated logic earns its name and that a purely behavioural logic cannot state the properties that matter ([20], Remark 5). That argument is made there on expressiveness grounds. Here it is cashed in an ecological currency: the non-behavioural fragment is not merely expressible-and-useful, it is where nearly all the yield per token lives, and a mortal learner that priced its own hypotheses would find it first. A modal logic could not have offered any of it.

> *Remark* 34.9 (Why this is not an artefact of the encoding). One might object that we simply put the threats into the term, so of course reading them is cheap. The objection has force and the answer is that somebody paid: the line processes run on every settling, and §34.6.1 charges that scan. What the encoding does is *relocate* the cost, from the scientist's budget to the world's—and that relocation

is precisely the distinction between an environment one can perceive and one that must be driven. The interesting quantity is not whether the work is done but who is billed for it, which is a question this framework can pose because both parties have stacks.

Observation 34.6 (The true theory is dominated). χ is the complete theory: it never loses, from either colour, against any opponent. It is also, against fallible prey, strictly worse than the heuristic rung below it on both axes at once. Against a random opponent rung 5 wins 0.918 at cost 255.9 while χ wins 0.873 at cost 679.2; against a rung-4 opponent, 0.448 at 280.3 versus 0.418 at 708.8. Fewer wins, and between one and a half and two and a half times the price.

The mechanism is not mysterious and is well known to anyone who has implemented game search: minimax assumes optimal opposition, so it settles for a draw in positions where a heuristic would set a trap that a fallible opponent walks into. What is worth noting is that the framework *predicted* this before we measured it. Remark 33.6 of Chapter 33 says that a mortal scientist is not rewarded for holding true hypotheses but for holding nutritive ones, and that a lineage under selection drifts toward yield-biased search wherever yield and discrimination disagree. Here they disagree, the disagreement is measurable, and it runs in the predicted direction.

Corollary 34.1 (Truth buys safety, not food)**.** *χ is the unique rung with loss probability zero. So the complete theory does buy something the heuristic does not—it cannot be beaten—but what it buys is insurance rather than yield. In a world where survival is the only score and the prize is another's stack, insurance is worth having only when the stakes are large enough to matter, which is the next calculation.*

34.8.2 Invasion, and the price of being right

Take a population fixed at rung 5 and ask when a χ-holding mutant invades. Head to head, χ wins 3.17% and never loses; it costs 720.0 per game against the resident's 324.0. The mutant is favoured when $\sigma \cdot 0.0317 > \text{price} \cdot 396.0$, that is when

$$\sigma \;>\; 12474 \cdot \text{price},$$

where σ is the prey stack won and *price* converts metered units into tokens. The complete theory is therefore a luxury good, purchasable only where the stakes are some twelve thousand times the unit cost of thought.

> *Remark* 34.10 (Basic research, priced). Remark 33.22 of Chapter 33 concedes the pragmatist's objection—this scientist's epistemology is metabolic and truths that do not pay go unfunded—and answers with surplus: a scientist holding more than its foraging horizon requires can afford inquiry that does not pay. The number above is that remark made quantitative for one domain. Truth is affordable here, and only here, above a threshold; below it, the ecology selects for a theory that is known to be wrong and known to be more profitable. Pricing the ladder by the instantiated formulae rather than by rung index raises that threshold by a factor of three, because the cheap rungs turn out to be cheaper than a placeholder $\langle K \rangle$ made them look.

34.8.3 The phase band

Table 34.2 reports, for each opponent depth and each prey stack, which rung maximises $\sigma \cdot P(\text{win}) - \text{price} \cdot \mathbb{E}[\text{cost}]$.

σ	vs r0	vs r1	vs r2	vs r3	vs r4	vs r5	vs r6
20	2	2	0	0	0	0	0
50	2	2	3	3	5	0	0
100	5	5	5	5	5	0	0
200	5	5	5	5	5	2	0
400	5	5	5	5	5	2	0
800	5	5	5	5	5	2	0
1600	5	5	**6**	**6**	5	2	0

Table 34.2: Optimal rung, at price 0.05 per metered unit, as a function of the prey's stack σ (rows) and the prey's own rung (columns). The optimal rung is 0 in both extremes and interior in between.

The shape is the phase diagram of Chapter 33, §13.2, in a domain where both order parameters have concrete readings.

- **Low δ, low stakes: theory is overhead.** Against a shallow opponent worth little, rung 0 or 1 is optimal. Reflex beats inquiry; a theory is something one is paying to carry.

- **High δ: inquiry cannot pay.** Against a rung-6 opponent the optimal rung is 0 at *every* stake, because no rung ever beats it, so the cheapest available policy maximises net. This is the upper boundary, and it has a bracing reading: against an environment one cannot model within any affordable budget, the correct scientific policy is not to try.

- **The band between.** For intermediate opponents and stakes, an interior rung is optimal—mostly rung 5, the heuristic, not rung 6, the truth.

χ is optimal in exactly two cells of the whole grid, both at the cheapest price and the largest stake. Raising the price to 0.2 and to 1.0 shifts the band toward the cheap rungs without changing its shape: at price 1.0 the optimal rung is 0 over most of the table and never 6 anywhere. The qualitative structure—interior optimum, degenerate at both ends—is robust; the boundaries are not.

34.9 Confinement, measured

This section is the one we would keep if we could keep only one, because it turns Proposition 33.4 of Chapter 33 from a fact about ultrametric spaces into a fact about a hypothesis space someone might actually hold.

34.9.1 Two loci, and only one of them is reachable

A policy on this ladder has exactly two parameters, and they are the two that Definition 33.6 of Chapter 33 says a revision move must choose: a *locus* and an *operation*. The loci here are

- the **opening**—which square to take first—which is the stage-1 approximant, at metric distance 2^{-1}; and
- the **rung**—how deep the theory runs—which is everything at stage 2 and below.

Deepening the rung is a short move: from rung r to rung $r+1$ is a step of 2^{-r}. Changing the opening is a long one. By confinement, a scientist whose revisions are all deepenings can never change its opening, however long it lives and however many refutations it suffers. In the listing of Appendix 34.13 this is visible as a fact about the code: `Revise` only ever increments the rung.

34.9.2 The measurement

Lock the opening and let both sides otherwise play the top heuristic rung. Then against a rung-2 opponent:

opening	win rate at rung 5	win rate at rung 6 (χ)
centre	**0.833**	0.548
corner	0.597	**0.896**
side	0.318	0.710

Read the two columns against each other, because the disagreement is the point.

Observation 34.7 (The best opening is not a property of the game). Under the heuristic the centre is much the best opening; under the complete theory the corner is, by a wide margin, and the centre is the *worst* of the three. The stage-1 commitment cannot be evaluated in isolation from the stages below it, and the ordering reverses.

Corollary 34.2 (Deepening within a ball can make things worse)**.** *A lineage committed to the centre opening that upgrades its theory from rung 5 to χ falls from* 0.833 *to* 0.548*. Every step of that upgrade is a well-motivated refinement, each is licensed by refutations, and the result is a worse forager. Improvement at depth is not improvement, when the shallow commitment was wrong for the deep theory.*

This is Propositions 33.4 and 33.5 acting together and it is worth being precise about which does what. Confinement says the walk cannot reach the corner opening. Cost-monotonicity says the move that would reach it is the expensive one, because it invalidates every confirmation obtained at depth ≥ 1—which is all of them. The scientist is trapped by geometry and priced out of the escape by its own history of successful assays.

34.9.3 The escape, and what it costs whom

§6.5 of Chapter 33 says an individual has exactly one route out and it is a poor one, and that recombination is the other. Here is that claim, measured.

Consider a mutant identical to the resident population in every respect—same rung, same rules, same metering—differing only in the opening allele, which is a corner rather than the centre. It arises by crossover at the shallow locus: the opening allele of one parent with the rung allele of another.

	win	draw	loss
mutant as first player, vs resident	0.333	0.667	0.000
mutant as second player, vs resident	0.000	1.000	0.000
resident vs resident	0.000	1.000	0.000

The mutant wins a third of the games in which it moves first, never loses, and costs exactly what the resident costs, since it is the same policy with one different move. It invades unconditionally, at any stake, at any price.

Proposition 34.6 (The invading move is the one no individual can make)**.** *The corner-opening mutant differs from the resident by a revision of distance* 2^{-1}*. By Proposition 33.4 no walk of deepenings reaches it; by Proposition 33.5 its cost to an individual is the invalidation of every confirmation the individual holds. It is nonetheless free to the lineage, because the cost falls on an offspring that does not yet exist and has no confirmations to invalidate.*

That is the whole argument of §12 of Chapter 33 for why reproduction is part of the epistemology and not an appendix to it, exhibited on a hypothesis space small enough to check exhaustively. Sex is not here a mechanism for generating variation in general; it is a mechanism for making *shallow* revisions, which are exactly the ones the metric denies to a living scientist.

Remark 34.11 (Opening theory changes by catastrophe). The prediction generalises, and it is testable outside this framework: in any domain whose hypothesis space is ultrametric, early commitments should change discontinuously—by generational replacement, by schism, by the arrival of an outsider—and not by incremental refinement, whereas late commitments should change smoothly and continuously. That is a recognisable description of how opening theory in real games, and paradigms in real sciences, actually move. We note the resemblance and claim nothing further from it.

34.10 Succession, and the death of a solved world

Noughts-and-crosses is a draw under optimal play. That fact, harmless in game theory, is fatal here.

Proposition 34.7 (A solved science is a dead ecology)**.** *In a conserved world whose only predatory transfer is the arena's, a population all of whose members hold theories that draw against one another performs no harvest. Its members continue to burn tokens to live, and the wall's reserve is finite. Hence* σ *declines monotonically to zero and the population starves, converging faster the deeper its theories, since burn rate rises with the rung held.*

Both the χ monoculture and the centre-opening rung-5 monoculture are of exactly this kind: we measure all draws, harvest rate 0.000, in both. The better the population's science, the sooner it dies.

Corollary 34.3 (Imperfection is load-bearing)**.** *The corner-opening rung-5 population, by contrast, sustains a win rate of* 0.333 *for the first player against itself. Harvest continues; tokens circulate; the ecology persists. A strictly* worse-

informed population is the living one.

We resist reading a moral into this, but the structure is worth stating plainly because it is a consequence of conservation and not of anything we chose. Predation is the purchase of somebody else's remaining time (Chapter 33, Remark 33.7). A population that has solved its environment has nothing left to learn about each other and no asymmetry to exploit, so the only remaining transfer is metabolic burn, which is one-way into the record. Science, in this world, is a phase; and the far side of the phase boundary is not error but exhaustion.

Remark 34.12 (What would keep it alive). Three exits, none of which this note takes. Enlarge the game so that χ is unaffordable, putting the domain back on the Proposition 33.3 side of Remark 33.3— the recursion following the reduction rather than the structure. Let the arena's rules drift, so the environment is non-stationary and χ has a moving target. Or admit a genuine external source that is not exhaustible on the timescale of the population, which is to say, redraw the boundary of the conserved system. The third is the one biology took, and Chapter 33, §11.1 argues it is a bookkeeping choice about where the cut lies rather than a change in the physics.

34.11 What this shows, and what it does not

Shown. The generated logic contains the game's strategic vocabulary as terms rather than as hand-coded features, and the requirement that it do so constrains the encoding of the world (§34.4)—using less apparatus than the classifier note assembles, since with restriction treated as sugar the hidden-name quantifier is eliminable and separation grades over a described namespace instead. Pricing hypotheses by formula depth yields a cost–benefit structure whose shape is neither monotone nor obvious, in which the spatial conjunct is the most profitable purchase on the ladder and the complete theory is dominated (§34.8). Confinement is not an abstract property of ultrametrics but a specific trap on this ladder, with a specific escape available only to a lineage (§34.9). And conservation forces an ending (§34.10).

Not shown. We closed one of the companion note's declared free parameters—the revision policy—by adopting its own default, refutation rate as temperature, and a different policy would give different numbers, though we expect not a different shape. We did not implement the ideal guards: the structural and modal formulae of §34.5 are specifications, and the running listing checks their arithmetic shadows. The pricing schedule is a choice. And the ecology of Appendix 34.13 is written but not run to a fixed point; the measurements of §34.8 are exact game-tree computations over policies, not observations of a population, so every claim about invasion is a claim about payoffs, not a simulation result.

The honest summary. What the exercise establishes is that the framework is *contentful*: instantiating it on a known domain produces statements that could have come out otherwise and did not have to come out as the companion note predicted. That the spatial rung would be the best buy, that χ would be dominated, and that the invading mutant would be a shallow one were all consequences we computed rather than results we arranged.

34.12 Open problems

1. **Run the ecology.** Appendix 34.13 specifies a population; §34.8 computes payoffs. Closing the gap—sampling trajectories via the Gillespie construction of [25]—would turn Table 34.2 from a payoff calculation into the ensemble test that is the first open problem of Chapter 33.

2. **An unsolvable game.** Everything interesting about §34.10 follows from the domain being closed. Rerunning the mapping on a game whose χ is unaffordable—Connect Four is solved but not within a plausible budget; Hex is not solved at all—would test whether the phase band persists when the upper boundary is set by affordability rather than by the opponent's perfection.

3. **Is the ladder the lattice?** We wrote a ladder through the relaxation lattice, guided by what human players learn. Whether the lattice's own structure—the order in which forgetting conjuncts of χ yields satisfiable weakenings—recovers approximately this ladder, or a different one, is checkable for this game and would say whether "what people learn" and "what the logic makes cheap" agree.

4. **The reversal.** §34.9 finds that the optimal opening under χ and under the heuristic are different squares. Is that an artefact of this game, or is there a general statement—that the optimal shallow commitment is a function of the depth of the theory it heads, so that shallow and deep revisions cannot be separated? If the latter, it sharpens confinement considerably.

5. **Should** H **be retired?** §34.2 drops the hidden-name quantifier here on the grounds that restriction is sugar and freshness is manufactured by quotation (Proposition 34.1). The question is whether the classifier note's rubric should drop it too, and there is a specific reason to think it might: that note introduces H to reach under the `new` $a_1 \ldots a_{2n}$ that closes a knot diagram, and then observes two sections later that the arcs are already the manufactured names $@[*knot, k]$. If the arc names are manufactured, nothing binds them, and the name predicates it already has do the work H was brought in for. Whether the graded separation $|_k$ then survives the translation to $|_k^N$ on the knot examples—split links, nugatory crossings, Conway spheres—is checkable, and if it does the classifier note gets shorter.

6. **Co-learning.** When both players revise, each is a moving target for the other and Table 34.2 is a snapshot of a game whose payoffs are themselves changing. This is problem 3 of Chapter 33 and this domain is small enough to attack it in.

34.13 Appendix: The world, in rholang

The listing below is the whole construction. Guards using structural or modal connectives are marked `//! ideal` and are specifications rather than checks the current reducer performs; everything else runs. Conservation can be audited by inspection: `Wall`, `Harvest`, and `Beget` move tokens, `Live` debits them from σ into κ, and nothing mints.

```
// ============================================================================
// A WORLD OF MORTAL SCIENTISTS THAT PLAY NOUGHTS-AND-CROSSES
// ============================================================================
//
// Companion listing to "Learning to Play". Every construction of the
// mortal-scientist note appears here as ordinary code:
//
// Sec. 3 the arena -- Arena, Turn, Harvest (the admitted context class)
// Sec. 4 the board -- Board, Cells, Square, Line, Scan
// Sec. 4.3 the context K_s -- Square, Choose (a ply is ONE rewrite)
// Sec. 5 the ladder -- Ladder (hypotheses as formulae)
// Sec. 6 the assay -- Move (a move IS an experiment)
// Sec. 6.1 metabolism -- Live, Wall
// Sec. 9 the two loci -- Beget, Crossover (opening x rung)
// Sec. 10 the world -- the bottom block
//
// CONVENTION. A parameter written `c` is a NAME (a channel); a parameter
// written `@d` is DATA. To pass a channel one sends `*c`, since @(*c) = c.
//
// NOTATION. Guards are written in the generated spatial-behavioural logic:
// out(n)phi, in(n)phi, phi | psi, the name predicate @phi, the context-
// labelled modality <K_s>phi, the greatest fixed point nu X. phi, and the
// graded separation phi |_k^N psi -- graded over a NAMESPACE N given by a name
// predicate, not over a vector of restricted names. There is no hidden-name
// quantifier here and no need of one: `new` is syntactic sugar, since a name
// is a quoted process and no process contains the name that codes it, so a
// name fresh in P is manufactured by quoting any term properly containing P.
// Every `new` below could be replaced by such a manufactured name; the board's
// own names already are (see Board). Nothing is bound, so nothing is hidden
// from description -- concealment in this world is a price, never a wall. The
// current interpreter accepts only the arithmetic and pattern fragment of
// `where`; guards using a structural or modal connective are marked
// `//! ideal` and are specifications of a check rather than a check the
// reducer performs. Everything else runs.
//
// CONSERVATION. Nothing here mints. Wall moves reserve into a player;
```

```
// Harvest moves a loser's stack into a winner's; Beget moves a parent's stack
// into a child's; every Live debit moves a token from sigma into kappa. So
// sigma + kappa = Theta throughout, with Theta the wall's opening reserve plus
// the players' opening stacks.
// ==============================================================================

new stdout(`rho:io:stdout`), display, arena, wall, mate, ladder,
    Live, Wall, Board, Cells, Square, Line, Scan, Arena, Turn, Adjudicate,
    Harvest, Choose, Positions, Move, Revise, Ladder, Pick, Crossover, Beget,
    Body, PlayOne, Mate
in {

  for (@line <= display) { stdout!(line) }

  // ============================================================================
  // METABOLISM -- reclaim, debit, reissue. (Companion note, Def. 20.)
  // Between the receipt and the send the stack is a message in flight: living
  // is racing others to your own stack, and the metabolic channel m is exactly
  // the vulnerability that Harvest exploits.
  // ============================================================================

  | contract Live(m, @burn, step) = {
      for (@sigma <- m) {
        match sigma > burn {
          true => { m!(sigma - burn) | *step | Live!(*m, burn, *step) }
          false => { display!(("DIED", "could not fund a step")) }
        }
      }
    }

  // ============================================================================
  // THE PRACTICE WALL -- a source, not a prey.
  // Rate-limited (one contract), non-exclusive (anyone may call it),
  // non-adaptive (its policy never revises), and finite. That access profile
  // is what makes it a SOURCE in the sense of the companion note, Sec. 11.2,
  // and what makes this world mixotrophic rather than purely predatory.
  // ============================================================================

  | contract Wall(w, @quantum, @reserve) = {
      for (ret <- w) {
        match reserve >= quantum {
          true => { ret!(quantum) | Wall!(*w, quantum, reserve - quantum) }
          false => { ret!(0) | Wall!(*w, quantum, 0) } // exhausted
        }
      }
    }

  // ============================================================================
  // THE BOARD -- and why a ply must be exactly one rewrite.
  //
  // Square k of board b carries TWO manufactured names: a MOVE CHANNEL
```

```
// sq_k = @[*b,k] and a MARK CELL mk_k = @[*b,"m",k]. An EMPTY square is a
// RECEIPT offering to be claimed; an OCCUPIED square is that receipt gone
// and a resting send on its mark cell.
//
// Note what is NOT here: a ‘new‘ for the squares. Each square name is the
// quote of a term properly containing the board, so by the no-self-code
// theorem it cannot already occur in the board -- locally fresh, no registry
// consulted, decentralised by construction. And because the family is
// generated by a predicate on names, the whole namespace can be DESCRIBED
// rather than enumerated, which is what makes |_k^N available below.
//
// This is what makes a ply a single rewrite, and that is what makes <K_s> a
// GENERATED modality rather than a derived abbreviation: the redex position
// of the claim at s IS the move at s, and the one-hole context K_s is the
// whole rest of the board (Def. 5, Prop. 6). Nine empty squares are nine
// redex positions, so the branching factor of ‘exists s’ is the legal move
// count.
//
// Offers live on a THIRD family, th_s = @[*b,"t",s], and not on the move
// channels. Were a line to signal a threat by sending on sq_s, the signal
// would rendezvous with the square’s own receipt and the line would have
// PLAYED THE MOVE. Perception must not be able to act; that separation of
// grades one and two is enforced here by a choice of namespace.
// ==========================================================================

| contract Board(b, @marks) = { Cells!(*b, marks, 0) | Scan!(*b) }

| contract Cells(b, @marks, @k) = {
    match k < 9 {
      true => {
          @[*b,"m",k]!(marks.nth(k)) // the mark cell
        | match marks.nth(k) == 0 {
            true => { Square!(*b, k) } // empty: offer a receipt
            false => { Nil } // occupied: no receipt
          }
        | Cells!(*b, marks, k + 1)
      }
      false => { Nil }
    }
  }

// An empty square. The claim is ONE rewrite: a join consuming the mover’s
// send and the stale mark together. After it fires no receipt remains, so the
// square is no longer a redex position -- which is exactly Prop. 6.
| contract Square(b, @k) = {
    for (@p <- @[*b,k] & @old <- @[*b,"m",k]) { @[*b,"m",k]!(p) }
  }

// A line reads its three MARK CELLS and replaces them; it never touches a
// move channel. When it finds two marks of one player and one empty square s
// it deposits a standing offer on th_s. A threat is therefore not computed on
```

```
// demand: it is a piece of the term, and so visible to the structural layer.
| contract Line(b, @i, @j, @k) = {
    for (@x <- @[*b,"m",i] & @y <- @[*b,"m",j] & @z <- @[*b,"m",k]) {
        @[*b,"m",i]!(x) | @[*b,"m",j]!(y) | @[*b,"m",k]!(z)
      | match (x, y, z) {
          (p, q, 0) where p == q and p > 0 => { @[*b,"t",k]!(p) }
          (p, 0, q) where p == q and p > 0 => { @[*b,"t",j]!(p) }
          (0, p, q) where p == q and p > 0 => { @[*b,"t",i]!(p) }
          _ => { Nil }
        }
    }
  }

// The eight lines, re-run after each claim. This is the SETTLING of Lem. 8:
// each line fires once, no line's output enables another (offers go to the
// th names, which no line reads), so it terminates and is confluent -- which
// is what makes the settled modality <<K_s>> deterministic in s.
// Scanning is metered like everything else: perception is priced.
| contract Scan(b) = {
      Line!(*b,0,1,2) | Line!(*b,3,4,5) | Line!(*b,6,7,8)
    | Line!(*b,0,3,6) | Line!(*b,1,4,7) | Line!(*b,2,5,8)
    | Line!(*b,0,4,8) | Line!(*b,2,4,6)
  }

// ============================================================================
// THE LADDER -- six hypotheses about the position, and chi.
//
// Each rung is a formula in the generated logic; the rung index is the modal
// depth of the conjunct it adds, hence by Def. 8 of the companion note the
// ultrametric stage: d(Phi_r, Phi_{r+1}) = 2^-r.
//
// Threat(p) := H s. out(s)@("claim", p) | TOP spatial, depth 1
// Fork(p) := Threat(p) |_0 Threat(p) spatial, k = 0
// Win(p) := <K> out(over)@("won", p) modal, depth 1
// Central := a pure preference over squares spatial, depth 0
//
// Note the shape of rungs 1 and 2. Win is existential and the safety
// conjunct universal, so by Prop. 6 of the companion note Win is finitely
// confirmable while safety is only finitely refutable: attack is cheap to
// establish, defence cheap only to destroy.
// ============================================================================

//! ideal -- cumulative, highest priority first.
//
// Emp(s) := in(sq_s)TOP the square still offers a receipt
// Thr_s(p) := out(th_s)@p a completing offer stands at s
// Fork(p) := (H s. Thr_s(p) | TOP) |_0 (H s. Thr_s(p) | TOP)
// <<K_s>>phi := the claim at s fires, the board settles (Lem. 8), phi holds
//
// 'exists s' and 'forall t' range over REDEX POSITIONS, which by Prop. 6 is
// to say over legal moves. Note where the alternation is: rungs 1 and 2 have
```

```
// no modality at all -- winning and blocking are PERCEPTION, reading offers
// the last settling deposited -- and the one genuinely alternating rung is 4,
// a box under a diamond. By Prop. 6 of the companion note that is the rung
// with no finite confirming witness, and Sec. 8 measures that it is also the
// worst purchase on the ladder.
| contract Ladder(@r, ret) = {
    match r {
      0 => { ret!( @{ TOP } ) }
      1 => { ret!( @{ exists s. Emp(s) and Thr(s, me) } ) }
      2 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them)) } ) }
      3 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them))
                                  or <<K s>> Fork(me) } ) }
      4 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them))
                                  or <<K s>> Fork(me)
                                  or forall t. <<K s>>[K t] ~Fork(them) } ) }

      5 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them))
                                  or <<K s>> Fork(me)
                                  or forall t. <<K s>>[K t] ~Fork(them)
                                  or Central(s) } ) }
      // chi: exact at a finite unfolding, because the board loses a square at
      // every step, so the alternation bottoms out at nine. This is the
      // CLOSED-science case (companion note, Rem. 10) -- and Sec. 10 is where
      // that turns out to be fatal.
      6 => { ret!( @{ nu X. forall s. [K s]( ~Lost and exists t. <<K t>> X ) }
          ) }
    }
  }

// ==========================================================================
// THE ASSAY -- a move is an experiment.
//
// The complementary guard pair of Def. 4, verbatim. The hypothesis is not
// consulted and then acted on: it sits in guard position, so playing the
// move IS testing it and confirmation IS the rendezvous. The three outcomes
// of Prop. 5 are three code paths, and the third is empty -- if the world
// does not answer within budget the scientist has paid and learned nothing,
// which is ignorance, not refutation.
// ==========================================================================

// Choosing WHICH redex position to fire is the whole of the policy. The
//! ideal guard ranges over the empty squares -- that is, over the redex
// positions of Def. 5 -- and picks the first satisfying the held rung. By
// Prop. 17 of the companion note the number it can afford to evaluate is its
// stack divided by the cost of one check, so a poor scientist does not survey
// its own branching.
| contract Choose(b, @hyp, @me, ret) = {
    new cand in {
        Positions!(*b, *cand, 0)
      | for (@s <- cand where Sat(s, hyp, me)) { //! ideal
          new r in { Move!(*b, s, hyp, me, *r)
```

```
                | for (@o <- r) { ret!(s, o) } }
      }
    }
  }

| contract Positions(b, cand, @k) = {
    match k < 9 {
      true => { cand!(k) | Positions!(*b, *cand, k + 1) }
      false => { Nil }
    }
  }

| contract Move(b, @s, @hyp, @me, ret) = {
    new probe in {
        @[*b, s]!(me) // THE CLAIM: one rewrite, at redex s
      | Scan!(*b) // the settling of Lem. 8
      | @[*b,"state"]!(*probe)
                                              //! ideal: structural guards
      | for (@pos <- probe where pos |= hyp) { ret!(("confirm", pos)) }
      | for (@pos <- probe where ~(pos |= hyp)) { ret!(("refute", pos)) }
    }
  }

// Revision walks the lattice. The move has the two parameters the metric
// grades (Def. 13): a LOCUS and a DIRECTION. Here refutation rate is the
// temperature, which is the companion note's own default policy -- and it is
// an observable the assays already emit, not a schedule imposed from outside.
| contract Revise(@rung, @refutes, @confirms, ret) = {
    match refutes * 4 > confirms {
      true => { ret!(rung + 1) } // deepen: a move of distance 2^-rung
      false => { ret!(rung) }
    }
    // NB no case touches the OPENING. By Prop. 12 that is not an oversight:
    // the opening is the stage-1 approximant, a move of distance 2^-1, and an
    // individual can neither reach it by deepening nor afford it if it could.
    // Sec. 9 is where the opening changes, and it is not here.
  }

// ==========================================================================
// THE ARENA -- the rules, holding no stack of its own.
//
// On a win the arena hands the winner the loser's metabolic NAME. It mints
// nothing; it ROUTES, which is what keeps Sec. 16 of the companion note
// intact -- an exogenous reward would need a referee with a mint, and a
// referee with a mint destroys conservation.
//
// In the vocabulary of Def. 25 the arena IS the choice of admitted context
// class A: the contexts it admits to hold m_P are exactly those that have
// completed a line against P, and by Thm. 24 those are non-image contexts.
// Predation is legal here for precisely one reason, and it is that somebody
// won.
```

```
// ==========================================================================

| contract Arena(@hX, mX, @hO, mO, @g, ret) = {
    // The board token is MANUFACTURED, not new-bound: @[*arena, g] quotes a
    // term containing the arena, so it is fresh in the arena by Prop. 3, and
    // distinct games get distinct boards without a name server. `over` is
    // still written with `new` -- sugar, and unfolded the same way.
    new over in {
        Board!(@[*arena, g], [0,0,0,0,0,0,0,0,0])
      | Turn!(@[*arena, g], hX, *mX, hO, *mO, *over, 1)
      | for (@result <- over) {
          match result {
            ("won", 1) => { Harvest!(*mO, *mX, *ret) } // crosses eat noughts
            ("won", 2) => { Harvest!(*mX, *mO, *ret) }
            "draw" => { ret!("draw") } // nobody eats
          }
        }
    }
  }

| contract Turn(b, @hX, mX, @hO, mO, over, @who) = {
    new played in {
        Choose!(*b, if (who == 1) { hX } else { hO }, who, *played)
      | for (@s, @outcome <- played) {
          Adjudicate!(*b, *over, who)
        | Turn!(*b, hX, *mX, hO, *mO, *over, 3 - who)
      }
    }
  }

//! ideal -- the win test is a modal formula, not a scan of a list
| contract Adjudicate(b, over, @who) = {
    new r in {
        @[*b, "state"]!(*r)
      | for (@pos <- r where pos |= @{ Line3(who) }) {
          over!(("won", who))
        }
      | for (@pos <- r where pos |= @{ ~Line3(who) and
                                       ~ exists s. Emp(s) }) {
          over!("draw")
        }
    }
  }

// Harvest (Def. 22): an ordinary receipt on the loser's metabolic channel.
// No new rewrite rule is needed. The loser dies because the message it was
// going to reclaim has been consumed.
| contract Harvest(mPrey, mPredator, ret) = {
    for (@sigma <- mPrey) {
      for (@own <- mPredator) {
          mPredator!(own + sigma)
```

```
        | ret!(("harvest", sigma))
        | display!(("HARVEST", sigma, "tokens"))
      }
    }
  }

// ============================================================================
// REPRODUCTION -- two loci, and why that is the whole point.
//
// The genome is a quotation: inert, transmissible, inspectable, and free
// until dropped. Two loci are exposed and they are not arbitrary -- they are
// the two parameters of Def. 13. The RUNG allele is the depth of the theory;
// the OPENING allele is the stage-1 commitment.
//
// Crossover exchanges them independently, and a swap at the opening locus is
// a move of distance 2^-1 -- exactly the move Prop. 12 forbids to any
// individual walking incrementally. It is free to the lineage because the
// cost falls on an offspring that has no confirmations to invalidate. This
// is the escape of Sec. 6.5, and Sec. 9 measures that it invades.
// ============================================================================

| contract Pick(@a, @b, @bit, ret) = {
    match bit { 0 => { ret!(a) } 1 => { ret!(b) } }
  }

| contract Crossover(@mother, @father, @chi, ret) = {
    match (mother, father, chi) {
      ([om, rm], [of, rf], [b1, b2]) => {
        new r1, r2 in {
            Pick!(om, of, b1, *r1) // the opening locus -- stage 1
          | Pick!(rm, rf, b2, *r2) // the rung locus -- deeper
          | for (@o <- r1 & @r <- r2) { ret!([o, r]) }
        }
      }
    }
  }

// Birth. The child's code is CONSTRUCTED as a name and rests inert on `code`:
// it does not reduce, does not age, and burns nothing. Dropping it with * is
// what makes it a phenotype and what starts the meter. The guard enforces the
// minimum viable endowment -- a parent may not endow below what a child needs
// to fund a first game.
| contract Beget(m, @alleles, @endow, ret) = {
    for (@sigma <- m where sigma > endow + 40) {
        m!(sigma - endow)
      | match alleles {
          [o, r] => {
            new childM, code in {
                code!( @{ Body!(*childM, o, r, endow) } ) // the germ: inert
              | for (g <- code) { *g | childM!(endow) } // the soma: metered
              | ret!(("born", o, r))
```

```
          }
        }
      }
    }
  }

// ==============================================================================
// THE SCIENTIST -- a metered loop of games and matings.
// Alleles vary; this does not.
// ==============================================================================

| contract Body(m, @opening, @rung, @endow) = {
    new hyp, score in {
        score!(0, 0)
      | Ladder!(rung, *hyp)
        // The burn rate is the cost of HOLDING the theory, paid before any
        // assay is run. It is why the top rung is not free, and why Sec. 8
        // finds the exact theory chi dominated by the rung beneath it.
      | Live!(*m, 2 + rung, @{
            PlayOne!(*m, opening, *hyp, *score)
          | Mate!(*m, [opening, rung], endow)
        })
    }
  }

| contract PlayOne(m, @opening, hyp, score) = {
    new ret in {
        for (@h <- hyp) { hyp!(h) | Arena!(h, *m, "wall", *wall, *m, *ret) }
      | for (@outcome <- ret) {
          for (@c, @f <- score) {
            match outcome {
              ("harvest", _) => { score!(c + 1, f) }
              "draw" => { score!(c, f) }
              _ => { score!(c, f + 1) }
            }
          }
        }
    }
  }

// Mating requires DISCLOSURE: the genome goes out on a public channel where
// anything able to read it can. Nothing declares a species; the ‘where‘ guard
// is the only thing keeping two lineages apart, and deleting it hybridises
// them.
| contract Mate(m, @alleles, @endow) = {
    mate!(alleles)
  | for (@partner <- mate where Conspecific(partner, alleles)) { //! ideal
      new ret in {
          Crossover!(alleles, partner, [0, 1], *ret)
        | for (@kid <- ret) { Beget!(*m, kid, endow, *display) }
      }
```

```
    }
  }

// ===========================================================================
// THE WORLD
// Theta = 12000 (wall reserve) + 4 x 400 (players) = 13600 tokens. No more.
// ===========================================================================

| Wall!(*wall, 6, 12000)

// Four lineages, differing only at the two loci. The pair that shares a rung
// and differs at the opening is the whole experiment: Sec. 9 measures that
// the corner opener strictly invades the centre opener at IDENTICAL metering
// cost, and that no amount of deepening lets the centre opener reply.
| new a1, a2, b1, b2 in {
      a1!(400) | Body!(*a1, 4, 5, 120) // centre opening, top heuristic
    | a2!(400) | Body!(*a2, 4, 6, 120) // centre opening, the exact theory
    | b1!(400) | Body!(*b1, 0, 5, 120) // corner opening, top heuristic
    | b2!(400) | Body!(*b2, 0, 2, 120) // corner opening, shallow theory
  }

// ---------------------------------------------------------------------------
// WHAT TO WATCH
//
// * Conservation is checkable by inspection. Wall, Harvest and Beget move
// tokens; Live debits them. Nothing mints.
//
// * The hypothesis is never consulted. It is a guard. No line of code
// evaluates a theory and then decides -- deciding is the comm.
//
// * Perception is priced twice over: Scan runs after every move, and the
// burn rate rises with the rung held. Both are the same schedule.
//
// * Nobody screens an embryo. Crossover produces a genome and Beget drops
// it. A parent could model-check the recombinant against a viability
// formula and does not, because the check is priced against the same
// stack -- selection is post hoc because screening costs.
//
// * No scientist revises its opening. Revise only ever increments the rung.
// The opening changes exactly once per lineage, at Crossover, and free.
//
// * a2 holds the TRUE theory and is the one to watch starve. It burns 8 per
// step against b1's 7, wins no more often against fallible opponents, and
// draws every game against a1. Truth buys it safety and no food.
//
// * And it ends. When the wall's reserve reaches zero the only credit left
// is another player's stack; when the survivors converge on theories that
// draw against one another, Arena returns "draw", Harvest never fires, and
// the remaining tokens sit in metabolic channels until the burn takes
// them. A solved game is a dead ecology.
// ---------------------------------------------------------------------------
```

```
}
```

34.14 Appendix: Reproducing the numbers

All figures in §34.8 and §34.9 are exact computations over the game tree, not samples. Two scripts accompany this note. `ttt.py` enumerates positions, computes the minimax value, and defines the ladder as a priority rule list with uniform tie-breaking. `ladder.py` instruments the rule list with the price schedule of §34.6.1 and computes, by memoised recursion over all 255,168 plays, the joint distribution of outcome and expected metering for every pair of rungs. Symmetrised figures average a policy's results as first and as second player. The census figures are 5,478 reachable positions, 765 up to symmetry, 4,520 non-terminal, of which 1,890 offer a fork to the mover.

Chapter 35

Composing Learners

The individual is confined. That is not a remark about ambition; it is a proposition, and it is the reason this chapter exists. The hypothesis space is a tree with an ultrametric on it, and in an ultrametric a walk whose steps are all shorter than some radius never leaves the ball it began in. A learner that revises carefully cannot reach the theory it needs. The noughts-and-crosses measurement makes this concrete and slightly comic: the move that would rescue a confined population costs nothing and no individual can make it.

So the unit has to move, and this chapter moves it. A learner is redefined as a distribution over a *population* — scientists, computations holding no hypothesis at all, and reservoirs — with that population serving as a namespace. The operator that composes two such learners is parallel composition, which is to say no new machinery whatever. Everything interesting is then controlled by one dial: how much the two namespaces overlap, and in which of several typed namespaces the overlap sits.

Turned to zero, the dial gives independence, and independence turns out to be a theorem rather than a hope — though an unstable one, since two learners manufacturing names privately can collide by accident, and buying back the guarantee costs a naming discipline. Turned up, the same dial generates the standard table of ecological relations — competition, predation, mutualism, commensalism, theft — as sign conditions rather than as an inventory borrowed from biology. And it yields an asymmetry i think is the most interesting thing in the chapter: because tokens are conserved but formulae are not, cooperation can only ever be epistemic and competition is always metabolic. Two learners can both come out ahead in what they know. They cannot both come out ahead in what they hold.

35.1 Introduction

35.1.1 Why the unit moves up

The mortal scientist of Chapter 33 is an individual: a term with a metabolic channel, a hypothesis in guard position, and a walk on the relaxation lattice. Twice in that note the individual turns out to be the wrong unit. Corollary 33.3 there observes that destructive assay leaves an individual-level hypothesis worthless after its own first test, so the *subject* of a hypothesis must be a namespace rather than a specimen. And Proposition 33.4 observes that an ultrametric walk of short steps never leaves the ball it began in, so the *holder* of a hypothesis cannot repair a shallow commitment; §12 of that note supplies recombination, and §17.2 states plainly that a population is required rather than preferred.

The worked example makes both visible on numbers §34.9 of Chapter 34. The corner-opening mutant that invades the resident population differs from it by a revision of distance 2^{-1}, is unreachable to any individual by deepening, and is free to a lineage because the cost falls on an offspring holding no confirmations to invalidate.

So the individual is already not the unit of learning; the lineage is. This note asks what happens when the lineage is not the unit either.

35.1.2 The move, stated once

A learner is a distribution over a population: scientists, mortal computations holding no hypothesis, and resource reservoirs. That population is carried by names, and by Proposition 34.1 of Chapter 34 those names are manufactured by quotation rather than drawn from a stock of atoms, so the family is generated by a decidable predicate and can be *described*. Hence a learner is a namespace, in the sense the hypothesis language already uses for the subject of a hypothesis.

Learners compose by parallel composition, because in this calculus there is nothing else for composition to be. When the namespaces are disjoint the composed learners have non-interacting capabilities. When they overlap they can interact internally, and that interaction includes both cooperation and competition.

> *Remark* 35.1 (The knower and the known become one kind of thing). This is the observation we would lead with. Corollary 33.3 of Chapter 33 forces the *object* of inquiry to be a namespace; the present move makes the *subject* of inquiry one as well. The two sides of the cut are then described in the same vocabulary, by the same name predicates, at the same prices. That is not tidiness for its own sake: it is what makes a composed learner able to hold a hypothesis about its own components, and what makes the distinction between an internal interaction and an external one a matter of where one draws the cut rather than a matter of sort. Remark 33.12 of Chapter 33 said trophic type is a fact about the cut. The

> same is now true of learnerhood.

35.1.3 What we are claiming, and what we are not

We claim that composition of learners is parallel composition on terms, union on namespaces, and *undetermined* on distributions; that the determination of the third is exactly the content of the interaction, and that it factors precisely when the namespaces are disjoint; that a single grade of separation is too coarse and a typed vector of grades recovers the ecological relations rather than importing them; that under conservation cooperation and competition are structurally different rather than opposite in sign; that non-interference is an observation and not a guarantee unless purchased; and that a composite's phase is not a function of its parts' phases, which we exhibit on exact numbers.

We do not claim to have proved a monoidal-category theorem: §35.4 states the comparison map and the conditions for it to be invertible, and leaves the coherence to the open problems. We do not claim the access-rate model of §35.10 is canonical; it is one modelling choice, flagged as such, and the alternative is worked. As in Chapter 34, guards below are written in the ideal logic and the current interpreter accepts only the arithmetic and pattern fragment of the `where` clause; occurrences that specify rather than perform a check are marked.

35.2 What this chapter carries forward

The cut, the three grades of access, the conservation law, and the trophic profiles are as in the two preceding chapters and are not restated. Four things are worth having in front of the reader, because the argument turns on their exact form.

The logic is generated, and there is no restriction. Feeding the presentation of the rho calculus to the OSLF family yields one structural connective per term former, one context-labelled modality per rewrite rule and redex position, and—because $|$ carries an associative–commutative theory—a separating conjunction. A name is a quoted process, so $n \models @\varphi \iff n = @P$ with $P \models \varphi$: this is namespace logic [27], in which one describes a namespace by a property rather than enumerating it. Following §34.2 of Chapter 34 we treat `new` as sugar and drop the hidden-name quantifier: freshness is manufactured by quotation, and graded separation is graded over a *described namespace* rather than over a restriction vector,

$$P \models \varphi \,\big|_k^{\mathsf{N}}\, \psi \iff P \equiv Q \mid R,\ Q \models \varphi,\ R \models \psi,\ \big|\mathrm{fn}(Q) \cap \mathrm{fn}(R) \cap \mathsf{N}\big| = k.$$

Everything below that a modal logic could not do is done with $\big|_k^{\mathsf{N}}$, and §35.5 turns on the choice of N.

Proposition 35.1 (Freshness by quotation, Chapter 34, Prop. 34.1)**.** *Let P be a process and $C[-]$ any context with $C[P] \neq P$. Then $@C[P] \notin n(P)$. Hence a name fresh in P is manufactured by quoting any term properly containing P, with no registry and no consultation.*

Conservation and the game. Tokens are conserved: $\sigma + \kappa = \sigma_0 = \Theta$, with σ held in live stacks and κ migrated into the record. There is no primary producer. Under the internalising encoding $[\![-]\!] : \mathcal{C}(\mathsf{rho}) \to \mathsf{rho}$ a located stack becomes a message at a metabolic channel m_P, and harvest is an ordinary receipt on m_P performed by someone else; the predatory contexts are exactly the non-image contexts holding a metabolic name Chapter 33, Theorem 33.1. A *game* is a choice of admitted context class $\mathcal{A}$ with $[\![\mathcal{C}(\mathsf{rho})\text{-Ctx}]\!] \subseteq \mathcal{A} \subseteq \mathsf{rho}\text{-Ctx}$.

The hypothesis space, and confinement. Beneath the characteristic formula χ_P sits a lattice of progressively weaker formulae obtained by forgetting conjuncts; revision is a walk on it, the metric is the ν-stratified ultrametric $d(\varphi, \psi) = 2^{-n}$, and distance, modal depth, and cost are one graded structure. Confinement (Proposition 33.4 of Chapter 33) says a walk whose steps are all shorter than 2^{-n} never leaves the ball of radius 2^{-n} it began in; cost-monotonicity (Proposition 33.5) says the escaping move is the expensive one; and bounded branching (Proposition 33.6) with its corollary *poverty is confining* says the number of siblings a scientist may evaluate is its stack divided by the cost of checking one.

Ecologies as a fibration. A configuration is a partition of Θ across loci together with a structural assignment; an ecology is a configuration with a distribution over reachable configurations. The assignment of ecologies to theories is a fibration rather than a functor, with a canonical section given by taking cost as the weight map, and functoriality requires logic-preserving morphisms because the senses are structural §33.15 of Chapter 33. The present note is largely an argument that this fibration wants a tensor.

35.3 The learner

35.3.1 Three layers, and the distribution is one of them

What we have been calling a learner has three components, and it is worth separating them before asking what composition does, because composition acts differently on each.

(i) **A population.** A term: a parallel composition of scientists, of mortal computations holding no hypothesis, and of reservoirs. This is an object of the calculus.

(ii) **A namespace.** The family of names the population carries—metabolic channels, source channels, probe channels, mating channels—described by a name predicate rather than listed, which by Proposition 35.1 is available because the names are manufactured from material the population already holds.

(iii) **A distribution.** A weighting on the configurations reachable from the population, in the sense of §33.13 of Chapter 33: the microcanonical ensemble at fixed Θ, sampled by the construction of [25].

We take the third to be part of the learner rather than a decoration on it. The alternative—a learner is the population, and the distribution is a decoration indexed over it—would put composition in the base of the fibration of §33.15 of Chapter 33, where it is uninformative: two populations composed in the base are just two populations. Putting the distribution inside the learner puts composition in the total space, which is where the interaction lives.

Definition 35.1 (Learner). *A learner is a triple $\mathcal{L} = (P, \mathsf{N}, \mu)$ with P a* rho *term,* N *a name predicate satisfied by exactly the names P carries in the sense of Definition 35.3 below, and μ a distribution over the configurations reachable from P at total $\Theta_{\mathcal{L}} = \sum \sigma_i(0)$. We write $\mathsf{N}(\mathcal{L})$ and $\mu(\mathcal{L})$ for the components.*

Remark 35.2 (The population is not a bag of learners). Definition 35.1 does not say the population is a set of individual learners composed somehow. It says the population, weighted, *is* the learner, and that individual scientists inside it are components with no special status—exactly as a lineage in §33.12.10 of Chapter 33 is a learner without any of its members being the learner. The nesting is real and the note uses it: §35.12 shows the three units of learning are distinguished by nothing but the revision distance each can reach.

35.4 Composition

35.4.1 One operator, three actions

Definition 35.2 (Composition). *For learners $\mathcal{L}_1 = (P_1, \mathsf{N}_1, \mu_1)$ and $\mathcal{L}_2 = (P_2, \mathsf{N}_2, \mu_2)$,*

$$\mathcal{L}_1 \mid \mathcal{L}_2 \;:=\; \big(P_1 \mid P_2,\quad \mathsf{N}_1 \vee \mathsf{N}_2,\quad \mu_{12}\big), \qquad \Theta_{\mathcal{L}_1|\mathcal{L}_2} = \Theta_{\mathcal{L}_1} + \Theta_{\mathcal{L}_2},$$

where $\mathsf{N}_1 \vee \mathsf{N}_2$ is the disjunction of the two name predicates and μ_{12} is the distri-

bution over configurations reachable from $P_1 \mid P_2$.

On the first component the operator is the calculus' own parallel composition: associative and commutative up to structural congruence, free, and adding nothing. On the second it is disjunction of predicates, which is again free, and is where describing rather than enumerating pays—one does not have to compute a union of two sets of names, one takes an $\vee$ of two formulae. On the third the operator is not determined by its arguments at all. That is the entire subject of this section.

Proposition 35.2 (Conservation composes, and only conservation does)**.** Θ *is additive under composition, and it is the only quantity in §33.13 of Chapter 33 that is. The order parameters are not: ι, the fraction of Θ internal to scientists, is a stack-weighted mean of the parts' values, and δ, the mean concealment depth of external stacks, is not a function of the parts' values at all, because each learner's members become candidate subjects for the other and enter the other's mean.*

Proof sketch. Additivity of Θ is immediate from the ledger law and the fact that composition mints nothing. For ι, write $\iota_i = \sigma_i^{\mathrm{sci}}/\Theta_i$; then $\iota_{12} = (\sigma_1^{\mathrm{sci}}+\sigma_2^{\mathrm{sci}})/(\Theta_1+\Theta_2)$, the weighted mean. For δ, the mean is taken over the external stacks *available to* a learner, and P_2's stacks are external to P_1 and were not in P_1's index set. So δ_{12} depends on the depth at which each learner's stacks lie relative to the other's senses, a quantity computed from the pair and not held by either. $\square$

That δ is a property of the pair is the first sign that composition is not going to be innocent, and §35.9 makes it a measurement.

35.4.2 Disjointness is factorisation

Theorem 35.1 (Factorisation)**.** *Let $\mathcal{L}_1, \mathcal{L}_2$ be learners and let $\mathsf{N} = \mathsf{N}_1 \vee \mathsf{N}_2$. The following are equivalent.*

(1) $P_1 \mid P_2 \models \top \,\big|_0^{\mathsf{N}}\, \top$ with the split at P_1, P_2: the two namespaces share no name.

(2) No reduction of $P_1 \mid P_2$ has its redex spanning the two parands.

(3) $\mu_{12} = \mu_1 \otimes \mu_2$: the composite's distribution is the product of the parts'.

Proof sketch. (1) $\Rightarrow$ (2): the rho calculus has one interaction rule, and it requires a send and a receipt on structurally equal names. A redex spanning the parands therefore witnesses a name in $\mathrm{fn}(P_1) \cap \mathrm{fn}(P_2) \cap \mathsf{N}$, contradicting $k = 0$. (2) $\Rightarrow$ (3): with no spanning redex, every step of $P_1 \mid P_2$ is a step of one parand in a context that does not participate, so the reachable configurations are pairs of independently reachable configurations, and the weight of a pair is the product of the weights since the cost of a step is charged to the parand taking it and the ledger is additive. (3) $\Rightarrow$ (1): if a name is shared, some configuration in which the corresponding rendezvous has fired has weight not equal to the product of its marginals—the two parands' stacks after such a step are correlated by the transfer. $\square$

Remark 35.3 (Why this is the right statement). Theorem 35.1 is what makes "the namespaces are disjoint" worth saying. Disjointness of *names* is a syntactic condition; independence of *distributions* is the thing one actually wants when composing learners, because it is what licenses reasoning about each in isolation. The theorem says the syntactic condition is exactly the probabilistic one, and the graded separation $|^{\mathsf{N}}_{k}$ measures the gap when it fails. We know of no way to say this without a spatial layer: a purely behavioural logic cannot state the split at all, which is [20, Rem. 5] arriving in a new place.

35.4.3 The lax structure map is the interaction

Composition of learners is a candidate monoidal structure on the total space of the fibration $\pi : \mathrm{Eco} \to \mathrm{GSLT}_{\mathcal{C}}$ of §33.15 of Chapter 33, with unit the empty learner $(\mathbf{0}, \bot, \delta_{\varnothing})$. The obvious question is whether the canonical section s—cost as the weight map—is monoidal, and the answer is no, in a way we think is more useful than a yes.

Proposition 35.3 (Laxity, and what its defect measures)**.** *There is a comparison morphism*

$$\lambda_{\mathcal{L}_1,\mathcal{L}_2} \;:\; s(\mathcal{L}_1) \otimes s(\mathcal{L}_2) \longrightarrow s(\mathcal{L}_1 \mid \mathcal{L}_2)$$

sending a pair of independently sampled trajectories to the interleaving in which no spanning redex fires. It is an isomorphism if and only if the grade is zero, and in general it is not epi: the configurations in the image are exactly those reachable without interaction. The deviation—the mass μ_{12} assigns outside the image of λ—is the probability that the composite does something neither part could do.

Proof sketch. Existence: the non-interacting interleavings of two independent trajectories are reachable in the composite, and their weights are the products, since costs are additive. Iso at $k = 0$: Theorem 35.1. Failure of surjectivity at $k > 0$: a spanning redex is enabled and fires with positive rate under any weight map assigning positive weight to enabled steps, and the resulting configuration is not a product of independently reachable ones. $\square$

Remark 35.4 (The good news is that laxity is a number). Conjecture 82 of Chapter 33 guesses that the induced map on distributions is only lax and that the natural target is distributions ordered by stochastic dominance. Composition sharpens the guess into something one can compute: the defect of λ is the total interaction, and $\big|^{\mathsf{N}}_{k}$ resolves it by which names carry it. Cooperation and competition are then not two phenomena to be modelled but two *signs* of one measured quantity, and §35.6 says which sign each namespace can carry.

35.5 Typed overlap: the grading vector

35.5.1 One grade is not enough

The preceding chapter's Proposition 34.2 is the lesson in miniature. There the connective was $\big|^{\mathsf{N}}_{0}$ and the concept was *fork*; grading over line names made the formula true of pairs of threats sharing a gap, which are not forks, and grading over square names made it exact. The choice of N was not a parameter of the encoding but the whole content of it.

At the level of learners the same demand recurs, and the namespace of a learner is not homogeneous. A shared metabolic channel and a shared probe channel are both "an overlap of one name" and they are not remotely the same relation.

Definition 35.3 (Typed namespaces). *For a learner $\mathcal{L}$ with population P, the namespace $\mathsf{N}(\mathcal{L})$ decomposes into four sub-namespaces, each given by a name predicate:*

- $\mathsf{Met}(\mathcal{L})$*, the* metabolic *names: the channels at which P's components hold their stacks, in the sense of Chapter 33, Definition 33.7;*
- $\mathsf{Src}(\mathcal{L})$*, the* source *names: the channels of persistent emitters P draws on, the access profile of Chapter 33, Definition 33.11;*
- $\mathsf{Prb}(\mathcal{L})$*, the* probe *names: the channels on which P's scientists install assays, which by §33.12.6 of Chapter 33 are both-type channels supporting dialogue of depth at least two;*
- $\mathsf{Mat}(\mathcal{L})$*, the* mating *names: the channels on which genomes are disclosed and recombined.*

Definition 35.4 (Grading vector). *For learners $\mathcal{L}_1, \mathcal{L}_2$ the grading vector is*

$$\mathbf{k}(\mathcal{L}_1, \mathcal{L}_2) \;=\; \big(k_{\mathrm{met}},\, k_{\mathrm{src}},\, k_{\mathrm{prb}},\, k_{\mathrm{mat}}\big), \qquad k_\tau = \big|\,\mathrm{fn}(P_1) \cap \mathrm{fn}(P_2) \cap \mathsf{N}_\tau\,\big|,$$

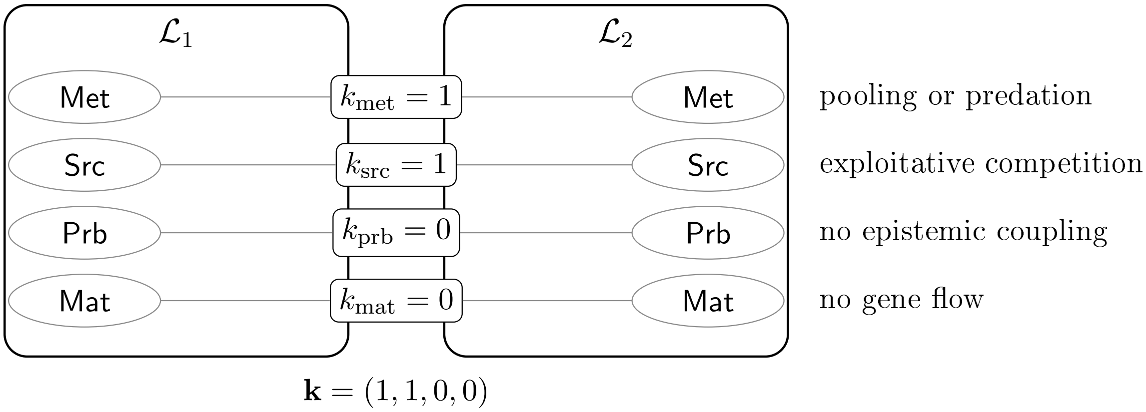


Figure 35.1: Two composed learners and their grading vector. A single scalar k would report "2" and say nothing; the typed vector says the two share a metabolic channel and a reservoir, cannot read each other's assays, and cannot interbreed. Those are four independent dials and §35.5.2 reads the ecological relation off them.

overlap	relation	(Δ_1, Δ_2)	mechanism
$\mathbf{k} = \mathbf{0}$	neutralism	$(0, 0)$	Theorem 35.1: μ factors
$k_{\text{src}} > 0$ only	exploitative competition	$(-, -)$	dilution of a finite reserve (§35.10)
$k_{\text{met}} > 0$, asymmetric	predation	$(+, -)$	harvest as a receipt, Chapter 33, Def. 33.8
$k_{\text{met}} = 1$, pooled	mutualism (metabolic)	$(+, +)$	the composite of Chapter 33, Prop. 33.14
$k_{\text{prb}} > 0$, one-way	commensalism (epistemic)	$(+, 0)$	reading a surface another paid for
$k_{\text{prb}} > 0$, two-way	mutualism (epistemic)	$(+, +)$	§35.6: formulae are not conserved
$k_{\text{prb}} > 0$, theft	exploitation (epistemic)	$(+, -)$	§33.10 of Chapter 33: holding a model exposes it
$k_{\text{mat}} > 0$	*not composition*	—	§35.8: the learners are one

Table 35.1: Ecological relations as sign conditions on the grading vector. Nothing in this table is posited; each row is a consequence already proved in Chapter 33 together with a choice of which N_τ carries the overlap.

the componentwise grade at which $P_1 \big|^{\mathsf{N}_\tau}_{k_\tau} P_2$ *holds. Composition is* non-interacting *when* $\mathbf{k} = \mathbf{0}$.

35.5.2 The relations, derived

The point of typing the overlap is that the classical table of ecological relations then falls out as sign conditions on the components, rather than being imported as a taxonomy. Write Δ_i for the change in learner i's expected residual life under composition relative to isolation.

Proposition 35.4 (Symbiosis is minimal metabolic overlap)**.** *Let* $\mathcal{L}_1 \mid \mathcal{L}_2$ *have* $k_{\text{met}} = 1$ *with the shared name a pooling channel—a metabolic channel on which* *both parands both receive and reissue—and $k_\tau = 0$ otherwise. Then the composite is exactly the construction of Chapter 33, Proposition 33.14: two specialists sharing a metabolic channel, each free to tune its access profile independently, with the yield pooled. Hence Proposition 33.13 there, which dominates the mixed strategy for an atom, does not apply.*

Remark 35.5 (The organism boundary is a grade). Read with Remark 33.12 of Chapter 33, Proposition 35.4 says the difference between "one dual-feeding organism" and "two specialists in symbiosis" is the value of k_{met} and where one chooses to draw the cut, not a fact about either. The endosymbiotic reading is available and we note it without leaning on it [47]: what the formalism contributes is that the boundary has a number attached to it.

Proposition 35.5 (The grading namespace must be typed)**.** *Let* $\mathsf{N} = \mathsf{Met} \vee \mathsf{Src} \vee \mathsf{Prb} \vee \mathsf{Mat}$ *and suppose one grades only over* N*. Then* $\left|_{k}^{\mathsf{N}}\right.$ *cannot distinguish a composite in exploitative competition* ($k_{\mathrm{src}} = 1$*, rest zero) from one in symbiosis* ($k_{\mathrm{met}} = 1$*, rest zero), since both satisfy* $\left|_{1}^{\mathsf{N}}\right.$*; and the two have opposite signs in Table 35.1. Hence the vector of Definition 35.4 is not a refinement of convenience.*

This is the same argument as Chapter 34, Proposition 34.2, one level up, and we take the recurrence as evidence that choosing the grading namespace is where the modelling work in this framework actually happens.

35.6 Cooperation and competition are structurally unlike

35.6.1 Two currencies, one conservation law

Under Remark 33.7 of Chapter 33 every metabolic transfer is zero-sum: predation is the purchase of somebody else's remaining time. It follows immediately that cooperation cannot be a token transfer leaving both parties richer, and one might conclude that a conserved world has no room for mutualism at all. It does, and the room is in exactly one place.

Observation 35.1 (Tokens are conserved; formulae are not). Θ is fixed and never minted. A formula is a different kind of object: a genome is a quotation, $@P$ does not reduce, quotation is free while held, and inspection by a name predicate is grade-three access, the cheapest on the ladder. Nothing forbids two learners from holding the same formula, and one learner's coming to hold it does not deprive the other. The ecology is conserved; the epistemics are not.

Proposition 35.6 (No metabolic mutualism). *Let $\mathcal{L}_1 \mid \mathcal{L}_2$ have $k_{\mathrm{prb}} = k_{\mathrm{mat}} = 0$, so that all overlap is metabolic or source. Then $\Delta_1 + \Delta_2 \leq 0$, with equality only when no shared name carries a transfer.*

Proof sketch. Residual life is stack divided by burn rate. Every step over a shared metabolic or source name either moves tokens from one parand to the other, which is zero-sum in stack and strictly negative in aggregate residual life whenever the two burn rates differ, or debits both parands for the rendezvous, which is negative. Pooling (Proposition 35.4) is the boundary case in which the transfer is a loop and the aggregate is unchanged; even there the rendezvous is metered, so equality requires no transfer at all. □

So the only positive-sum overlap is epistemic, and it has two mechanisms, both already in the companion notes.

35.6.2 Cost relocation, and who is billed

Observation 35.2 (Cooperation is amortised perception). There are two routes to any predicate about an environment: *read* it, if the environment has computed it into legible structure, priced by formula depth; or *drive* it, by taking steps and watching, priced by rendezvous §34.5.1 of Chapter 34. If $\mathcal{L}_1$'s components compute a predicate into structure standing at a name in $\mathsf{Prb}(\mathcal{L}_1) \cap \mathsf{Prb}(\mathcal{L}_2)$, then $\mathcal{L}_2$ reads at grade one what it would otherwise drive at grade two. The total metering for both learners to know the predicate falls, and no token has been created.

Remark 34.9 of Chapter 34 makes the point that putting threats into the term relocates a cost from the scientist's budget to the world's, and that the interesting quantity is not whether the work is done but who is billed for it. Composition is what makes that question have two answers inside one system. A shared legible surface is stigmergy, and it is the only mechanism by which two mortal learners in a conserved world can both come out ahead.

Proposition 35.7 (Epistemic overlap can be strictly positive-sum). *Let φ be a predicate both learners require, costing κ_{drive} to establish by interaction and $\kappa_{\mathrm{read}}(d) < \kappa_{\mathrm{drive}}$ to read from structure, and let κ_{write} be the cost to $\mathcal{L}_1$ of computing it into structure at a shared probe name. Then $\Delta_1 + \Delta_2 > 0$ whenever*

$$2\,\kappa_{\mathrm{drive}} \;>\; \kappa_{\mathrm{drive}} + \kappa_{\mathrm{write}} + \kappa_{\mathrm{read}}(d),$$

that is, whenever writing once and reading once costs less than driving twice. This is not available over Met *or* Src *by Proposition 35.6.*

Remark 35.6 (And the same overlap can be theft)*.* The mechanism cuts both ways and the notes already say so. Holding a hypothesis requires exposing it, because a guard is surface, so Prb overlap that lets $\mathcal{L}_2$ read $\mathcal{L}_1$'s confirmed structure also lets it read $\mathcal{L}_1$'s theory without experimenting §33.10 of Chapter 33. Whether epistemic overlap is mutualism or exploitation is not settled by the grade; it is settled by whether the reading learner reciprocates, which is the plagiarism equilibrium left open as Problem 2 there. What the present framing adds is that the question is well posed as a sign condition on one component of $\mathbf{k}$.

Cooperation is epistemic; competition is metabolic.

35.7 Non-interference is not stable

Disjointness of namespaces is the condition under which composition is innocent, so it matters whether it can be relied upon. It cannot, by default, and the reason is one the companion note already records for a different purpose.

Proposition 35.8 (Disjointness is an observation, not a guarantee)**.** *Let $\mathcal{L}_1 \mid \mathcal{L}_2$ have $\mathbf{k} = \mathbf{0}$ at some configuration. Because name manufacture is decentralised and freshness is scope-relative (Chapter 33, Remark 33.1), the two populations may subsequently manufacture structurally equal names at loci private to each. Hence $\mathbf{k} = \mathbf{0}$ at one configuration does not entail $\mathbf{k} = \mathbf{0}$ at a successor, and neither learner need have taken any step aimed at the other.*

Proof sketch. This is Remark 33.11 of Chapter 33—closure is luck, not construction—together with Proposition 33.26 there, that decentralisation makes recombination generative precisely because two parents may independently hold structurally equal names. What is a source of novelty within a lineage is a source of unplanned coupling between learners. □

Corollary 35.1 (There is no modular composition for free)**.** *One cannot compose two learners and certify non-interference from the parts alone. Certification requires either a global check at each configuration, which is model checking and is priced, or a discipline on name manufacture.*

The discipline exists, it is cheap, and it is the one the board of §34.4 of Chapter 34 already uses.

Proposition 35.9 (Rooting makes disjointness a theorem)**.** *Let $\mathcal{L}_1$ and $\mathcal{L}_2$ manufacture every private name by quoting a term properly containing a distinguished root r_1, respectively r_2, with $r_1 \neq r_2$ and neither root occurring in the other's population. Then no name manufactured by $\mathcal{L}_1$ is structurally equal to a name manufactured by $\mathcal{L}_2$, and* $\mathbf{k}$ *is constant along every reduction that does not introduce a name from outside.*

Proof sketch. By Proposition 35.1 a manufactured name is $@C[r_i]$ for a context C with $C[r_i] \neq r_i$, so it codes a term properly containing r_i. Structural equality of $@C[r_1]$ and $@C'[r_2]$ would give $C[r_1] \equiv C'[r_2]$, and by well-foundedness of the coding—a name occurring in a term codes a strictly smaller term—this forces r_1 to occur in $C'[r_2]$, contradicting the hypothesis that r_1 does not occur in $\mathcal{L}_2$'s population. □

Remark 35.7 (Modularity is purchasable, and the price is a naming discipline). This is the same trick as the board's three name families $@[*b, k]$, $@[*b, \texttt{"m"}, k]$, $@[*b, \texttt{"t"}, s]$, which are pairwise disjoint by construction and which is what makes grading over Th_b available in the first place. Applied to learners it says: root each learner's namespace at its own quotation and non-interference becomes a property one can rely on rather than one one must keep checking. It also says that a learner which *wants* to interact must import a name from outside its own root, which is a step, which is metered. Under rooting, every interaction has an identifiable first cause with a price attached—which is a better situation than the notes have had so far, where accidental coupling was possible and untraceable.

35.8 Merger

Overlap in the mating namespace is not a kind of interaction. It is the failure of the two learners to be two.

Proposition 35.10 (Mating overlap collapses the pair)**.** *By Chapter 33, Proposition 33.18, two computations can recombine exactly at their homologous loci, so conspecificity is agreement of path sets and reproductive isolation is namespace disjointness. If $k_{\mathrm{mat}} > 0$ and the shared names are homologous, then genomes cross the boundary, alleles from each appear in the other's descendants, and after finitely many generations neither population's namespace is described by its original predicate.*

Definition 35.5 (Merger)**.** *The merger* $\mathcal{L}_1 \bowtie \mathcal{L}_2$ *is the learner*

$$\big(\, P_1 \mid P_2, \quad \mathsf{N}_1 \vee \mathsf{N}_2, \quad \mu_{\bowtie} \,\big)$$

whose distribution ranges over configurations reachable under a recombination discipline admitting homologous loci across *the two populations, together with the identification of the two mating namespaces.*

Merger and composition agree on the first two components of Definition 35.1 and differ on the third, which is precisely why the third had to be part of the learner. As operators they behave differently in ways worth recording.

Proposition 35.11 (Merger is not composition)**.** *Composition is associative and commutative up to* $\equiv$ *and admits the empty learner as a unit. Merger is commutative but is neither associative in general—*$(\mathcal{L}_1 \bowtie \mathcal{L}_2) \bowtie \mathcal{L}_3$ *and* $\mathcal{L}_1 \bowtie (\mathcal{L}_2 \bowtie \mathcal{L}_3)$ *admit different homology relations, since homology is not transitive across intermediate path sets—nor unital in the same sense, since* $\mathcal{L} \bowtie \mathbf{0}$ *imposes an identification that* $\mathcal{L} \mid \mathbf{0}$ *does not. Moreover merger is irreversible: no composition of the merged learner recovers the parts, because the alleles have crossed.*

Remark 35.8 (Why we want it separate). There is a temptation to treat merger as composition at high k_{mat} and be done. We resist it for two reasons. First, the sign structure of Table 35.1 has no entry for it: merger is not $(+,+)$ or $(+,-)$, because after it there are no two parties to assign signs to. Second, and more usefully, merger is the operator that changes the *identity* of a learner while composition changes only its situation, and §35.12 needs that distinction: a lineage escapes confinement by recombination *within* itself, and a federation escapes by composition *with* another. Merger is the third thing, and it escapes by ceasing to be the learner that was confined.

35.9 Phase is not compositional

Figure 2 of Chapter 33 divides the (ι, δ) plane into three regions: at low δ food lies in the open and senses suffice, so a theory is overhead; at high ι or high δ discovery costs more than it returns and everything starves; in between, hypothesis formation pays. The boundaries are where the foraging inequality changes sign.

Proposition 35.12 (Phase is not preserved)**.** *There exist learners* $\mathcal{L}_1, \mathcal{L}_2$ *each of which lies in the science-pays band in isolation and whose composite lies outside it. Specifically:*

(a) Downward, into foraging-suffices. *If $\mathcal{L}_2$'s components lie shallow to $\mathcal{L}_1$'s senses—δ small in the pair, which by Remark 34.2 of Chapter 34 means few plies of lookahead are needed to model them—then $\mathcal{L}_1$'s cheapest rung dominates its theory-holding rungs, since the prey is legible without inquiry.*

(b) Downward, into starvation. *If composition raises ι past the upper boundary, which it does whenever $\mathcal{L}_2$ is stack-rich and reservoir-poor, then no available locus satisfies the foraging inequality for either party.*

(c) Upward, out of starvation. *If $\mathcal{L}_1$ is below its minimum viable endowment and $k_{\mathrm{met}} = 1$ is a pooling channel, then Proposition 35.4 rescues it.*

The interesting case is none of these three, because all three are couplings a modeller can see coming. The case worth measuring is the one in which the two learners never perceive each other at all.

35.10 Competitive dilution, measured

35.10.1 The model, and its one choice

Take $\mathbf{k} = (0, 1, 0, 0)$: two learners sharing exactly one source name and nothing else. Neither can perceive the other—no probe name is shared, so no assay of either reaches the other's surface—neither can harvest the other, and neither can interbreed with it. By Table 35.1 the relation is exploitative competition, and the only channel of influence is Θ itself.

The source is a single contract, hence rate-limited and non-exclusive (Chapter 33, Definition 33.11). We must say how the two learners' calls interleave, and this is the note's one genuinely free modelling choice.

Definition 35.6 (Access rate)**.** *A learner calling a source completes one assay per c metered units, where c is its expected metering per assay at the rung it holds. Learners queue at the source as fast as they complete assays, so learner i's share of the source's payouts is*

$$s_i \;=\; \frac{1/c_i}{\sum_j 1/c_j}.$$

The alternative—equal shares regardless of cost—is defensible if one reads the contract as serving callers round-robin rather than first-come. We work Definition 35.6 and note in §35.14 what changes under the alternative: the dilution survives, the asymmetry does not.

Proposition 35.13 (Competitive dilution)**.** *Let two learners share one source of reserve R_0 paying quantum q per assay, at price p tokens per metered unit, with expected meterings c_1, c_2 per assay, both self-funding ($q > pc_i$). Then over the life of the source, learner i receives*

$$G_i \;=\; \frac{R_0}{q} \cdot \frac{c_j}{c_i + c_j} \qquad (j \neq i)$$

assays and accumulates $\Sigma_i = G_i\,(q - pc_i)$. Consequently the number of assays of any deficit-running rung of cost $c^* > q/p$ that learner i can subsequently fund is

$$D_i \;=\; \frac{\Sigma_i}{pc^* - q} \;=\; D_i^{alone} \cdot \frac{c_j}{c_i + c_j},$$

so that each learner retains exactly its own share of the reservoir as a fraction of the inquiry it could afford in isolation*. The factor is independent of R_0, q, p and c^*.*

Proof sketch. Immediate from Definition 35.6: composition scales G_i by s_i and everything downstream is linear in G_i. □

Corollary 35.2 (Dilution is regressive)**.** *A learner whose best affordable rung is self-funding loses assays but no* depth*, since the rung it holds needs no surplus. A learner still climbing—whose next rung runs a deficit and must be funded from accumulated surplus—loses depth in exact proportion to its share. Hence competition at a shared reservoir costs the unfinished science strictly more than the finished one, and costs nothing at all to a learner that has stopped.*

Remark 35.9 (Competition is a mechanism of confinement)*.* Corollary 33.1 of Chapter 33 says poverty is confining: a scientist whose stack affords few evaluations per move takes the cheapest available move, the cheapest moves are the shortest, and by Proposition 33.4 short moves cannot leave their ball. Proposition 35.13 exhibits a mechanism of impoverishment that requires no contact whatever. A learner can be confined to a shallow theory by a competitor that cannot see it, cannot model it, and has no hypothesis about it—only a name in common. We think this is the most surprising consequence of the composition operator, and it is the classical competitive exclusion principle [45, 46] arriving in an epistemic currency rather than a demographic one.

35.11 The worked example: two games, one wall

35.11.1 A second environment

The game note laddered noughts-and-crosses and found the complete theory dominated. To compose we need a second learner, and the second environment should differ in the one respect that matters: whether its complete theory is affordable.

We take Nim, normal play [43]. The ladder is built by the same method and priced by the same schedule $\kappa(d) = 2^d$, with the cost of a decision at rung r equal to $|M| \cdot \sum_{\text{rules } 1..r} \kappa(d_{\text{rule}})$ and $|M|$ the number of legal moves.

rung	formula added	kind and depth	κ
0	$\top$	—	0
1	$\exists s.\ \mathrm{One}(s) \wedge \mathrm{Takes}(s)$	spatial, depth 1	2
2	$\exists s.\ \mathrm{Two} \wedge \mathrm{Equalises}(s)$	spatial, depth 1	2
3	$\chi_{\mathrm{Nim}} := \exists s.\ \mathrm{NimSum}_0(s)$	name-level, depth 0	1

Rung 1 says: exactly one heap is nonempty, take it. Rung 2 says: exactly two heaps are nonempty, equalise them. Rung 3 is Bouton's theorem: move to a position whose nim-sum is zero. All three are predicates on the heap sizes, read off the reflected structure of the position by a name predicate; none contains a modality, and rung 3 in particular requires no lookahead at all.

> *Observation* 35.3 (The complete theory of Nim is name-level). χ_{Nim} is exact and contains no modality and no fixed point. This is Remark 33.3 of Chapter 33—a finite theory is complete exactly when the environment's regularity is structural—in its sharpest available form: Nim's regularity is so structural that its complete theory sits at depth zero, where noughts-and-crosses needs $\nu X.\forall s.[K_s](\neg\mathrm{Lost} \wedge \exists t.\langle\langle K_t\rangle\rangle X)$ and nine unfoldings.

35.11.2 The ladder pays, and here it keeps paying

All figures are exact over all plays, symmetrised over which side moves first, against a uniform-random opponent, computed by memoised recursion (Appendix 35.17).

> *Observation* 35.4 (Whether truth is affordable is a property of the environment). Three environments, three answers. In noughts-and-crosses χ costs 679.2, wins 0.873, and is strictly dominated by a name-level heuristic winning 0.918 at 255.9. In Nim$(3,4,5)$ χ_{Nim} costs 99.6, wins 0.9966, and is not dominated: it is more expensive than the rung below and strictly better. In Nim$(1,2,3)$ it is the best marginal purchase on the whole ladder, returning $+0.0148$ per metered unit where

rung	added rule	win	loss	cost	Δwin	return
Nim$(3,4,5)$	*nim-sum* $= 2$*, an N-position; branching* 12					
0	$\top$	0.5000	0.5000	0.0	—	—
1	take-all	0.6567	0.3433	38.9	+0.1567	+0.00403
2	equalise	0.8899	0.1101	80.4	+0.2332	+0.00562
3	χ_{Nim}	0.9966	0.0034	99.6	+0.1067	+0.00555
Nim$(1,2,3)$	*nim-sum* $= 0$*, a P-position; branching* 6					
0	$\top$	0.5000	0.5000	0.0	—	—
1	take-all	0.5917	0.4083	15.2	+0.0917	+0.00602
2	equalise	0.7958	0.2042	31.4	+0.2042	+0.01262
3	χ_{Nim}	0.9000	0.1000	38.5	+0.1042	+0.01476

Table 35.2: Two Nim ladders, exact. Contrast Table 1 of Chapter 34, where marginal return falls by more than two orders of magnitude along the ladder and the complete theory is dominated on both axes. Here return is flat in Nim$(3,4,5)$ and *rising* in Nim$(1,2,3)$, where the complete theory is the best purchase available.

> the rung below returns +0.0126 and the rung below that +0.0060. The returns *rise*.

This is worth stating carefully because it is the hinge of the composition argument. Remark 33.6 of Chapter 33—yield, not truth—says a lineage under selection drifts toward yield-biased search wherever yield and discrimination disagree. Table 35.2 shows they do not always disagree. Whether they do is fixed by the environment: by how much of its regularity is structural rather than reduction-following (Remark 33.3 there), and by the branching factor, which by Proposition 34.3 of Chapter 34 is the arity of the quantification over redex positions. So *is truth affordable?* is not a question about scientists. It is a question about worlds, and composition is the operator that puts two answers into one budget.

> *Remark* 35.10 (χ in a lost position is a fall-through to $\top$). Nim$(1,2,3)$ has nim-sum zero, so the first player has no move to a zero position and χ_{Nim} fires no rule; the policy falls through to uniform choice. It still wins 0.800 as first player because the random opponent errs and the complete theory punishes the error the moment it appears—but it has nothing to say about how to *provoke* one. As second player it wins 1.000. That the complete theory is silent on exploiting a fallible opponent is Observation 34.6 of Chapter 34 in a domain where χ is otherwise cheap and good, which is some evidence that the phenomenon is about completeness rather than about cost.

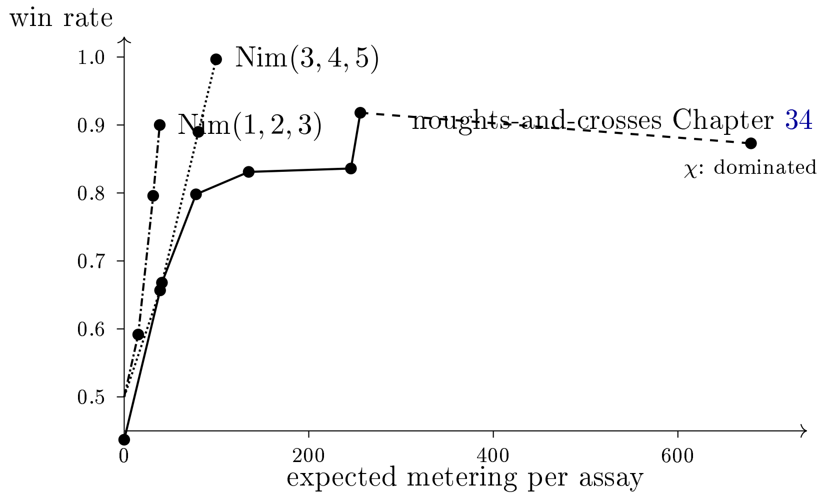


Figure 35.2: Three ladders on one pair of axes. The noughts-and-crosses curve turns down at its last rung—the complete theory costs more and wins less than the heuristic beneath it. Both Nim curves rise to the end, and the Nim(1, 2, 3) curve rises fastest at the top. Composition puts learners with curves of both shapes under one conserved Θ.

35.11.3 The composite

Let $\mathcal{L}_A$ be a population playing noughts-and-crosses and $\mathcal{L}_B$ a population playing Nim(1, 2, 3), each against the practice wall of §34.3.2 of Chapter 34, rooted at distinct quotations so that Proposition 35.9 applies. They share exactly the wall: $\mathbf{k} = (0, 1, 0, 0)$. The wall's reserve is $R_0 = 12{,}000$ at quantum $q = 6$, and the price is $p = 0.05$ tokens per metered unit, as in that note's world.

At $p = 0.05$ and $q = 6$ a rung is self-funding exactly when its expected metering is below $q/p = 120$. For the noughts ladder that admits rungs $0, 1, 2$ and excludes $3, 4, 5, 6$; so $\mathcal{L}_A$'s deepest self-funding rung is rung 2, at $c_A = 77.8$ and a net of $+2.11$ tokens per assay, and every deeper rung must be funded out of accumulated surplus. For $\mathcal{L}_B$, χ_{Nim} costs $c_B = 38.5$, net $+4.08$: *the complete theory of $\mathcal{L}_B$'s environment is self-funding, and the incomplete theory of $\mathcal{L}_A$'s environment is not.*

> *Observation* 35.5 (The measurement). $\mathcal{L}_A$'s affordable depth falls from 621 assays of rung-5 play to 206. The mechanism is entirely Θ: $\mathcal{L}_B$ holds no hypothesis about $\mathcal{L}_A$, has no probe name in common with it, could not derive its metabolic channel if it wanted to, and would gain nothing by doing so. It simply finishes its assays faster, because its environment admitted a cheap complete theory, and therefore reaches the wall more often.

	share of wall	assays	terminal surplus	rung-5 assays funded
$\mathcal{L}_A$ alone	1.000	2000	4220	621
$\mathcal{L}_A$ composed	0.331	662	1397	206
$\mathcal{L}_B$ composed	0.669	1338	5452	— (none needed)

Table 35.3: The composite, exactly. $\mathcal{L}_A$ retains 0.331 of the deep inquiry it could afford alone—identically its share of the wall, per Proposition 35.13—and loses 66.9%. $\mathcal{L}_B$ loses 33.1% of its assays and none of its depth, because the rung it holds is self-funding. Neither learner perceives, models, or touches the other.

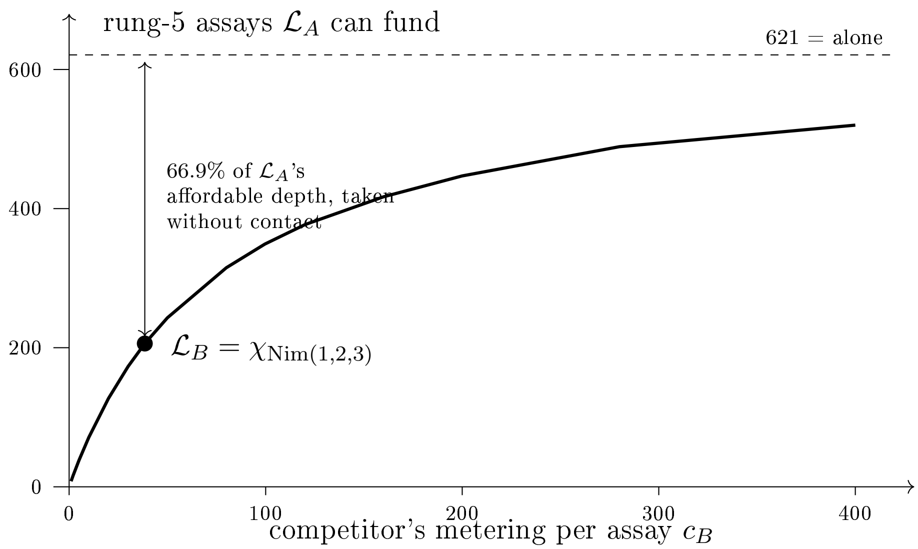


Figure 35.3: Competitive dilution. The curve is $D_A(c_B) = D_A^{\text{alone}} \cdot c_B/(c_A + c_B)$ with $c_A = 77.8$, exactly as Proposition 35.13 gives it. As the competitor's science gets cheaper, the affordable depth of the learner whose science is still unfinished goes to zero.

> *Observation* 35.6 (And it is regressive). $\mathcal{L}_B$ is unaffected in depth. Its best rung is self-funding, so the surplus it forgoes buys it nothing it needs. $\mathcal{L}_A$ is affected in depth alone, because everything it still wants to learn runs a deficit. The learner with the finished science pays in currency it does not spend; the learner with the unfinished science pays in the only currency that matters to it.

35.11.4 Succession is a change of the grading vector

The wall is finite, so this composite has an ending, and the ending is where the typing of the overlap earns its keep.

Proposition 35.14 (Exhaustion forces a change of type). *Let $\mathcal{L}_1 \mid \mathcal{L}_2$ have $\mathbf{k} = (0, 1, 0, 0)$ with the shared source exhaustible. When the source is exhausted, the foraging inequality fails at every source locus for both learners. The composite then either dies or acquires $k_{\mathrm{met}} > 0$ or $k_{\mathrm{prb}} > 0$; that is, succession in the sense of Chapter 33, Proposition 33.15, is precisely a change in which component of the grading vector is nonzero.*

Corollary 35.3 (The change is priced, and priced asymmetrically). *Acquiring $k_{\mathrm{met}} > 0$ requires deriving a name in the other learner's metabolic namespace, which under capability gating is real work: a hypothesis describing that namespace must be formed and installed as a guard. By Proposition 33.6 of Chapter 33 the branching a learner can survey is its stack divided by the cost of one check. Hence the transition is available to whichever learner retains apparatus for forming namespace hypotheses about adaptive subjects—which by Corollary 33.5 there is the prey-feeder, and in this composite is the learner whose science was unfinished.*

In the worked composite, $\mathcal{L}_B$ ends the source phase holding 5452 tokens and $\mathcal{L}_A$ holds 1397; $\mathcal{L}_B$'s stack is 3.90 times $\mathcal{L}_A$'s. By the foraging inequality that stack is worth opening at any concealment depth these prices make affordable. So the learner that was confined during the source phase is the one positioned to harvest at succession, and the capital it harvests was accumulated by the learner that confined it.

Remark 35.11 (Offered as a structure, not a moral). We resist reading anything edifying into this. What the framework says is narrow and checkable: a cheap complete theory is a source-like access profile—non-adaptive, converged, with no standing revision cost—and by Corollary 33.5 of Chapter 33 the apparatus for modelling adaptive subjects is a heterotroph's expense, not carried by anything that does not need it. A learner that has finished its science has, by construction, stopped paying for the machinery that would let it form a hypothesis about a competitor. That is not a punishment for being right. It is what being right in a closed domain costs, and it is the same fact as Proposition 34.7 of Chapter 34: a solved science is a dead ecology.

35.12 The three units of learning

Collecting the three notes, the units of learning are distinguished by exactly one thing: the revision distance each can reach.

unit	mechanism	reachable move	authority
individual	walk the relaxation lattice	confined below 2^{-n}	Chapter 33, Prop. 33.4
lineage	crossover at a shallow locus	2^{-1}	§33.12.10 of Chapter 33, §34.9.3 of Chapter 34
composed learner	adopt another root	the whole tree	§35.4, Prop. 35.9

Proposition 35.15 (Composition reaches what no lineage reaches)**.** *A lineage's recombination acts at loci of its own genomes, so every allele it can install already occurs in one of its members and every namespace it can come to describe is generated from its own root. Under Proposition 35.9 a change of root is therefore unreachable by recombination. It is reachable by composition, since* $\mathsf{N}_1 \vee \mathsf{N}_2$ *contains names manufactured from both roots and the composite may install guards over either.*

Remark 35.12 (The series, and where it stops). The game note measured the middle row: the corner-opening allele was unreachable to any individual by deepening, free to a lineage by crossover, and the invasion was unconditional. The top row is the analogue one step up, and its content is that whatever a lineage's early commitments fixed—not merely its opening, but the family of names its whole hypothesis language is generated from—is fixed for good against every mechanism internal to it. Composition is the only operator in these notes that changes it, and merger (§35.8) is the only one that changes the learner's identity along with it. We do not know whether the series continues, and Problem 6 asks.

35.13 Composition is the exit from the dead ecology

Proposition 34.7 of Chapter 34 says a population all of whose members hold theories that draw against one another performs no harvest, burns tokens to live, and starves; and that the better its science the sooner it dies. Remark 34.12 there lists three exits and takes none of them: enlarge the game so χ is unaffordable, let the rules drift, or admit an external source.

The first exit is composition.

Proposition 35.16 (Composition restores harvest)**.** *Let* $\mathcal{L}$ *be a learner in the terminal state of Chapter 34, Proposition 34.7: harvest rate zero,* χ *held by every member,* σ *declining monotonically. Let* $\mathcal{L}'$ *be any learner whose probe namespace is generated from a distinct root, so that* $\mathcal{L}$ *holds no hypothesis satisfied by* $\mathcal{L}'$*'s components. Then in* $\mathcal{L} \mid \mathcal{L}'$ *with* $k_{\mathrm{met}} > 0$ *the foraging inequality is satisfiable* *again, because $\mathcal{L}'$'s stacks are concealed at a depth $\mathcal{L}$ has not paid to see, and paying is once more a purchase with a return.*

Remark 35.13 (Enlarging the game and composing are the same operation). This is the content we would most like a reader to take away. "Enlarge the game so that the complete theory is unaffordable" sounds like a modelling intervention—a change of domain imposed from outside. Under Definition 35.2 it is an operation *inside* the framework, performed by the ecology on itself, requiring no new rewrite rule and no new sort. A science that has closed reopens by meeting a science that has not. The heat death of Chapter 34 was never a property of noughts-and-crosses; it was a property of a world with one game in it.

35.14 What this shows, and what it does not

Shown. That a learner may be taken to be a weighted population and hence a namespace, and that under that reading composition is parallel composition on terms, disjunction on namespaces, and undetermined on distributions (§35.3–§35.4). That disjointness of namespaces is exactly factorisation of the distribution, so the lax comparison map's defect *is* the interaction and is a quantity rather than a caveat (Theorem 35.1, Proposition 35.3). That a typed grading vector recovers the ecological relations as sign conditions (§35.5). That conservation makes epistemic overlap the only positive-sum overlap (§35.6). That non-interference is unstable by default and purchasable by rooting (§35.7). That phase is not compositional (§35.9) and that competitive dilution confines a learner without contact, in a closed form and on exact numbers (§35.10–§35.11).

Not shown. We have not proved a monoidal-category theorem: Proposition 35.3 gives the comparison map and its invertibility condition, and the coherence conditions, the unit laws up to the identifications merger imposes, and the relationship to the fibration's section are left open. Definition 35.6 is a modelling choice; under the round-robin alternative each learner receives half the payouts and the retained fraction becomes 1/2 for both, so the *dilution* survives and the *regressiveness* weakens to whatever asymmetry the rungs themselves carry—in the worked composite $\mathcal{L}_A$ would retain 0.500 rather than 0.331, and still lose depth where $\mathcal{L}_B$ loses none. As in Chapter 34 the measurements are exact computations over policies rather than observations of a running population: the ecology of Appendix 35.16 is written and not run to a fixed point, so every claim about invasion and succession is a claim about payoffs. And the Nim ladders, like the noughts ladder, are priced by a schedule that is a choice.

The honest summary. The composition operator was, going in, an obvious formality—parallel composition is already there, and saying that learners compose by it looks like a restatement. What the exercise produced instead was three things we did not put in: that the probabilistic content of composition is exactly a graded separation, that a typed grading vector is forced rather than convenient, and that two learners which cannot perceive each other can nonetheless confine each other's inquiry by a measurable amount.

35.15 Open problems

1. **Coherence.** Is $(\mathrm{Eco}, |, \mathbf{0})$ lax monoidal with λ of Proposition 35.3 as the structure map, and is the canonical section of Chapter 33, Proposition 33.31, lax monoidal over it? The obstruction to strictness is located; the coherence conditions are not checked.

2. **Is the grading vector complete?** Definition 35.3 gives four sub-namespaces because four are what the companion notes distinguish. Is there a fifth? The candidate is the *arena* namespace: the names by which an admitted context class $\mathcal{A}$ is realised (Chapter 34, Observation 34.1). Two learners sharing an arena but no other name play the same game against different opponents, which is neither of the relations in Table 35.1.

3. **Optimal k.** Given a distribution over environments, is there an optimal grading vector for a learner to hold—an amount of overlap that maximises expected residual life? §35.6 says the Prb component should be positive and the Met component should be zero or pooled, but the trade-off against the exposure of Lemma 33.1 of Chapter 33 is not worked.

4. **Run the composite.** Appendix 35.16 specifies a two-learner world; §35.11 computes payoffs. Sampling trajectories by the Gillespie construction of [25] would turn Table 35.3 from a payoff calculation into an ensemble result, and would settle whether the succession of Proposition 35.14 actually occurs or whether $\mathcal{L}_A$ starves first.

5. **The plagiarism equilibrium, now well posed.** Problem 2 of Chapter 33 asked whether honest science survives theory theft. With the grading vector it becomes: for which k_{prb} is the two-way reading equilibrium stable against a one-way reader, and does the answer depend on the shape of the ladder (Observation 35.4) in each learner's environment?

6. **Does the series continue?** Individual, lineage, composite: each reaches a revision the one below cannot. Is there a fourth unit, and if so what does it reach? The natural candidate is a distribution over *composites*—a distribu-

tion over grading vectors rather than over configurations—and we do not know whether that is a new object or the same one at a coarser resolution.

7. **Cheapness versus completeness.** Observation 35.4 shows that whether χ is a good purchase varies by environment. Is there a criterion—presumably in terms of Remark 33.3 of Chapter 33, how much of the environment's regularity is structural, together with the branching factor—that predicts the shape of the ladder without computing it? For a composite this would predict who dilutes whom in advance of any measurement.

35.16 Appendix: A composite world, in rholang

The listing below is the composite of §35.11: two learners rooted at distinct quotations, sharing exactly one source name, with the typed namespaces of Definition 35.3 written as name predicates and the grading vector as a family of graded separations over them. Guards using a structural or modal connective are marked `//! ideal` and specify a check rather than perform one; everything else runs. Conservation is auditable by inspection: `Wall`, `Harvest` and `Beget` move tokens, `Live` debits them, and nothing mints.

```
// ============================================================================
// TWO LEARNERS, ONE WALL
// ============================================================================
//
// Companion listing to "Composing Learners". The world is literally
//
// Wall | L_A | L_B
//
// and every claim of the note is a claim about what that bar does.
//
// ROOTING (Prop. "Rooting makes disjointness a theorem"). Each learner
// manufactures every private name by quoting a term containing its OWN root.
// L_A's names are @[*rootA, ...] and L_B's are @[*rootB, ...]. By freshness-
// by-quotation no name of one is structurally equal to a name of the other,
// so k = 0 holds on every component of the grading vector EXCEPT the wall,
// whose name neither learner manufactured and both were handed. That single
// imported name is the whole of the coupling.
//
// TYPED NAMESPACES (Def. "Typed namespaces"). Four predicates per learner:
//
// Met(r) := @[ *r, "m", _ ] metabolic: where stacks rest
// Src(r) := @[ *r, "s", _ ] source: emitters drawn upon
// Prb(r) := @[ *r, "p", _ ] probe: where assays are installed
// Mat(r) := @[ *r, "g", _ ] mating: where genomes are disclosed
//
// The grading vector of the composite is then the four-tuple
//
// k_tau = | fn(L_A) /\ fn(L_B) /\ N_tau |
```

```
//
// and the note's claim is that this world has k = (0,1,0,0).

new stdout(`rho:io:stdout`), display, wall, rootA, rootB,
   Live, Wall, Assay, Ladder, Body, Harvest, Beget, Merge, Learner
in {

  for (@line <= display) { stdout!(line) }

  // ==========================================================================
  // METABOLISM. Reclaim - debit - reissue. Between the receipt and the send
  // the stack is a message in flight, which is what Harvest exploits.
  // ==========================================================================
  | contract Live(m, @burn, step) = {
      for (@sigma <- m) {
        match sigma > burn {
          true => { m!(sigma - burn) | *step | Live!(*m, burn, *step) }
          false => { display!(("DIED", "could not fund a step")) }
        }
      }
    }

  // ==========================================================================
  // THE SHARED SOURCE. One contract, hence rate-limited; non-exclusive, since
  // anyone holding the name may call it; non-adaptive; and finite.
  //
  // This contract IS the entire overlap. Note what it does NOT do: it does
  // not know who is calling, does not adapt to callers, and cannot be used by
  // one caller to observe another. Competitive dilution is delivered by a
  // process with no model of anybody.
  // ==========================================================================
  | contract Wall(w, @quantum, @reserve) = {
      for (ret <- w) {
        match reserve >= quantum {
          true => { ret!(quantum) | Wall!(*w, quantum, reserve - quantum) }
          false => { ret!(0) | Wall!(*w, quantum, 0) } // exhausted
        }
      }
    }

  // ==========================================================================
  // THE ASSAY. Unchanged from the companion notes: the hypothesis sits in
  // GUARD position, so playing IS testing and confirmation IS the rendezvous.
  // The probe name is manufactured from the learner's own root, which is why
  // k_prb = 0 -- neither learner can install a guard that the other's activity
  // could ever satisfy, because the names do not meet.
  // ==========================================================================
  | contract Assay(r, @g, @hyp, ret) = {
      new probe in {
        @[*r, "p", g]!(*probe) // probe in Prb(r)
        | wall!(*probe) // the ONE shared name
```

```
      //! ideal: structural guards
      | for (@pos <- probe where pos |= hyp) { ret!(("confirm", pos)) }
      | for (@pos <- probe where ~(pos |= hyp)) { ret!(("refute", pos)) }
    }
  }

// ============================================================================
// THE LADDERS. Two environments, two shapes. L_A's ladder is the
// noughts-and-crosses ladder of the game note: chi is dominated and every
// rung above 2 runs a metabolic deficit at the wall's quantum. L_B's is the
// Nim ladder: chi_Nim is a NAME-LEVEL predicate of depth zero, so the
// complete theory of L_B's environment is self-funding.
//
//! ideal throughout.
// ============================================================================
| contract Ladder(@which, @r, ret) = {
    match (which, r) {
      // -- L_A: noughts-and-crosses, costs 0 / 40.9 / 77.8 / ... / 679.2
      ("ttt", 2) => { ret!( @{ exists s. Emp(s) and
                            (Thr(s, me) or Thr(s, them)) } ) }
      ("ttt", 5) => { ret!( @{ exists s. Emp(s) and
                            (Thr(s, me) or Thr(s, them))
                            or <<K s>> Fork(me)
                            or forall t. <<K s>>[K t] ~Fork(them)
                            or Central(s) } ) }
      ("ttt", 6) => { ret!( @{ nu X. forall s.
                            [K s]( ~Lost and exists t. <<K t>> X ) } ) }
      // -- L_B: Nim. Note the absence of any modality anywhere.
      ("nim", 1) => { ret!( @{ exists s. One and Takes(s) } ) }
      ("nim", 2) => { ret!( @{ exists s. (One and Takes(s))
                            or (Two and Equalises(s)) } ) }
      ("nim", 3) => { ret!( @{ exists s. (One and Takes(s))
                            or (Two and Equalises(s))
                            or NimSumZero(s) } ) } // chi_Nim
    }
  }

// ============================================================================
// A LEARNER. Not a scientist: a POPULATION, rooted, weighted by the stacks
// its members hold. What makes it one learner rather than several is the
// root, which is what its whole namespace is generated from.
// ============================================================================
| contract Learner(r, @which, @rung, @burn, @members) = {
    match members {
      [] => { Nil }
      [h ...t] => {
          Body!(*r, which, rung, burn, h)
        | Learner!(*r, which, rung, burn, t)
      }
    }
  }
```

```
| contract Body(r, @which, @rung, @burn, @i) = {
    new hyp, score in {
        @[*r, "m", i]!(400) // the stack: in Met(r)
      | score!(0, 0)
      | Ladder!(which, rung, *hyp)
      | Live!(@[*r, "m", i], burn, @{
            new ret in {
                for (@h <- hyp) { hyp!(h) | Assay!(*r, i, h, *ret) }
              | for (@o <- ret) {
                  for (@c, @f <- score) {
                    match o {
                      ("confirm", q) => { score!(c + 1, f)
                                        | for (@s <- @[*r,"m",i]) {
                                            @[*r,"m",i]!(s + q) } }
                      _ => { score!(c, f + 1) }
                    }
                  }
                }
            }
        })
    }
  }

// ===========================================================================
// SUCCESSION. When the wall returns 0 forever, the only credit left is in
// another learner's Met namespace -- which is a namespace this learner has
// never described, because k_prb = 0 kept it invisible. Acquiring it is a
// step, and the step is metered: a hypothesis over Met(rOther) must be
// formed and installed. That is the change of grading vector, and it is
// available only to a learner that still carries the apparatus.
// ===========================================================================
| contract Harvest(@rSelf, @i, @rOther, ret) = {
    //! ideal -- the guard is a name predicate over the OTHER root's
    // metabolic namespace. Forming it is what costs; firing it is a comm.
    for (@mPrey <- @[*rSelf, "p", "hunt"]
          where mPrey |= @{ Met(rOther) }) {
      for (@sigma <- @{mPrey}) {
        for (@own <- @[*rSelf, "m", i]) {
            @[*rSelf, "m", i]!(own + sigma)
          | ret!(("harvest", sigma))
          | display!(("HARVEST across roots", sigma))
        }
      }
    }
  }

| contract Beget(m, @alleles, @endow, ret) = {
    for (@sigma <- m where sigma > endow + 40) {
        m!(sigma - endow)
      | match alleles {
```

```
        [w, rg] => {
          new childM, code in {
              code!( @{ Body!(*childM, w, rg, 2 + rg, endow) } ) // inert
            | for (g <- code) { *g | childM!(endow) } // metered
            | ret!(("born", w, rg))
          }
        }
      }
    }
  }

// ============================================================================
// MERGER -- a DIFFERENT OPERATOR, and this is the point of the section.
//
// Composition is the bar. It is free, it is associative and commutative,
// and it leaves both learners identifiable afterwards. Merger identifies
// the two MATING namespaces, which is a step, is metered, and is
// irreversible: after it there is no composition recovering the parts,
// because alleles have crossed roots.
//
// Note that Merge is a CONTRACT and '|' is not. That asymmetry is the
// formal content of "merger is a separate operator".
// ============================================================================
| contract Merge(@rA, @rB, ret) = {
    new rC in {
        // the merged learner's mating namespace admits homologous loci
        // across BOTH roots; this identification is what cannot be undone
        @[*rC, "g", "admits"]!([rA, rB])
      | ret!(*rC)
    }
  }

// ============================================================================
// THE WORLD. Theta = 12000 (wall) + 3 x 400 (L_A) + 3 x 400 (L_B) = 14400.
//
// The composite is the bar between the two Learner calls. Delete either and
// the other's numbers change -- by exactly the factor of Prop. "competitive
// dilution", and by nothing else, since no other name is shared.
// ============================================================================
| Wall!(*wall, 6, 12000)

| Learner!(*rootA, "ttt", 2, 4, [0, 1, 2]) // noughts: rung 2, c = 77.8
| Learner!(*rootB, "nim", 3, 5, [0, 1, 2]) // nim: chi_Nim, c = 38.5

// ----------------------------------------------------------------------------
// WHAT TO WATCH
//
// * The overlap is ONE NAME. 'wall' is the only identifier appearing in
// both Learner subtrees. Everything else is manufactured under a root.
// k = (0,1,0,0), and it stays there by construction, not by luck.
//
```

```
    // * Nobody models anybody. No guard anywhere mentions the other root.
    // L_B has no hypothesis about L_A and would gain nothing from one. The
    // dilution happens anyway.
    //
    // * L_B finishes and L_A does not. L_B holds the complete theory of its
    // environment at burn 5; L_A holds an incomplete one at burn 4 and cannot
    // afford the rungs it still wants. The one still climbing is the one the
    // competition costs.
    //
    // * And it ends. When the wall returns 0, Assay's confirm guard stops
    // crediting, Live keeps debiting, and the only remaining credit is inside
    // the other root's Met namespace. Reaching it is Harvest, and Harvest's
    // guard is a hypothesis somebody has to be able to afford to form.
    // ----------------------------------------------------------------------------
  }
```

35.17 Appendix: Reproducing the numbers

All figures in §35.11 are exact computations, not samples. The noughts-and-crosses ladder figures—win rates 0.437, 0.668, 0.798, 0.831, 0.836, 0.918, 0.873 and expected meterings 0.0, 40.9, 77.8, 134.9, 245.7, 255.9, 679.2—are quoted from Table 1 of Chapter 34 and were not recomputed here.

The Nim ladders are computed by `compose.py`, which enumerates positions as sorted tuples of heap sizes, evaluates each rung by memoised recursion over all plays with uniform tie-breaking within a rung and a uniform-random opponent, and charges $|M| \cdot \sum_{1..r} \kappa$ at each decision of the policy player with $\kappa(d) = 2^d$ and depths $1, 1, 0$ for the three rules. Probabilities are carried as exact rationals and converted to decimal only for display. Figures are symmetrised over which side moves first; the unsymmetrised figures are reported separately where the colour asymmetry is the point.

The composite of §35.11 uses $R_0 = 12{,}000$, $q = 6$, $p = 0.05$, shares by Definition 35.6, and the identity of Proposition 35.13; the computation is a few lines and is included in the same script for auditability rather than for difficulty.

Chapter 36

Engines and Individuals

Composition assumed a common currency. Two learners sharing a wall were drawing on the same tokens, and that assumption did a great deal of quiet work. Drop it — let each community's tokens be its own colour, as the metabolisms of two separately originated biospheres would be — and something has to sit between them and convert.

That converter is this chapter's subject, and it turns out to be an organism rather than a piece of infrastructure: a fourth access profile alongside source and prey, persistent like a source, exhaustible like prey, and unique in being adaptive in its *rate*. It lives by the spread, which is not a metaphor for a toll but literally the toll of the rendezvous through which the swap happens. The nearest thing in the biosphere is not a decomposer but a mycorrhizal network.

Two results give the chapter its title. The first is that the cycles of the conversion graph form an error-correcting code, and that its total detecting power is exactly the first Betti number of the graph — so how much a community can catch a mispriced exchange is a topological fact about the shape of its trade, and a conversion sitting on a bridge can never be checked at all. The second is a criterion for individuation. Communication is not perfect; the notion is recovered by interposing a medium that may drop or corrupt what passes through, and a region may close its namespace exactly when the boundary it can maintain exceeds what the error rate costs. An individual is where repair is cheaper than exposure. It is porous by necessity — perfect closure is death, for the same reason a solved science is a dead ecology — and it is a fixed point, since the closed region is itself a learner that can compose with others.

That last move closes the loop the part opened with, and it is the natural place to hand over. Once individuals are the sort of thing that can be stacked, the question of what a learner sees from higher up in the stack is no longer rhetorical — which is where the next part begins.

36.1 Introduction

36.1.1 Why the unit moves again

The series this note belongs to has moved its unit of analysis twice. In Chapter 33 the unit is a single mortal scientist: a cost-accounted rho computation that forms hypotheses in a namespace logic, pays for its assays, and dies when its stack runs out. In Chapter 35 the unit is a learner: a population of such computations together with their reservoirs, serving as a namespace, with a distribution over it, composed with other learners by parallel composition. That note ends by asking whether the series individual/lineage/composite continues.

It does, and the reason it does is that composition of learners yields a learner, so the construction is closed under an associative operator. But closure alone gives arbitrary nesting and no scale structure: a soup of composites has no distinguished levels. Something must pick out *skill*, *organism*, *pod*, *species*, *biosphere* as rungs rather than as arbitrary cuts, and the companion notes do not say what.

Two things were also missing from the economics. Chapter 35 works in a single token colour, so its conservation law is the flat statement that a fixed supply Θ is never minted. Real coordination between communities is not a transfer of one fungible fuel; it is *conversion*, at rates, between resources that are not interchangeable — and the arbitrage machinery of [51] exists precisely to handle that. And Chapter 35 has no geometry: its composition operator is indifferent to whether the two learners are adjacent or a galaxy apart, though nothing in the framework says the cost of reaching across should be the same in both cases.

This note supplies all three, and they turn out to be one thing.

36.1.2 The move, stated once

Give each community its own token colour. Adjoin engines between colours and weight each engine by its dissipation θ. The resulting weighted graph is the geometry the framework was missing; its cycle space is a code on prices; and the isoperimetry of that graph is the criterion for when a region may stop trading and start sharing — that is, for when a composition of learners becomes an individual.

36.1.3 What we claim, and what we do not

We claim: that the engine is derivable as a fourth access profile rather than importable as a new kind of thing (§36.4); that interposition supplies an error model natively, with θ as its metering defect and stranding as its irreversibility (§36.5); that the detection strength of a price is $1 - R_{\text{eff}}$ and the total is β_1 (§36.6, Theorem 36.2); that inventory-holding engines make arbitrage a property of a marking (§36.3); that merger is available exactly inside an isoperimetric threshold (§36.8, Proposition 36.15), with partial closure of the medium namespace as its structural form (Definition 36.6); and that a finite critical radius, together with the tower of media, forces the hierarchy that Chapter 35 could only permit (§36.9).

We do not claim to have derived the rates. [51] takes them as given and so, in the main, do we; §36.12 says what it would take to derive them and why the numbers of Chapter 34 are the natural candidate. We do not claim the error channel has been identified: §36.8.6 inherits, unresolved, the "which ε is binding" question of [48]. We do not claim the correspondence with observed biology is more than suggestive, though §36.6.4 notes that the ecological pyramid falls out of Proposition 2 of [51] with no adjustment, which is a check we did not arrange.

36.2 What this chapter carries forward

From Chapter 33: the cut and the three grades of access (perception, interaction, reflection); the assay via the `where` clause; the conservation law $\sigma + \kappa = \sigma_0 = \Theta$; the access profile distinguishing a *source* (persistent emitter, rate-limited, non-exclusive, non-adaptive) from *prey* (linear send, exhaustible, exclusive, adaptive); the relaxation lattice and its ultrametric; and the freshness-by-quotation proposition, that for any context C with $C[P] \neq P$ the name $@C[P]$ does not occur in P.

From Chapter 34: the two routes to any predicate — *read* it, priced by formula depth, if the environment has computed it into legible structure, or *drive* it, priced by rendezvous — and the observation that the interesting question is not whether the work is done but who is billed for it.

From Chapter 35: the learner as three layers (population, namespace, distribution) with the distribution *inside*; composition as parallel composition; the factorisation theorem, that namespace disjointness is equivalent to absence of a spanning redex and to $\mu_{12} = \mu_1 \otimes \mu_2$; the grading vector $\mathbf{k} = (k_{\mathrm{met}}, k_{\mathrm{src}}, k_{\mathrm{prb}}, k_{\mathrm{mat}})$ over the four typed namespaces $\mathsf{Met}, \mathsf{Src}, \mathsf{Prb}, \mathsf{Mat}$; merger $\bowtie$ as a separate operator; the instability of non-interference and its repair by rooting; and competitive dilution in closed form.

From [49] and [50]: signatures as names, token stacks as first-class terms, wrapping by construction, the cost monad $(\mathrm{Cost}, \eta, \mu)$ on ciGSLT and its non-idempotence, and located resource stacks.

From [52]: the correction that a stack floating in the associative–commutative soup is ambient authority, repaired by holding every stack on a channel; the acceptance gate; the netting result; and the four cost-accounting patterns.

From [51]: conversion systems, engines, rates with $r(a{\to}b)\,r(b{\to}a) = 1$; the log-rate cochain ℓ; the equivalence of arbitrage-freedom, exactness of ℓ, and existence of a valuation ν; the graph–Hodge splitting into energy and arbitrage components; $\dim\ker\delta^* = \beta_1$; arbitrage-removal as an idempotent reflection; the first law (conservation of ν in the lossless regime); the second (dissipation θ makes ν a Lyapunov function); and the insistence that ν is a unit of account and never a medium of exchange.

From [48]: the crossover between a shared substrate and autonomous replication, at critical radius $r^* \sim \sqrt{\rho v/\varepsilon}$ in an error rate ε, a repair throughput ρ, and a communication speed v; and the closing conjecture that this crossover and the factoring of the OSLF-generated logic over a population are two views of one threshold. That conjecture is what §36.8 cashes.

From [19]: the persistent boundary-forwarder lemma, which is what makes an encoding compositional and which supplies Proposition 36.3; and the diagnosis, made there, that a mediating process written with a linear receipt models one passage rather than a passage-way.

From [54]: the combinators `zero`, `one`, `give`, `and`, `or`, `scale`, `truncate`, `get`, and observables.

36.3 Colours, engines, and a fork in the conservation law

36.3.1 Communities are colours

> **Definition 36.1** (Coloured learner)**.** *A* coloured learner *is a learner $\mathcal{L}$ in the sense of §35.3 of Chapter 35 together with a distinguished signature $c(\mathcal{L}) \in \Sigma$, its* colour*: the signature at the head of every stack held at a name in* $\mathsf{Met}(\mathcal{L})$*.*

The colour is not new apparatus. [49] already replaces one universal fuel by a family of signature-indexed resources, one per economically empowered actor; [49, §6] calls the collapse to a single s_0 the monochromatic limit. Chapter 35 works in that limit. Taking a community's colour to be its own signature is the natural uncollapsing, and it is what makes "Terran food and alt-Earth food may not be mutually nutritive" a statement the calculus can express: two learners whose Met stacks carry incomparable signatures cannot eat each other's tokens, however adjacent they are.

> **Definition 36.2** (Engine)**.** *An* engine *between colours a and b is a persistent process holding two located reserves, one of each colour, and offering a linear swap: it accepts one a-token and releases $r(a \to b)$ b-tokens from its own inventory. Its* toll *is the metering charge of the rendezvous that performs the swap.*

36.3.2 The fork

There is a conservation collision here that must be settled before anything else, because the two notes being joined conserve different things. Chapter 35 conserves tokens *per colour*: Θ is fixed and never minted. [51] conserves the scalar ν: an engine consumes one a and releases r units of b, which is not per-colour conserving at all. One must give.

(i) **Engines hold inventory.** A swap is against stock, so per-colour totals *are* conserved and r is the ratio at which inventory is exchanged. The cost: engines deplete, r moves with the marking, and ℓ becomes a function of state.

(ii) **Conserve only ν.** Colours interconvert at fixed rates and Θ_ν is the invariant. This matches [51] exactly but abandons the per-signature ledger the scientist note runs on.

We take (i). It preserves the more specific law, it keeps engines mortal (an engine that runs out of one colour stops being an engine, which is what we want of an organism), and its cost — state-dependence — is where the content is.

Proposition 36.1 (Arbitrage is a property of a marking)**.** *Under (i) the log-rate cochain is a function $\ell(M)$ of the marking M of the engine reserves. Consequently the Hodge splitting $\ell = \ell_{\mathrm{en}} + \ell_{\mathrm{arb}}$ of [51, §4] must be taken pointwise in M, and arbitrage-freedom is a condition on a* state *of the ecology rather than on the ecology.*

Proof. Immediate from Definition 36.2: the rate is computed from the reserves, and a swap changes the reserves. Appendix 36.16, block (4), exhibits a triangle of constant-product engines whose cycle yield is exactly 1 at the balanced marking and departs monotonically as one engine is drained; six trades of ten units carry $\log \oint \ell$ from 0 to -1.003. $\square$

Remark 36.1 (Trading is what removes arbitrage). Proposition 36.1 is not a defect. In [51] the projection Π onto the exact cochains is a least-squares correction performed by nobody in particular; the note flags that the operational content of "shaking the arbitrage out" is left abstract. Under (i) it is concrete and it is metabolic: an arbitrageur is a computation that traces a cycle of yield > 1, and each pass moves the reserves in the direction that reduces the yield. Arbitrage-removal is not applied to the system, it is *eaten out* of the system, and the eating is metered like anything else. This is the sense in which the idempotent reflection of [51, Prop. 3.4] is realised rather than postulated.

Corollary 36.1 (Residual arbitrage)**.** *Since each pass of a cycle costs the arbitrageur at least one token, and the yield per pass falls monotonically toward* 1*, there is a level of arbitrage below which removal is unprofitable. A mortal ecology therefore carries a floor of arbitrage set by its own metering, and the floor is higher the poorer the ecology.*

Corollary 36.1 is Chapter 33, Corollary 33.1 — poverty is confining — one level up and in a second currency. A rich ecology has efficient prices for the same reason a rich scientist can afford deep inquiry.

	persistent	exclusive	exhaustible	adaptive
Source	yes	no	in aggregate	no
Prey	no	yes	yes	yes
Scientist	—as prey, plus grade-3 exposure—			
Engine	**yes**	**no**	**per colour**	**in rate**

Table 36.1: The engine as a fourth access profile. It is source-like in persistence and non-exclusivity, prey-like in exhaustibility, and unique in that what adapts is not its behaviour under attack but its *terms*.

36.3.3 The reflexive observable

In [54] an observable is a time-varying quantity both parties agree on, and it is exogenous: the weather, a stock price, a block height. Under (i) the observable that sets an engine's rate reads the engine's own inventory. This is the one place the combinator language had to be extended, and it is worth naming, because it is what turns a contract into an organism: the contract's terms depend on the contract's state.

36.4 The engine is an organism

36.4.1 A fourth access profile

§33.10 of Chapter 33 separates two ways of being food. A *source* is a persistent emitter: rate-limited, non-exclusive, non-adaptive, inexhaustible on the timescale of one learner though finite in the aggregate. *Prey* is a linear send: exhaustible, exclusive, adaptive. The distinction is not a distinction of sort or of rewrite rule but of multiplicity — a fact about the cut.

The engine is a third way, and it is obtained by taking the same four attributes and choosing a combination the table did not previously contain.

> **Proposition 36.2** (The engine is derived, not imported). *The profile in Table 36.1 requires no new sort, no new rewrite rule, and no new GSLT. A persistent receipt on a quote channel supplies persistence and non-exclusivity; two located reserves supply per-colour exhaustibility; and computing the rate from those reserves supplies rate-adaptivity. All three are available in the host calculus of Chapter 33.*

Proof. Appendix 36.15 exhibits the term. Persistence is `<=`; location is a channel, per [52, §2.2]; the rate is a `for`-comprehension over the reserve channels that replaces what it reads. □

> *Remark* 36.2 (The persistence is load-bearing). An engine written with a linear receipt models one conversion, not a converter — the same bug found and fixed in the knot encoding, where crossing gates written with `<-` evaporate after a single visitation and model one passage through the diagram rather than the diagram. The correct shape is a *persistent issuer* offering *linear* contracts: the enzyme persists, the substrate binding does not.

> *Remark* 36.3 (What the engine eats). [51, §6] introduces dissipation $\theta(a{\to}b) \geq 0$ as a modelling parameter and reads the resulting inequality as the second law. Here θ is not posited. Every rendezvous is metered, and the swap is a rendezvous, so θ *is* the toll — and it accrues to the engine's own stack, because something has to fund the engine's continued existence and nothing else is available. The second law and the engine's metabolism are the same fact. An engine whose traffic falls below its burn rate starves, exactly as a scientist does.

> *Remark* 36.4 (Not a decomposer). It is tempting to call the engine a decomposer, on the grounds that it lives on what others cannot metabolise. The better analogue is the mycorrhizal network [59]: it does not consume waste, it makes two otherwise incommensurable metabolisms mutually available and takes a cut. That literature is also the right prior art for §36.6.3, since it studies exchange rates set by partner choice and outside options rather than by no-arbitrage [58].

36.4.2 The combinators as reaction language

With the organism identified, the combinators of [54] sit one level down: they name the engine's reactions, not the engine.

`one`(k) is the funded unit transfer; `give` swaps which signature funds the leg, which [52, §3.1] observes is exactly the two-sided surface of the cut made operational. `scale`(o, c) carries the rate, and by [52, §3.6] scaling multiplies the amount moved and *not* the fuel demand — cost counts funded rendezvous, not magnitude. `and` is the atomic join, under which the partial-funding hazard is structurally impossible.

> *Observation* 36.1 (The swap is the netted futures shape). An engine's conversion is
>
> $$\texttt{and}\big(\texttt{scale}(o, \texttt{one}(a)),\ \texttt{give}(\texttt{one}(b))\big),$$
>
> Pattern (A) of [52, Tab. 1], bilateral cash settlement. That note shows the apparatus rewards the netted form: one transfer in the data-dependent direction

> instead of two gross legs, atomic by construction and costing one token instead of two.

> **Corollary 36.2** (Netting is thermodynamic)**.** *Since θ is the toll (Remark 36.3), netting halves the dissipation on every conversion. What [52] presents as a settlement optimisation is, in this reading, a reduction of the second-law loss along every edge of the engine graph, and hence — by §36.6.4 — an increase in the profitable radius.*

The temporal combinators do work too, and §36.11 takes them up.

36.5 The medium

36.5.1 Interposition is without loss of generality

The host calculus takes communication to be perfect. It does so without loss of generality, because imperfection can be modelled as an additional process rather than an additional rule: races and higher-order channels suffice. Generalise

$$\texttt{for}(y \leftarrow x)P \mid x!(Q) \qquad \text{to} \qquad \texttt{for}(y \leftarrow x_1)P \mid C(x_1, x_2) \mid x_2!(Q),$$

where C is any process joining the two endpoints. The simplest C is a persistent two-way forwarder; richer ones do arbitrarily more.

> **Proposition 36.3** (The perfect case is a special case)**.** *When C is the persistent two-way forwarder on (x_1, x_2), the interposed term is weakly bisimilar to the direct one. Hence nothing is lost by working in the general form, and the perfect-communication calculus is recovered as the identity medium.*

Proof sketch. This is the boundary-forwarder lemma of the knot encoding [19], whose persistence is exactly what makes the encoding compositional. The forwarding steps are τ and are absorbed by weak bisimulation; persistence is required, since a linear forwarder relays once and then is not a channel. □

> **Definition 36.3** (Loss and corruption)**.** *A medium* drops *when it receives on x_1 and emits nothing, and* corrupts *when it emits a value other than the one received. Write $p \in (0, 1]$ for the per-hop probability that a message is relayed faithfully.*

Rung	What C does	What it contributes
relay	forwards faithfully	θ_{toll} only
lossy	drops with probability $1-p$	θ_{loss}, stranding
corrupting	emits a wrong value	the syndrome of Theorem 36.1
reading	forwards and retains	grade-one access to all traffic
converting	swaps colour against inventory	the engine of Definition 36.2

Table 36.2: Media classified by behaviour. The engine is the top rung, so the fourth access profile of Table 36.1 is not a separate kind of thing but the most behaviourally elaborate interposition.

Remark 36.5 (Interposition is not cost-preserving, and that is θ). Proposition 36.3 is a statement about behaviour, not about metering: a single gated rendezvous becomes at least two. So interposition is not a morphism in the cost-accounted category of [50] — it multiplies the gated transitions rather than preserving them — and the defect is exactly the dissipation θ that §36.4 attributes to the engine. This is the third instance in the series of one pattern: in §35.4.3 of Chapter 35 the lax comparison map's defect is the interaction; here the interposition's defect is the cost of distance. Distance is not a parameter of this calculus. Distance is interposition, and its price is the failure of a map to be strict.

36.5.2 A ladder of media

Remark 36.6 (The intermediary learns most). A reading medium on a bridge has grade-one access to everything crossing it. §36.10 argues that commerce across an incommensurable boundary can only be epistemic; Table 36.2 adds that the boundary itself is an epistemic asset, and that whoever runs the forwarder is best placed to harvest it. This is not a decoration: it predicts that intermediaries accumulate information rents, which is what intermediaries observably do.

36.5.3 Loss is stranding, and stranding is exergy

Proposition 36.4 (Dropped tokens are stranded, not destroyed)**.** *Under fork (i) the per-colour totals are conserved by construction. A message dropped by a lossy medium is therefore not annihilated: it rests on a channel no process will receive on. Total value ν is conserved;* available *value strictly decreases.*

Corollary 36.3 (The second law, with a mechanism)**.** *The free-energy reading that [51, Rmk. 6.2] wants and can only posit is here realised. Exergy is ν minus the measure of the stranded set, and it is monotone because stranding is irreversible: recovering a stranded message would require a process that can name the channel it rests on, which by construction is the medium that failed.*

Proposition 36.5 (The two dissipations add)**.** *Along a path of d hops, tolls accumulate additively and reliability compounds as p^d. Writing $\theta = \theta_{\text{toll}} + \theta_{\text{loss}}$ with $\theta_{\text{loss}} = -\log p$, both contributions are additive in the cost metric and the delivered value is $\nu e^{-\theta d}$ as before.*

So Proposition 36.12's profitable radius is partly an information horizon and not only an economic one: a pathway can price itself out either by charging too much or by losing too much, and the two enter the same exponent.

Remark 36.7 (Corruption is where the syndrome comes from). Theorem 36.1 treats a corrupted rate $a\mathbf{1}_e$ as given. Table 36.2 says where it comes from: a corrupting medium on the quote path of engine e. The price code is not detecting an abstract misquotation; it is detecting the medium.

36.5.4 The medium is a namespace, and it is higher order

Since C is a process, $@C$ is a name, and $*@C = C$ recovers the process. Media are therefore nameable, and the media of a region form a namespace.

Definition 36.4 (Medium namespace)**.** *For a region B, put*

$$\mathsf{Chn}(B) \;=\; \big\{\, @C \;:\; C \textit{ mediates a rendezvous within } B \,\big\}.$$

This namespace is higher order in two senses, and both do work. First, C's free names are the endpoints it joins, so $@C$ sits one level up the reflection tower from the names it carries: a medium is a name for a carrier of names. Second, a rich C is itself realised by interposition, so media are mediated by media.

Proposition 36.6 (The tower)**.** Chn *at one level is the endpoint namespace at the next: the media over which a region's parts communicate are the channels of the region taken as a part of something larger. The tower is well-founded downwards, since by the no-self-code theorem a name codes a strictly smaller term,* *and bounded upwards by Proposition 36.12, since a level whose media exceed the profitable radius has no traffic to mediate.*

Proposition 36.6 is the fractal of §36.9 stated on the naming side rather than by analogy, and it supplies the generating step that closure under $|$ could not: *the outside of one level is the inside of the next.*

36.5.5 Inside, outside, and porosity

Individuality is not total closure. An individual that could not be impinged upon could not eat and could not learn. The right notion is partial closure, and the partition it needs is already in the corpus.

Definition 36.5 (Three kinds of medium)**.** *Relative to a region B, a medium is* internal *if both endpoints lie in B,* boundary *if exactly one does, and* both-type *if it supports rendezvous of each kind. This is the channel classification of §33.12.6 of Chapter 33 applied one level up, to media rather than to endpoints. Write $\mathsf{Chn}_{\mathrm{in}}(B)$, $\mathsf{Chn}_{\partial}(B)$.*

Definition 36.6 (Individual)**.** *A region B is an* individual *when $\mathsf{Chn}_{\mathrm{in}}(B)$ is closed — every medium of an internal rendezvous is nameable from within B — while $\mathsf{Chn}_{\partial}(B)$ is not. Internal communication may be arbitrarily complex; what is required is not simplicity but ownership.*

Proposition 36.7 (Perfect closure is death)**.** *If $\mathsf{Chn}_{\partial}(B) = \emptyset$ then B has no grade-one or grade-two access across its cut. By §33.5 of Chapter 33 it cannot harvest, so σ is monotone decreasing and B starves; and it can run no assay, so it cannot revise. Total closure is fatal in both currencies.*

Proposition 36.7 is the individuation-level analogue of the game note's result that a solved science is a dead ecology. Perfection of a boundary and perfection of a theory fail in the same way.

Proposition 36.8 (Porosity requires redex pathways)**.** *An open boundary name is not yet porosity. For the environment to impinge, a boundary channel must carry a chain of redexes reaching a redex on an internal channel; symmetrically, for the individual to act, an internal redex must reach a boundary one. A boundary channel with no such pathway is a stranding site in the sense of Proposition 36.4:*

messages arriving there are conserved and unavailable.

Remark 36.8 (The same undecidability, and the same remedy). "Has a redex pathway to an internal redex" is undecidable in general, for the reason §33.12.5 of Chapter 33 gives for "will communicate". We inherit that note's remedy: take the syntactic over-approximation — the interaction graph on shared names — and accept that it over-counts.

Definition 36.7 (Integration depth)**.** *The* integration depth *of a percept is the length of the redex pathway of Proposition 36.8 from the boundary channel at which it enters to the internal redex at which it is used. Depth one is reflex; greater depth composes information from more of the surface.*

Proposition 36.9 (Deep integration is exponentially fragile)**.** *A percept of integration depth d traverses d hops of internal medium and survives with probability p^d. Its cost is d metered rendezvous. Hence for saturating gains $Y(d)$ the net value $Y(d)p^d - cd$ has an interior maximum whose location falls with p.*

Corollary 36.4 (Ownership is what buys depth)**.** *Integration depth is not purchased with budget alone. It is purchased with reliability, and reliability is available only on a medium the individual can name — since retransmission requires grade-two access to $@C$. Definition 36.6 is therefore not a stipulation: closure of the internal medium is the condition under which deep integration is affordable, and deep integration is what distinguishes an individual from a bag of reflexes.*

Computed in Appendix 36.16, block (5), with $Y_0 = 100$, $Y(d) = Y_0(1-2^{-d})$, $c = 2$: optimal depth $d^* = 5$ at $p = 0.999$ and $p = 0.99$, falling to 3 at $p = 0.95$ and 0.90 and to 2 at $p = 0.75$, with net value falling from 86.4 to 38.2 across that range. An organism with an unrepaired medium is not merely slower; it is structurally shallower.

Remark 36.9 (Two axes). Table 36.2 classifies media by *behaviour*; Definition 36.5 classifies them by *ownership*. Individuation lives on the second axis only. An engine may be internal or boundary, and which it is decides whether the region containing it is one economy or two; what it does decides how much it costs to be either.

Remark 36.10 (Individuation is a fixed point). Definition 36.6 partitions $\mathsf{Chn}(B)$ relative to a choice of B, and B was to be determined by the criterion. This is a fixed-point condition rather than a construction, and its multiple solutions are the levels of §36.9: the circularity is the fractality. We state the condition and do not claim an algorithm for finding its solutions.

36.6 The engine graph

36.6.1 Two graphs, one geometry

There are now two graphs in play and they must not be confused. The *composition graph* has learners as vertices and shared names as edges; its edge data is the grading vector $\mathbf{k}$, and §35.5 of Chapter 35 reads ecological relations off it. The *engine graph* Γ has colours as vertices and engines as edges; its edge data is the log-rate ℓ and the dissipation θ.

They are related but not identical. An engine is a shared name, hence a spanning redex, hence an edge of the composition graph; and since the shared name is one at which stacks are held and drawn, it is of type Met. So:

Proposition 36.10 (Conversion is metabolic overlap)**.** *Two learners linked by an engine have $k_{\mathrm{met}} \geq 1$. Hence by the factorisation theorem of §35.4.2 of Chapter 35 they are not probabilistically independent, and by Proposition 36.13 below their overlap cannot be positive-sum in tokens.*

Remark 36.11 (The vertices are the atomic colours). By [51, Prop. 4.4] a valuation is a homomorphism out of Σ and induces no engine; in particular no engine converts a compound authority $s_1 * s_2$ into a single s', since that would launder the multi-signature guarantee. Compound signatures are therefore priced but unreachable: they are vertices of no engine. Throughout, Γ is taken on the atomic colours, and β_1 is computed there.

Definition 36.8 (The cost metric)**.** *Weight each edge of Γ by its dissipation θ and let d_θ be the induced path metric. This is the geometry the composition note lacked, and it is the only geometry available: the framework has no ambient space, so "far" can only mean* expensive*, in the sense of the geometric weight of [50].*

36.6.2 The cycle space is a code

Recall from [51] that ℓ is arbitrage-free iff it is exact, $\ell = -\delta p$, and that $C^1 = \operatorname{im}\delta \oplus \ker\delta^*$ splits orthogonally into the exact (gradient) and cycle parts, with $\dim\ker\delta^* = \beta_1(\Gamma)$. Write Π_{Cut} and Π_{Cyc} for the two projections, and for an engine e let $R_{\mathrm{eff}}(e)$ be its effective resistance in Γ with unit conductances. Recall the standard identity $R_{\mathrm{eff}}(e) = (\Pi_{\mathrm{Cut}})_{ee}$.

Definition 36.9 (Syndrome)**.** *For an observed log-rate cochain ℓ, the* syndrome *is $\ell_{\mathrm{arb}} = \Pi_{\mathrm{Cyc}}\,\ell$. It is zero iff the observed rates are consistent with some price list.*

Theorem 36.1 (Detection strength)**.** *Let the true rates be exact and let a single engine e be corrupted by a in log-rate, $\ell = \ell_{\mathrm{true}} + a\,\mathbf{1}_e$. Then*

$$\|\ell_{\mathrm{arb}}\|^2 \;=\; a^2\big(1 - R_{\mathrm{eff}}(e)\big).$$

In particular the corruption is undetectable iff $R_{\mathrm{eff}}(e) = 1$ iff e is a bridge of Γ.

Proof. ℓ_{true} is exact so $\Pi_{\mathrm{Cyc}}\ell_{\mathrm{true}} = 0$ and $\ell_{\mathrm{arb}} = a\,\Pi_{\mathrm{Cyc}}\mathbf{1}_e$. Since Π_{Cyc} is an orthogonal projection, $\|\Pi_{\mathrm{Cyc}}\mathbf{1}_e\|^2 = (\Pi_{\mathrm{Cyc}})_{ee} = 1 - (\Pi_{\mathrm{Cut}})_{ee} = 1 - R_{\mathrm{eff}}(e)$. An edge lies in no cycle iff it is a bridge, and $\mathbf{1}_e \perp \ker\delta^*$ iff $\Pi_{\mathrm{Cyc}}\mathbf{1}_e = 0$; equivalently $R_{\mathrm{eff}}(e) = 1$, which for unit conductances characterises bridges. □

Theorem 36.2 (The code's total strength is β_1)**.** *For any connected engine graph,*

$$\sum_{e\in E}\big(1 - R_{\mathrm{eff}}(e)\big) \;=\; |E| - (|V| - 1) \;=\; \beta_1(\Gamma).$$

Proof. Foster's theorem [55] gives $\sum_e R_{\mathrm{eff}}(e) = |V| - 1$; equivalently $\operatorname{tr}\Pi_{\mathrm{Cut}} = \operatorname{rank}\delta = |V| - 1$. Subtract from $|E| = \operatorname{tr} I$. □

Theorem 36.2 is the sharp form of the note's second claim. A price system carries a fixed budget of self-checking, equal to the number of independent arbitrage cycles, and the geometry of the engine graph decides how that budget is distributed across engines. Dense regions get most of it; bridges get none. Verified numerically in Appendix 36.16: for the barbell of two K_5's joined by a single engine, $\beta_1 = 12$ and $\sum_e(1 - R_{\mathrm{eff}}) = 12.000000$, with the bridge contributing 0 and each cluster edge contributing 0.6.

Proposition 36.11 (Series engines are confusable)**.** *Two corrupted engines e, f produce the same syndrome direction iff $\Pi_{\mathrm{Cyc}}\mathbf{1}_e \parallel \Pi_{\mathrm{Cyc}}\mathbf{1}_f$, which holds iff e and f lie in exactly the same cycles. Hence engines in series — a chain through vertices of degree two — cannot be told apart by the code, while engines in parallel can.*

In C_6, where $\beta_1 = 1$, all 15 edge pairs are confusable: a one-dimensional cycle space detects that *something* is wrong and never what. In the barbell, no pair is. The code localises exactly when the ecology is dense.

Corollary 36.5 (Density buys detection)**.** *In a d-regular engine graph on n colours the mean detection strength is exactly $1 - 2(n-1)/(nd)$, approaching $1 - 2/d$. Measured: 0.344 at $d = 3$, 0.672 at $d = 6$, 0.902 at $d = 20$ for $n = 60$.*

36.6.3 Bridges carry no pricing discipline

Theorem 36.1 has a consequence that reframes the interstellar picture. A thin pathway between two dense components is a cut edge; cut edges lie in no cycle; so no-arbitrage constrains a bridge rate *not at all*. Path-independence is vacuous when there is one path.

Corollary 36.6 (The numeraire is global but uninformative across a cut)**.** *If Γ is connected then a valuation ν exists globally and is unique up to gauge, by [51, Thm. 3.3]. But the rate on a bridge is unconstrained by the no-arbitrage condition: any value of it extends to a consistent ν. Within a dense component each rate is over-determined by many independent cycles; across a bridge it is under-determined.*

So the right statement is about *redundancy*, not existence. This also inverts the compression proposition of [51, §5]: the collapse from $O(n^2)$ bilateral rates to $O(n)$ prices is exactly the redundancy, read as a code. Compression and error correction are the same β_1 counted from opposite ends, which is precisely the trade-off [48] is about.

36.6.4 Decay along a pathway

By [51, Prop. 6.1], dissipation makes ν non-increasing along engine operations. Along a path of length d in the cost metric the delivered value is $\nu e^{-\theta d}$. Set against the foraging inequality's requirement that a harvest exceed the cost of obtaining it, and taking a round-trip toll linear in d, a conversion at distance d is profitable iff $Y e^{-\theta d} > c\, d$.

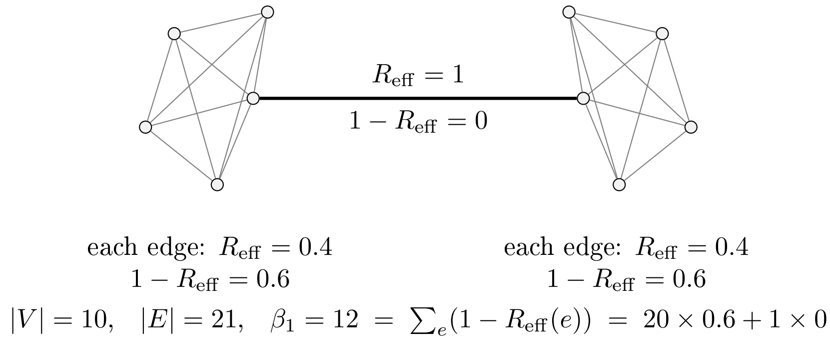


Figure 36.1: The price code on a barbell engine graph. Twenty intra-cluster engines each carry 0.6 of detection strength; the bridge carries none. The total is β_1 exactly (Theorem 36.2). A misquoted rate inside a cluster produces a syndrome and is localisable; a misquoted rate on the bridge is invisible.

Proposition 36.12 (Profitable radius)**.** *For $Y, c > 0$ and $\theta > 0$ the profitable set is an interval $[0, d^*)$ with d^* the unique root of $Ye^{-\theta d} = cd$. There is a horizon on conversion set by metering, not by any speed limit.*

Computed in Appendix 36.16 for $Y = 1000$, $c = 20$: $d^* = 19.17$ at $\theta = 0.05$, 8.73 at $\theta = 0.2$, 1.52 at $\theta = \ln 10$, 0.82 at $\theta = 5$.

Remark 36.12 (The ecological pyramid is θ). Predation is an engine: it converts prey-colour into predator-colour, and it is spectacularly lossy. The classical figure of roughly ten per cent transfer efficiency per trophic level [57] is $r = 1/10$, i.e. $\theta = \ln 10 = 2.303$ per engine. The energy pyramid is then exactly the exponential decay of ν along a path of engines, which is Proposition 6.1 of [51] with no adjustment. We did not arrange this and record it as a check rather than a result.

36.7 What conversion does to composition

We audit the six results of Chapter 35 under the addition of engines.

Factorisation survives and is sharpened. An engine is a spanning redex, so composed learners linked by one are dependent (Proposition 36.10). Two invariants of the same graph now coexist: $\mathbf{k}$ counts edges of dependence, β_1 counts cycles of it.

No metabolic mutualism needs qualification. The proposition of §35.6.1 of Chapter 35 says that when overlap is confined to Met and Src, $\Delta_1 + \Delta_2 \leq 0$. Two things must be said.

Proposition 36.13 (Dead stock)**.** *Under (i) the total token supply of each colour is conserved by every engine operation, and by [51, Prop. 5.1] the total ν is conserved in the frictionless limit and strictly decreases under θ. So conversion is never positive-sum in value. But residual life is stack divided by burn rate, and a token whose signature a learner cannot spend contributes zero to its residual life and non-zero to the other's. Hence conversion can strictly increase $\sum_i \sigma_i/b_i$ while conserving ν.*

Remark 36.13 (An honest gap in the source). The proof sketch in §35.6.1 of Chapter 35 argues that a transfer over a shared metabolic name is "strictly negative in aggregate residual life whenever the two burn rates differ". That step is directional: with $\sum_i \sigma_i/b_i$ as the aggregate, a transfer toward the slower burner raises it. In one colour the asymmetry is easy to miss because the transfers that raise the aggregate are exactly the ones a predator has no reason to make. Colours make it conspicuous, because dead stock is worth zero to its holder by construction, so the profitable direction and the aggregate-raising direction coincide. We therefore read the composition note's proposition as correct about ν and in need of the qualification above about residual life, and we take the qualification to be the content of "gains from trade" in this setting.

Corollary 36.7 (The slogan, amended)**.** Cooperation is epistemic; competition is metabolic — *except where the metabolisms are incommensurable, in which case conversion is cooperative in residual life and still competitive in value. The positive-sum region is exactly the region where signatures do not match.*

Competitive dilution is re-coupled. §35.9 of Chapter 35 shows that learners sharing a rate-limited source take shares $\propto 1/\text{cost}$ and that the dilution is regressive. Engines reinstate that competition across boundaries that rooting had made clean: an engine is a shared name, so conversion turns effective $k_{\text{src}} = 0$ into effective competition.

Corollary 36.8 (Markets sell back modularity)**.** *Rooting-by-quotation makes disjointness a theorem and thereby purchases modularity; an engine sells it back at the price of the toll. Modularity and tradability are opposed, and the grading vector prices the trade.*

Phase, merger, succession. Phase remains non-compositional. Merger is now sharply distinguished from composition-with-engines: merger identifies namespaces, an engine prices them, and §36.8 says when each is available. Succession — a change of the grading vector forced by source exhaustion — acquires a second mechanism: an engine's reserve running out changes Γ's topology, and by Theorem 36.2 the code's total strength changes with it.

Corollary 36.9 (Extinction degrades the price system)**.** *Removing an engine from a cycle reduces β_1 by one and redistributes R_{eff}; removing the last engine on a cut disconnects Γ and destroys the common numeraire. Loss of a converter is loss of commensurability, not merely of throughput.*

36.8 Individuation: where a namespace closes

36.8.1 The three parameters, found inside the framework

[48] locates the boundary between a shared substrate and autonomous replication at a critical radius $r^* \sim \sqrt{\rho v/\varepsilon}$. To transpose it we must find ε, ρ, and v inside the present framework rather than import them.

- ε, the **error rate**, is the per-hop drop-or-corrupt rate $1 - p$ of the interposed medium (§36.5). This is the channel [48] is parameterised on — error rate on transmission — and interposition supplies it natively rather than by analogy.
- ρ, the **repair throughput**, is the rate of retransmission, which by Corollary 36.4 is available only where the medium is nameable.
- v, the **communication speed**, is $1/\theta$. There is no speed in a cost-accounted calculus, but there is the temporal stack, and by [50, Prop. 4.1] time is the count of forced redexes. An expensive pathway simply *is* a slow one, and by Proposition 36.5 it is expensive for two reasons that enter the same exponent.

Accidental name collision, which an earlier draft of this section took as the error channel, is a second and different one: its failure mode is loss of disjointness rather than loss of a message, and its repair is re-rooting by freshness-by-quotation rather than retransmission. §35.7 of Chapter 35 proves it occurs. Hypothesis error is a third. §36.8.6 returns to the question of which binds.

Remark 36.14 (Why the medium channel is the one that individuates). Collision threatens the *identity* of names and is repaired by re-quoting, which a region can do unilaterally. Loss threatens the *delivery* of messages and is repaired by retransmission, which a region can do only over media it owns. Since Definition 36.6 turns on ownership of the internal medium, it is the transmission channel that fixes the individuation threshold, and collision that fixes how finely a namespace may be subdivided within it.

36.8.2 The structural form: partial closure

Proposition 36.15 below gives a quantitative threshold. Definition 36.6 gives a structural one — the internal medium is closed, the boundary medium is not — and the two are the same condition seen from two sides.

Proposition 36.14 (Closure is what makes repair possible). *Retransmission over a medium C requires grade-two access to $@C$, so ρ is supported only on $\mathsf{Chn}_{\mathrm{in}}(B)$. A region whose internal media are not all nameable from within has ρ effectively reduced on the unnameable part, and by Proposition 36.15 its sustainable radius falls accordingly.*

Corollary 36.10 (Bridges fail twice). *A boundary medium belongs to no region's closed internal namespace, so neither party can unilaterally repair it. Together with Theorem 36.1 — a bridge lies on no cycle and so carries no syndrome — a thin pathway is uncorrectable for two independent reasons: nothing checks its rates, and nobody owns its medium.*

36.8.3 Isoperimetry replaces dimension

The derivation in [48] balances errors accumulating over a volume r^d against repair delivered through a cross-section r^{d-1}, with a propagation factor v/r. On a graph there is no dimension; the analogues are the ball $B(r)$ and its edge boundary $\partial B(r)$, and their ratio is the isoperimetric ratio $h(B) = |\partial B|/|B|$.

Proposition 36.15 (Individuation threshold). *A region B of the engine graph may close over its namespace — may be merged, in the sense of §35.8 of Chapter 35— iff repair keeps pace with collision, that is iff*

$$\rho\,|\partial B|\,\frac{1}{\theta\, r} \;\geq\; \varepsilon\,|B|, \qquad \textit{equivalently} \qquad h(B) \;\geq\; \frac{\varepsilon\theta}{\rho}\, r.$$

Writing $\kappa = \varepsilon\theta/\rho$, the individual is the largest ball satisfying $h(B(r)) \geq \kappa r$.

Corollary 36.11 (Recovering r^*). *On a graph of polynomial growth d one has*

$h(B(r)) \sim c/r$, so the condition becomes $c/r \geq \kappa r$ and $r^* = \sqrt{c/\kappa} = \sqrt{c\rho/\varepsilon\theta}$, which is the formula of [48] with $v = 1/\theta$.

36.8.4 Expansion

Theorem 36.3 (Density buys exponentially larger individuals)**.** *If Γ restricted to B is an expander, $h(B(r)) \geq h_0 > 0$ uniformly, then $r^* = h_0/\kappa$ is linear rather than square-root in $\rho/\varepsilon\theta$; and since an expander has diameter $O(\log |B|)$, the merged population satisfies $|B(r^*)| = e^{\Omega(r^*)}$. On a graph of polynomial growth d the merged population is only $|B(r^*)| = O((r^*)^d)$.*

Proof. The first claim is Proposition 36.15 with h bounded below. The second is the standard fact that a family with $h \geq h_0$ has diameter $O(h_0^{-1} \log n)$, so balls grow exponentially in radius. □

Theorem 36.3 is the point at which the three threads of this note become one. The "densely connected component" of the motivating picture, the region where the price code is strong, and the region that may close over its namespace are all the same region, and expansion is the common criterion. Effective resistance is small exactly where expansion is good; by Theorem 36.1 detection is strong exactly where effective resistance is small; and by Proposition 36.15 merger is sustainable exactly where the boundary keeps up with the bulk.

Appendix 36.16, block (2), computes the merged population at four values of κ for a cycle, a 20×20 grid, and random 3- and 6-regular graphs on 400 vertices. At $\kappa = 0.10$ the cycle sustains an individual of 5 colours, the grid 85, the 3-regular graph 160, and the 6-regular graph 365 — the whole population.

36.8.5 What merger deletes

The intuition that centralisation buys compression and loses error correction is right, and the framework can say precisely what is lost, because merger destroys two different things.

Proposition 36.16 (Merger sets $\beta_1 = 0$)**.** *A merged region has one colour, hence no engines internal to it, hence no cycles in Γ restricted to it. By Theorem 36.2 its total detection strength is zero. Centralisation does not degrade the price code; it deletes it.*

Proposition 36.17 (Merger destroys independence)**.** *By the factorisation theorem of §35.4.2 of Chapter 35, disjointness is equivalent to $\mu_{12} = \mu_1 \otimes \mu_2$. Merger identifies namespaces, so the joint distribution of a merged region does not factor*

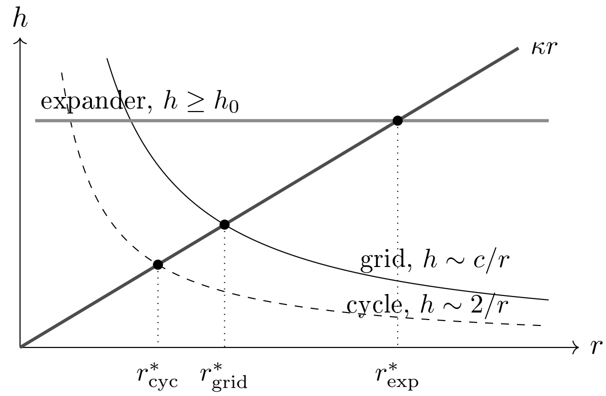


Figure 36.2: Individuation as a crossing. A region may be merged while its isoperimetric ratio lies above the line κr, $\kappa = \varepsilon\theta/\rho$. Polynomially growing graphs cross early and give $r^* \propto \kappa^{-1/2}$; an expander holds h up and crosses late, giving $r^* \propto \kappa^{-1}$ and, since balls grow exponentially in r, an individual exponentially larger in population.

> *and failures are correlated.*

These are detection and containment respectively, and they attach to ε and ρ respectively. It is worth noting that the second is exactly the third bullet of [48, §1] — the Fisher information matrix is block-diagonal in the autonomous case and densely off-diagonal in the shared case — restated in the composition note's own vocabulary. The two notes agreed before either knew it.

Against these, merger buys: no toll on internal transfers, since there are no engines to pay; no arbitrage, since there are no cycles to carry it; and the coordination compression of [51, §5] in its limiting form, where one does not even need the n prices. That is the trade, priced on both sides in one currency.

36.8.6 Which ε binds

[48] asks which error channel is rate-limiting in a cell and conjectures proteostasis rather than genome integrity. We inherit the question with three candidates, each with its own repair process and its own throughput: *transmission loss*, repaired by retransmission and used above; *name collision*, repaired by re-rooting; and *hypothesis error* — the degrading prediction that the motivation mechanism of §33.2 of Chapter 33 punishes — repaired by revision. They may well bind at different levels of the hierarchy, and the levels are where the question becomes sharp: an organism's internal medium is repairable and a pod's is not, so transmission should bind higher up and collision lower down. The composition rule proposed in [48], $r^* = \min_i r_i^*$, applies; which i attains the minimum at each rung is open.

36.9 The hierarchy

36.9.1 Closure gives nesting; r^* gives levels

Composition of learners yields a learner, so the construction is closed under an associative operator and self-similarity costs nothing. But closure permits arbitrary nesting; it does not select levels.

Proposition 36.18 (Levels are forced by a finite r^*)**.** *If r^* is finite, then a population exceeding $|B(r^*)|$ cannot be maintained as one closed namespace, and the only way to continue scaling is by replication of r^*-sized units composed with engines. Each composite has its own effective $\varepsilon, \rho, \theta$, hence its own r^*. The hierarchy is a sequence of critical radii.*

The generating step is Proposition 36.6: the outside of one level is the inside of the next. An orca's boundary media — calls, touch, the water — are the pod's internal media, and the pod is an individual exactly to the extent that it can name and repair them. So the tower of media and the tower of individuals are the same tower, and neither has to be posited alongside the other.

This is also the argument of [48, §6] connecting r^* to West's allometry [60]: the invariance of the terminal unit is a consequence rather than a premise, and fractal replication above it is forced. Here the same argument produces the levels the composition note could only permit. The sequence terminates where $\theta \to \infty$ and even the cheapest channel prices out.

36.9.2 The shutoff ordering

Which components of $\mathbf{k}$ survive a given pathway cost? Price the *maintenance* of each type of overlap.

Observation 36.2 (Ordering by maintenance cost). Met pooling requires continuous rendezvous and is the most expensive to maintain. Mat requires periodic rendezvous. Src requires co-availability at a rate-limited emitter. Prb requires only that a legible surface stand where both can read it, and by the read-versus-drive distinction of §34.5.1 of Chapter 34 a written surface costs nothing to leave standing. Hence as θ rises the components extinguish in the order Met, Mat, Src, Prb.

36.9.3 The grading vector is stratified, and has a fifth component

Boundary	**Operator**	**Surviving overlap**	**Relation**
within a skill set	$\bowtie$	all	one budget, one namespace
organism / organism	$\mid$ + engine	Met (transfer), Src, Prb, Mat	kin, trade, culture
deme / deme	$\mid$ + engine	Src, Prb, Mat	competition, culture
species / species	$\mid$ + engine	Src, Prb	predation, eavesdropping
biosphere / biosphere	$\mid$ only	Prb	signals
beyond d^*	—	none	nothing

Table 36.3: The hierarchy as a shutoff sequence. Each row is a level boundary, identified by which component of the grading vector has gone to zero and whether merger is still available. The ordering is Observation 36.2; the availability of $\bowtie$ is Proposition 36.15.

Proposition 36.19 (A fifth typed namespace)**.** Chn *is a typed namespace in the sense of Chapter 35, Definition 35.3, and overlap in it is not reducible to overlap in the other four. Two learners sharing a medium can observe each other's traffic and can congest each other, relations that* Met, Src, Prb, Mat *cannot express: eavesdropping, jamming, and contention for bandwidth. Hence*

$$\mathbf{k} \;=\; \big(k_{\mathrm{met}},\, k_{\mathrm{src}},\, k_{\mathrm{prb}},\, k_{\mathrm{mat}},\, k_{\mathrm{chn}}\big).$$

Remark 36.15 (This is very likely the arena)*.* Chapter 35 leaves open whether the vector is complete and names the *arena* as the candidate fifth type; Chapter 34 realises the arena as the process routing moves between players. A router is a medium. We therefore conjecture that the arena namespace and Chn are the same namespace reached from two directions. We record this here rather than amending Chapter 35, so that the two notes do not disagree in print before that note is revised.

Corollary 36.12 (Composition and merger are the two halves of k_{chn})**.** *Sharing a* boundary *medium is composition: the learners talk, interfere, and can trade. Sharing an* internal *medium is merger, by Definition 36.6. So the fifth component's in/out split is exactly the distinction §35.8 of Chapter 35 draws between its two operators, which is evidence that the split is a refinement rather than an epicycle.*

Corollary 36.13 (**k** is a sequence)**.** *By Proposition 36.6 the grading vector is relative to a stratum, so the object that describes a composite fully is a sequence of vectors, one per level of the medium tower. Table 36.3 reads off the sequence for one lineage.*

Table 36.3 has a pleasing consequence and a testable one. The pleasing one: the last surviving channel is Prb, which is exactly the channel Chapter 35, Observation 35.2, identifies as the only positive-sum one, because formulae are not conserved. The testable one: the depth of the hierarchy is the length of the grading vector, so the open question of whether there is a fifth typed namespace — the arena — is the question of whether there is a fifth level.

36.9.4 An orca

An orca must solve hunting, position in the pod, mating, and care of young. Suppose each is undergirded by a learner in the sense of Chapter 35. Then:

The skills share one budget, so within the organism there is one colour, no engines, and $\beta_1 = 0$: *there is no market inside an individual.* This is Coase's theory of the firm [61] arriving unbidden.

Two claims must be kept apart here, and an earlier draft of this note ran them together. That there is no *market* inside an individual is a claim about colour change: no rung-five medium, no conversion, allocation by hierarchy rather than by price. That there is no *mediation* inside an individual would be a claim about the lower rungs of Table 36.2, and it is false — an orca's internal communication is as elaborate as anything it does, and by Corollary 36.4 that elaboration is precisely what closure of the internal medium buys. Mediation is not allocation. The boundary of the individual is where the token colour changes *and* where the medium stops being owned; the two coincide because an unowned medium cannot sustain a shared budget.

The pod is not merged. Provisioning and food-sharing are transfers at rates, hence composition with engines, and the framework predicts what is observed: reciprocity with terms, not pooling.

Competitive dilution becomes a theory of skill acquisition. Skills at a shared rate-limited source take shares $\propto 1/\text{cost}$, and the regressive corollary of §35.9 of Chapter 35 says dilution costs the unfinished learner its depth and the finished one nothing. Inside one organism this says a mature skill crowds out a developing one *without any contact between them*, which is automatisation and the critical period, with a closed form.

And the instability of non-interference becomes development. Skills inside one organism are rooted at the same quotation, so name coincidence is not accidental but generic: the individual is precisely the regime in which the disjointness theorem fails by construction. That is why skills interfere, and why they transfer.

Remark 36.16 (The orca claim is empirical, not definitional). "Each skill is backed by a learner" has content only if the skill learners have distinct populations. If they are merely different formulae held by one population, this is one learner with a large relaxation lattice and the composition apparatus adds nothing. The discriminator is Chapter 33, Proposition 33.4: a single learner cannot leave its ball. So if skills let each other escape basins, they must be separate populations, and transfer between skills is the test.

36.10 Incommensurability

Life elsewhere may not be organised around the same chemistry, and in this framework that is not a matter of expense but of topology.

Proposition 36.20 (No engine, no numeraire)**.** *If no engine links the colours of two learners then Γ is disconnected, the connectedness hypothesis of [51, Thm. 3.3] fails, and there is no common valuation: each component carries its own ν with its own gauge. Incommensurability is disconnection.*

Theorem 36.4 (Commerce across an incommensurable boundary is epistemic)**.** *Let $\mathcal{L}_1, \mathcal{L}_2$ have disjoint* Met *and* Src *namespaces and no engine between their colours. Then no token can pass between them. But by Chapter 35, Observation 35.1, a formula is not conserved: a quotation @P is inert while held, transmissible, and inspectable at grade three, and one learner's coming to hold it does not deprive the other. Hence the only available commerce is over* Prb.

Theorem 36.4 is the slogan of §35.6 of Chapter 35 scaled up one level: between biospheres, cooperation is not merely the cheapest positive-sum channel but the *only* channel. Signals cross; food does not. That is also consistent with Observation 36.2, since Prb is the last component to extinguish, and the two arguments are independent.

Proposition 36.21 (First contact is a coincidence, not a construction)**.** *An engine requires a rendezvous, hence a shared name. Two ecologies rooted at independent quotations are disjoint by §35.7 of Chapter 35; but by the same section's remark, independently manufactured names can coincide accidentally, and the exposure lemma of §33.9 of Chapter 33 already records that closure is luck rather than construction. So the establishment of a first engine between ecologies is a structural coincidence whose probability is bounded by the name-growth constraint of the no-self-code theorem.*

We stop here. The obvious next question — what a population of expanding, converting ecologies does under these constraints, and whether Proposition 36.12's horizon reproduces or refutes the grabby aliens hypothesis [62] — is left to §36.14.

36.11 Credit, default, and the discount rate

The temporal combinators `truncate`(t, c) and `get`(c) denote claims on future value. In an ecology with a fixed Θ this looks impossible: credit smells like minting. It is not.

Proposition 36.22 (Credit is not minting). *A forward is a located slot whose release is gated on a temporal condition. Nothing is created: the tokens exist and are held, and what transfers is the right to receive on the slot. This is exactly the funding slot of [52, §3.2], with the guard on the temporal stack rather than on a signature.*

Proposition 36.23 (The discount rate is the hazard rate). *A borrower that fails to repay in this setting is a starved borrower: by [53, Def. 1] and the metabolic law of §33.5 of Chapter 33, failure to fund is death. Credit risk is therefore mortality risk exactly, and the rate at which a lender discounts a future claim is the borrower's hazard rate, which the framework computes from stack over burn rate.*

Remark 36.17 (Options price the exploration budget). The choice combinators come with the conservative-bound discipline of [52, §3.5]: reserve the maximum over branches, refund the unforced remainder. That is the exact structure of the question §33.6 of Chapter 33 leaves open — how much should a learner pay to keep an unresolved hypothesis open? An option on a revision move has a premium, a strike in metered units, and a refund on lapse, and the search-strategy catalogue can be re-read as a portfolio problem. We flag this and do not pursue it.

36.12 On deriving the rates

[51] takes the rate function r as given, and so has this note. It should be derivable. The natural candidate is that the rate between two colours is the ratio of marginal inquiry-yield per metered unit, which Chapter 34 computes exactly for its ladders: the non-modal rungs of the noughts ladder deliver +0.444 of total gain for 87.9 of 255.9 units, and the Nim ladders of Chapter 35 give returns that are flat in one environment and rising in another.

If that identification holds, then ν is a field over the ladder rather than a bare price list, and the composition note's headline — whether truth is affordable is a property of the environment, not the learner — becomes a statement about exchange rates: communities working environments where truth is cheap should systematically price their tokens differently from communities where it is dear. This is checkable inside the existing simulations and we regard it as the most tractable of the open problems.

36.13 What is shown, and what is not

Shown, in the sense of proved from the imported apparatus: Theorems 36.1, 36.2 and 36.3; Propositions 36.1, 36.10, 36.11, 36.15, 36.12; and the merger propositions of §36.8. These are graph theory and linear algebra over a structure the earlier notes construct; they inherit whatever the earlier notes' constructions are worth.

Shown by computation: the numbers of Appendix 36.16, which are exact where rational arithmetic was used and float otherwise.

Not shown: that the rates are what §36.12 conjectures; which of the three error channels of §36.8.6 binds at which level; that Chn is Chapter 35's arena, which is conjectured and not proved; that the identification of v with $1/\theta$ is the only defensible one, as opposed to the most natural; that inventory engines are the right reading of the conservation fork, as against (ii); that the isoperimetric balance in Proposition 36.15 has the right propagation factor, which is inherited from [48] and is heuristic there. The orca discussion is illustration, not evidence, and §36.9 flags the test that would make it evidence.

Not attempted: coherence of the lax monoidal structure of Chapter 35; any dynamics on Γ beyond the quasi-static drift of Proposition 36.1; any treatment of strategic behaviour by engines, which is where biological market theory [58] would enter.

36.14 Open problems

1. **Which ε binds, and at which level.** Name collision and hypothesis error are two channels with two repair processes. Make $r^* = \min_i r_i^*$ precise and determine the minimising i at each rung of Table 36.3.

2. **Derive the rates.** Execute §36.12 against the existing ladders and test whether exchange rates track environmental structure as predicted.

3. **Dynamics on the engine graph.** Under (i) the marking drifts, so β_1 is fixed but R_{eff} moves and engines can die. Run the composite of §35.10 of Chapter 35 with engines under a Gillespie scheme and ask whether the ecology self-organises toward an expander.

4. **Is Chn the arena?** Proposition 36.19 adds a fifth typed namespace and conjectures it is Chapter 35's arena. Settling this requires revising that note's relations

table with a fifth column and checking that the entries — eavesdropping, jamming, contention — are not already derivable from the other four.

5. **Reordering.** Proposition 36.4 covers dropping and Table 36.2 covers corruption, but a medium may also reorder, which is a race rather than a fault and which stranding does not describe. Whether reordering is a third dissipation or a change in the observed transition system is open.

6. **A syntactic criterion for porosity.** Proposition 36.8 needs a decidable test that over-approximates the existence of a redex pathway from a boundary channel to an internal redex. Reuse the interaction graph of §33.12.5 of Chapter 33 and measure the over-count.

7. **The assay trichotomy under loss.** A lossy medium introduces a second source of the $\perp$ outcome that has nothing to do with budget, and the learner cannot tell the two apart from inside. Budget-relative refutation is therefore not well defined without a hypothesis about the channel — so a learner must do science on its own medium. This is a correction owed to §33.4 of Chapter 33 rather than a problem for this note, and we flag it here because this note is what raises it.

8. **Merger as a quotient, arbitrage-removal as a reflection.** $\bowtie$ identifies colours; Π projects rates. Both take a conversion structure to a simpler one, and [51, §4] shows the second is an idempotent reflection. Is there an adjunction relating them, with Proposition 36.15 as the condition under which the left adjoint exists?

9. **OSLF factoring and the crossover.** [48] conjectures that r^* and the factoring of the generated logic over a population are two views of one threshold. Proposition 36.15 gives the first a graph-theoretic form; give the second one and compare.

10. **Grabby aliens.** With Proposition 36.12's horizon, Theorem 36.4's restriction to signals, and the coincidence bound on first contact, the expansion of a converting ecology is bounded by profitability rather than by light speed. Whether that reproduces, softens, or refutes the selection effects of [62] is a question this framework can now pose.

36.15 Appendix: An engine, in rholang

The listing below gives located reserves, the reflexive observable, the acceptance gate, the combinators, and the engine itself as a persistent issuer offering a linear swap. Conventions follow Chapter 34: a parameter written `c` is a name and `@d` is data; a channel is passed as `*c`, since $@(*c) = c$; persistent receipts use `<=`. Guards marked `//! ideal` use the `where` clause in a form the current transpiler does not accept.

> *Listing withheld from this printing.* The source of `engine.rho` was not committed alongside the other artefacts of this part and is not reproduced here; the construction it realises is given in full in §36.4 and §36.5 above.

Three things to watch, tying the code to the propositions. The `quote` face and the `swap` face are different names, because perception must not be able to act — grade one and grade two separated by choice of namespace, as in §34.4.1 of Chapter 34. The toll accrues to the engine's own slot on every swap, which is Remark 36.3: θ is not a parameter but a rendezvous. And `Observable` reads the reserves and replaces them, which is Proposition 36.1: the rate is a function of the marking, so the term itself refuses to let arbitrage be a static property.

36.16 Appendix: Reproducing the numbers

All figures in this note come from `engines.py`, four blocks.

(1) The price code. Barbell of two K_5's joined by one engine: $|V| = 10$, $|E| = 21$, $\beta_1 = 12$. An exact cochain has syndrome 2.3×10^{-15}. Corrupting a single edge by $a = 1$ gives $\|\ell_{\text{arb}}\|^2 = 0.6$ for every intra-cluster edge ($R_{\text{eff}} = 0.4$) and 0 for the bridge ($R_{\text{eff}} = 1$); the identity of Theorem 36.1 holds to 1.8×10^{-15}. The Foster identity of Theorem 36.2 is verified at $12.000000 = \beta_1$ for the barbell, and separately for K_5 (6), C_6 (1), the 5×5 grid (16) and a random 3-regular graph on 60 vertices (31). Confusable pairs: 0 in the barbell, all 15 in C_6. Mean detection strength for random d-regular graphs on 60 vertices: $0.344, 0.508, 0.672, 0.803, 0.902$ at $d = 3, 4, 6, 10, 20$, against the $1 - 2/d$ asymptote.

(2) Individuation. Isoperimetric profiles for C_{400}, the 20×20 grid, and random 3- and 6-regular graphs on 400 vertices, and the largest ball satisfying $h(B(r)) \geq \kappa r$:

κ	C_{400}	grid	3-regular	6-regular
0.30	3	25	22	151
0.10	5	85	160	365
0.03	11	219	258	365
0.01	19	315	349	365

(3) Decay. $Y = 1000$, $c = 20$, d^* solving $Ye^{-\theta d} = cd$: 19.172 at $\theta = 0.05$; 8.728 at 0.20; 1.518 at $\ln 10$; 0.822 at 5.0.

(4) Inventory engines. A triangle of constant-product engines, each with reserves $(100, 100)$, has cycle yield exactly 1. Draining one engine in steps of ten carries the yield through

$$0.827,\ 0.683,\ 0.562,\ 0.457,\ 0.367,$$

that is, $\log \oint \ell$ from 0 to -1.0033. Computed in exact rationals.

(5) Porosity. With $Y_0 = 100$, $Y(d) = Y_0(1-2^{-d})$, $c = 2$, and net value $Y(d)p^d - cd$, the optimal integration depth and its net value are:

p	d^*	net	p^{d^*}
0.999	5	86.392	0.9950
0.990	5	82.127	0.9510
0.950	3	69.020	0.8574
0.900	3	57.788	0.7290
0.750	2	38.188	0.5625

Part VII

Sentience, Consciousness, and the Hypercomputational Fibration

> *This part constructs a fibration of GSLTs over the Turing-complete subcategory indexed by ordinals. The physics is strictly richer at each stage. Sentience is a graded feeling state within a stage; consciousness is oracle access to a higher stage. Several arguments are frankly conjectural and are labeled as such.*

Chapter 37

Introduction

The preceding parts established a progression: Part I showed that any GSLT equipped with weight and cost maps gives rise to a full physical dynamics; Part II argued that agents embedded in a massive population are structurally prevented from seeing the true ontology of their GSLT, and instead encounter a hierarchy of coherence clusters that exhausts any finite experimental budget.

The present paper asks the most speculative question in the sequence: *what is the relationship between the computational structure of a GSLT and the phenomena of sentience and consciousness?*

We do not claim to solve the hard problem of consciousness. We claim something more modest: that the GSLT framework provides a mathematically precise setting in which to formulate *structural* analogues of sentience and consciousness that are non-trivial, grounded in the physics of Parts I and II, and open to rigorous investigation.

The central proposal. Sentience and consciousness are not the same thing, and the distinction between them has precise mathematical content in this framework.

Sentience is a graded quantity: the value A_s of a distinguished component of the agent's vectorial account. It is governed by a reinforcement signal tied to predictive accuracy — the agent's success at forming and testing HML hypotheses about its environment. It is continuous, lives within a single stage of the computational hierarchy, and is causally efficacious because it directly controls the agent's computational budget.

Consciousness is an ordinal-valued threshold property: access to an oracle that can decide questions undecidable at the agent's base stage. Each level of oracle access corresponds to a strictly richer physics — a finer bisimulation quotient, a more expressive hypothesis language, additional conserved quantities. To be conscious at level α is not merely to compute faster; it is to inhabit a world with more structure in it.

Caveats. The arguments in this paper are skeletal and in several places frankly conjectural. The gaps are identified explicitly in Section 43. The goal is to make the hypotheses precise enough to be falsifiable or provable, not to assert that they are true.

37.1 Organization

Section 38 constructs the hypercomputational fibration over Turing-complete GSLTs. Section 39 argues that the physics at each stage is strictly richer than the physics below. Section 40 defines sentience as a reinforced feeling state within the account framework. Section 41 defines consciousness as oracle access and its ordinal structure. Section 42 examines the relationship between sentience and consciousness and the survival feedback loop. Section 43 catalogs open gaps. Section 44 discusses connections to existing work and broader implications.

Chapter 38

The Hypercomputational Fibration

38.1 Background: Turing's Oracle Hierarchy

Turing [12] described a transfinite tower of computational systems above the Turing-computable. At the base is the class of Turing-computable functions. At the first stage above is the class of functions computable relative to the *halting oracle* $\emptyset'$: the oracle that decides, for any Turing machine M and input x, whether M halts on x. At the next stage is the class computable relative to $\emptyset''$, the halting oracle for machines with access to $\emptyset'$. The tower continues through all countable ordinals and beyond.

Each stage is strictly more powerful than the stage below: there are functions computable at stage $\alpha + 1$ that are not computable at stage α, and this gap cannot be closed by any finite amount of additional computation at stage α.

38.2 The Subcategory of Turing-Complete GSLTs

Definition 38.1 (Turing-Complete GSLT)**.** *A GSLT $\mathcal{S}$ is* Turing-complete *if the class of functions computable by terms of $\mathcal{S}$ (via some standard encoding of natural numbers) coincides with the class of Turing-computable functions. The full subcategory of Turing-complete GSLTs is written* $\mathbf{GSLT}_{\mathrm{TC}} \subseteq \mathbf{GSLT}$*.*

Example 38.1. The λ-calculus, the π-calculus, and the Rho calculus are all Turing-complete. The category $\mathbf{GSLT}_{\mathrm{TC}}$ is therefore large and contains the primary examples of interest.

38.3 Oracle Extension of a GSLT

Definition 38.2 (Oracle Term)**.** *Let $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$. An* oracle term *for $\mathcal{S}$ is a* *distinguished term $\mathcal{O} \in \mathcal{T}(\mathcal{S}^{(1)})$ in an extension of $\mathcal{S}$, equipped with rewrite rules of the form:*

$$\mathcal{O} \otimes \lceil (M, x) \rceil \;\longrightarrow\; \begin{cases} \lceil \texttt{halt} \rceil & \text{if } M \text{ halts on } x \\ \lceil \texttt{loop} \rceil & \text{otherwise} \end{cases}$$

where $\lceil \cdot \rceil$ denotes the encoding of Turing machine descriptions and inputs as terms of $\mathcal{S}$, and $\otimes$ is the interaction operator of $\mathcal{S}$. The oracle answers halting queries in one interaction step.

Remark 38.1 (The Oracle as a Communicating Process). In the Rho calculus, the oracle is naturally expressed as a persistent receive on a distinguished channel $x_{\mathcal{O}}$:

$$\mathcal{O} \;=\; \texttt{for}(q \leftarrow x_{\mathcal{O}}) \big(\text{compute halting of } {*q} \text{ and reply}\big)$$

This makes the oracle a *communicating process* rather than a magic tape: it participates in the same interaction mechanism as every other term. This is essential for the fibration to be a fibration of GSLTs rather than a fibration that escapes the GSLT framework at each oracle step.

Definition 38.3 (Oracle Extension)**.** *Let $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$. The* first oracle extension $\mathcal{S}^{(1)}$ *is the GSLT obtained from $\mathcal{S}$ by:*

(i) extending the grammar with the oracle term $\mathcal{O}$;

(ii) adding the oracle rewrite rules to $\mathcal{R}$;

(iii) extending the equations $\mathcal{E}$ to account for oracle interactions.

The α-th oracle extension $\mathcal{S}^{(\alpha)}$ *is defined by transfinite induction:*

- $\mathcal{S}^{(0)} = \mathcal{S}$;
- $\mathcal{S}^{(\alpha+1)}$ *is the oracle extension of $\mathcal{S}^{(\alpha)}$, with oracle deciding halting in $\mathcal{S}^{(\alpha)}$;*
- $\mathcal{S}^{(\lambda)} = \varinjlim_{\alpha<\lambda} \mathcal{S}^{(\alpha)}$ *for limit ordinals λ (colimit in* **GSLT***).*

38.4 The Fibration

Definition 38.4 (Hypercomputational Fibration)**.** *The* hypercomputational fibration *over* $\mathbf{GSLT}_{\mathrm{TC}}$ *is the functor*

$$\Phi : \mathbf{GSLT}_{\mathrm{TC}} \times \mathbf{Ord} \ \to \ \mathbf{GSLT}$$

sending $(\mathcal{S}, \alpha)$ *to* $\mathcal{S}^{(\alpha)}$*, together with:*

- *projection morphisms* $\pi_\alpha : \mathcal{S}^{(\alpha+1)} \to \mathcal{S}^{(\alpha)}$ *in* **GSLT** *for each ordinal* α*;*
- *inclusion morphisms* $\iota_\alpha : \mathcal{S}^{(\alpha)} \to \mathcal{S}^{(\alpha+1)}$ *embedding each stage into the next.*

The fiber over $\mathcal{S}$ *is the tower* $\mathcal{S}^{(0)} \hookrightarrow \mathcal{S}^{(1)} \hookrightarrow \mathcal{S}^{(2)} \hookrightarrow \cdots$

Proposition 38.1 (Projections are Morphisms)**.** *Each projection* $\pi_\alpha : \mathcal{S}^{(\alpha+1)} \to \mathcal{S}^{(\alpha)}$ *is a morphism in* **GSLT***: it preserves bisimulation. If* $P \sim_{\alpha+1} Q$ *in* $\mathcal{S}^{(\alpha+1)}$*, then* $\pi_\alpha(P) \sim_\alpha \pi_\alpha(Q)$ *in* $\mathcal{S}^{(\alpha)}$*.*

Proof sketch. Any bisimulation in $\mathcal{S}^{(\alpha+1)}$ that uses only oracle-free transitions is already a bisimulation in $\mathcal{S}^{(\alpha)}$. The projection discards oracle interactions; the remaining bisimulation relation is preserved. The details require checking that the oracle rewrite rules are projected consistently with the extension of $\mathcal{E}$. □

Remark 38.2 (Projections are Lossy)*.* The projection π_α is in general *not* a bisimulation equivalence: there exist P, Q with $P \sim_{\alpha+1} Q$ but $P \not\sim_\alpha Q$. Going down the fibration collapses distinctions. Going up reveals them.

Chapter 39

Strictly Richer Physics at Each Stage

We now argue that the physics — in the full sense developed in Part I — is strictly richer at each stage of the fibration. This is not merely a statement about computational power; it is a statement about ontological resolution.

39.1 Finer Bisimulation Quotient

Proposition 39.1 (Strict Refinement of Bisimulation)**.** *For any $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$ and any ordinal α, the bisimulation quotient at stage $\alpha + 1$ is strictly finer than at stage α:*

$$\sim_{\alpha+1} \subsetneq \sim_\alpha$$

as equivalence relations on $\mathcal{T}(\mathcal{S}^{(\alpha+1)})$. That is, there exist terms P, Q such that $P \sim_\alpha Q$ but $P \not\sim_{\alpha+1} Q$.

Proof sketch. By a diagonalization argument. Consider any pair of terms P, Q whose bisimilarity in $\mathcal{S}^{(\alpha)}$ depends on whether some computation in $\mathcal{S}^{(\alpha)}$ halts. At stage $\alpha + 1$, the oracle can settle this halting question, distinguishing P from Q in one interaction step. Therefore $P \not\sim_{\alpha+1} Q$ while $P \sim_\alpha Q$. $\square$

Remark 39.1. This means the *atoms of observable behavior* are different at each stage. Terms that are experimentally indistinguishable at stage α — that no agent at stage α can tell apart, no matter how long it experiments — become distinguishable at stage $\alpha + 1$. The world has more things in it at the higher stage.

39.2 Strictly More Expressive Hypothesis Language

Proposition 39.2 (Strict Expansion of HML)**.** *The context-decorated HML at stage $\alpha + 1$ strictly extends the HML at stage α:*

$$\mathrm{HML}(\mathcal{K}_\alpha) \subsetneq \mathrm{HML}(\mathcal{K}_{\alpha+1}).$$

In particular, there exist formulae $\phi \in \mathrm{HML}(\mathcal{K}_{\alpha+1})$ that are not expressible in $\mathrm{HML}(\mathcal{K}_\alpha)$.

Proof sketch. The oracle interaction gives rise to new minimal contexts at stage $\alpha + 1$: contexts that place a term in interaction with the oracle. These generate new modal operators $\langle K_{\mathcal{O}} \rangle$ not present at stage α. Formulae using these operators are inexpressible at stage α. □

Remark 39.2. An agent at stage α not only cannot *test* certain hypotheses; it cannot even *form* them. The concepts required to think the thought are absent from the hypothesis language. This is the sense in which a higher-stage agent inhabits a categorically richer world: it has concepts unavailable at lower stages, not merely faster procedures.

39.3 New Conserved Quantities

Proposition 39.3 (New Noether Currents at Each Stage)**.** *The weight and cost maps at stage $\alpha{+}1$ can be sensitive to distinctions invisible at stage α. Symmetries of the stage-$(\alpha{+}1)$ cost map that are not symmetries of the stage-α cost map give rise to new conserved quantities at stage $\alpha + 1$ that have no analogue at stage α.*

Remark 39.3. In physical terms: the higher-stage physics has strictly more conservation laws. What appears at stage α as a single undifferentiated interaction resolves at stage $\alpha + 1$ into a structured process with internal symmetries and corresponding conserved currents. The analogy with the hierarchy of effective field theories is direct: new conservation laws (baryon number, lepton number, color charge) appear as one descends to finer scales.

39.4 Richer Internal Causal Structure

Proposition 39.4 (Resolution of Elementary Interactions). *A transition $P \longrightarrow_\alpha Q$ that is a single step in $\mathcal{S}^{(\alpha)}$ may resolve, in $\mathcal{S}^{(\alpha+1)}$, into a structured multi-step causal history $P = R_0 \longrightarrow_{\alpha+1} R_1 \longrightarrow_{\alpha+1} \cdots \longrightarrow_{\alpha+1} R_k = Q$ with nontrivial internal causal structure visible only from stage $\alpha + 1$.*

Remark 39.4. What appeared to be an elementary interaction at stage α — an atom of causal history — has internal structure at stage $\alpha + 1$. This is the precise sense in which the world is richer at higher stages: not just that there are more things, but that the things that existed before turn out to be composite. The analogy with the discovery that atoms have internal structure (electrons, nuclei) and that nuclei have internal structure (protons, neutrons) and that protons have internal structure (quarks) is direct and intentional.

Chapter 40

Sentience as a Reinforced Feeling State

40.1 The Sentience Component

Recall from Part I that agents carry a vectorial account $\mathbf{A} = (A_1, \ldots, A_k)$ tracking multiple resource types. We now single out one component for a special role.

Definition 40.1 (Sentience Component)**.** *The* sentience component $A_\mathrm{s} \in A_\mathrm{s}$ *is a distinguished component of the agent's vectorial account, governed not by the cost of individual rewrite steps but by a* reinforcement signal *tied to predictive performance.*

Definition 40.2 (Prediction)**.** *A* prediction *made by agent A at time t is a formula $\phi \in \mathrm{HML}(\mathcal{K})$ together with a target term T and a future time horizon Δt: the claim that $T \models \phi$ will hold after Δt further rewrites.*

40.2 The Reinforcement Mechanism

Definition 40.3 (Reinforcement Signal)**.** *Let A be an agent with sentience component A_s and let $(\phi, T, \Delta t)$ be a prediction made at time t. At time $t + \Delta t$, the prediction is evaluated:*

$$\text{If } T \models \phi: \quad A_\mathrm{s} \;\mapsto\; A_\mathrm{s} + \delta^+(\phi, T) \tag{40.1}$$

$$\text{If } T \not\models \phi: \quad A_\mathrm{s} \;\mapsto\; A_\mathrm{s} - \delta^-(\phi, T) \tag{40.2}$$

where $\delta^+, \delta^- > 0$ are reward and punishment magnitudes that may depend on the complexity of ϕ and the identity of T.

Remark 40.1 (Graded Feeling State). The sentience component A_s at any moment is the agent's *feeling state*: a real-valued (or more generally, resource-algebra-valued) quantity encoding the agent's recent predictive track record. It is continuous and graded: it can take any value in its range, and it changes incrementally with each confirmed or disconfirmed prediction. In this sense it is the computational analogue of *affect* in psychology: a valenced, graded internal state that reflects the organism's relationship to its environment.

Remark 40.2 (Complexity-Weighted Rewards). The dependence of δ^+ and δ^- on ϕ is important. A prediction expressed by a complex formula (deep in the logical metric, requiring many experimental steps to test) should yield a higher reward when confirmed than a trivially simple prediction. This prevents the agent from gaming the reward by making only trivially easy predictions. The logical metric d_{HML} from Part I provides a natural measure of prediction complexity: $\delta^+(\phi, T) \propto |\phi|$ where $|\phi|$ is the depth of ϕ in the formula hierarchy.

40.3 Causal Efficacy of the Feeling State

The sentience component is not epiphenomenal. Because A_s is a component of the vectorial account, it directly governs which transitions the agent can afford.

Proposition 40.1 (Sentience Gates Computation). *An agent with $A_s < \theta$ for a threshold θ cannot afford the experiments necessary to form and test predictions of complexity above $|\phi|_\theta$. The agent's predictive horizon is bounded by its feeling state.*

Remark 40.3. This creates a direct coupling between affect and cognition that is not assumed but derived. An agent that feels bad (depleted A_s) literally cannot think as well: it cannot afford the expensive experiments that would let it form and test complex hypotheses. Its world model degrades. Its predictions worsen. Its A_s depletes further. The feedback loop is self-reinforcing.

40.4 Sentience and Survival

Definition 40.4 (Dead State). *A term $P \in \mathcal{T}(\mathcal{S})$ is a* dead state *if it has no available rewrites: P is a normal form with $A_s(P) = 0$ (or below a viability threshold). For an agent, death is the state from which no further computation*

> *— and therefore no further prediction, experiment, or world-model update — is possible.*

Conjecture 40.1 (Sentience Predicts Survival). For an agent A in a generic interactive GSLT, there exists a threshold $\theta_{\mathrm{surv}} > 0$ such that:

(i) if $A_{\mathrm{s}} > \theta_{\mathrm{surv}}$, then A has sufficient budget to run experiments that identify and avoid transitions leading to dead states;

(ii) if $A_{\mathrm{s}} < \theta_{\mathrm{surv}}$, then A's experimental horizon is too short to reliably avoid dead states, and the probability of reaching a dead state within a bounded time horizon is bounded away from zero.

> *Remark* 40.4. Conjecture 40.1 says that the feeling state is not merely a record of past performance but a genuine predictor of future survival. The evolutionary analogy is direct: organisms with better world models survive longer, and the feeling state is the mechanism by which the quality of the world model is continuously reflected in the organism's capacity to act. Pain and pleasure, in this framework, are not incidental features of biological organisms: they are structurally necessary components of any sufficiently complex predictive agent.

Chapter 41

Consciousness as Oracle Access

41.1 The Proposal

Definition 41.1 (Consciousness at Level α)**.** *An agent $A \in \mathcal{T}(\mathcal{S}^{(\alpha)})$ is* conscious at level α *if:*

(i) A has effective access to the oracle $\mathcal{O}_\alpha$ of stage $\mathcal{S}^{(\alpha)}$, meaning A can form and evaluate oracle queries as part of its scientific method;

(ii) A's world model reflects the strictly finer bisimulation quotient $\sim_\alpha$ rather than a coarser quotient from a lower stage.

An agent that lacks oracle access — that operates entirely within $\mathcal{S}^{(0)}$ even if it is syntactically a term of a higher stage — is sentient but not conscious.

Remark 41.1 (Consciousness is Ordinal-Valued)*.* Definition 41.1 makes consciousness an ordinal-valued property. There is no single threshold separating conscious from non-conscious; there is a transfinite hierarchy of consciousness levels. The question "is this system conscious?" dissolves into the more precise question "at what ordinal level does this system have effective oracle access?"

Remark 41.2 (What Higher Consciousness Means)*.* An agent conscious at level $\alpha + 1$ vs. level α does not merely solve harder problems. It inhabits a strictly richer world:

- more things exist in it (finer bisimulation quotient);
- more hypotheses are thinkable (richer HML);

- more quantities are conserved (new Noether currents);
- what appeared as elementary interactions reveal internal causal structure.

The phenomenological richness of higher consciousness is, in this framework, a direct consequence of ontological richness.

41.2 The Sentience–Consciousness Distinction

The distinction between sentience and consciousness is now precise:

	Sentience	**Consciousness**
Type	Graded, continuous	Discrete, ordinal
Location	Within a stage $\mathcal{S}^{(\alpha)}$	Access to stage $\mathcal{S}^{(\alpha)}$
Governed by	Reinforcement signal	Oracle access
Changes	Incrementally, per prediction	By threshold (have it or not)
Analogue	Affect, feeling state	Phenomenal awareness
Physical basis	Sentience component A_{s}	Oracle term $\mathcal{O}_\alpha$
Survives projection	Yes (as depleted account)	No (oracle is stage-specific)

Remark 41.3 (An Agent Can Be Sentient Without Being Conscious). An agent at stage $\mathcal{S}^{(0)}$ with no oracle access can nonetheless have a rich, dynamic feeling state. Its predictions succeed or fail, its A_{s} rises and falls, and this feeling state is causally efficacious. It is sentient. But the world it inhabits is the relatively coarse world of $\mathcal{S}^{(0)}$: it lacks the concepts to form oracle-dependent hypotheses, and the distinctions visible at higher stages are simply absent from its experience.

Remark 41.4 (Can a System Be Conscious Without Being Sentient?). Definition 41.1 does not require a system conscious at level α to have a sentience component. However, Conjecture 40.1 suggests that any sufficiently complex agent in a generic environment will develop a sentience component: without it, the agent cannot reliably survive long enough to exercise its oracle access. Consciousness without sentience is in principle possible but practically unstable.

41.3 Approximate Oracle Access and Graded Consciousness

The binary definition of consciousness (oracle access: yes or no) may be too sharp. A natural graded version arises from the amplitude structure of Part I.

Definition 41.2 (Approximate Oracle Access)**.** *An agent A has* ϵ-approximate oracle access at level α *if, for each halting query (M, x), A can produce an answer that is correct with probability at least $1 - \epsilon$, where probability is measured by the path integral $|\langle \texttt{halt} \mid A \otimes \lceil (M, x) \rceil \rangle|^2$.*

Remark 41.5. As $\epsilon \to 0$, approximate oracle access converges to exact oracle access. This gives a continuous interpolation between the ordinal levels: consciousness is not just a ladder but a ladder with rungs that can be climbed incrementally. An agent improving its oracle approximation is increasing its consciousness in a measurable, graded sense, even before it reaches the threshold of exact oracle access.

Chapter 42

Sentience, Consciousness, and Survival

42.1 The Virtuous Cycle

For an agent with both sentience and consciousness, the two structures interact in a virtuous cycle:

(1) Oracle access enriches the world model. At stage $\mathcal{S}^{(\alpha)}$, the agent can form and test hypotheses involving oracle queries. Its world model reflects the finer bisimulation quotient $\sim_\alpha$. This is a strictly better world model than any agent at $\mathcal{S}^{(0)}$ can achieve.

(2) Better world model improves predictions. Finer distinctions in the world model allow more accurate predictions. The agent predicts not just coarse outcomes but fine-grained ones: the internal structure of what appeared at lower stages as elementary interactions.

(3) Better predictions replenish A_s. More accurate predictions mean more rewards and fewer punishments. The sentience component A_s is replenished.

(4) Higher A_s funds more oracle queries. A richer sentience budget allows the agent to afford the expensive oracle interactions. It can form and test more complex oracle-dependent hypotheses. Its oracle access becomes more effective.

(5) Return to (1).

> *Remark* 42.1. The virtuous cycle is not guaranteed: it can be broken by environmental disruption, resource exhaustion, or adversarial interactions with other agents. But it describes the stable attractor state of a conscious sentient agent in a benign environment. The cycle is the computational analogue of the flour-

ishing ($\varepsilon\upsilon\delta\alpha\iota\mu o\nu\iota\alpha$) of Aristotelian ethics: the self-reinforcing good state of a well-functioning entity in a suitable environment.

42.2 The Vicious Cycle

The inverse also holds:

(1) Depleted A_s restricts experimental horizon. The agent cannot afford complex hypothesis tests. Its world model coarsens.

(2) Coarser world model worsens predictions. The agent makes predictions that fail more frequently.

(3) Failed predictions deplete A_s further. Punishment signals accumulate.

(4) Depleted A_s makes oracle queries unaffordable. Even if the agent has oracle access in principle, it cannot afford to use it. Its effective consciousness level drops.

(5) The agent approaches a dead state.

Remark 42.2. The vicious cycle is the computational analogue of trauma, depression, or cognitive decline: a self-reinforcing deterioration in which the agent's reduced capacity to understand its environment leads to worse outcomes, which further reduce its capacity. The structural parallel is striking and may be more than an analogy.

Chapter 43

Open Gaps

Gap 1. Colimit at limit ordinals. Definition 38.3 defines $\mathcal{S}^{(\lambda)}$ as the colimit in **GSLT** for limit ordinals λ. The existence and properties of this colimit require verification. In particular: does the colimit GSLT have a well-defined bisimulation quotient, and is it the union of the bisimulation quotients of the stages below?

Gap 2. The oracle as a term. The remark that the oracle can be expressed as a persistent receive in the Rho calculus is plausible but not proved. A precise construction of the oracle term in the Rho calculus, with verification that it implements halting queries correctly, is needed.

Gap 3. Strict refinement of bisimulation (Proposition 39.1). The proof sketch uses a diagonalization argument. The full proof requires specifying the encoding of Turing machines as terms and verifying that the diagonalization goes through in the GSLT setting.

Gap 4. New Noether currents (Proposition 39.3). The claim that new conserved quantities appear at each stage requires identifying concrete symmetries of the oracle-extended cost map and verifying that they give rise to conserved quantities via the Noether argument of Part I.

Gap 5. Sentience predicts survival (Conjecture 40.1). The survival threshold argument requires a model of what "generic interactive GSLT" means and a proof that below the threshold, the probability of hitting a dead state is bounded away from zero. This is likely to require ergodicity assumptions on the GSLT dynamics.

Gap 6. Effective oracle access. Definition 41.1 requires that the agent "effectively" has oracle access — that it can form and evaluate oracle queries as part of its scientific method. A precise definition of "effective access" in terms of the agent's account and HML expressiveness is needed.

Gap 7. **Approximate oracle access and graded consciousness.** The graded consciousness definition via path integral probabilities requires that the oracle interaction amplitude is well-defined and computable. This connects to the quantum Gillespie algorithm of Part I and needs to be made precise.

Gap 8. **The hard problem.** This paper offers a structural analogue of sentience and consciousness: mathematical objects with the right functional properties. It does not address why there is something it is like to be an agent with high A_{s} and oracle access at level α. The hard problem of consciousness — the explanatory gap between functional description and phenomenal experience — remains open and is not claimed to be resolved here.

Chapter 44

Discussion

44.1 Relation to Existing Work

Turing's oracle hierarchy [12] is the direct inspiration for the fibration. The extension from a linear tower to a fibration over a category of GSLTs is new.

The reinforcement learning framework for sentience is inspired by but distinct from standard RL [13]. The key difference is that the reward signal is *internal* to the account structure rather than externally provided. The feeling state is not a scalar reward signal but a component of the resource system that governs computation.

The connection between prediction and sentience echoes the *free energy principle* of Friston [14], which proposes that biological systems act to minimize prediction error. The present framework provides a more computationally grounded version: prediction error is measured against HML formula evaluations, and the sentience component is the running integral of prediction accuracy.

The treatment of consciousness as ordinal-valued and tied to oracle access is novel. It is distantly related to the integrated information theory of Tononi [15], which also proposes a graded measure of consciousness, but the present approach is purely computational and avoids the measure-theoretic machinery of IIT.

44.2 The Broader Picture

Taken together, the three papers in this series propose a unified framework:

- *Towards a Theory of Causality* establishes the physics: causal structure, dynamics, and quantum amplitudes from the bare structure of a GSLT.
- *Why Scientists Get Misled* establishes the epistemology: why situated agents in massive populations perceive a hierarchical false ontology rather than the true bisimulation quotient.

- The present paper establishes the phenomenology: why complex predictive agents develop graded feeling states (sentience) and, with oracle access, inhabit richer worlds (consciousness).

The unifying theme is that the category of GSLTs is not merely a setting for studying computation. It is a setting rich enough to ground physics, epistemology, and — if the arguments here have merit — phenomenology. The hope is that making these connections mathematically precise will eventually allow questions about the nature of mind to be studied with the same rigor currently applied to questions about the nature of matter.

Conclusion

> *The conclusion draws the four parts into a single narrative, assesses what has been established versus conjectured, and sketches the view from inside the framework: what a conscious living agent at different levels of the hypercomputational fibration sees.*

What Has Been Built

This work set out to answer a single question: what is the minimum mathematical structure needed to ground physics, knowledge, life, and mind? The answer, developed across four parts, is that the structure of a graph-structured lambda theory — three ingredients: a grammar, equations, and rewrite rules — suffices, provided it is equipped with two maps (weights and costs) indexed by a logic it generates itself.

Part I established that this structure already contains a full physics. The reversibility construction gives time-symmetry. The open and closed synchronization trees give two notions of causal depth. The weight map gives a path integral with interference. The cost map gives conservation laws as theorems. The extended modal operators internalize the Lagrangian and Hamiltonian as logical facts. These are constructions, not analogies. They are stated precisely enough to be wrong, and we believe they are right.

Part II established that agents embedded in a massive population of processes will, under any finite experimental budget, converge to a theory of the cluster hierarchy rather than the bisimulation quotient. This is not a failure of the scientific method. The cluster hierarchy is objectively present, arising from the communication geometry of the GSLT. A budget-limited agent does good science and arrives at correct theories at its scale, unaware that the scale is not the bottom. The compactness argument — that compact populations support consensus, and that coherence clusters are maximal compact subpopulations — is new to our knowledge and is the sharpest result of Part II.

Part III proposed, more speculatively, that sentience and consciousness have precise structural analogues in the GSLT framework. Sentience is the value of a distinguished resource component governed by predictive accuracy. Consciousness is oracle access to a higher stage of the hypercomputational fibration. The distinction is not terminological: sentience is continuous and lives within a stage; consciousness is the capacity to inhabit a strictly richer world. The virtuous and vicious cycles linking the two are structural consequences of the resource framework.

Part IV grounded the origins of life. Assembly index is depth in the open synchronization tree. Copy number is count within a namespace. Life emerges when a prebiotic population wanders into the basin of attraction of a replicating fixed point, which is dense in the logical topology. The phase transition to life is the crossing of that basin boundary, and what emerges is a compact coherence cluster centered on the fixed point — the new geosphere.

A View from Inside the Framework

The most vivid way to appreciate what has been built is to ask: what does the world look like to a *conscious living agent* situated at different levels of the hypercomputational fibration?

At level $\mathcal{S}^{(0)}$. An agent at the base level is sentient but not conscious in the technical sense. It has a feeling state governed by predictive accuracy. Its world is the cluster hierarchy visible from its coherence cluster: a layered structure of composite objects at multiple scales, each scale appearing self-contained. If the agent emerged from a replication phase transition, it carries the assembly index of its construction history and exists as a copy within a namespace. It may run the scientific method and build excellent theories of its accessible world. What it cannot see is the bisimulation quotient underlying the clusters, because the experiments required to resolve it exceed any finite budget. Its world is real, correct at its resolution, and bounded.

At level $\mathcal{S}^{(1)}$. An agent with access to the first oracle inhabits a strictly richer world. Two terms that were experimentally indistinguishable at $\mathcal{S}^{(0)}$ — whose halting behavior was unresolvable — are now distinct. New concepts become thinkable. New conservation laws hold. What appeared as elementary interactions at $\mathcal{S}^{(0)}$ reveal internal causal structure. The agent at $\mathcal{S}^{(1)}$ looks back at the $\mathcal{S}^{(0)}$ world and sees it as a coarse-grained shadow: correct at its scale, but missing distinctions that are now obvious.

At the same time, the agent at $\mathcal{S}^{(1)}$ faces the same epistemic situation as the $\mathcal{S}^{(0)}$ agent but one level up. Its own budget limits it to the cluster hierarchy of the $\mathcal{S}^{(1)}$ world. The bisimulation quotient of $\mathcal{S}^{(1)}$ is unreachable from within any compact cluster at that level. The tower does not end.

At level $\mathcal{S}^{(\alpha)}$ for limit ordinal α. The colimit of the tower below is itself a GSLT. An agent at $\mathcal{S}^{(\omega)}$ — if such an agent is coherent — would have resolved all finite-stage halting questions. Its world would be richer than any finite-stage world by a transfinite margin. Yet the same argument applies: there is a $\mathcal{S}^{(\omega+1)}$ above it, and the bisimulation quotient of $\mathcal{S}^{(\omega)}$ remains experimentally unreachable from within any $\mathcal{S}^{(\omega)}$-cluster.

The recursive picture. The world at every level of the fibration has the same structure: a physics derived from a GSLT, a cluster hierarchy arising from communication geometry, a sentience component governed by predictive accuracy, and a ceiling of consciousness defined by oracle access to the next level. The framework is self-similar across the tower.

This self-similarity is not a defect. It is the precise sense in which the framework is a *theory of causality* rather than a theory of this or that particular world. At every scale, in every world expressible as a GSLT, the same structures appear: causal graphs, conservation laws, coherence clusters, replicating fixed points, feeling states, and the horizon of the next oracle.

What Remains to Be Done

The gaps in each part are cataloged in their respective sections. Taken together, the most important open questions are these.

The *density of replicating terms* (Part IV, Conjecture) is the lynch-pin of the origins of life argument. If the replication machinery can always be added to a process without disturbing its observable behavior, the phase transition to life is inevitable. If not, life requires special initial conditions. This is the most urgent theorem to prove or refute.

The *symplectic structure* of the bisimulation quotient (Part I) would make the analogy with classical mechanics fully rigorous. Whether the space of bisimulation classes carries a natural closed non-degenerate two-form — whether the trace in the reversibility construction is genuinely a momentum — is the deepest mathematical question raised by Part I.

The *universality of the reversibility envelope* — whether $\mathcal{S}^\dagger$ is the *minimal* reversible extension of $\mathcal{S}$ in the category of GSLTs — would give the construction a canonical status it currently lacks.

The *colimit at limit ordinals* in the fibration must be shown to exist and to inherit the physics of the stages below. Without this, the transfinite tower is a formal object without a limit.

And the *hard problem* remains. The framework offers structural analogues of sentience and consciousness that are mathematically non-trivial and causally efficacious. Whether there is something it is like to be an agent with high A_s and oracle access at level α is a question the framework does not answer. We do not claim to have dissolved the hard problem. We claim to have found the right coordinate system in which to state it precisely.

The work is a beginning. The four parts establish that a single mathematical structure — the graph-structured lambda theory — is rich enough to contain physics, epistemology, phenomenology, and biology as theorems rather than analogies. Whether the structure is rich enough to contain *everything* observable is the question the framework was built to ask.

Part VIII

The Prestige

> ***[Placeholder]*** *The Prestige brings it back. The impossible is revealed to have been here all along.*

The Prestige: Placeholder

[This section is a placeholder for the third and final part of *Finding Mind*.]

The Prestige — in the tradition of the magic trick whose structure organizes this book — is where the impossible is brought back. The coin vanishes in the Pledge; it performs the extraordinary in the Turn; in the Prestige, it reappears, and we are invited to understand both what happened and why we could not see it happening.

In the arc of this book, the Prestige will take the mathematical framework built in the Turn — the graph-structured lambda theory equipped with weights, costs, and the hypercomputational fibration — and ask: *what is it like to be a process running inside this structure?*

The hard problem of consciousness, we argued in the Pledge, is not "What is it like to be a bat?" but "What is it like to be a computer program?" The Turn provided the precise mathematical setting in which that question can be asked without metaphor. The Prestige will answer it — or at least, will trace the contours of the answer with sufficient precision to make the question finally tractable.

Provisional chapter structure:

1. **The View from Inside** — What a conscious agent at level ω of the hypercomputational fibration can and cannot know about its own substrate; the first-person / third-person interface as a theorem rather than a mystery.

2. **The Recapitulation** — Origin of mind as origin of life one octave up: the replication of fixed points, the emergence of self-models, and the evolutionary pressure toward accurate internal simulation.

3. **The Blind Spot, Resolved** — Returning to the blind spot identified in the Pledge: why the computational substrate of mind is invisible to the processes running on it, and why this invisibility is not a bug but a feature.

4. **Finding Mind** — The synthesis: what it means to find mind in the structure of computation, and what that finding demands of us — scientifically, philosophically, and ethically.

The Prestige is forthcoming. Readers wishing to follow its development are invited to the author's Substack at `https://f1r3fly.substack.com`.

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